Estimating Heterogeneous Teacher Effects with Grouped Three-Way Interactive Fixed Effects

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Abstract

We develop a grouped three-way interactive fixed-effects model for estimating heterogeneous teacher effects in multidimensional panel data. The model represents unobserved heterogeneity through the low-rank trilinear component $\sum_{k=1}^{\ell} u_{g(i),k} v_{jk} w_{tk}$, linking observed student types, teachers, and years. Grouping and separability pool information across these dimensions and reduce the number of parameters in the fixed effects relative to a flattened two-way interactive fixed-effects model with unrestricted loadings. We establish aggregate consistency of the global least-squares estimator, derive stochastic expansions and limiting distributions for the slope estimator and aligned factor estimates, and characterize three mode-specific incidental-parameter bias terms. Under the maintained identification and growth conditions, aggregate consistency in balanced panels permits one dimension to remain fixed when the other two diverge. We also develop a consistent information criterion for selecting the number of trilinear components. An application to matched student–teacher achievement data reveals heterogeneity of teacher effects across prior-achievement groups. Under common reference weights, teacher–group matching accounts for approximately 27% of fitted relative teacher value added variation. A randomized-roster robustness comparison shows a strong descriptive agreement for teacher-level effects.

JEL: C33, C55, I21

1 Introduction

Conventional teacher value-added specifications often summarize a teacher's effectiveness using a scalar coefficient, even though effectiveness may differ across student types and years. One teacher may be especially effective with students who enter the year with low prior achievement, while another may be more effective with initially high-achieving students. Recent research explicitly studies such heterogeneous teacher productivity, including the multidimensional value-added framework of Ahn et al. (2025). Estimating this heterogeneity remains challenging because teacher–student-type–year cells often contain few observations. This paper develops a grouped three-way interactive fixed-effects model that pools information across observed student types, teachers, and years through a parsimonious multiplicative structure.

The general model is

$$y_{ijt} = x_{ijt}'\delta + \pi_{ijt} + \varepsilon_{ijt}, \qquad \pi_{ijt} = \sum_{k=1}^{\ell} u_{g(i),k} v_{jk} w_{tk},$$

for $i = 1, \ldots, N$, $j = 1, \ldots, M$, and $t = 1, \ldots, T$. The known function $g : \{1, \ldots, N\} \to \{1, \ldots, G\}$ assigns first-mode units or positions to observed groups. The latent component has a low-rank canonical polyadic (CP) structure whose first-mode loading matrix has repeated rows determined by group membership. Grouping imposes a common latent response within each observed type, while covariates and idiosyncratic errors allow outcomes to vary across observations within that type. The specification contains the ungrouped rank-ℓ CP model as the special case when $G = N$ and $g(i) = i$. When $G < N$, grouping reduces the number of first-mode loading parameters from $N\ell$ to $G\ell$.

The distinction between units and positions is important for matched student–teacher data. Let (j, t) denote a teacher–year cell, and let i index a position within a common ordering of student groups rather than a persistent student identity. In the balanced benchmark, each cell contains N_g students from group g, and its records are arranged into the same group blocks. Students occupying a given position may differ across teachers and years. Consequently, $u_{g(i),k}$ describes an observed student type rather than an individual student. The group loading links otherwise incomparable local student positions i through observed prior-achievement groups.

The proposed structure differs from flattening (i, j) into a single cross-sectional index and applying the two-way interactive fixed-effects framework of Bai (2009). The flattened specification is

$$y_{ijt} = x'_{ijt}\delta + \lambda'_{ij} f_t + \varepsilon_{ijt},$$

with unrestricted loadings across the NM pairs. Our specification imposes $\lambda_{ij,k} = u_{g(i),k} v_{jk}$. For fixed rank, the number of parameters therefore scales with $G + M + T$, compared with $NM+T$ in the unrestricted flattened model. The restriction also links the fitted effects across economically meaningful dimensions: u_{gk} describes group exposure to latent dimension k, v_{jk} describes teacher exposure, and w_{tk} governs their interaction over time.

This paper contributes to the literature on regression models with multidimensional interactive effects. Freeman (2026) studies multidimensional interactive fixed-effects regression and develops transformation-based slope estimators. Our focus is joint least-squares estimation under a known grouped first mode and a separable CP restriction, including the estimation of the latent component itself. We characterize the consequences of estimating the three factor spaces for slope and factor inference. The empirical application connects this econometric framework to the heterogeneous-value-added literature by linking teacher–student-type effects across school years.

The grouped separable structure changes the number of observations contributing to each loading estimate. In the balanced benchmark, the relevant sizes are $N_g MT$ for u_g, NT for v_j, and NM for w_t. These counts motivate the mode-specific estimation rates under the paper's factor-strength, design, and dependence conditions. In contrast, an unrestricted flattened loading λ_{ij} is estimated using observations along the time dimension alone. The grouped separable structure can therefore be particularly useful when the panel has a short time dimension.

The paper makes four contributions. First, we develop the grouped separable specification in the fixed effects and establish aggregate consistency of the global least-squares estimator under sufficient identification and growth conditions that account for its structure. The balanced-panel growth condition is

$$\min\{NMT/G,\ NT,\ NM\} \longrightarrow \infty.$$

Because $G \leqslant N$, this condition holds whenever any two of N, M, and T diverge, allowing the remaining dimension to stay fixed. In particular, aggregate consistency does not require

$T \to \infty$ when $N, M \to \infty$. We also extend the consistency analysis to the inclusion of additive effects and unbalanced observation patterns, distinguishing convergence on the observed sample from recovery of the full latent component.

Second, we derive stochastic expansions and limiting distributions for the slope estimator and aligned factor estimates. The slope expansion identifies three mode-specific incidental-parameter bias components, with bounds $O_p(G/\sqrt{NMT})$, $O_p(M/\sqrt{NMT})$, and $O_p(T/\sqrt{NMT})$. Grouping and separability replace the unrestricted pair-loading bias bound $O_p(NM/\sqrt{NMT})$ with separate group- and teacher-side bounds, while the time-side bound remains of the same order. We construct a feasible bias correction under additional error and moment conditions. The inferential results require stronger rate conditions than aggregate consistency. We provide an alternating-least-squares implementation and assess its finite-sample performance through Monte Carlo experiments.

Third, we propose an information criterion for selecting the number of trilinear components, following the penalized-fit approach of Bai and Ng (2002). The stochastic improvement from overfitting is controlled at the rate $(G + M + T)/(NMT)$, reflecting the grouped nuisance complexity. We establish consistency of the selected rank when the positive penalty satisfies

$$g(G, N, M, T) \to 0, \qquad \frac{NMT}{G + M + T} g(G, N, M, T) \to \infty.$$

The criterion thus balances approximation and complexity using the grouped separable structure.

Finally, we apply the method to matched student–teacher achievement data from the National Center for Teacher Effectiveness. The empirical specification combines additive teacher effects with a grouped interactive component that permits effectiveness to vary across prior-achievement groups and school years. We decompose the total fitted teacher-effect surface under common reference weights into teacher-level effectiveness and teacher–group matching. Matching accounts for approximately 27% of fitted relative-value-added variation under the chosen weights. A comparison with a final-year randomized-roster benchmark shows a strong descriptive agreement for teacher-level estimates.

The framework also applies to other settings with observed types and three-dimensional data, including worker-type–firm, product-type–market, and patient-type–hospital panels. Its usefulness depends on whether grouping and separability provide an appropriate representation of the latent heterogeneity.

Related literature. This paper is related to four strands of literature. The first studies interactive fixed effects and multifactor error structures in panel data. Pesaran (2006) proposes the common correlated effects estimator for heterogeneous panels with unobserved common factors, while Bai (2009) develops least-squares estimation and inference for large-(N), large-(T) panels in which the factors and loadings may be correlated with the observed regressors. Moon and Weidner (2015) study inference when the number of factors used in estimation may exceed the true number. More recently, Freeman (2026) develops transformation-based slope estimators for panels with three or more dimensions under separable interactive fixed effects, treating the latent effects as nuisance parameters. Our focus is the grouped separable restriction $\lambda_{ij,k} = u_{g(i),k} v_{jk}$ and the joint least-squares estimation. We establish asymptotic theory for both the slope coefficients and the fixed effects. Our analysis can naturally be extended to models with more dimensions as in Freeman (2026).

The second strand studies grouped and structured heterogeneity. Bonhomme and Manresa (2015) assign units to latent groups whose time-varying fixed effects are common within

groups but differ across groups. Our model likewise pools observations through a group structure, but differs in two respects. First, group membership is observed and treated as known in our analysis. Second, the grouped heterogeneity enters multiplicatively through a low-rank tensor component rather than through group-specific additive time effects.

The third strand is the statistical literature on tensor decompositions and tensor factor models. Kolda and Bader (2009) provide a general survey of tensor decompositions, while Chen et al. (2026) and Chang et al. (2026) develop CP tensor factor methods for high-dimensional tensor data. Our contribution differs by embedding the CP structure in a regression model with observed covariates, allowing the covariates to be correlated with the latent component, and focusing on estimation and inference for the slope coefficient as well as the grouped interactive effects.

The fourth strand studies heterogeneous teacher effectiveness and student–teacher matching. Biasi et al. (2026) divide students into low- and high-prior-achievement groups according to whether lagged achievement lies below or above the grade-specific state median. They estimate two teacher-specific value-added measures using OLS and empirical Bayes shrinkage and use them as teacher attributes in a structural model of district wage setting and teacher sorting. Ahn et al. (2025) do not impose a single mutually exclusive grouping. Instead, they allow teacher value added to depend on continuous and discrete student attributes, including prior achievement, gender, race, free or reduced-price lunch status, limited English proficiency, and their interactions. They estimate the population distribution of the resulting teacher-specific coefficients using a two-step fixed-effects and Gaussian-mixture procedure. Bates et al. (2025) separate economically disadvantaged and advantaged students, construct best linear predictions of group-specific teacher value added from prior-year outcomes, and incorporate these measures into a two-sided matching model. Delgado (2025) estimates reliability-weighted teacher effects for Black and non-Black students, with extensions based on race-by-income and race-by-gender groups. Eastmond et al. (2026) consider partitions based on gender, race and ethnicity, quantiles of lagged achievement, and their interactions, combining shrinkage and regularization to determine the degree of heterogeneity useful for teacher assignment.

These studies estimate subgroup-specific teacher effects or teacher-specific coefficients on observed student attributes. In contrast, we represent the full student-group–teacher–year effect as $\sum_{k=1}^{\ell} u_{g(i),k} v_{j,k} w_{t,k}$. The low-rank structure pools information jointly across student groups, teachers, and years, while reducing the dimension of the heterogeneous-effect component. We estimate this component jointly with the slope coefficients and provide asymptotic and rank-selection theory for the resulting grouped three-way panel estimator.

The remainder of the paper is organized as follows. Section 2 introduces the grouped three-way interactive fixed effects model. Section 3 defines the least-squares estimator and describes its alternating-least-squares implementation. Sections 4 and 5 state the assumptions and develop the large-sample theory. Section 6 proposes an information criterion for selecting the number of trilinear components. Section 7 extends the model to the inclusion of additive fixed effects and unbalanced panels. Section 8 compares finite-sample performance with the ungrouped CP estimator, the estimator of Freeman (2026), and the unrestricted flattened two-way interactive fixed-effects estimator. Section 9 presents the teacher-value-added application, and Section 10 concludes.

Notation. We use $\circ$, $*$, $\otimes$, $\odot$ to denote the outer, Khatri-Rao, Kronecker, and Hadamard products, respectively. In particular, for

$$A = (a_1, \ldots, a_R) \in \mathbb{R}^{I \times R} \quad \text{and} \quad B = (b_1, \ldots, b_R) \in \mathbb{R}^{J \times R},$$

the Khatri-Rao product is defined as

$$A * B = (a_1 \otimes b_1, \ldots, a_R \otimes b_R) \in \mathbb{R}^{IJ \times R}.$$

For arbitrary matrix $A \in \mathbb{R}^{m \times r}$, define the projection and annihilator matrices

$$P_A = A(A'A)^+ A' = AA^+, \qquad M_A = I_m - P_A,$$

where A^+ is the Moore-Penrose inverse. We use $\|\cdot\|_F$ and $\|\cdot\|$ to denote the Frobenius and spectral norms, respectively, for vectors, matrices, and tensors. For tensors $A, B \in \mathbb{R}^{N_1 \times \cdots \times N_d}$, define the inner product

$$\langle A, B \rangle = \sum_{i_1=1}^{N_1} \cdots \sum_{i_d=1}^{N_d} a_{i_1 \ldots i_d} b_{i_1 \ldots i_d},$$

so that $\|A\|_F = \langle A, A \rangle^{1/2}$. The spectral norm of A is defined as

$$\|A\| = \max_{\substack{\|u^{(m)}\|_2 = 1 \\ m=1,\ldots,d}} \left| \left\langle A, u^{(1)} \circ \cdots \circ u^{(d)} \right\rangle \right|,$$

where $u^{(m)} \in \mathbb{R}^{N_m}$ for $m = 1, \ldots, d$. For vectors, the Frobenius and spectral norms both coincide with the Euclidean norm, while for matrices the above definition of $\|\cdot\|$ coincides with the usual matrix spectral norm. In particular, for vectors $u^{(m)} \in \mathbb{R}^{N_m}$, $m = 1, \ldots, d$,

$$\left\| u^{(1)} \circ \cdots \circ u^{(d)} \right\| = \left\| u^{(1)} \circ \cdots \circ u^{(d)} \right\|_F = \prod_{m=1}^{d} \|u^{(m)}\|_2.$$

For $Y \in \mathbb{R}^{N_1 \times \ldots \times N_d}$, define the ℓ-th generalized singular value of a tensor as

$$\delta_\ell(Y) = \inf_{(a_s^{(i)})_{i,k=1}^{d,\ell-1}} \left\| Y - \sum_{s=1}^{\ell-1} (a_s^{(1)} \circ \cdots \circ a_s^{(d)}) \right\|.$$

For matrices, this residual equals the ℓ-th singular value by the Eckart–Young theorem.

The mode-n product of a tensor $Y \in \mathbb{R}^{N_1 \times \cdots \times N_d}$ with a matrix $A \in \mathbb{R}^{J \times N_n}$ is denoted by

$$Y \times_n A \in \mathbb{R}^{N_1 \times \cdots \times N_{n-1} \times J \times N_{n+1} \times \cdots \times N_d}$$

and has entries

$$(Y \times_n A)_{i_1 \cdots i_{n-1} j i_{n+1} \cdots i_d} = \sum_{r=1}^{N_n} y_{i_1 \cdots i_{n-1} r i_{n+1} \cdots i_d} a_{jr}.$$

The mode-n matricization of Y, denoted by

$$Y_{(n)} \in \mathbb{R}^{N_n \times \prod_{r \neq n} N_r},$$

is the matrix whose columns are the mode-n fibers of Y. We order the mode-n fibers by backward, or last-index-fastest, enumeration: among the modes other than n, the highest-numbered mode varies fastest. Formally, let $(r_1, \dots, r_{d-1})$ be the increasing sequence obtained by removing n from $(1, \dots, d)$. Then

$$(Y_{(n)})_{i_n,k} = y_{i_1,\dots,i_d}, \qquad k = 1 + \sum_{q=1}^{d-1} (i_{r_q} - 1) \prod_{h=q+1}^{d-1} N_{r_h},$$

where an empty product is understood to equal one.[1]

Likewise, $\mathrm{vec}(Y)$ denotes the vectorization of Y induced by backward enumeration. Specifically,

$$\big[\mathrm{vec}(Y)\big]_k = y_{i_1,\dots,i_d}, \qquad k = 1 + \sum_{q=1}^{d} (i_q - 1) \prod_{h=q+1}^{d} N_h.$$

For matrices, $\mathrm{vec}(\cdot)$ therefore coincides with row-major vectorization.

The main reason we adopt backward enumeration is that it preserves the natural ordering of the modes in the Kronecker and Khatri–Rao representations of outer-product tensors. In particular,

$$\mathrm{vec}\left(a^{(1)} \circ \cdots \circ a^{(d)}\right) = a^{(1)} \otimes \cdots \otimes a^{(d)}.$$

Moreover, let

$$Y = \sum_{s=1}^{\ell} (a_s^{(1)} \circ \cdots \circ a_s^{(d)}), \qquad A^{(m)} = (a_1^{(m)}, \dots, a_\ell^{(m)}) \in \mathbb{R}^{N_m \times \ell}.$$

Then its mode-n matricization is

$$Y_{(n)} = A^{(n)} \left(A^{(r_1)} * \cdots * A^{(r_{d-1})}\right)',$$

where $(r_1, \dots, r_{d-1})$ is the increasing sequence of mode indices other than n.

For a tensor $Y \in \mathbb{R}^{N_1 \times \cdots \times N_d}$ and $i, j \in \{1, \dots, d\}$, let $K_{i,j} \in \{0,1\}^{\mathcal{N} \times \mathcal{N}}$, where $\mathcal{N} = \prod_{m=1}^{d} N_m$, denote the mode-permutation matrix satisfying

$$K_{i,j}\, \mathrm{vec}\left(Y_{(j)}\right) = \mathrm{vec}\left(Y_{(i)}\right).$$

More explicitly, let $(r_1^{(n)}, \dots, r_{d-1}^{(n)})$ be the increasing sequence obtained by removing n from $(1, \dots, d)$. Then, for arbitrary vectors $a^{(m)} \in \mathbb{R}^{N_m}$, $m = 1, \dots, d$,

$$K_{i,j}\left(a^{(j)} \otimes a^{(r_1^{(j)})} \otimes \cdots \otimes a^{(r_{d-1}^{(j)})}\right) = a^{(i)} \otimes a^{(r_1^{(i)})} \otimes \cdots \otimes a^{(r_{d-1}^{(i)})}.$$

The mode-permutation matrices satisfy

$$K_{i,i} = I_{\mathcal{N}}, \qquad K_{i,j}^{-1} = K_{j,i} = K_{i,j}', \qquad K_{i,k} = K_{i,j} K_{j,k}.$$

Since $\mathrm{vec}(Y) = \mathrm{vec}(Y_{(1)})$ under our backward-enumeration convention,

$$K_{1,j}\, \mathrm{vec}\left(Y_{(j)}\right) = \mathrm{vec}(Y), \qquad j = 1, \dots, d.$$

[1] This ordering differs from the convention of Kolda and Bader (2009), under which the lowest-numbered remaining mode varies fastest, i.e., first-index-fastest.

2 Model

We consider a balanced three-way array indexed by $i = 1, \ldots, N$, $j = 1, \ldots, M$, and $t = 1, \ldots, T$. The indices may represent units, locations, products, cohorts, or other observational positions, depending on the application. The observed outcome satisfies

$$y_{ijt} = \delta' x_{ijt} + \pi_{ijt} + \varepsilon_{ijt}, \tag{1}$$

where $x_{ijt} \in \mathbb{R}^p$ is a vector of observed covariates, $\delta \in \mathbb{R}^p$ is the corresponding coefficient vector, ε_{ijt} is an idiosyncratic error, and π_{ijt} is an unobserved interactive fixed-effect component. We specify the interactive component as

$$\pi_{ijt} = \sum_{k=1}^{\ell} u_{g(i),k} v_{jk} w_{tk}, \tag{2}$$

where $g : \{1, \ldots, N\} \to \{1, \ldots, G\}$ assigns each first-mode unit i to one of G groups. Group membership is observed and treated as known throughout the analysis.[2] Accordingly, units belonging to the same group share a common vector of first-mode factor loadings. This restriction reduces the number of first-mode loading parameters from $N\ell$ to $G\ell$, yielding substantial parsimony when $G \ll N$. The specification in (2) is multiplicatively separable across the three modes and therefore admits a CP representation with at most ℓ rank-one components. The grouping restriction imposes repeated rows in the first-mode loading matrix. In the special case $G = N$ and $g(i) = i$ for every i, each first-mode unit has its own loading vector, so the grouping restriction disappears and (2) reduces to $\pi_{ijt} = \sum_{k=1}^{\ell} u_{ik} v_{jk} w_{tk}$.

Let

$$Y = (y_{ijt}), \qquad \Pi = (\pi_{ijt}), \qquad E = (\varepsilon_{ijt}) \in \mathbb{R}^{N \times M \times T},$$

and let

$$X = (x_{ijtq}) \in \mathbb{R}^{N \times M \times T \times p}, \qquad x_{ijt} = (x_{ijtq})_{q=1}^{p}.$$

Suppressing the singleton fourth mode, the mode-4 product of X and δ' satisfies

$$X \times_4 \delta' := (\delta' x_{ijt})_{i,j,t=1}^{N,M,T} \in \mathbb{R}^{N \times M \times T}.$$

Define the membership matrix $D \in \{0, 1\}^{N \times G}$ by

$$D_{ih} = 1\{g(i) = h\}, \qquad i = 1, \ldots, N, \quad h = 1, \ldots, G.$$

Let

$$N_g = \sum_{i=1}^{N} 1\{g(i) = g\}, \qquad g = 1, \ldots, G.$$

We assume that the group membership is exclusive, i.e.,

$$D\iota_G = \iota_N, \qquad D'D = \operatorname{diag}(N_1, \ldots, N_G),$$

[2] The analysis conditions on the group assignments. When group membership is latent, it may instead be estimated jointly with the remaining parameters using a grouped fixed-effects procedure such as that of Bonhomme and Manresa (2015). Analyzing the asymptotic consequences of estimating group membership is beyond the scope of this paper.

where ι_G, ι_N are G and N dimensional vectors of ones. For $k = 1, \ldots, \ell$, collect the unobserved fixed effects in each mode as

$$\begin{aligned} u_k &= (u_{1k}, \ldots, u_{Gk})' \in \mathbb{R}^G \\ v_k &= (v_{1k}, \ldots, v_{Mk})' \in \mathbb{R}^M \\ w_k &= (w_{1k}, \ldots, w_{Tk})' \in \mathbb{R}^T. \end{aligned}$$

The model (1) can be written in tensor form as

$$Y = X \times_4 \delta' + \sum_{k=1}^{\ell} (Du_k \circ v_k \circ w_k) + E, \tag{3}$$

where fixed effects are written as outer products, reflecting a multiplicatively separable structure across the three modes. Collect the component-specific vectors into the factor matrices

$$U = (u_1, \ldots, u_\ell) \in \mathbb{R}^{G \times \ell}, \qquad V = (v_1, \ldots, v_\ell) \in \mathbb{R}^{M \times \ell}, \qquad W = (w_1, \ldots, w_\ell) \in \mathbb{R}^{T \times \ell}.$$

Under the backward-enumeration convention, the mode-1, mode-2, and mode-3 matricizations of Π are

$$\begin{aligned} \Pi_{(1)} &= DU(V * W)' \in \mathbb{R}^{N \times MT}, \\ \Pi_{(2)} &= V(DU * W)' \in \mathbb{R}^{M \times NT}, \\ \Pi_{(3)} &= W(DU * V)' \in \mathbb{R}^{T \times NM}. \end{aligned} \tag{4}$$

Example: Matched student–teacher data. Let $j = 1, \ldots, M$ index teachers and $t = 1, \ldots, T$ index academic periods. We interpret each teacher–period cell (j, t) as a classroom taught by teacher j in period t. In the balanced benchmark, each classroom contains N students, indexed locally by $i = 1, \ldots, N$. The index i denotes a within-classroom position rather than a persistent student identifier. Consequently, the student occupying position i may differ across classrooms and periods, and the $N \times M \times T$ array does not require the same students to be followed across teachers or over time. The grouped specification is particularly useful in this setting because it links otherwise incomparable local student positions through observed prior-achievement groups.

We group the students into G observed groups according to prior achievement. Note that the group assignment $g : \{1, \ldots, N\} \to \{1, \ldots, G\}$ does not depend on (j, t). Let N_g denote the number of students in group g within each classroom, where $\sum_{g=1}^{G} N_g = N$. The homogeneous group assignment assumes that every teacher–period classroom contains exactly N_g students from prior-achievement group g.[3]

Under this interpretation, the loading vector u_g governs how the latent teacher component varies across prior-achievement groups, v_j captures persistent teacher-specific heterogeneity, and w_t allows the interactive component to vary systematically over time. The model therefore permits a given teacher to have different latent effects on students with different levels of prior achievement, while retaining a parsimonious low-dimensional structure across groups, teachers, and periods.

[3]When classroom sizes or group compositions vary across teacher–period cells, the resulting sample is unbalanced. While the benchmark model assumes a balanced sample, we discuss the unbalanced case in Section 7.2.

3 Estimation

For a given number ℓ of interactive components, we estimate the model by minimizing the average squared residual

$$\begin{aligned} S_{NMT}(\delta, U, V, W) &= \frac{1}{NMT} \left\| Y - X \times_4 \delta' - \sum_{k=1}^{\ell} (Du_k \circ v_k \circ w_k) \right\|_F^2 \\ &= \frac{1}{NMT} \sum_{i=1}^{N} \sum_{j=1}^{M} \sum_{t=1}^{T} \left(y_{ijt} - x_{ijt}'\delta - \sum_{k=1}^{\ell} u_{g(i),k} v_{jk} w_{tk} \right)^2 \end{aligned} \tag{5}$$

with respect to (δ, U, V, W). The model admits the vectorized representation

$$\operatorname{vec}(Y) = X_{(4)}'\delta + \sum_{k=1}^{\ell} (Du_k \otimes v_k \otimes w_k) + \operatorname{vec}(E), \tag{6}$$

where $\operatorname{vec}(X \times_4 \delta') = X_{(4)}'\delta$.

Provided that the relevant Gram matrices are nonsingular, the first-order condition with respect to δ can be written as

$$\hat{\delta} = \left(X_{(4)} X_{(4)}' \right)^{-1} X_{(4)} \left(\operatorname{vec}(Y) - \sum_{k=1}^{\ell} (D\hat{u}_k \otimes \hat{v}_k \otimes \hat{w}_k) \right). \tag{7}$$

Similarly, the first-order conditions with respect to the factor matrices can be expressed as

$$\begin{aligned} \hat{U} &= (D'D)^{-1} D' \left(Y - X \times_4 \hat{\delta}' \right)_{(1)} (\hat{V} * \hat{W}) \left((\hat{V} * \hat{W})'(\hat{V} * \hat{W}) \right)^{-1}, \\ \hat{V} &= \left(Y - X \times_4 \hat{\delta}' \right)_{(2)} (D\hat{U} * \hat{W}) \left((D\hat{U} * \hat{W})'(D\hat{U} * \hat{W}) \right)^{-1}, \\ \hat{W} &= \left(Y - X \times_4 \hat{\delta}' \right)_{(3)} (D\hat{U} * \hat{V}) \left((D\hat{U} * \hat{V})'(D\hat{U} * \hat{V}) \right)^{-1}. \end{aligned} \tag{8}$$

Equations (7) and (8) jointly characterize a stationary point of the least-squares criterion.

We compute a solution to these simultaneous first-order conditions using an alternating least squares (ALS) algorithm. Starting from initial values, ALS successively applies the blockwise solution maps in (7) and (8), using the most recently available values of the remaining parameter blocks. Because each block update minimizes the criterion while holding the other blocks fixed, the sequence of objective values is nonincreasing and therefore converges. Under suitable boundedness and regularity conditions, every accumulation point of the parameter sequence is a stationary point of the least-squares criterion. Because the criterion is nonconvex, global optimality is not guaranteed. We therefore use multiple initializations and retain the solution with the smallest residual sum of squares.

Projected representation. Since the mode-3 matricization of the interactive component is $W(DU * V)'$, concentrating over W gives

$$\begin{aligned} S^W_{NMT}(\delta, U, V) &= \min_{W \in \mathbb{R}^{T\times \ell}} S_{NMT}(\delta, U, V, W) \\ &= \frac{1}{NMT} \left\| (Y - X \times_4 \delta')_{(3)} M_{DU*V} \right\|_F^2 \\ &= \frac{1}{NMT} \left(\mathrm{vec}(Y) - X'_{(4)}\delta \right)' (M_{DU*V} \otimes I_T) \left(\mathrm{vec}(Y) - X'_{(4)}\delta \right). \end{aligned} \tag{9}$$

Consequently, provided that the displayed regressor Gram matrix is nonsingular, the joint normal equations for δ and W imply

$$\hat{\delta} = \left[X_{(4)} \left(M_{D\hat{U}*\hat{V}} \otimes I_T \right) X'_{(4)} \right]^{-1} X_{(4)} \left(M_{D\hat{U}*\hat{V}} \otimes I_T \right) \mathrm{vec}(Y). \tag{10}$$

This is a Frisch–Waugh–Lovell-type representation conditional on the estimated loadings $D\hat{U}, \hat{V}$: the outcome vector and the columns of $X'_{(4)}$ are projected onto the orthogonal complement of $\mathrm{col}(D\hat{U} * \hat{V}) \otimes \mathbb{R}^T$.

A representation incorporating all three factor normal equations can be obtained. For given (U, V, W), define

$$\begin{aligned} \mathcal{T}(U, V, W) = \Bigg\{ &\sum_{k=1}^{\ell} (Da_k \otimes v_k \otimes w_k + Du_k \otimes b_k \otimes w_k + Du_k \otimes v_k \otimes c_k) : \\ &a_k \in \mathbb{R}^G, \quad b_k \in \mathbb{R}^M, \quad c_k \in \mathbb{R}^T, \quad k = 1, \ldots, \ell \Bigg\}. \end{aligned} \tag{11}$$

This is the image of the differential of the grouped CP parametrization and is the tangent space of the grouped CP model. An orthogonal projector onto this space can be constructed in grouped coordinates. Since D has full column rank, let

$$Q_D = D(D'D)^{-1/2}, \qquad \bar{U} = (D'D)^{1/2}U, \qquad Q = Q_D \otimes I_M \otimes I_T. \tag{12}$$

Then $DU = Q_D\bar{U}$ and $Q'Q = I_{GMT}$. Let $K^G_{i,j}$ denote the mode-permutation matrices defined in the notation section for tensors of dimensions $G \times M \times T$. Suppressing dependence on (U, V, W), define the grouped block projectors

$$\begin{aligned} P^G_1 &= I_G \otimes P_{V*W}, \\ P^G_2 &= K^G_{1,2} \left(I_M \otimes P_{\bar{U}*W} \right) K^G_{2,1}, \\ P^G_3 &= K^G_{1,3} \left(I_T \otimes P_{\bar{U}*V} \right) K^G_{3,1}. \end{aligned} \tag{13}$$

These project onto the grouped perturbation spaces generated by changes in U, V, and W, respectively. Set

$$S^G = P^G_1 + P^G_2 + P^G_3, \qquad \mathcal{T}^G = \mathrm{range}(S^G). \tag{14}$$

Because the P^G_m are orthogonal projectors,

$$\mathcal{T}^G = \mathrm{range}(P^G_1) + \mathrm{range}(P^G_2) + \mathrm{range}(P^G_3), \qquad \mathcal{T}(U, V, W) = Q\mathcal{T}^G.$$

The block spaces generally overlap, so S^G need not itself be an orthogonal projector. The orthogonal projectors onto $\mathcal{T}^G$ and $\mathcal{T} = \mathcal{T}(U, V, W)$ are instead

$$P_{\mathcal{T}^G} = S^G (S^G)^+, \qquad P_{\mathcal{T}} = Q P_{\mathcal{T}^G} Q'. \tag{15}$$

The corresponding annihilator in the original observation space is

$$\begin{aligned} M_{\mathcal{T}} &= I_{NMT} - Q P_{\mathcal{T}^G} Q' \\ &= M_D \otimes I_M \otimes I_T + Q \left(I_{GMT} - P_{\mathcal{T}^G} \right) Q'. \end{aligned} \tag{16}$$

In particular, this annihilator retains variation orthogonal to the group-membership space.

Evaluate the tangent space at the joint estimates and write

$$\widehat{\mathcal{T}} = \mathcal{T}(\hat{U}, \hat{V}, \hat{W}), \qquad \hat{\pi} = \sum_{k=1}^{\ell} (D\hat{u}_k \otimes \hat{v}_k \otimes \hat{w}_k), \qquad \hat{e} = \operatorname{vec}(Y) - X'_{(4)} \hat{\delta} - \hat{\pi}.$$

The normal equations for the three factor blocks imply that $\hat{e}$ is orthogonal to every perturbation in (11), whereas the normal equation for δ gives $X_{(4)} \hat{e} = 0$. Thus,

$$P_{\widehat{\mathcal{T}}} \hat{e} = 0, \qquad X_{(4)} \hat{e} = 0. \tag{17}$$

Moreover, $\hat{\pi} \in \widehat{\mathcal{T}}$, as follows by taking $a_k = \hat{u}_k$ and $b_k = c_k = 0$ in (11). Consequently,

$$\hat{e} = M_{\widehat{\mathcal{T}}} \left(\operatorname{vec}(Y) - X'_{(4)} \hat{\delta} \right).$$

Combining this identity with $X_{(4)} \hat{e} = 0$ yields

$$\hat{\delta} = \left[X_{(4)} M_{\widehat{\mathcal{T}}} X'_{(4)} \right]^{-1} X_{(4)} M_{\widehat{\mathcal{T}}} \operatorname{vec}(Y), \tag{18}$$

provided that $X_{(4)} M_{\widehat{\mathcal{T}}} X'_{(4)}$ is nonsingular. Equations (10) and (18) are exact, implicit representations of the same joint least-squares estimator, not additional estimators.

4 Assumptions

Let

$$\mathcal{F} = \sigma(X, D, U, V, W), \qquad \mathcal{I} = \{1, \ldots, N\} \times \{1, \ldots, M\} \times \{1, \ldots, T\}.$$

For $a = (i_a, j_a, t_a) \in \mathcal{I}$, write $\varepsilon_a = \varepsilon_{i_a j_a t_a}$. Suppose that the following conditions hold.

Assumption 1 (Asymptotic Framework, Normalization, and Estimation)**.** *Along an asymptotic sequence satisfying*

$$\min\{NMT/G,\ NT,\ NM\} \to \infty,$$

the dimensions p and ℓ remain fixed. The membership matrix $D \in \{0,1\}^{N \times G}$ has full column rank, and

$$1 \leqslant \ell \leqslant \min\{G, M, T\}.$$

For a constant $c_U > 0$ independent of the panel dimensions, let Θ_ℓ be the set of triples $(U, V, W) \in \mathbb{R}^{G \times \ell} \times \mathbb{R}^{M \times \ell} \times \mathbb{R}^{T \times \ell}$ satisfying

$$\operatorname{diag}\left(\frac{1}{M} V'V \right) = \operatorname{diag}\left(\frac{1}{T} W'W \right) = \iota_\ell, \qquad \lambda_{\max}\left(\frac{1}{N} U'D'DU \right) \leqslant c_U, \tag{19}$$

where ι_ℓ is the ℓ-dimensional vector of ones. The true factor matrices admit a representation in Θ_ℓ. The least-squares estimator satisfies

$$(\hat{\delta}, \hat{U}, \hat{V}, \hat{W}) \in \underset{\delta \in \mathbb{R}^p,\ (U,V,W)\in\Theta_\ell}{\operatorname{argmin}} S_{NMT}(\delta, U, V, W).$$

Assumption 2 (Strong Factors and Stochastic Bounds)**.** *The regressor x_{ijt} has uniformly bounded fourth moment conditional on group assignment, i.e., there exists a constant $c_x < \infty$ independent of the panel dimensions,*

$$\sup_{(i,j,t)\in\mathcal{I}} \mathbb{E}[\|x_{ijt}\|^4 | D] \leqslant c_x \tag{20}$$

almost surely, uniformly along the asymptotic sequence. The grouped-error satisfies

$$\|E^G\| = O_p(\sqrt{G+M+T}),$$

where

$$E^G = E \times_1 (D'D)^{-1/2} D' = \left(\frac{1}{\sqrt{N_g}} \sum_{\substack{i=1 \\ g(i)=g}}^{N} \varepsilon_{ijt} \right)_{g,j,t=1}^{G,M,T}.$$

For constants $0 < c_0 \leqslant 1 \leqslant c_1 < c_U < \infty$, with probability approaching one,

$$c_0 I_\ell \preceq \frac{1}{N} U'D'DU \preceq c_1 I_\ell$$
$$c_0 I_\ell \preceq \frac{1}{M} V'V \preceq c_1 I_\ell$$
$$c_0 I_\ell \preceq \frac{1}{T} W'W \preceq c_1 I_\ell.$$

Assumption 3. *The errors have uniformly finite fourth moments and $\mathbb{E}[\varepsilon_a \mid \mathcal{F}] = 0$. Let $\Gamma_{ab} = \mathbb{E}[\varepsilon_a \varepsilon_b \mid \mathcal{F}]$ for $a, b \in \mathcal{I}$. For some finite constant c_ε, with probability approaching one,*

$$\sup_{a\in\mathcal{I}} \sum_{b\in\mathcal{I}} |\Gamma_{ab}| \leqslant c_\varepsilon.$$

Define

$$\mathcal{R}^D = \{(a,b) \in \mathcal{I}^2 : g(i_a) = g(i_b) \text{ or } j_a = j_b \text{ or } t_a = t_b\}, \qquad z_{ab} = \varepsilon_a \varepsilon_b - \Gamma_{ab}.$$

For some finite constant c_4, with probability approaching one,

$$\sup_{(a,b)\in\mathcal{R}^D} \sum_{(c,d)\in\mathcal{R}^D} |\operatorname{Cov}(z_{ab}, z_{cd} \mid \mathcal{F})| \leqslant c_4.$$

Assumption 4 (Identification of δ)**.** *There exists a constant $c_\delta > 0$ such that, with probability approaching one,*

$$\inf_{\widetilde{U}\in\mathbb{R}^{G\times\ell},\, \widetilde{V}\in\mathbb{R}^{M\times\ell}} \lambda_{\min}\left[\frac{1}{NMT} X_{(4)} \left(M_{D\widetilde{U}*\widetilde{V}} \otimes M_W \right) X'_{(4)} \right] \geqslant c_\delta.$$

Assumption 1 describes the asymptotic framework. In particular, not all N, M, T need to diverge as long as $\min\{NMT/G,\ NT,\ NM\} \to \infty$. Additional rate conditions will be imposed when necessary. We define the estimator as a global minimizer satisfying the constraints in Θ_ℓ. We may show that the constraints do not bind so that the first-order conditions (8) hold with probability approaching one.

Assumption 2 implies that the factors are strong and the within-group averaged idiosyncratic errors are non-pervasive. Assumption 3 imposes weak serial and cross-sectional correlation. For conditionally centered Gaussian errors, the Gaussian fourth-moment identity, e.g., fourth-order case of Isserlis' formula, gives

$$\mathrm{Cov}(z_{ab}, z_{cd} \mid \mathcal{F}) = \Gamma_{ac}\Gamma_{bd} + \Gamma_{ad}\Gamma_{bc}$$

It follows that

$$\begin{aligned}\sum_{(c,d)\in\mathcal{R}^D} |\mathrm{Cov}(z_{ab}, z_{cd} \mid \mathcal{F})| &\leqslant \sum_{c,d}\big(|\Gamma_{ac}\Gamma_{bd}| + |\Gamma_{ad}\Gamma_{bc}|\big) \\ &\leqslant 2c_\varepsilon^2.\end{aligned}$$

Thus, for this class, the fourth-order condition follows from the second-order row-sum bound.

Assumption 4 requires sufficient regressor variation to remain after projecting the (i,j)-mode away from an arbitrary candidate structured loading space $(D\widetilde{U} * \widetilde{V})$ and projecting the time mode away from the true factor space W. Equivalently, with probability approaching one,

$$\frac{1}{NMT}\left\| M_W \left(X \times_4 a'\right)_{(3)} M_{D\widetilde{U}*\widetilde{V}} \right\|_F^2 \geqslant c_\delta \|a\|^2$$

uniformly over $a \in \mathbb{R}^p$, $\widetilde{U} \in \mathbb{R}^{G\times\ell}$, and $\widetilde{V} \in \mathbb{R}^{M\times\ell}$. Thus, the assumption rules out changes in δ that can be absorbed by changing the candidate interactive component. Uniformity over $(\widetilde{U}, \widetilde{V})$ is required because the corresponding loading space is estimated jointly with δ.

Suppose instead that the indices (i,j) are enumerated into a single cross-sectional index and the model is estimated as the unrestricted two-way interactive fixed-effects model of Bai (2009). Because the two-way least-squares problem is symmetric in its cross-sectional and time dimensions, the corresponding identification condition, i.e., Assumption A of Bai (2009), may be written in the following form:

$$\inf_{H\in\mathbb{R}^{NM\times\ell}} \lambda_{\min}\left[\frac{1}{NMT} X_{(4)} \left(M_H \otimes M_W\right) X_{(4)}'\right] > 0$$

with probability approaching one. This unrestricted two-way condition is stronger than Assumption 4. Indeed, every structured loading matrix $D\widetilde{U} * \widetilde{V}$ is an element of $\mathbb{R}^{NM\times\ell}$, whereas an unrestricted $H \in \mathbb{R}^{NM\times\ell}$ need not have separable or grouped columns. To illustrate, consider a scalar regressor of the form

$$x_{ijt} = q_{ij}\tau_t.$$

Let

$$Q = (q_{ij})_{i,j=1}^{N,M} \in \mathbb{R}^{N\times M}, \qquad q = \mathrm{vec}(Q) \in \mathbb{R}^{NM}, \qquad \tau = (\tau_1, \ldots, \tau_T)' \in \mathbb{R}^T.$$

For any candidate cross-sectional loading matrix H,

$$\frac{1}{NMT} X_{(4)} \left(M_H \otimes M_W\right) X_{(4)}' = \left(\frac{q' M_H q}{NM}\right)\left(\frac{\tau' M_W \tau}{T}\right).$$

If $H = D\widetilde{U} * \widetilde{V}$, every vector in the column space of H, when reshaped as an $N \times M$ matrix, has matrix rank at most ℓ. Therefore, by the Eckart–Young inequality,

$$\begin{aligned} q' M_{D\widetilde{U}*\widetilde{V}} q &= \left\| q - P_{D\widetilde{U}*\widetilde{V}} q \right\|_F^2 \\ &= \inf_{a \in \mathbb{R}^\ell} \left\| Q - \sum_{k=1}^{\ell} a_k (D\widetilde{u}_k) \widetilde{v}_k' \right\|_F^2 \\ &\geqslant \inf_{\operatorname{rank}(A) \leqslant \ell} \|Q - A\|_F^2 \\ &= \sum_{r=\ell+1}^{\min(N,M)} \sigma_r^2(Q). \end{aligned}$$

Consequently, Assumption 4 holds in this example if

$$\liminf_{N,M \to \infty} \frac{1}{NM} \sum_{r=\ell+1}^{\min(N,M)} \sigma_r^2(Q) > 0$$

and

$$\liminf_{T \to \infty} \frac{\tau' M_W \tau}{T} > 0.$$

By contrast, under the unrestricted two-way condition, one may choose $H \in \mathbb{R}^{NM \times \ell}$ such that

$$M_H q = 0$$

and hence

$$\inf_{H \in \mathbb{R}^{NM \times \ell}} \frac{1}{NMT} X_{(4)} \left(M_H \otimes M_W \right) X_{(4)}' = 0.$$

Thus, the grouped separable structure can identify δ for regressors that would not be identified after flattening the (i, j) indices and allowing an unrestricted $NM \times \ell$ loading matrix.

5 Asymptotic Theory

The following lemma gives the consistency and convergence rate of the interactive fixed effect components (U, V, W).

Lemma 1 (Normalized factor alignment and asymptotic inactivity of the inequality constraint)**.** *Suppose that Assumptions 1, 2, 3, and 4 hold. Let (U, V, W) denote the true factor matrices in the normalized representation satisfying* (19)*. Let $(\hat{\delta}, \hat{U}, \hat{V}, \hat{W})$ be the global least-squares estimator defined in Assumption 1. There exist possibly random $\ell \times \ell$ permutation and diagonal sign matrices Π_ℓ, S_v, S_w, with the diagonal entries of S_v and S_w belonging to $\{-1, 1\}$, such that*

$$\frac{\|D(\hat{U}\Pi_\ell S_v S_w - U)\|_F}{\sqrt{N}} + \frac{\|\hat{V}\Pi_\ell S_v - V\|_F}{\sqrt{M}} + \frac{\|\hat{W}\Pi_\ell S_w - W\|_F}{\sqrt{T}} = O_p\left(\sqrt{\frac{G + M + T}{NMT}} \right). \tag{21}$$

Define the residual tensor

$$\hat{E} = Y - X \times_4 \hat{\delta}' - \sum_{k=1}^{\ell} (D\hat{u}_k \circ \hat{v}_k \circ \hat{w}_k).$$

With probability approaching one, the estimator satisfies the unrestricted normal equations

$$\begin{aligned} X_{(4)} \operatorname{vec}(\hat{E}) = 0, \quad D'\hat{E}_{(1)}(\hat{V} * \hat{W}) = 0, \\ \hat{E}_{(2)}(D\hat{U} * \hat{W}) = 0, \quad \hat{E}_{(3)}(D\hat{U} * \hat{V}) = 0. \end{aligned} \tag{22}$$

Lemma 1 shows that the regression equations (8) hold with probability approaching one. Also, the estimated fixed effect components $(\hat{U}, \hat{V}, \hat{W})$ are consistent up to permutations and sign changes. This is consistent with the inherent indeperminacies of the CP decomposition as studied by Kruskal (1977). The following Proposition 1 gives the stochastic expansion for $\hat{\delta}$ needed to apply the central limit theorem.

Proposition 1. *Suppose that Assumptions 1, 2, 3, and 4 hold. Let $\mathcal{T}$ be the tangent space defined in* (11) *evaluated at the true parameters (U, V, W). Let*

$$\Gamma_{NMT} = \frac{1}{NMT} X_{(4)} M_{\mathcal{T}} X'_{(4)}, \qquad \mathcal{Z}_{NMT} = \frac{1}{\sqrt{NMT}} X_{(4)} M_{\mathcal{T}} \operatorname{vec}(E).$$

The least squares estimator $\hat{\delta}$ defined in Assumption 1 satisfies

$$\Gamma_{NMT}\sqrt{NMT}(\hat{\delta} - \delta) = \mathcal{Z}_{NMT} - \mathcal{B}_1 - \mathcal{B}_2 - \mathcal{B}_3 + O_p\left(\sqrt{\frac{G+M+T}{NMT}} + \frac{(G+M+T)^{3/2}}{NMT}\right), \tag{23}$$

where

$$\begin{aligned} \mathcal{B}_1 &= \frac{1}{\sqrt{NMT}} X_{(4)} P_{\mathcal{T}} Q \operatorname{vec}\left(\bar{U}(\bar{U}'\bar{U})^{-1}((V*W)'(V*W))^{-1}(V*W)'\Omega_1^G M_{V*W}\right) \\ \mathcal{B}_2 &= \frac{1}{\sqrt{NMT}} X_{(4)} P_{\mathcal{T}} Q K_{1,2}^G \operatorname{vec}\left(V(V'V)^{-1}((\bar{U}*W)'(\bar{U}*W))^{-1}(\bar{U}*W)'\Omega_2^G M_{\bar{U}*W}\right) \\ \mathcal{B}_3 &= \frac{1}{\sqrt{NMT}} X_{(4)} P_{\mathcal{T}} Q K_{1,3}^G \operatorname{vec}\left(W(W'W)^{-1}((\bar{U}*V)'(\bar{U}*V))^{-1}(\bar{U}*V)'\Omega_3^G M_{\bar{U}*V}\right). \end{aligned}$$

Here, $\bar{U} = (D'D)^{1/2}U$ and $\Omega_m^G = \mathbb{E}[(E_{(m)}^G)'E_{(m)}^G \mid \mathcal{F}]$ for $m = 1, 2, 3$. The matrices Q and $P_{\mathcal{T}}$ are as defined in (12) *and* (15)*, respectively, and $K_{1,m}^G$ denotes the mode-permutation matrix for tensors of dimensions $G \times M \times T$. All factor-dependent matrices and projectors in the displayed bias terms are evaluated at the true parameters. Moreover,*

$$\mathcal{B}_1 = O_p(G/\sqrt{NMT}), \qquad \mathcal{B}_2 = O_p(M/\sqrt{NMT}), \qquad \mathcal{B}_3 = O_p(T/\sqrt{NMT}).$$

Remark: Bias reduction from the grouped separable structure The three bias components in (23) are incidental-parameter bias terms associated with estimation of the three sets of factor parameters. More precisely, first-order perturbations of the mode-specific projectors M_{V*W}, $M_{\bar{U}*W}$, and $M_{\bar{U}*V}$ generate quadratic error terms whose conditional expectations produce the displayed covariance-based biases. Their bounds imply

$$\mathcal{B}_1 + \mathcal{B}_2 = O_p\left(\frac{G+M}{\sqrt{NMT}}\right), \qquad \mathcal{B}_3 = O_p\left(\frac{T}{\sqrt{NMT}}\right).$$

All three bias components are $o_p(1)$ in the $\sqrt{NMT}$-normalized expansion if

$$\frac{G^2}{NMT} \to 0, \qquad \frac{M}{NT} \to 0, \qquad \frac{T}{NM} \to 0.$$

These sufficient conditions are equivalent to $(G + M + T)^2/(NMT) \to 0$. They hold, in particular, when $N \asymp M \asymp T \to \infty$, because $G \leqslant N$.

For comparison, stack (i, j) into a cross-sectional index of dimension NM. The true loading matrix in the resulting two-way representation is $A = DU * V$. Let $\hat{\delta}_{\text{flat}}$ denote the estimator obtained by allowing an unrestricted $NM \times \ell$ loading matrix. This estimator differs from $\hat{\delta}$, which imposes both grouping and separability. The conventional loading-side and time-factor-side bias scales associated with Bai (2009) are, respectively,

$$O_p\left(\sqrt{\frac{NM}{T}}\right) = O_p\left(\frac{NM}{\sqrt{NMT}}\right), \qquad O_p\left(\sqrt{\frac{T}{NM}}\right) = O_p\left(\frac{T}{\sqrt{NMT}}\right).$$

Relative to these scales, the grouped separable structure reduces the loading-side dimensional bias bound from $NM/\sqrt{NMT}$ to $(G+M)/\sqrt{NMT}$, while retaining the same dimensional bound for the time-factor-side bias.

Remark: Dimension conditions for negligible expansion remainders Write $n = NMT$ and $s = G + M + T$. The remainder in (23) has the bound

$$R_{NMT} = O_p\left(\sqrt{\frac{s}{n}} + \frac{s^{3/2}}{n}\right).$$

We only need $\frac{s^{3/2}}{n} \to 0$ for the remainder terms to be asymptotically negligible. In particular, this allows one dimension to remain fixed, as long as the other two dimensions diverge at the same rate. Let any two of N, M, T have the same order as $L \to \infty$ while the remaining dimension is $O(L)$. Since $G \leqslant N$ and all dimensions are positive integers,

$$s = G + M + T \leqslant N + M + T = O(L), \qquad n = NMT \gtrsim L^2.$$

Consequently,

$$\frac{s}{n} = O(L^{-1}) \to 0, \qquad \frac{s^{3/2}}{n} = O(L^{-1/2}) \to 0.$$

This is regardless of which two dimensions diverge.

To derive the stochastic expansion of the estimated factors and factor loadings, we impose the following assumption.

Assumption 5 (Slice-averaged weighted covariance summability)**.** *Let u_h, v_j, and w_t denote the transposes of the corresponding rows of U, V, and W, respectively. For $b = (i_b, j_b, t_b) \in \mathcal{I}$, define*

$$\omega_b = \|u_{g(i_b)}\|^2\|v_{j_b}\|^2 + \|u_{g(i_b)}\|^2\|w_{t_b}\|^2 + \|v_{j_b}\|^2\|w_{t_b}\|^2.$$

Define the source slices

$$\begin{aligned}
\mathcal{I}_\tau^{(t)} &= \{a \in \mathcal{I} : t_a = \tau\}, & |\mathcal{I}_\tau^{(t)}| &= NM, \\
\mathcal{I}_h^{(g)} &= \{a \in \mathcal{I} : g(i_a) = h\}, & |\mathcal{I}_h^{(g)}| &= N_h MT, \\
\mathcal{I}_{j_0}^{(j)} &= \{a \in \mathcal{I} : j_a = j_0\}, & |\mathcal{I}_{j_0}^{(j)}| &= NT.
\end{aligned}$$

For every fixed admissible τ, h, and j_0, suppose that

$$\frac{1}{NM}\sum_{a\in\mathcal{I}_\tau^{(t)}}\sum_{b\in\mathcal{I}}|\Gamma_{ab}|\omega_b = O_p(1), \tag{24}$$

$$\frac{1}{N_hMT}\sum_{a\in\mathcal{I}_h^{(g)}}\sum_{b\in\mathcal{I}}|\Gamma_{ab}|\omega_b = O_p(1), \tag{25}$$

and

$$\frac{1}{NT}\sum_{a\in\mathcal{I}_{j_0}^{(j)}}\sum_{b\in\mathcal{I}}|\Gamma_{ab}|\omega_b = O_p(1). \tag{26}$$

These conditions are pointwise in the slice indices; no uniform bound over τ, h, or j_0 is imposed.

Remark: Role of slice-averaged weighted summability Assumption 5 controls covariance magnitudes weighted by the factor norms that enter higher-order error contractions. Its additional content is control on the selected source slices. To see the distinction from global averaging, write $a_U = \mathrm{tr}(U'D'DU/N)$, $a_V = \mathrm{tr}(V'V/M)$, and $a_W = \mathrm{tr}(W'W/T)$. By covariance symmetry, Assumption 3, and the Gram bounds in Assumption 2, with probability approaching one,

$$\frac{1}{NMT}\sum_{a\in\mathcal{I}}\sum_{b\in\mathcal{I}}|\Gamma_{ab}|\omega_b \leqslant c_\varepsilon\frac{1}{NMT}\sum_{b\in\mathcal{I}}\omega_b = c_\varepsilon(a_Ua_V + a_Ua_W + a_Va_W) = O_p(1).$$

Thus global weighted summability already follows from the existing assumptions, whereas Assumption 5 prevents the weighted covariance mass from concentrating on the particular period, group, or teacher under consideration. A stronger sufficient condition is $\max_{b\in\mathcal{I}}\omega_b = O_p(1)$, which follows, for example, from uniformly bounded factor row norms. Together with Assumption 3, this stronger condition bounds all source-slice averages simultaneously. Assumption 5 requires only the stated pointwise slice averages and does not impose independence of the errors.

Proposition 2 (Fixed-index expansions with mode-specific remainders)**.** *Suppose that Assumptions 1, 2, 3, 4, and 5 hold. Let $(\hat\delta, \hat U, \hat V, \hat W)$ be the global least-squares estimator defined in Assumption 1, and let H_u, H_v, H_w be the alignment matrices defined as*

$$\begin{aligned}
H_u' &= (V * W)'(\hat V * \hat W)\left[(\hat V * \hat W)'(\hat V * \hat W)\right]^{-1},\\
H_v' &= (DU * W)'(D\hat U * \hat W)\left[(D\hat U * \hat W)'(D\hat U * \hat W)\right]^{-1},\\
H_w' &= (DU * V)'(D\hat U * \hat V)\left[(D\hat U * \hat V)'(D\hat U * \hat V)\right]^{-1}
\end{aligned}$$

and let $\Sigma_{VW} = \left(\frac{1}{M}V'V\right)\odot\left(\frac{1}{T}W'W\right)$, $\Sigma_{UW} = \left(\frac{1}{N}U'D'DU\right)\odot\left(\frac{1}{T}W'W\right)$, and $\Sigma_{UV} = \left(\frac{1}{N}U'D'DU\right)\odot\left(\frac{1}{M}V'V\right)$.

Write

$$n = NMT, \qquad s = G + M + T, \qquad \rho = \sqrt{\frac{s}{n}}, \qquad \theta_{NMT} = \sqrt{n}(\hat\delta - \delta).$$

For each fixed admissible group g, teacher j, and period t, define

$$S_{u,g} = \frac{1}{\sqrt{N_g MT}} \sum_{\substack{1 \leqslant i \leqslant N \\ g(i)=g}} \sum_{j'=1}^{M} \sum_{t'=1}^{T} \varepsilon_{ij't'} (v_{j'} \odot w_{t'}),$$

$$S_{v,j} = \frac{1}{\sqrt{NT}} \sum_{i=1}^{N} \sum_{t'=1}^{T} \varepsilon_{ijt'} (u_{g(i)} \odot w_{t'}),$$

$$S_{w,t} = \frac{1}{\sqrt{NM}} \sum_{i=1}^{N} \sum_{j'=1}^{M} \varepsilon_{ij't} (u_{g(i)} \odot v_{j'}),$$

and the $\ell \times p$ cross-moment matrices

$$Q_{u,g} = \frac{1}{N_g MT} \sum_{\substack{1 \leqslant i \leqslant N \\ g(i)=g}} \sum_{j'=1}^{M} \sum_{t'=1}^{T} (v_{j'} \odot w_{t'}) x'_{ij't'},$$

$$Q_{v,j} = \frac{1}{NT} \sum_{i=1}^{N} \sum_{t'=1}^{T} (u_{g(i)} \odot w_{t'}) x'_{ijt'},$$

$$Q_{w,t} = \frac{1}{NM} \sum_{i=1}^{N} \sum_{j'=1}^{M} (u_{g(i)} \odot v_{j'}) x'_{ij't}.$$

Then

$$\sqrt{NM} H_w^{-1} (\hat{w}_t - H_w w_t) = \Sigma_{UV}^{-1} \left[S_{w,t} - \frac{1}{\sqrt{T}} Q_{w,t} \theta_{NMT} \right] + R_{w,t,NMT}, \tag{27}$$

$$\sqrt{NT} H_v^{-1} (\hat{v}_j - H_v v_j) = \Sigma_{UW}^{-1} \left[S_{v,j} - \frac{1}{\sqrt{M}} Q_{v,j} \theta_{NMT} \right] + R_{v,j,NMT}, \tag{28}$$

$$\sqrt{N_g MT} H_u^{-1} (\hat{u}_g - H_u u_g) = \Sigma_{VW}^{-1} \left[S_{u,g} - \sqrt{\frac{N_g}{N}} Q_{u,g} \theta_{NMT} \right] + R_{u,g,NMT}. \tag{29}$$

The remainder vectors satisfy

$$\|R_{w,t,NMT}\| = O_p\left(\rho + \frac{\sqrt{N}}{T\sqrt{M}} + \frac{\sqrt{M}}{T\sqrt{N}} \right), \tag{30}$$

$$\|R_{v,j,NMT}\| = O_p\left(\rho + \frac{\sqrt{N}}{M\sqrt{T}} + \frac{\sqrt{T}}{M\sqrt{N}} \right), \tag{31}$$

$$\|R_{u,g,NMT}\| = O_p\left(\rho + \frac{\sqrt{N_g M}}{N\sqrt{T}} + \frac{\sqrt{N_g T}}{N\sqrt{M}} + \sqrt{N_g MT}\, \rho^4 \right). \tag{32}$$

If $N_g G / N = O_p(1)$, we have

$$\|R_{u,g,NMT}\| = O_p\left(\rho + \frac{\sqrt{N_g M}}{N\sqrt{T}} + \frac{\sqrt{N_g T}}{N\sqrt{M}} \right). \tag{33}$$

The group-size condition holds, in particular, when G remains fixed or when $N_g = O_p(N/G)$ for the group under consideration.

Remark: When slope estimation affects the leading expansion The normalizations in Proposition 2 reflect the numbers of observations entering the respective normal equations: NM for a period factor, NT for a teacher loading, and N_gMT for a group loading. The upper group-size condition $N_gG/N = O_p(1)$ used to simplify the remainder is weaker than two-sided balance: it does not impose a lower bound on N_g.

The normalized Gram bounds and the regressor moment bound imply $Q_{u,g} = O_p(1)$, $Q_{v,j} = O_p(1)$, and $Q_{w,t} = O_p(1)$ by the Cauchy–Schwarz inequality. If, in addition, $\theta_{NMT} = O_p(1)$, the slope-estimation terms satisfy

$$\frac{1}{\sqrt{T}}Q_{w,t}\theta_{NMT} = O_p(T^{-1/2}), \qquad \frac{1}{\sqrt{M}}Q_{v,j}\theta_{NMT} = O_p(M^{-1/2}), \qquad \sqrt{\frac{N_g}{N}}Q_{u,g}\theta_{NMT} = O_p\left(\sqrt{\frac{N_g}{N}}\right).$$

Consequently, when the respective remainders are $o_p(1)$, the period and teacher expansions have the same first-order score representations as if the slope were known whenever $T \to \infty$ and $M \to \infty$, respectively. The group expansion has this oracle property if $N_g/N \to_p 0$. By contrast, if G is fixed and $N_g/N \to_p \pi_g > 0$, the slope correction generally remains in the leading group expansion unless the cross-moment or its product with the slope error vanishes. The limiting distribution must then account jointly for $S_{u,g}$ and θ_{NMT}, including their covariance.

Remark: Mode-specific dimension conditions In addition to $\rho \to 0$, sufficient conditions for the displayed remainder bounds to be $o_p(1)$ are

$$\begin{aligned}
&\frac{N}{MT^2} + \frac{M}{NT^2} \to 0 && \text{for the period expansion,}\\
&\frac{N}{M^2T} + \frac{T}{M^2N} \to 0 && \text{for the teacher expansion,}\\
&\frac{N_gM}{N^2T} + \frac{N_gT}{N^2M} \to_p 0 && \text{for the group expansion under the upper group-size condition.}
\end{aligned}$$

These restrictions are mode-specific and need not follow from $\rho \to 0$. For example, if N is fixed and $M \asymp T \to \infty$, the period and teacher remainder bounds vanish, whereas the group remainder bound need not vanish.

Under $N \asymp M \asymp T \asymp L \to \infty$, the restriction $G \leqslant N$ gives $\rho = O(L^{-1})$, and

$$\|R_{w,t,NMT}\| + \|R_{v,j,NMT}\| = O_p(L^{-1}), \qquad \|R_{u,g,NMT}\| = O_p\left(L^{-1} + \frac{\sqrt{N_g}}{L}\right).$$

In particular, under $N_g \asymp N/G$, the group remainder is $O_p(L^{-1}+(LG)^{-1/2})$ and is negligible for both fixed and growing G. These are sufficient conditions derived from the stated upper bounds, rather than necessary conditions for the actual remainder vectors.

Assumption 6 (Joint fixed-index and slope-score central limit theorem)**.** *Let $S_{u,g}, S_{v,j}, S_{w,t}$, the cross-moment matrices $Q_{u,g}, Q_{v,j}, Q_{w,t}$, and the normalized loading Gram matrices $\Sigma_{VW}, \Sigma_{UW}, \Sigma_{UV}$ be as defined in Proposition 2. Let $\mathcal{Z}_{NMT}$, Γ_{NMT}, and $\mathcal{B}_m$, $m = 1, 2, 3$, be as defined in Proposition 1. All scores, cross-moment matrices, Gram matrices, and bias terms are evaluated at the same normalized representation of the true factors. Along the asymptotic sequence in Assumption 1, the following conditions hold. All limiting matrices, vectors, and scalars specified below are finite and nonrandom.*

The slope Gram matrix and bias terms satisfy

$$\Gamma_{NMT} \xrightarrow{p} \Gamma_\delta \succ 0, \qquad \mathcal{B}_m \xrightarrow{p} b_m, \quad m = 1, 2, 3, \tag{34}$$

where $\Gamma_\delta \in \mathbb{R}^{p\times p}$ *and* $b_m \in \mathbb{R}^p$*. Define*

$$b_\delta = b_1 + b_2 + b_3.$$

The normalized loading Gram matrices satisfy

$$\begin{aligned} \Sigma_{VW} &\xrightarrow{p} \overline{\Sigma}_{VW} \succ 0, \\ \Sigma_{UW} &\xrightarrow{p} \overline{\Sigma}_{UW} \succ 0, \\ \Sigma_{UV} &\xrightarrow{p} \overline{\Sigma}_{UV} \succ 0, \end{aligned} \tag{35}$$

where $\overline{\Sigma}_{VW}, \overline{\Sigma}_{UW}, \overline{\Sigma}_{UV} \in \mathbb{R}^{\ell\times\ell}$*.*

For every fixed admissible group g*, teacher* j*, and period* t*, the group share satisfies*

$$\frac{N_g}{N} \xrightarrow{p} \pi_g, \qquad \pi_g \in [0, 1], \tag{36}$$

and the cross-moment matrices satisfy

$$Q_{u,g} \xrightarrow{p} \overline{Q}_{u,g}, \qquad Q_{v,j} = O_p(1), \qquad Q_{w,t} = O_p(1), \tag{37}$$

where $Q_{u,g}, Q_{v,j}, Q_{w,t} \in \mathbb{R}^{\ell\times p}$ *and* $\overline{Q}_{u,g} \in \mathbb{R}^{\ell\times p}$*. These matrices need not be zero. Define the limiting group slope coefficient*

$$C_{u,g} = \sqrt{\pi_g}\,\overline{Q}_{u,g}.$$

Consequently,

$$\sqrt{\frac{N_g}{N}} Q_{u,g} \xrightarrow{p} C_{u,g}. \tag{38}$$

No probability limits are required for the unscaled teacher and period cross-moment matrices. In particular, along sequences with $M, T \to \infty$*, their effective coefficients satisfy*

$$\frac{1}{\sqrt{M}} Q_{v,j} = o_p(1), \qquad \frac{1}{\sqrt{T}} Q_{w,t} = o_p(1).$$

For every fixed admissible triple (g, j, t)*, the loading scores and the oracle slope score satisfy the joint central limit theorem*

$$\begin{pmatrix} S_{u,g} \\ S_{v,j} \\ S_{w,t} \\ \mathcal{Z}_{NMT} \end{pmatrix} \xrightarrow{d} \mathcal{N}(0, \Sigma_{gjt}), \tag{39}$$

where $\Sigma_{gjt} \in \mathbb{R}^{(3\ell+p)\times(3\ell+p)}$ *is symmetric and positive semidefinite, with block representation*

$$\Sigma_{gjt} = \begin{pmatrix} \Gamma_g^{vw} & \Xi_{uv,gj} & \Xi_{uw,gt} & \Xi_{gZ} \\ \Xi'_{uv,gj} & \Gamma_j^{uw} & \Xi_{vw,jt} & \Xi_{jZ} \\ \Xi'_{uw,gt} & \Xi'_{vw,jt} & \Gamma_t^{uv} & \Xi_{tZ} \\ \Xi'_{gZ} & \Xi'_{jZ} & \Xi'_{tZ} & \Omega_Z \end{pmatrix}.$$

The diagonal covariance blocks satisfy

$$\Gamma_g^{vw}, \Gamma_j^{uw}, \Gamma_t^{uv} \in \mathbb{R}^{\ell \times \ell}, \qquad \Gamma_g^{vw} \succ 0, \quad \Gamma_j^{uw} \succ 0, \quad \Gamma_t^{uv} \succ 0,$$

and

$$\Omega_Z \in \mathbb{R}^{p \times p}, \qquad \Omega_Z \succ 0.$$

The cross-covariance blocks have dimensions

$$\Xi_{uv,gj}, \Xi_{uw,gt}, \Xi_{vw,jt} \in \mathbb{R}^{\ell \times \ell}, \qquad \Xi_{gZ}, \Xi_{jZ}, \Xi_{tZ} \in \mathbb{R}^{\ell \times p},$$

and are not required to vanish. The full matrix Σ_{gjt} need not be positive definite, so the joint Gaussian limit in (39) *may be degenerate.*

All index-specific convergence statements and stochastic bounds are pointwise in the fixed admissible indices. No uniform convergence over groups, teachers, or periods is imposed, and (39) *concerns one fixed triple at a time.*

Theorem 1 (Joint asymptotic normality of the slope and aligned factor errors)**.** *Suppose that Assumptions 1, 2, 3, 4, 5, and 6 hold. Let $(\hat{\delta}, \hat{U}, \hat{V}, \hat{W})$ be the global least-squares estimator defined in Assumption 1. Write*

$$n = NMT, \qquad s = G + M + T, \qquad \rho = \sqrt{\frac{s}{n}},$$

and suppose that

$$\frac{s^{3/2}}{n} \to 0. \tag{40}$$

Let $b_\delta = b_1 + b_2 + b_3$, and use the limiting matrices and vectors defined in Assumption 6.

The estimated factors are expressed in the true-factor column labeling and sign orientation. Specifically, among the equivalent least-squares triples

$$(\hat{U}\Pi_\ell S_v S_w,\ \hat{V}\Pi_\ell S_v,\ \hat{W}\Pi_\ell S_w),$$

where Π_ℓ ranges over $\ell \times \ell$ permutation matrices and S_v, S_w range over diagonal matrices with entries in $\{-1, 1\}$, choose a representative minimizing

$$\frac{1}{M}\|\hat{V}\Pi_\ell S_v - V\|_F^2 + \frac{1}{T}\|\hat{W}\Pi_\ell S_w - W\|_F^2.$$

Retain the hat notation for the selected representative, and define H_u, H_v, H_w from it as in Proposition 2. The followings hold.

1. *(Identity limits of the alignment matrices.)*
   The matrices H_u, H_v, H_w are nonsingular with probability approaching one and satisfy

$$H_u \xrightarrow{p} I_\ell, \qquad H_v \xrightarrow{p} I_\ell, \qquad H_w \xrightarrow{p} I_\ell.$$

2. *(Slope central limit theorem.)*
   The slope estimator satisfies

$$\sqrt{n}(\hat{\delta} - \delta) \xrightarrow{d} \mathcal{N}\left(-\Gamma_\delta^{-1} b_\delta,\ \Gamma_\delta^{-1} \Omega_Z \Gamma_\delta^{-1}\right). \tag{41}$$

3. *(Joint fixed-index central limit theorem.)*
Fix an admissible triple (g, j, t) and suppose additionally that

$$\frac{N+M}{T\sqrt{NM}} \to 0, \qquad \frac{N+T}{M\sqrt{NT}} \to 0, \qquad \frac{\sqrt{N_g}(M+T)}{N\sqrt{MT}} \xrightarrow{p} 0. \tag{42}$$

Let

$$C_{u,g} = \sqrt{\pi_g}\,\overline{Q}_{u,g}.$$

Then the aligned factor errors satisfy

$$\begin{pmatrix} \sqrt{N_g MT}(\hat{u}_g - H_u u_g) \\ \sqrt{NT}(\hat{v}_j - H_v v_j) \\ \sqrt{NM}(\hat{w}_t - H_w w_t) \\ \sqrt{n}(\hat{\delta} - \delta) \end{pmatrix} \xrightarrow{d} \mathcal{N}\big(\mu_{gjt},\ \mathcal{A}_{gjt}\Sigma_{gjt}\mathcal{A}'_{gjt}\big), \tag{43}$$

where

$$\mathcal{A}_{gjt} = \begin{pmatrix} \overline{\Sigma}_{VW}^{-1} & 0 & 0 & -\overline{\Sigma}_{VW}^{-1} C_{u,g} \Gamma_\delta^{-1} \\ 0 & \overline{\Sigma}_{UW}^{-1} & 0 & 0 \\ 0 & 0 & \overline{\Sigma}_{UV}^{-1} & 0 \\ 0 & 0 & 0 & \Gamma_\delta^{-1} \end{pmatrix}$$

and

$$\mu_{gjt} = \begin{pmatrix} \overline{\Sigma}_{VW}^{-1} C_{u,g} \Gamma_\delta^{-1} b_\delta \\ 0_\ell \\ 0_\ell \\ -\Gamma_\delta^{-1} b_\delta \end{pmatrix}.$$

For completeness, define

$$\begin{aligned} V_{u,g} &= \overline{\Sigma}_{VW}^{-1}\Big[\Gamma_g^{vw} + C_{u,g}\Gamma_\delta^{-1}\Omega_Z\Gamma_\delta^{-1}C'_{u,g} - \Xi_{gZ}\Gamma_\delta^{-1}C'_{u,g} - C_{u,g}\Gamma_\delta^{-1}\Xi'_{gZ}\Big]\overline{\Sigma}_{VW}^{-1}, \\ V_{v,j} &= \overline{\Sigma}_{UW}^{-1}\Gamma_j^{uw}\overline{\Sigma}_{UW}^{-1}, \\ V_{w,t} &= \overline{\Sigma}_{UV}^{-1}\Gamma_t^{uv}\overline{\Sigma}_{UV}^{-1}. \end{aligned}$$

The marginal fixed-index limits are

$$\begin{aligned} \sqrt{N_g MT}(\hat{u}_g - H_u u_g) &\xrightarrow{d} \mathcal{N}\Big(\overline{\Sigma}_{VW}^{-1} C_{u,g}\Gamma_\delta^{-1} b_\delta,\ V_{u,g}\Big), \\ \sqrt{NT}(\hat{v}_j - H_v v_j) &\xrightarrow{d} \mathcal{N}(0, V_{v,j}), \\ \sqrt{NM}(\hat{w}_t - H_w w_t) &\xrightarrow{d} \mathcal{N}(0, V_{w,t}). \end{aligned} \tag{44}$$

The joint covariance matrix in (43) *may be singular, as may $V_{u,g}$. All fixed-index conclusions are pointwise in (g, j, t), and no additional condition on $N_g G/N$ is required. The slope conclusion in part (ii) does not require* (42)*. The factor conclusions retain the finite-sample alignment centers $H_u u_g, H_v v_j, H_w w_t$; they do not assert the same limits after replacing those centers by u_g, v_j, w_t.*

Remark: Oracle limits and the group slope correction The teacher and period expansions have the same first-order score representations as if the slope were known, because their effective slope coefficients vanish when $M, T \to \infty$. The group expansion has this oracle property when $C_{u,g} = 0$. In particular, if $\pi_g = 0$, boundedness of $Q_{u,g}$ implies $C_{u,g} = 0$, and

$$V_{u,g} = \overline{\Sigma}_{VW}^{-1} \Gamma_g^{vw} \overline{\Sigma}_{VW}^{-1}.$$

The group limit is then centered even when the slope limit has a nonzero mean.

When $C_{u,g} \neq 0$, slope estimation enters the leading group expansion and must be accounted for jointly with the group score. Its effect on the group variance includes the covariance between $S_{u,g}$ and $\mathcal{Z}_{NMT}$, and need not increase the variance relative to the oracle expression. Moreover, an oracle marginal representation does not imply asymptotic independence from the slope estimator. The limiting teacher–slope and period–slope cross-covariance matrices are, respectively,

$$\overline{\Sigma}_{UW}^{-1} \Xi_{jZ} \Gamma_\delta^{-1}, \qquad \overline{\Sigma}_{UV}^{-1} \Xi_{tZ} \Gamma_\delta^{-1},$$

which need not vanish.

Remark: Bias limits for groups with a positive limiting share If $\pi_g > 0$, the group loading rate condition can be written as

$$\sqrt{\frac{N_g}{N}} \frac{M+T}{\sqrt{n}} \to_p 0.$$

It therefore implies $M/\sqrt{n} \to 0$ and $T/\sqrt{n} \to 0$. The bias bounds in Proposition 1 yield $\mathcal{B}_2, \mathcal{B}_3 = o_p(1)$, so $b_2 = b_3 = 0$. If G is also fixed, $\mathcal{B}_1 = O_p(G/\sqrt{n}) = o_p(1)$, and hence $b_\delta = 0$. All limiting means in part (iii) then vanish. Nevertheless, $C_{u,g}$ may remain nonzero, so slope estimation can affect the group variance even when it produces no asymptotic mean shift.

Remark: Alignment matrices and the centering of factor errors The identity limits of H_u, H_v, H_w allow their inverse premultipliers to be removed from the normalized expansions because the corresponding normalized errors are stochastically bounded. For example,

$$\sqrt{N_g MT}(\hat{u}_g - H_u u_g) = \sqrt{N_g MT} H_u^{-1}(\hat{u}_g - H_u u_g) + o_p(1).$$

This argument does not justify replacing $H_u u_g$ by u_g. Such a replacement would require the additional condition

$$\sqrt{N_g MT}(H_u - I_\ell) u_g = o_p(1),$$

with analogous conditions for the teacher and period factors. Convergence of the alignment matrices to the identity alone does not provide these stronger centering conditions. The theorem therefore retains the finite-sample alignment centers even after selecting the true-factor column labeling and sign orientation.

Remark: Degenerate limits and the scope of joint inference The transformation matrix $\mathcal{A}_{gjt}$ is invertible because it is block triangular with nonsingular diagonal blocks. Consequently,

$$\operatorname{rank}(\mathcal{A}_{gjt} \Sigma_{gjt} \mathcal{A}_{gjt}') = \operatorname{rank}(\Sigma_{gjt}).$$

Allowing a positive-semidefinite score covariance therefore permits a degenerate joint Gaussian limit. The teacher, period, and slope marginal covariance matrices are positive definite under Assumption 6, while the group marginal covariance can be singular through cancellation between its score and the slope correction.

The joint conclusion concerns one fixed admissible triple (g, j, t). Neither the pointwise assumptions nor the stated score central limit theorem imply joint Gaussian convergence for a larger collection containing several groups, teachers, or periods. Such a conclusion requires a joint central limit theorem for the corresponding larger score vector. Simultaneous inference over a growing collection additionally requires suitable uniform control.

5.1 Bias Corrected Estimator

In this section, we show that the three asymptotic bias terms in (23) can be bias-corrected, as in the framework of Hahn and Kuersteiner (2002) and Bai (2009).

Suppose the errors are conditionally uncorrelated, i.e., $\mathbb{E}[\varepsilon_a \varepsilon_b \mid \mathcal{F}] = 0$ for $a \neq b$. We define

$$\hat{\varepsilon}_{ijt} = y_{ijt} - x_{ijt}'\hat{\delta} - \sum_{k=1}^{\ell} \hat{u}_{g(i),k}\hat{v}_{jk}\hat{w}_{tk}, \qquad \widehat{\sigma}^2_{gjt} = \frac{1}{N_g} \sum_{i:g(i)=g} \hat{\varepsilon}^2_{ijt}.$$

The grouped conditional variance is $\sigma^{G,2}_{gjt} = N_g^{-1} \sum_{i:g(i)=g} \mathbb{E}[\varepsilon^2_{ijt} \mid \mathcal{F}]$, and each Ω^G_m is diagonal. Corresponding residual-based estimators are

$$\begin{aligned}
\widehat{\Omega}^G_1 &= \operatorname{diag}\left(\left(\sum_{g=1}^{G} \widehat{\sigma}^2_{gjt} \right)_{(j,t)} \right), \\
\widehat{\Omega}^G_2 &= \operatorname{diag}\left(\left(\sum_{j=1}^{M} \widehat{\sigma}^2_{gjt} \right)_{(g,t)} \right), \\
\widehat{\Omega}^G_3 &= \operatorname{diag}\left(\left(\sum_{t=1}^{T} \widehat{\sigma}^2_{gjt} \right)_{(g,j)} \right),
\end{aligned} \tag{45}$$

with the pair indices ordered according to the corresponding unfolding convention.

Proposition 3. *Suppose that Assumptions 1, 2, 3, and 4 hold. Suppose also that* $\mathbb{E}[\varepsilon_a \varepsilon_b \mid \mathcal{F}] = 0$ *for all distinct* $a, b \in \mathcal{I}$. *The true factor rows satisfy*

$$\frac{1}{N} \sum_{g=1}^{G} N_g \|u_g\|^4 + \frac{1}{M} \sum_{j=1}^{M} \|v_j\|^4 + \frac{1}{T} \sum_{t=1}^{T} \|w_t\|^4 = O_p(1)$$

and the group sizes satisfies

$$\sum_{g=1}^{G} \frac{1}{N_g} = O_p\left(\frac{G^2}{N} \right). \tag{46}$$

Let $\mathcal{T}$ *be the tangent space defined in* (11) *evaluated at the true parameters* (U, V, W). *Let*

$$\Gamma_{NMT} = \frac{1}{NMT} X_{(4)} M_{\mathcal{T}} X_{(4)}', \qquad \mathcal{Z}_{NMT} = \frac{1}{\sqrt{NMT}} X_{(4)} M_{\mathcal{T}} \operatorname{vec}(E).$$

Set $\widehat{\tilde{U}} = (D'D)^{1/2}\hat{U}$ *and*

$$(\widehat{F}_1, \widehat{F}_2, \widehat{F}_3) = (\widehat{\tilde{U}}, \hat{V}, \hat{W}), \qquad (\widehat{C}_1, \widehat{C}_2, \widehat{C}_3) = (\hat{V} * \hat{W}, \widehat{\tilde{U}} * \hat{W}, \widehat{\tilde{U}} * \hat{V}).$$

Let $\widehat{\Omega}_m^G$, $m = 1, 2, 3$, *be data-based estimators of the conditional covariance defined in* (45), *of dimensions* $MT \times MT$, $GT \times GT$, *and* $GM \times GM$, *respectively. Put* $K_{1,1}^G = I_{GMT}$ *and define*

$$\widehat{\mathcal{B}}_m = \frac{1}{\sqrt{n}} X_{(4)} P_{\widehat{\mathcal{T}}} Q K_{1,m}^G \operatorname{vec}\left(\widehat{F}_m (\widehat{F}_m' \widehat{F}_m)^{-1} (\widehat{C}_m' \widehat{C}_m)^{-1} \widehat{C}_m' \widehat{\Omega}_m^G M_{\widehat{C}_m}\right), \qquad m = 1, 2, 3. \tag{47}$$

The bias-corrected estimator $\hat{\delta}_{bc}$ *is defined as*

$$\hat{\delta}_{\mathrm{bc}} = \hat{\delta} + \frac{1}{\sqrt{NMT}} \widehat{\Gamma}_{NMT}^{-1} \left(\widehat{\mathcal{B}}_1 + \widehat{\mathcal{B}}_2 + \widehat{\mathcal{B}}_3\right). \tag{48}$$

Then, we have

$$\Gamma_{NMT} \sqrt{NMT} (\hat{\delta}_{bc} - \delta) = \mathcal{Z}_{NMT} + O_p\left(\sqrt{\frac{G+M+T}{NMT}} + \frac{(G+M+T)^{3/2}}{NMT}\right). \tag{49}$$

Remark: Centered asymptotic normality. Write $n = NMT$ and $s = G + M + T$. Suppose additionally that

$$\frac{s^{3/2}}{n} \to 0, \qquad \Gamma_{NMT} \xrightarrow{p} \Gamma_\delta \succ 0, \qquad \mathcal{Z}_{NMT} \xrightarrow{d} \mathcal{N}(0, \Omega_Z).$$

The bias-corrected expansion and Slutsky's theorem imply

$$\sqrt{n}(\hat{\delta}_{\mathrm{bc}} - \delta) \xrightarrow{d} \mathcal{N}\left(0, \Gamma_\delta^{-1} \Omega_Z \Gamma_\delta^{-1}\right).$$

The limiting covariance is determined by the oracle slope score. This conclusion does not require the individual bias terms $\mathcal{B}_m$ to converge to finite limits: the correction estimates and removes their sum at the required rate.

Remark: Heteroskedasticity and vanishing population biases. Conditional uncorrelatedness allows the conditional variances to differ across observations. The residual-based correction estimates the resulting aggregate bias terms and does not require consistent estimation of every individual group–teacher–period variance. If the errors are additionally conditionally homoskedastic, with $\mathbb{E}[\varepsilon_{ijt}^2 \mid \mathcal{F}] = \sigma^2$ for all (i, j, t), then

$$\Omega_1^G = G\sigma^2 I_{MT}, \qquad \Omega_2^G = M\sigma^2 I_{GT}, \qquad \Omega_3^G = T\sigma^2 I_{GM}.$$

Since $C_m' M_{C_m} = 0$, the displayed population bias formulas give

$$\mathcal{B}_1 = \mathcal{B}_2 = \mathcal{B}_3 = 0.$$

Thus conditional homoskedasticity eliminates these leading population bias terms. The estimated correction can still be nonzero because the residual variance estimates fluctuate across cells.

Remark: Aggregate control of group sizes. The reciprocal-size condition can be written as

$$\frac{1}{G}\sum_{g=1}^{G}\frac{N/G}{N_g}=\frac{N}{G^2}\sum_{g=1}^{G}\frac{1}{N_g}=O_p(1).$$

Cauchy–Schwarz implies that this quantity is at least one, with equality for equal group sizes. The condition therefore controls the average reciprocal group size relative to the equal-size benchmark. A uniform lower bound $N_g \geqslant cN/G$ is sufficient, but the aggregate condition allows some substantially smaller groups. For example, if $N = L^2$, $G = L$, $N_1 = 1$, and $N_g = L + 1$ for $g \geqslant 2$, then

$$\frac{N}{G^2}\sum_{g}\frac{1}{N_g}=1+\frac{L-1}{L+1}\to 2,$$

although $N_1/(N/G) = 1/L \to 0$.

6 Selecting the Number of Interactive Components

Let ℓ_0 denote the true number of interactive components, and let $\ell \in \{0, \ldots, \ell_{\max}\}$ denote a candidate number, where $\ell_{\max}$ is fixed and $1 \leqslant \ell_0 \leqslant \ell_{\max}$. Assumptions 1 and 2 concerning the true factor matrices are understood with $\ell = \ell_0$. For each candidate rank, we jointly minimize the residual sum of squares over the slope coefficient and the grouped interactive component. Following the penalized-fit approach of Bai and Ng (2002), we select the rank by adding a complexity penalty to the resulting criterion.

For $\ell \geqslant 1$, let Θ_ℓ impose the normalization and upper bound in Assumption 1, with ℓ columns and the same constant c_U for every candidate rank:

$$\begin{aligned}\Theta_\ell = \Big\{&(\widetilde{U},\widetilde{V},\widetilde{W}) \in \mathbb{R}^{G\times\ell}\times\mathbb{R}^{M\times\ell}\times\mathbb{R}^{T\times\ell}: \\ &\operatorname{diag}\left(\frac{1}{M}\widetilde{V}'\widetilde{V}\right)=\operatorname{diag}\left(\frac{1}{T}\widetilde{W}'\widetilde{W}\right)=\iota_\ell, \lambda_{\max}\left(\frac{1}{N}\widetilde{U}'D'D\widetilde{U}\right)\leqslant c_U\Big\}.\end{aligned} \tag{50}$$

Define the admissible grouped common-component classes by

$$\begin{aligned}\mathcal{C}^D_{\ell,c_U} &= \left\{\sum_{k=1}^{\ell}(D\widetilde{u}_k\circ\widetilde{v}_k\circ\widetilde{w}_k):(\widetilde{U},\widetilde{V},\widetilde{W})\in\Theta_\ell\right\}, \qquad \ell\geqslant 1,\\ \mathcal{C}^D_{0,c_U} &= \{0\}.\end{aligned} \tag{51}$$

Every tensor in $\mathcal{C}^D_{\ell,c_U}$ has CP rank at most ℓ, but the class also incorporates the parameter bound used in estimation. Appending a zero column to $\widetilde{U}$ and columns with norms $\sqrt{M}$ and $\sqrt{T}$ to $\widetilde{V}$ and $\widetilde{W}$, respectively, preserves both the fitted tensor and the normalization restrictions. Therefore,

$$\mathcal{C}^D_{0,c_U}\subseteq\mathcal{C}^D_{1,c_U}\subseteq\cdots\subseteq\mathcal{C}^D_{\ell_{\max},c_U}.$$

For each candidate ℓ, define

$$\mathcal{V}(\ell)=\min_{\delta\in\mathbb{R}^p,\ \Pi\in\mathcal{C}^D_{\ell,c_U}}\frac{1}{NMT}\left\|Y-X\times_4\delta'-\Pi\right\|_F^2. \tag{52}$$

Concentrating the criterion over δ gives

$$\mathcal{V}(\ell) = \min_{\Pi \in \mathcal{C}^D_{\ell, c_U}} \frac{1}{NMT} \left\| M_{X'_{(4)}} \{\mathrm{vec}(Y) - \mathrm{vec}(\Pi)\} \right\|^2.$$

Let $g(G, N, M, T) > 0$ be a deterministic penalty. Define

$$PC(\ell) = \mathcal{V}(\ell) + \ell g(G, N, M, T), \qquad \widehat{\ell} = \min \operatorname*{argmin}_{0 \leqslant \ell \leqslant \ell_{\max}} PC(\ell). \tag{53}$$

The outer minimum specifies the tie-breaking rule.

Assumption 7 (Uniform identification over candidate ranks)**.** *There exists a constant $c_{\mathrm{rank}} > 0$ such that, with probability approaching one,*

$$\inf_{\substack{\widetilde{U} \in \mathbb{R}^{G \times \ell_{\max}} \\ \widetilde{V} \in \mathbb{R}^{M \times \ell_{\max}}}} \lambda_{\min} \left[\frac{1}{NMT} X_{(4)} \left(M_{D\widetilde{U} * \widetilde{V}} \otimes M_W \right) X'_{(4)} \right] \geqslant c_{\mathrm{rank}}. \tag{54}$$

Here W is the true $T \times \ell_0$ factor matrix. The candidate matrices in the infimum are unrestricted, and the projectors are defined even when those matrices are rank deficient. Padding matrices with zero columns leaves their column spaces unchanged, so (54) *also holds for every candidate rank $0 \leqslant \ell \leqslant \ell_{\max}$, with the same constant. For $\ell = 0$, the first annihilator is I_{NM}.*

Proposition 4 (Consistency of the information criterion)**.** *Suppose that Assumptions 1, 2, 3, and 7 hold, with the true-factor restrictions evaluated at rank ℓ_0. Let $\ell_{\max}$ be fixed, with $1 \leqslant \ell_0 \leqslant \ell_{\max}$, and define $\widehat{\ell}$ by* (53)*. If the positive deterministic penalty satisfies*

$$g(G, N, M, T) \to 0, \qquad \frac{NMT}{G + M + T} g(G, N, M, T) \to \infty, \tag{55}$$

then, along the asymptotic sequence in Assumption 1,

$$\mathbb{P}\{\widehat{\ell} = \ell_0\} \to 1.$$

Remark: Penalty choice. The conditions in (55) require a penalty that vanishes but dominates the order $(G + M + T)/(NMT)$ of the improvement from fitting additional components. One admissible choice is

$$\begin{aligned} g_1(G, N, M, T) &= c \frac{G + M + T}{NMT} \log\left(\frac{NMT}{G + M + T} \right) \\ g_2(G, N, M, T) &= c \frac{G + M + T}{NMT} \log\left(C^2_{GNMT} \right) \\ g_3(G, N, M, T) &= c \frac{\log(C^2_{GNMT})}{C^2_{GNMT}}, \end{aligned} \tag{56}$$

where $C^2_{GNMT} = \min(NMT/G, NT, NM)$ and $c > 0$. This choice remains admissible throughout the asymptotic framework of Assumption 1, without requiring the three panel dimensions to grow at comparable rates.

Remark: Zero true rank. The result also extends to $\ell_0 = 0$. In that case, set $\Pi = 0$, regard U, V, W as matrices with no columns, and set $M_W = I_T$. The positive-rank dimension restriction and true-factor normalization and Gram conditions in Assumptions 1 and 2 are then omitted; their remaining dimensional, membership, regressor, and error conditions are retained. There are no underfitted candidates, and the uniform criterion bound and overfitting argument prove $\mathbb{P}\{\widehat{\ell} = 0\} \to 1$.

Remark: Scale-invariant information criterion The additive criterion

$$PC(\ell) = \mathcal{V}(\ell) + \ell g(G, N, M, T)$$

depends on the scale of the outcome variable because $\mathcal{V}(\ell)$ has the units of a residual variance, whereas $g(G, N, M, T)$ is dimensionless. A scale-invariant alternative is

$$IC(\ell) = \log \mathcal{V}(\ell) + \ell g(G, N, M, T).$$

Suppose, in addition, that

$$\mathcal{V}(\ell_0) \xrightarrow{p} \sigma_\varepsilon^2 > 0.$$

For overfitted models,

$$\mathcal{V}(\ell_0) - \mathcal{V}(\ell) = O_p(\rho^2),$$

and hence a first-order expansion of the logarithm gives

$$\log \mathcal{V}(\ell) - \log \mathcal{V}(\ell_0) = O_p(\rho^2).$$

For underfitted models, the residual-variance gap remains bounded away from zero. Therefore, the same penalty conditions imply consistency of

$$\widehat{\ell}_{IC} = \underset{0 \leqslant \ell \leqslant \ell_{\max}}{\operatorname{argmin}} \; IC(\ell).$$

Another scale-adjusted alternative is

$$PC^*(\ell) = \mathcal{V}(\ell) + \ell \widehat{\sigma}^2 g(G, N, M, T),$$

where

$$\widehat{\sigma}^2 = \mathcal{V}(\ell_{\max})$$

or any other consistent estimator of the limiting idiosyncratic variance.

7 Additive Fixed Effects and Unbalanced Panels

7.1 Consistency with Additive Effects

Consider the model with both additive and interactive fixed effects

$$y_{ijt} = x_{ijt}'\delta + \kappa + \mu_{g(i)} + \nu_j + \xi_t + \pi_{ijt} + \varepsilon_{ijt}, \qquad \pi_{ijt} = \sum_{k=1}^{\ell} u_{g(i),k} v_{jk} w_{tk}. \tag{57}$$

To select a unique intercept–group–teacher–period representation of the additive component, impose

$$\iota_N' D\mu = \sum_{g=1}^{G} N_g \mu_g = 0, \qquad \iota_M'\nu = 0, \qquad \iota_T'\xi = 0. \tag{58}$$

Define the additive tensor by

$$\mathcal{A}(\kappa,\mu,\nu,\xi) = \kappa(\iota_N \circ \iota_M \circ \iota_T) + (D\mu \circ \iota_M \circ \iota_T) + (\iota_N \circ \nu \circ \iota_T) + (\iota_N \circ \iota_M \circ \xi).$$

Let $y = \mathrm{vec}(Y)$, $e = \mathrm{vec}(E)$, $a(\kappa,\mu,\nu,\xi) = \mathrm{vec}(\mathcal{A}(\kappa,\mu,\nu,\xi))$, and $\pi(U,V,W) = \mathrm{vec}(\Pi(U,V,W))$. Suppressing parameter arguments at the true values, the tensor and vector representations are

$$Y = X \times_4 \delta' + \mathcal{A} + \Pi + E, \qquad y = X'_{(4)}\delta + a + \pi + e.$$

Define a global least-squares estimator by

$$(\hat\delta, \hat\kappa, \hat\mu, \hat\nu, \hat\xi, \hat U, \hat V, \hat W) \in \underset{\substack{\delta\in\mathbb{R}^p,\ \kappa\in\mathbb{R},\ \mu\in\mathbb{R}^G,\ \nu\in\mathbb{R}^M,\ \xi\in\mathbb{R}^T \\ (U,V,W)\in\Theta_\ell \\ \iota_N'D\mu=0,\ \iota_M'\nu=0,\ \iota_T'\xi=0}}{\operatorname{argmin}} \frac{1}{NMT}\left\| y - X'_{(4)}\delta - a(\kappa,\mu,\nu,\xi) - \pi(U,V,W)\right\|^2. \tag{59}$$

To derive the profiled objective function from (59), let

$$\mathcal{A}_D = \begin{pmatrix} \iota_N\otimes\iota_M\otimes\iota_T & D\otimes\iota_M\otimes\iota_T & \iota_N\otimes I_M\otimes\iota_T & \iota_N\otimes\iota_M\otimes I_T \end{pmatrix}$$

Although $\mathcal{A}_D$ contains redundant columns, its column space is precisely the space of additive fitted vectors satisfying the representation above. Conditional on (δ, U, V, W), minimization over the normalized additive coefficients is therefore orthogonal projection of $y - X'_{(4)}\delta - \pi(U,V,W)$ onto $\mathcal{A}_D$. Since $\iota_N \in \mathrm{col}(D)$, the matrix $P_D - P_{\iota_N}$ is the orthogonal projector onto the group-constant vectors with observation-weighted mean zero. Consequently,

$$P_{\mathcal{A}_D} = P_{\iota_n} + (P_D - P_{\iota_N}) \otimes P_{\iota_M} \otimes P_{\iota_T} + P_{\iota_N} \otimes M_{\iota_M} \otimes P_{\iota_T} + P_{\iota_N} \otimes P_{\iota_M} \otimes M_{\iota_T}, \tag{60}$$

where $\iota_n = \iota_N \otimes \iota_M \otimes \iota_T$. The four summands are mutually orthogonal projectors onto the intercept, normalized group-effect, normalized teacher-effect, and normalized period-effect spaces. In particular,

$$\mathrm{rank}(P_{\mathcal{A}_D}) = 1 + (G-1) + (M-1) + (T-1) = G + M + T - 2. \tag{61}$$

Let $M_{\mathcal{A}_D} = I_n - P_{\mathcal{A}_D}$. Profiling the additive coefficients out of (59) gives

$$(\hat\delta, \hat U, \hat V, \hat W) \in \underset{\substack{\delta\in\mathbb{R}^p \\ (U,V,W)\in\Theta_\ell}}{\operatorname{argmin}} \frac{1}{NMT}\left\| M_{\mathcal{A}_D}\{y - X'_{(4)}\delta - \pi(U,V,W)\}\right\|^2. \tag{62}$$

Conversely, any global minimizer of (62), completed with its normalized additive least-squares coefficients, is a global minimizer of (59). Writing $\hat\Pi = \Pi(\hat U, \hat V, \hat W)$ and $\hat{\mathcal{A}} = \mathcal{A}(\hat\kappa, \hat\mu, \hat\nu, \hat\xi)$, the fitted additive component satisfies

$$\mathrm{vec}(\hat{\mathcal{A}}) = P_{\mathcal{A}_D}\{y - X'_{(4)}\hat\delta - \mathrm{vec}(\hat\Pi)\}. \tag{63}$$

The individual additive estimates have explicit expressions. For an arbitrary residual tensor $R = (r_{ijt})$, define

$$\bar r_{\cdots} = \frac{1}{NMT}\sum_{i=1}^N\sum_{j=1}^M\sum_{t=1}^T r_{ijt}, \qquad \bar r_{g\cdot\cdot} = \frac{1}{N_g MT}\sum_{i:g(i)=g}\sum_{j=1}^M\sum_{t=1}^T r_{ijt},$$

$$\bar r_{\cdot j\cdot} = \frac{1}{NT}\sum_{i=1}^N\sum_{t=1}^T r_{ijt}, \qquad \bar r_{\cdot\cdot t} = \frac{1}{NM}\sum_{i=1}^N\sum_{j=1}^M r_{ijt}.$$

The projection and residual-making formulas, indexed according to the original tensor entries, are

$$\begin{aligned}[P_{\mathcal{A}_D}\operatorname{vec}(R)]_{ijt} &= \bar{r}_{g(i)\cdot\cdot} + \bar{r}_{\cdot j\cdot} + \bar{r}_{\cdot\cdot t} - 2\bar{r}_{\cdots},\\ [M_{\mathcal{A}_D}\operatorname{vec}(R)]_{ijt} &= r_{ijt} - \bar{r}_{g(i)\cdot\cdot} - \bar{r}_{\cdot j\cdot} - \bar{r}_{\cdot\cdot t} + 2\bar{r}_{\cdots}.\end{aligned} \tag{64}$$

Thus, for $\hat{r}_{ijt} = y_{ijt} - x_{ijt}'\hat{\delta} - \hat{\pi}_{ijt}$, the normalized additive coefficients associated with the selected profiled minimizer are

$$\begin{aligned}\hat{\kappa} &= \bar{\hat{r}}_{\cdots}, & \hat{\mu}_g &= \bar{\hat{r}}_{g\cdot\cdot} - \bar{\hat{r}}_{\cdots},\\ \hat{\nu}_j &= \bar{\hat{r}}_{\cdot j\cdot} - \bar{\hat{r}}_{\cdots}, & \hat{\xi}_t &= \bar{\hat{r}}_{\cdot\cdot t} - \bar{\hat{r}}_{\cdots}.\end{aligned} \tag{65}$$

The means in (65) are computed after subtracting both the fitted regression component and the fitted interactive component.

Assumption 8 (Additive fixed effects)**.** *The following two assumptions hold.*

(i) Strong centered factors. *Define*

$$\bar{u}' = \frac{1}{N}\iota_N' DU, \qquad \bar{v}' = \frac{1}{M}\iota_M' V, \qquad \bar{w}' = \frac{1}{T}\iota_T' W,$$

$$U^c = U - \iota_G\bar{u}', \qquad V^c = V - \iota_M\bar{v}', \qquad W^c = W - \iota_T\bar{w}'.$$

Assume $\ell \leqslant \min\{G-1, M-1, T-1\}$. For a constant $c_c > 0$ independent of the panel dimensions, with probability approaching one,

$$\begin{aligned} c_c I_\ell &\preceq \frac{1}{N}(U^c)'D'DU^c \preceq c_1 I_\ell,\\ c_c I_\ell &\preceq \frac{1}{M}(V^c)'V^c \preceq c_1 I_\ell,\\ c_c I_\ell &\preceq \frac{1}{T}(W^c)'W^c \preceq c_1 I_\ell. \end{aligned} \tag{66}$$

Here c_1 is as in Assumption 2. These conditions concern the true centered factors; neither the true nor the estimated raw factor matrices are required to have zero column means.

(ii) Identification after accounting for additive and interactive effects. *For arbitrary $\widetilde{U} \in \mathbb{R}^{G\times\ell}$ and $\widetilde{V} \in \mathbb{R}^{M\times\ell}$, define*

$$\widetilde{C}_A(\widetilde{U},\widetilde{V}) = \left[\iota_N \otimes \iota_M,\ D\widetilde{U} * \widetilde{V}\right], \qquad W_A = [\iota_T, W].$$

There exists a constant $c_{\delta,A} > 0$ such that, with probability approaching one,

$$\inf_{\substack{\widetilde{U}\in\mathbb{R}^{G\times\ell}\\ \widetilde{V}\in\mathbb{R}^{M\times\ell}}} \lambda_{\min}\left[\frac{1}{NMT}X_{(4)}\left(M_{\widetilde{C}_A(\widetilde{U},\widetilde{V})} \otimes M_{W_A}\right)X_{(4)}'\right] \geqslant c_{\delta,A}. \tag{67}$$

The projectors are defined using Moore–Penrose inverses. This condition extends Assumption 4 for the benchmark model to the inclusion of additive fixed effects.

Lemma 2 (Consistency with group, teacher, and period additive effects)**.** *Suppose Assumptions 1, 2, 3, and 8 hold. Both the true and candidate additive effects satisfy the observation-weighted normalizations in* (58)*. Define the global least-squares estimator by* (59)*. Then the slope and the interactive component satisfy*

$$\|\hat{\delta} - \delta\| + \frac{\|\hat{\Pi} - \Pi\|_F}{\sqrt{n}} = O_p\left(\sqrt{\frac{G+M+T}{NMT}}\right). \tag{68}$$

The uncentered joint loading spaces satisfy

$$\begin{aligned}
\frac{1}{\sqrt{MT}}\|M_{\hat{V}*\hat{W}}(V*W)\|_F &= O_p\left(\sqrt{\frac{G+M+T}{NMT}}\right), \\
\frac{1}{\sqrt{NT}}\|M_{D\hat{U}*\hat{W}}(DU*W)\|_F &= O_p\left(\sqrt{\frac{G+M+T}{NMT}}\right), \\
\frac{1}{\sqrt{NM}}\|M_{D\hat{U}*\hat{V}}(DU*V)\|_F &= O_p\left(\sqrt{\frac{G+M+T}{NMT}}\right).
\end{aligned} \tag{69}$$

Moreover, the additive-effect estimators satisfy

$$\begin{aligned}
|\hat{\kappa} - \kappa| &= O_p\left(\frac{1}{\sqrt{NMT}} + \sqrt{\frac{G+M+T}{NMT}}\right), \\
\frac{1}{\sqrt{N}}\|D(\hat{\mu} - \mu)\| = \left[\frac{1}{N}\sum_{g=1}^{G} N_g(\hat{\mu}_g - \mu_g)^2\right]^{1/2} &= O_p\left(\sqrt{\frac{G}{NMT}} + \sqrt{\frac{G+M+T}{NMT}}\right), \\
\frac{1}{\sqrt{M}}\|\hat{\nu} - \nu\| &= O_p\left(\frac{1}{\sqrt{NT}} + \sqrt{\frac{G+M+T}{NMT}}\right), \\
\frac{1}{\sqrt{T}}\|\hat{\xi} - \xi\| &= O_p\left(\frac{1}{\sqrt{NM}} + \sqrt{\frac{G+M+T}{NMT}}\right).
\end{aligned} \tag{70}$$

Remark: The interactive component after profiling. Additive profiling removes the intercept and the group, teacher, and period additive effects. To describe what remains in the interactive effects, let u_k^c, v_k^c, w_k^c denote the columns of U^c, V^c, W^c, and let $\bar{u}_k, \bar{v}_k, \bar{w}_k$ denote the corresponding column means. Expanding each rank-one component gives

$$\Pi^{\mathcal{A}_D} = \sum_{k=1}^{\ell}\Big\{Du_k^c \circ v_k^c \circ w_k^c + \bar{w}_k(Du_k^c \circ v_k^c \circ \iota_T) + \bar{v}_k(Du_k^c \circ \iota_M \circ w_k^c) + \bar{u}_k(\iota_N \circ v_k^c \circ w_k^c)\Big\}.$$

The last three terms are, respectively, group–teacher interactions constant across periods, group–period interactions constant across teachers, and teacher–period interactions constant across groups. Each is orthogonal to the additive space because its two varying factors have zero means under the corresponding observation weights. Thus additive profiling retains these two-way interactions together with the fully centered three-way interaction; full three-way centering retains only the first term. In general,

$$M_{\mathcal{A}_D} \neq M_{\iota_N} \otimes M_{\iota_M} \otimes M_{\iota_T}.$$

Fitting an unrestricted rank-ℓ CP tensor to the outcome and regressors transformed by $M_{\iota_N} \otimes M_{\iota_M} \otimes M_{\iota_T}$ would generally define a different estimator. The fully centered sample alone does not identify the raw factor means or the additive coefficients.

Consistency of the fully centered slope estimator requires identification from the regressor variation surviving $M_{\iota_N} \otimes M_{\iota_M} \otimes M_{\iota_T}$, which is not generally implied by Assumption 8(ii). The additional contrasts retained by profiling may strengthen identification and improve precision.

Remark: Average consistency and the stochastic expansions of the model with additive fixed effects as in Lemma 1 and Propositions 1 and 2 can be derived from Lemma 2 using the same arguments as in the corresponding proofs; the details are therefore omitted.

7.2 Unbalanced Panel

In matched student–teacher data, only a subset of the potential student–teacher–period observations is available. Let

$$\mathcal{I} = \{1, \dots, N\} \times \{1, \dots, M\} \times \{1, \dots, T\}, \qquad \Omega \subseteq \mathcal{I}, \qquad n_\Omega = |\Omega| \geqslant 1.$$

The observation pattern may be random. For $(i, j, t) \in \Omega$, consider

$$y_{ijt} = x_{ijt}'\delta + \pi_{ijt} + \varepsilon_{ijt}, \qquad \pi_{ijt} = \sum_{k=1}^{\ell} u_{g(i),k} v_{jk} w_{tk}. \tag{71}$$

Write $\Pi(U, V, W) = \sum_{k=1}^{\ell} (Du_k \circ v_k \circ w_k)$ and $\pi(U, V, W) = \operatorname{vec}(\Pi(U, V, W))$. Let $R_\Omega \in \{0, 1\}^{n_\Omega \times NMT}$ select the distinct observed entries in the order induced by vec, so $R_\Omega R_\Omega' = I_{n_\Omega}$. Define

$$y_\Omega = R_\Omega \operatorname{vec}(Y), \qquad X_\Omega = R_\Omega X_{(4)}', \qquad e_\Omega = R_\Omega \operatorname{vec}(E), \qquad z_\Omega = R_\Omega \operatorname{vec}(Z),$$

for any third-order tensor Z of relevant dimensions. We have

$$y_\Omega = X_\Omega \delta + \pi_\Omega(U, V, W) + e_\Omega,$$

where

$$\pi_\Omega(U, V, W) = R_\Omega \pi(U, V, W).$$

Only the selected coordinates are used in estimation. We distinguish convergence of the fitted interactive component on Ω from recovery of the full latent tensor, including its unobserved entries.

For a constant $c_U > 0$ independent of the panel dimensions, use the normalized factor space

$$\begin{aligned} \Theta_\ell = \Big\{ &(\widetilde{U}, \widetilde{V}, \widetilde{W}) \in \mathbb{R}^{G \times \ell} \times \mathbb{R}^{M \times \ell} \times \mathbb{R}^{T \times \ell} : \\ &\operatorname{diag}\left(\frac{1}{M}\widetilde{V}'\widetilde{V}\right) = \operatorname{diag}\left(\frac{1}{T}\widetilde{W}'\widetilde{W}\right) = \iota_\ell, \lambda_{\max}\left(\frac{1}{N}\widetilde{U}' D' D \widetilde{U}\right) \leqslant c_U \Big\}. \end{aligned} \tag{72}$$

The true factor matrices admit a representation in Θ_ℓ. Define

$$S_\Omega(\delta, U, V, W) = \frac{1}{n_\Omega} \sum_{(i,j,t)\in\Omega} \left[y_{ijt} - x'_{ijt}\delta - \sum_{k=1}^{\ell} u_{g(i),k} v_{jk} w_{tk} \right]^2 = \frac{1}{n_\Omega} \|y_\Omega - X_\Omega \delta - \pi_\Omega(U, V, W)\|^2, \tag{73}$$

and let

$$(\hat{\delta}, \hat{U}, \hat{V}, \hat{W}) \in \operatorname*{argmin}_{\delta\in\mathbb{R}^p,\ (U,V,W)\in\Theta_\ell} S_\Omega(\delta, U, V, W) \tag{74}$$

be a global minimizer. Concentrating over δ gives

$$\min_{\delta\in\mathbb{R}^p} \|y_\Omega - X_\Omega\delta - \pi_\Omega(U, V, W)\|^2 = \|M_{X_\Omega}\{y_\Omega - \pi_\Omega(U, V, W)\}\|^2.$$

Whenever $X'_\Omega X_\Omega$ is nonsingular, the first order condition for $\hat{\delta}$ is given by

$$\hat{\delta} = \left(\sum_{(i,j,t)\in\Omega} x_{ijt} x'_{ijt} \right)^{-1} \sum_{(i,j,t)\in\Omega} x_{ijt} \left[y_{ijt} - \sum_{k=1}^{\ell} \hat{u}_{g(i),k} \hat{v}_{jk} \hat{w}_{tk} \right]. \tag{75}$$

Define the observed slices

$$\Omega_g^{(u)} = \{(i,j,t) \in \Omega : g(i) = g\}, \qquad \Omega_j^{(v)} = \{(i,j',t) \in \Omega : j' = j\}, \qquad \Omega_t^{(w)} = \{(i,j,t') \in \Omega : t' = t\}.$$

At a minimizer for which $\lambda_{\max}(\hat{U}'D'D\hat{U}/N) < c_U$, compensating column rescalings permit unrestricted factor perturbations while maintaining the normalizations. If the corresponding observed block Gram matrices are nonsingular, the factor normal equations consequently yield

$$\begin{aligned}
\hat{u}_g &= \left[\sum_{(i,j,t)\in\Omega_g^{(u)}} (\hat{v}_j \odot \hat{w}_t)(\hat{v}_j \odot \hat{w}_t)' \right]^{-1} \sum_{(i,j,t)\in\Omega_g^{(u)}} (\hat{v}_j \odot \hat{w}_t)(y_{ijt} - x'_{ijt}\hat{\delta}), \\
\hat{v}_j &= \left[\sum_{(i,j,t)\in\Omega_j^{(v)}} (\hat{u}_{g(i)} \odot \hat{w}_t)(\hat{u}_{g(i)} \odot \hat{w}_t)' \right]^{-1} \sum_{(i,j,t)\in\Omega_j^{(v)}} (\hat{u}_{g(i)} \odot \hat{w}_t)(y_{ijt} - x'_{ijt}\hat{\delta}), \\
\hat{w}_t &= \left[\sum_{(i,j,t)\in\Omega_t^{(w)}} (\hat{u}_{g(i)} \odot \hat{v}_j)(\hat{u}_{g(i)} \odot \hat{v}_j)' \right]^{-1} \sum_{(i,j,t)\in\Omega_t^{(w)}} (\hat{u}_{g(i)} \odot \hat{v}_j)(y_{ijt} - x'_{ijt}\hat{\delta}).
\end{aligned} \tag{76}$$

The same formulas define unrestricted conditional least-squares updates when the other blocks are held fixed.

Assumption 9 (Uniform slope identification in the observed sample)**.** *For $\widetilde{U} \in \mathbb{R}^{G\times\ell}$ and $\widetilde{V} \in \mathbb{R}^{M\times\ell}$, define*

$$B_\Omega(\widetilde{U}, \widetilde{V}) = R_\Omega \Big[(D\widetilde{U} * \widetilde{V}) \otimes I_T,\ D \otimes I_M \otimes W \Big], \tag{77}$$

where W is the true time-factor matrix. The observed regressors satisfy

$$\frac{1}{n_\Omega}\|X_\Omega\|_F^2 = O_p(1).$$

There exists a constant $c_{\delta,\Omega} > 0$, independent of the panel dimensions, such that, with probability approaching one,

$$\inf_{\substack{\widetilde{U}\in\mathbb{R}^{G\times\ell}\\ \widetilde{V}\in\mathbb{R}^{M\times\ell}}} \lambda_{\min}\left[\frac{1}{n_\Omega}X_\Omega' M_{B_\Omega(\widetilde{U},\widetilde{V})}X_\Omega\right] \geqslant c_{\delta,\Omega}. \tag{78}$$

The annihilator is defined using the Moore–Penrose inverse, so candidate loading matrices need not have full column rank.

Assumption 10 (Observed-sample exogeneity and error non-pervasiveness)**.** *Let*

$$\mathcal{F}_\Omega = \sigma(\Omega, D, U, V, W, X_\Omega).$$

The observed errors satisfy $\mathbb{E}[e_\Omega \mid \mathcal{F}_\Omega] = 0$. For a constant $c_\varepsilon < \infty$ independent of the panel dimensions, with probability approaching one,

$$\left\|\mathbb{E}[e_\Omega e_\Omega' \mid \mathcal{F}_\Omega]\right\| \leqslant c_\varepsilon. \tag{79}$$

For a positive integer r, define

$$\mathcal{L}_r^D = \left\{\sum_{k=1}^r (Da_k \circ b_k \circ c_k) : a_k \in \mathbb{R}^G,\ b_k \in \mathbb{R}^M,\ c_k \in \mathbb{R}^T\right\}, \quad \|E\|_{\Omega,r} = \sup_{\substack{Z\in\mathcal{L}_r^D\\ \|z_\Omega\|>0}} \frac{|e_\Omega' z_\Omega|}{\|z_\Omega\|},$$

where $z_\Omega = R_\Omega \operatorname{vec}(Z)$. For the fixed rank ℓ,

$$\|E\|_{\Omega,2\ell} = O_p\left(\sqrt{G+M+T}\right). \tag{80}$$

Assumption 9 supplies global slope separation from candidate interactive components in the observed sample. Assumption 10 controls a supremum in the observed norm and therefore applies to an estimated low-rank difference that depends on the errors. Conditioning on Ω makes the exogeneity requirement conditional on the realized observation pattern. The conditional mean-zero restriction remains an assumption on the selection mechanism. Neither assumption alone identifies interactive-component values outside the observed sample.

Assumption 11 (Sampling stability for the full interactive component)**.** *There exists a constant $c_\Pi > 0$, independent of the panel dimensions, such that, with probability approaching one,*

$$\frac{1}{n_\Omega}\|R_\Omega(\pi(\widetilde{U},\widetilde{V},\widetilde{W}) - \pi(U,V,W))\|^2 \geqslant \frac{c_\Pi}{NMT}\|\pi(\widetilde{U},\widetilde{V},\widetilde{W}) - \pi(U,V,W)\|^2 \tag{81}$$

simultaneously for all $(\widetilde{U},\widetilde{V},\widetilde{W}) \in \Theta_\ell$.

Assumption 11 is imposed only for conclusions about the full latent component. It rules out feasible components that differ substantially from the truth but are indistinguishable, or nearly indistinguishable, on the observed matches. The condition concerns differences from the true component, not every tensor of a given rank.

Lemma 3 (Convergence with an unbalanced sample)**.** *Maintain the model, membership conditions, normalized parameter space, and true-parameter feasibility specified above. Let p, ℓ be fixed, and suppose that*

$$\rho_\Omega = \sqrt{\frac{G+M+T}{n_\Omega}}, \quad \rho_\Omega \to_p 0. \tag{82}$$

Let $(\hat{\delta}, \hat{U}, \hat{V}, \hat{W})$ be a global minimizer in (74)*, and suppose that Assumptions 9 and 10 hold.*

(i) Observed-sample convergence. *The slope and observed interactive component satisfy*

$$\|\hat{\delta} - \delta\| + \frac{\|R_\Omega(\hat{\pi} - \pi)\|}{\sqrt{n_\Omega}} = O_p(\rho_\Omega) = o_p(1). \tag{83}$$

(ii) Full-component convergence. *If Assumption 11 also holds, then*

$$\frac{\|\hat{\Pi} - \Pi\|_F}{\sqrt{NMT}} = O_p(\rho_\Omega) = o_p(1). \tag{84}$$

(iii) Full joint loading-space convergence. *Suppose Assumption 11 holds and, for a constant $c_f > 0$ independent of the panel dimensions, with probability approaching one,*

$$\frac{1}{N}U'D'DU \succeq c_f I_\ell, \qquad \frac{1}{M}V'V \succeq c_f I_\ell, \qquad \frac{1}{T}W'W \succeq c_f I_\ell. \tag{85}$$

Then

$$\begin{aligned} \frac{1}{\sqrt{MT}}\|M_{\hat{V}*\hat{W}}(V*W)\|_F &= O_p(\rho_\Omega), \\ \frac{1}{\sqrt{NT}}\|M_{D\hat{U}*\hat{W}}(DU*W)\|_F &= O_p(\rho_\Omega), \\ \frac{1}{\sqrt{NM}}\|M_{D\hat{U}*\hat{V}}(DU*V)\|_F &= O_p(\rho_\Omega). \end{aligned} \tag{86}$$

The observed-sample conclusion requires neither Assumption 11 nor the strong-factor bounds in (85)*. All conclusions are aggregate-norm statements, not uniform fixed-index consistency results.*

Remark: Observed fit and full-tensor recovery. Part (i) establishes convergence of the interactive fitted values on the observed matches. Recovery of the full latent tensor additionally requires Assumption 11, which ensures that feasible departures from the true component remain detectable in the observed sample. This condition concerns the geometry of the matches as well as their number. Observing every group, teacher, and period does not by itself establish sampling stability, since disconnected match patterns can leave unobserved interactions unidentified.

Remark: Random matching and conditional exogeneity. The observation pattern may depend on the regressors and latent factors, provided $\mathbb{E}[e_\Omega \mid \mathcal{F}_\Omega] = 0$. Conditioning on Ω makes this exogeneity requirement explicit for the realized matches. The low-rank error supremum is a separate stochastic restriction: it controls error-dependent fitted tensor differences in the observed norm and must be imposed or established in addition to the conditional covariance bound.

8 Monte Carlo Simulations

We assess the finite-sample performance of the proposed grouped three-way estimator in a model with a grand mean, additive fixed effects, and grouped trilinear interactive effects. We compare four estimators: grouped three-way least squares, ungrouped three-way least squares, Freeman's linear-projection estimator, and a flattened matrix interactive-fixed-effects estimator. The three least-squares specifications form a nested sequence of correctly specified conditional-mean models, whereas Freeman's estimator provides a projection-based alternative. The comparison examines the consequences of exploiting the observed grouping, imposing the trilinear structure, and estimating the slope through joint least squares rather than projection.

8.1 Data-Generating Process

The observations satisfy

$$y_{ijt} = x'_{ijt}\delta_0 + \kappa_0 + \mu_{0,g(i)} + \nu_{0j} + \xi_{0t} + \pi_{0,ijt} + \varepsilon_{ijt}, \qquad \pi_{0,ijt} = \sum_{k=1}^{\ell} u_{0,g(i),k} v_{0,jk} w_{0,tk}. \tag{87}$$

We set $p = 2$ and $\ell = 2$. We assume that G divides N and construct equally sized groups:

$$N_g = \frac{N}{G}, \qquad g(i) = g \quad \text{for} \quad i = (g-1)N_g + 1, \ldots, gN_g, \qquad g = 1, \ldots, G.$$

We independently draw the primitive additive effects as

$$\mu^{\circ}_{0g},\ \nu^{\circ}_{0j},\ \xi^{\circ}_{0t} \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0,1)$$

and define

$$\mu_{0g} = \mu^{\circ}_{0g} - \frac{1}{G}\sum_{h=1}^{G} \mu^{\circ}_{0h},$$

$$\nu_{0j} = \nu^{\circ}_{0j} - \frac{1}{M}\sum_{j'=1}^{M} \nu^{\circ}_{0j'},$$

$$\xi_{0t} = \xi^{\circ}_{0t} - \frac{1}{T}\sum_{t'=1}^{T} \xi^{\circ}_{0t'}.$$

Consequently,

$$\sum_{g=1}^{G} N_g \mu_{0g} = \sum_{j=1}^{M} \nu_{0j} = \sum_{t=1}^{T} \xi_{0t} = 0.$$

The grand mean is $\kappa_0 = 1$.

For each $k = 1, \ldots, \ell$, we draw the primitive factor variables independently:

$$u^{\circ}_{0,gk},\ v^{\circ}_{0,jk},\ w^{\circ}_{0,tk} \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0,1).$$

To impose the scale normalizations without changing the generated interactive component, let

$$s_{v,k} = \left(\frac{1}{M}\sum_{j=1}^{M} (v^{\circ}_{0,jk})^2\right)^{1/2}, \qquad s_{w,k} = \left(\frac{1}{T}\sum_{t=1}^{T} (w^{\circ}_{0,tk})^2\right)^{1/2},$$

and set

$$u_{0,gk} = s_{v,k}s_{w,k}u^{\circ}_{0,gk}, \qquad v_{0,jk} = \frac{v^{\circ}_{0,jk}}{s_{v,k}}, \qquad w_{0,tk} = \frac{w^{\circ}_{0,tk}}{s_{w,k}}.$$

Then

$$\frac{1}{M}\sum_{j=1}^{M} v^2_{0,jk} = \frac{1}{T}\sum_{t=1}^{T} w^2_{0,tk} = 1, \qquad u_{0,gk}v_{0,jk}w_{0,tk} = u^{\circ}_{0,gk}v^{\circ}_{0,jk}w^{\circ}_{0,tk}.$$

No sample mean-zero restriction is imposed on the columns of U_0, V_0, or W_0.

The regressors are generated according to

$$x_{ijt,q} = m_q + c_q\pi_{0,ijt} + \eta_{ijt,q}, \qquad q = 1, 2, \tag{88}$$

where $\eta_{ijt,q} \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0,1)$. We set

$$\delta_0 = \begin{pmatrix} 1 \\ 3 \end{pmatrix}, \qquad m_1 = m_2 = c_1 = c_2 = 1.$$

Thus, the regressors are correlated with the interactive fixed effects but contain independent variation through their idiosyncratic components.

We consider two error specifications. In the first specification,

$$\varepsilon_{ijt} \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0,1).$$

In the second specification, we independently draw $\widetilde{\varepsilon}_{ijt} \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0,1)$ for $t = 0, \ldots, T$ and define

$$\varepsilon_{ijt} = \frac{1}{\sqrt{2}}\left(\widetilde{\varepsilon}_{ijt} + \widetilde{\varepsilon}_{ij,t-1}\right), \qquad t = 1, \ldots, T. \tag{89}$$

This moving-average specification satisfies

$$\operatorname{Var}(\varepsilon_{ijt}) = 1, \qquad \operatorname{Corr}(\varepsilon_{ijt}, \varepsilon_{ij,t-1}) = \frac{1}{2}, \qquad t = 2, \ldots, T,$$

with zero autocorrelations at lags greater than one and independence across distinct pairs (i, j). Both error specifications therefore have the same marginal variance. All primitive Gaussian draws used to generate the additive effects, interactive factors, regressor innovations, and error innovations are mutually independent.

8.2 Four Estimators

The first two estimators impose $\ell = 2$ trilinear components, and the flattened estimator imposes matrix rank at most $\ell = 2$ on its interactive component. Freeman's estimator retains two loading directions in each centered unfolding. Under the Gaussian DGP, these are the true centered unfolding ranks whenever $G, M, T \geqslant 3$. All four estimators are applied to the same observations in each replication.

1. **Grouped three-way least squares.**

   The proposed estimator uses the observed group assignment and jointly fits

$$y_{ijt} = x'_{ijt}\delta + \kappa + \mu_{g(i)} + \nu_j + \xi_t + \sum_{k=1}^{\ell} a_{g(i),k}b_{jk}f_{tk} + e_{ijt}. \tag{90}$$

We denote the slope component of the joint least-squares estimator by $\widehat{\delta}_{\mathrm{G}}$. The first-mode loading matrix has dimension $G \times \ell$, and the additive effects satisfy

$$\sum_{g=1}^{G} N_g \mu_g = \sum_{j=1}^{M} \nu_j = \sum_{t=1}^{T} \xi_t = 0.$$

The factor scales are normalized so that

$$\frac{1}{M}\sum_{j=1}^{M} b_{jk}^2 = \frac{1}{T}\sum_{t=1}^{T} f_{tk}^2 = 1, \qquad k = 1, \ldots, \ell,$$

with the corresponding scales absorbed into the first-mode loadings. No mean-zero restrictions are imposed on the factor columns. The additive effects and trilinear component are estimated jointly, or equivalently the additive effects are profiled from the joint least-squares criterion.

2. **Ungrouped three-way least squares.**

   The second estimator ignores the group restriction and jointly fits

$$y_{ijt} = x_{ijt}'\delta + \kappa + \mu_i + \nu_j + \xi_t + \sum_{k=1}^{\ell} a_{ik} b_{jk} f_{tk} + e_{ijt}. \tag{91}$$

   We denote its slope estimator by $\widehat{\delta}_{\mathrm{U}}$. This model contains the DGP under

$$\mu_i = \mu_{0,g(i)}, \qquad a_{ik} = u_{0,g(i),k}, \qquad b_{jk} = v_{0,jk}, \qquad f_{tk} = w_{0,tk}.$$

   The estimator uses the same factor scale normalizations as the grouped estimator, while the additive effects satisfy

$$\sum_{i=1}^{N} \mu_i = \sum_{j=1}^{M} \nu_j = \sum_{t=1}^{T} \xi_t = 0.$$

   Relative to the grouped specification, this model estimates $N\ell$ rather than $G\ell$ first-mode loadings and $N-1$ rather than $G-1$ first-mode additive effects. It therefore introduces $(N-G)(\ell+1)$ additional first-mode nuisance parameters. The comparison between $\widehat{\delta}_{\mathrm{G}}$ and $\widehat{\delta}_{\mathrm{U}}$ measures the gains from imposing the observed grouping on both the additive effects and the interactive loadings.

3. **Freeman linear projection.**

   The third estimator implements the linear-projection construction in Section 5.3 of Freeman (2026), using separate preliminary matrix interactive fixed effects regressions to estimate the three loading spaces. It does not use the group assignment or jointly fit a trilinear decomposition.

   Let $X_q \in \mathbb{R}^{N\times M\times T}$ denote the tensor containing regressor q. We first center the outcome and each regressor in all three modes:

$$\begin{aligned} Y^c &= Y \times_1 M_{\iota_N} \times_2 M_{\iota_M} \times_3 M_{\iota_T}, \\ X_q^c &= X_q \times_1 M_{\iota_N} \times_2 M_{\iota_M} \times_3 M_{\iota_T}, \qquad q = 1, \ldots, p. \end{aligned} \tag{92}$$

This removes the additive effects while preserving a trilinear representation of the remaining interactive component:

$$\pi^c_{0,ijt} = \sum_{k=1}^{\ell} (u_{0,g(i),k} - \bar{u}_{0k})(v_{0,jk} - \bar{v}_{0k})(w_{0,tk} - \bar{w}_{0k}),$$

where $\bar{u}_{0k}$, $\bar{v}_{0k}$, and $\bar{w}_{0k}$ are the averages over groups, the second mode, and the third mode, respectively.

Write $n = NMT$, $d_1 = N$, $d_2 = M$, and $d_3 = T$. For each $m = 1, 2, 3$, let $Y^c_{(m)}$ and $X^c_{q,(m)}$ denote the mode-m unfoldings, with dimension $d_m \times (n/d_m)$. The preliminary estimates solve

$$(\widetilde{\delta}_m, \widehat{B}_m) \in \underset{\theta \in \mathbb{R}^p,\ B:\mathrm{rank}(B) \leqslant \ell}{\operatorname{argmin}} \frac{1}{n} \left\| Y^c_{(m)} - \sum_{q=1}^{p} \theta_q X^c_{q,(m)} - B \right\|_F^2, \tag{93}$$

where $B \in \mathbb{R}^{d_m \times (n/d_m)}$. For each mode, form

$$\widehat{R}_m = Y^c_{(m)} - \sum_{q=1}^{p} \widetilde{\delta}_{m,q} X^c_{q,(m)}.$$

Let $\widehat{L}_m \in \mathbb{R}^{d_m \times \ell}$ contain the leading ℓ left singular vectors of $\widehat{R}_m$, normalized by $\widehat{L}'_m \widehat{L}_m = I_\ell$, and define

$$\widehat{M}_m = I_{d_m} - \widehat{L}_m \widehat{L}'_m, \qquad m = 1, 2, 3.$$

The singular vectors are extracted before subtracting the fitted low-rank component $\widehat{B}_m$.

We then transform the centered data:

$$\begin{aligned} \check{Y} &= Y^c \times_1 \widehat{M}_1 \times_2 \widehat{M}_2 \times_3 \widehat{M}_3, \\ \check{X}_q &= X^c_q \times_1 \widehat{M}_1 \times_2 \widehat{M}_2 \times_3 \widehat{M}_3, \qquad q = 1, \ldots, p. \end{aligned} \tag{94}$$

Writing $\check{x}_{ijt} = ((\check{X}_1)_{ijt}, \ldots, (\check{X}_p)_{ijt})'$, the final slope estimator is

$$\widehat{\delta}_{\mathrm{FW}} = \left(\sum_{i=1}^{N} \sum_{j=1}^{M} \sum_{t=1}^{T} \check{x}_{ijt} \check{x}'_{ijt} \right)^{-1} \sum_{i=1}^{N} \sum_{j=1}^{M} \sum_{t=1}^{T} \check{x}_{ijt} \check{y}_{ijt}. \tag{95}$$

No intercept is included in this final regression.

This comparator is the full-sample linear-projection point estimator, without kernel smoothing or sample splitting. The preliminary centering in (92) also removes pairwise additive variation that is constant in one mode and is therefore stronger than profiling only the additive effects specified in (90) or (91). The two joint three-way least-squares estimators are fitted to the original data and do not use this preliminary three-mode centering.

4. **Flattened matrix IFE.**

The fourth estimator treats (i, j) as a single cross-sectional index and fits

$$y_{ijt} = x'_{ijt}\delta + \kappa + \alpha_{ij} + \xi_t + \sum_{k=1}^{\ell} \lambda_{ij,k} f_{tk} + e_{ijt}. \tag{96}$$

The model permits unrestricted pair-specific intercepts and loadings, while retaining a rank-ℓ factor structure over time. It contains the DGP under

$$\alpha_{ij} = \mu_{0,g(i)} + \nu_{0j}, \qquad \lambda_{ij,k} = u_{0,g(i),k} v_{0,jk}, \qquad f_{tk} = w_{0,tk}.$$

Unlike the ungrouped trilinear specification, it does not impose the additive decomposition of α_{ij} or the separability of $\lambda_{ij,k}$ across i and j.

Let $\mathcal{Y}, \mathcal{X}_q \in \mathbb{R}^{NM \times T}$ denote the outcome and regressor matrices with (i, j) indexing rows. Profiling out the pair-specific intercepts and time effects gives the slope estimator

$$\widehat{\delta}_{\mathrm{F}} \in \operatorname*{argmin}_{\theta \in \mathbb{R}^p} \min_{\operatorname{rank}(B) \leqslant \ell} \frac{1}{NMT} \left\| M_{\iota_{NM}} \left(\mathcal{Y} - \sum_{q=1}^{p} \theta_q \mathcal{X}_q \right) M_{\iota_T} - B \right\|_F^2. \tag{97}$$

Here $B \in \mathbb{R}^{NM \times T}$ represents the interactive component after the additive transformation.

The grouped, ungrouped, and flattened specifications impose successively weaker restrictions on the conditional mean. Freeman's estimator differs from these joint least-squares estimators in both its nuisance estimation and the variation retained by its transformations. Consequently, differences in finite-sample performance reflect not only the imposed grouping and separability restrictions but also the estimation procedure.

8.3 Numerical Implementation and Performance Measures

We consider the sample-size configurations

$$(N, M, T, G) \in \{(24, 24, 8, 4), (48, 24, 8, 6), (48, 48, 8, 8), (48, 48, 12, 8)\}.$$

For each configuration and each error specification, we generate $R = 1{,}000$ Monte Carlo replications. All four estimators are applied to the same simulated observations within each replication.

The grouped and ungrouped three-way estimators are computed by alternating least squares. Conditional on the current slope and additive effects, we update the three loading matrices sequentially by linear least squares. Conditional on the updated interactive component, we jointly update the slope and additive effects by profiling out the appropriate additive-effect space. The second- and third-mode loading columns are rescaled to have sample mean square one, with compensating scales absorbed into the first-mode loadings. These normalizations preserve the fitted interactive component, and no sample mean-zero restrictions are imposed on the CP loading columns.

Each CP regression is computed from 20 starting values. For the first start, we fit the regression containing only the slope and the specified additive effects and extract the leading ℓ left singular vectors from the second- and third-mode unfoldings of its residual tensor.

These vectors initialize the corresponding loading matrices, while the first-mode loadings are initialized by conditional least squares. The remaining starts use orthonormalized independent Gaussian draws for the second- and third-mode loading matrices, followed by the same conditional first-mode update. Only the grouped estimator uses the observed group assignment.

The flattened matrix IFE estimator and each of Freeman's three preliminary matrix IFE regressions alternate between a slope update and a rank-ℓ truncated singular value decomposition of the regression residual matrix. Each matrix regression also uses 20 starts: the first uses the ordinary least-squares slope from the relevant transformed data, the second uses a zero slope vector, and the remaining starts use Gaussian perturbations of the ordinary least-squares slope, scaled by the residual and regressor root mean squares. Freeman's final slope is obtained by a single least-squares regression after applying the estimated loading-space projections; no kernel bandwidth or sample splitting is used.

Slope-estimation performance. Let $\widehat{\delta}_a^{(r)}$ denote estimator $a \in \{\mathrm{G}, \mathrm{U}, \mathrm{FW}, \mathrm{F}\}$ in replication r. For coefficient $q = 1, 2$, let

$$\overline{\delta}_{a,q} = \frac{1}{R} \sum_{r=1}^{R} \widehat{\delta}_{a,q}^{(r)}.$$

We report the empirical bias, standard deviation, and root mean squared error:

$$\begin{aligned}
\mathrm{Bias}_{a,q} &= \frac{1}{R} \sum_{r=1}^{R} \left(\widehat{\delta}_{a,q}^{(r)} - \delta_{0,q} \right), \\
\mathrm{SD}_{a,q} &= \left\{ \frac{1}{R-1} \sum_{r=1}^{R} \left(\widehat{\delta}_{a,q}^{(r)} - \overline{\delta}_{a,q} \right)^2 \right\}^{1/2}, \\
\mathrm{RMSE}_{a,q} &= \left\{ \frac{1}{R} \sum_{r=1}^{R} \left(\widehat{\delta}_{a,q}^{(r)} - \delta_{0,q} \right)^2 \right\}^{1/2}.
\end{aligned} \tag{98}$$

Table 1: Monte Carlo performance of the slope estimators

Estimator	$\widehat{\delta}_1$				$\widehat{\delta}_2$				R_{use}
	Bias	SD	RMSE	Ratio	Bias	SD	RMSE	Ratio	
i.i.d. errors; $N = 24$, $M = 24$, $T = 8$, $G = 4$									
Grouped CP	-0.0004	0.0146	0.0146	1.0000	-0.0002	0.0148	0.0148	1.0000	1000
Ungrouped CP	-0.0001	0.0148	0.0147	1.0114	0.0001	0.0149	0.0149	1.0056	1000
Freeman linear	-0.0007	0.0218	0.0218	1.4952	0.0005	0.0214	0.0214	1.4425	1000
Flattened IFE	0.0013	0.0195	0.0195	1.3382	0.0021	0.0192	0.0193	1.3026	1000
i.i.d. errors; $N = 48$, $M = 24$, $T = 8$, $G = 6$									
Grouped CP	0.0001	0.0105	0.0105	1.0000	0.0003	0.0106	0.0106	1.0000	1000
Ungrouped CP	0.0002	0.0105	0.0105	1.0041	0.0003	0.0106	0.0106	1.0037	1000
Freeman linear	-0.0003	0.0148	0.0148	1.4131	0.0001	0.0145	0.0145	1.3653	1000
Flattened IFE	0.0003	0.0132	0.0132	1.2606	0.0006	0.0132	0.0132	1.2416	1000
i.i.d. errors; $N = 48$, $M = 48$, $T = 8$, $G = 8$									
Grouped CP	0.0003	0.0074	0.0074	1.0000	-0.0008	0.0075	0.0075	1.0000	1000
Ungrouped CP	0.0003	0.0074	0.0074	1.0038	-0.0007	0.0075	0.0075	1.0030	1000
Freeman linear	0.0001	0.0099	0.0099	1.3360	-0.0007	0.0099	0.0100	1.3258	1000
Flattened IFE	0.0004	0.0093	0.0093	1.2598	-0.0006	0.0096	0.0096	1.2842	1000
i.i.d. errors; $N = 48$, $M = 48$, $T = 12$, $G = 8$									
Grouped CP	-0.0002	0.0060	0.0060	1.0000	-0.0001	0.0061	0.0061	1.0000	1000
Ungrouped CP	-0.0001	0.0060	0.0060	1.0045	-0.0001	0.0061	0.0061	1.0042	1000
Freeman linear	-0.0002	0.0076	0.0076	1.2608	0.0000	0.0075	0.0075	1.2310	1000
Flattened IFE	-0.0001	0.0072	0.0072	1.1900	0.0001	0.0071	0.0071	1.1647	1000

Notes: Bias, SD, and RMSE are computed from converged replications with finite slope vectors. R_{use} is the number of usable replications; failures and finite nonconverged fits are reported in the numerical diagnostics and retained in the raw results. Ratio is the estimator RMSE divided by the grouped-estimator RMSE, recomputed on their common usable replications. True values: $\delta_{0,1} = 1$, $\delta_{0,2} = 3$.

Results under i.i.d. errors. Table 1 shows that grouped CP has the lowest RMSE for both slope coefficients in every reported design. Its bias is small relative to its standard deviation, and its RMSE declines approximately at the $1/\sqrt{NMT}$ rate: the values of $\sqrt{NMT}$ times RMSE lie between approximately 0.99 and 1.02. Ungrouped CP performs almost identically, with RMSE only 0.3%–1.1% larger. Thus, in these designs, the additional gain from grouping is modest.

The larger precision gains arise when comparing CP estimation with the Freeman linear and flattened IFE estimators. Their RMSEs are approximately 23%–50% and 16%–34% larger than those of grouped CP, respectively. The flattened IFE estimator has lower RMSE than the Freeman linear estimator in every i.i.d. design. Because biases are small relative to standard deviations, these differences primarily reflect differences in sampling variability. The results support the precision benefits of exploiting the separable three-way structure.

Table 2: Monte Carlo performance of the slope estimators

Estimator	$\widehat{\delta}_1$				$\widehat{\delta}_2$				R_{use}
	Bias	SD	RMSE	Ratio	Bias	SD	RMSE	Ratio	
MA(1) errors; $N = 24$, $M = 24$, $T = 8$, $G = 4$									
Grouped CP	-0.0006	0.0144	0.0145	1.0000	-0.0003	0.0147	0.0147	1.0000	1000
Ungrouped CP	0.0010	0.0247	0.0247	1.7060	0.0014	0.0241	0.0241	1.6334	1000
Freeman linear	0.0010	0.0184	0.0184	1.2745	0.0017	0.0186	0.0186	1.2647	1000
Flattened IFE	0.0721	0.0914	0.1164	8.0500	0.0726	0.0915	0.1168	7.9213	1000
MA(1) errors; $N = 48$, $M = 24$, $T = 8$, $G = 6$									
Grouped CP	0.0003	0.0101	0.0101	1.0000	0.0000	0.0104	0.0104	1.0000	1000
Ungrouped CP	0.0004	0.0102	0.0102	1.0019	0.0002	0.0104	0.0104	1.0037	1000
Freeman linear	0.0004	0.0130	0.0130	1.2798	0.0002	0.0121	0.0121	1.1661	1000
Flattened IFE	0.0723	0.0971	0.1210	11.9342	0.0724	0.0967	0.1208	11.6442	1000
MA(1) errors; $N = 48$, $M = 48$, $T = 8$, $G = 8$									
Grouped CP	0.0003	0.0076	0.0076	1.0000	-0.0005	0.0074	0.0074	1.0000	1000
Ungrouped CP	0.0003	0.0076	0.0076	0.9994	-0.0004	0.0074	0.0074	1.0015	1000
Freeman linear	0.0004	0.0088	0.0088	1.1541	-0.0002	0.0087	0.0087	1.1837	1000
Flattened IFE	0.0634	0.0873	0.1079	14.2169	0.0627	0.0887	0.1086	14.7104	1000
MA(1) errors; $N = 48$, $M = 48$, $T = 12$, $G = 8$									
Grouped CP	-0.0001	0.0058	0.0058	1.0000	-0.0001	0.0061	0.0061	1.0000	1000
Ungrouped CP	0.0000	0.0058	0.0058	1.0003	-0.0001	0.0061	0.0061	1.0049	1000
Freeman linear	0.0000	0.0069	0.0069	1.1798	0.0000	0.0069	0.0069	1.1254	1000
Flattened IFE	0.0139	0.0150	0.0204	3.5015	0.0139	0.0152	0.0206	3.3736	1000

Notes: Bias, SD, and RMSE are computed from converged replications with finite slope vectors. R_{use} is the number of usable replications; failures and finite nonconverged fits are reported in the numerical diagnostics and retained in the raw results. Ratio is the estimator RMSE divided by the grouped-estimator RMSE, recomputed on their common usable replications. True values: $\delta_{0,1} = 1$, $\delta_{0,2} = 3$.

Results under MA(1) errors. Table 2 shows that grouped CP maintains small bias and declining RMSE under serial dependence. Grouped and ungrouped CP have essentially identical RMSEs in the three larger designs; ungrouped CP has a slightly lower RMSE for the first coefficient when $N = M = 48$ and $T = 8$. In the smallest design, however, ungrouped CP RMSE is approximately 63%–71% larger than grouped CP RMSE. Freeman linear RMSE is approximately 13%–28% larger than grouped CP RMSE across the reported MA(1) designs.

The flattened IFE estimator exhibits substantially larger bias and dispersion, with RMSE approximately 3.4–14.7 times that of grouped CP. Squared bias accounts for approximately 33%–46% of its MSE, so both bias and sampling variability contribute to the deterioration. At $T = 8$, increasing the cross-sectional dimensions produces little reduction in its bias. Increasing T from 8 to 12, holding $N = M = 48$ and $G = 8$, sharply reduces both bias and standard deviation. This pattern is consistent with difficulties associated with estimating unrestricted interactive effects under serial dependence when the time dimension is short. The Freeman linear estimator outperforms flattened IFE in every MA(1) design, reversing their ranking under i.i.d. errors.

9 Empirical Application: Heterogeneous Teacher Value Added

We use student-level administrative records from the National Center for Teacher Effectiveness Main Study, ICPSR 36095. The study collected data on elementary mathematics instruction during the 2010–2011, 2011–2012, and 2012–2013 school years. The sample consists of fourth- and fifth-grade mathematics teachers and their students in four public school districts. The student controls x_{ijt} include prior standardized test score and dummies for grade, gender, race, free or reduced-price lunch eligibility, special education status, and limited English proficiency status.

We use the following model with additive teacher effects and grouped three-way interactive fixed effects,

$$y_{ijt} = x_{ijt}'\delta + \nu_j + \pi_{g(i)jt} + \varepsilon_{ijt}, \tag{99}$$

where

$$\pi_{g(i)jt} = \sum_{k=1}^{\ell} u_{g(i),k} v_{j,k} w_{t,k}. \tag{100}$$

The parameter $\nu_j + \pi_{gjt}$ is the value added of teacher j in year t for students in prior-achievement group g.[4] This object allows teacher effectiveness to differ across the distribution of incoming student achievement and over time.

The group loading $u_{g(i),k}$ captures how strongly students in prior-achievement group g load on the latent teacher-effect dimension. The teacher loading $v_{j,k}$ captures teacher j's position on that dimension. For example, if $u_{g(i),k}$ is large for low-achievement groups, then teachers with high $v_{j,k}$ have especially large predicted effects for students who entered with low prior achievement. If $u_{g(i),k}$ is large for high-achievement groups, then teacher differences matter more for initially high-achieving students. The year factor $w_{t,k}$ allows the magnitude of these teacher-student-type interactions to change across school years.

We plot the estimated teacher components $(\hat{u}_g, \hat{v}_j, \hat{w}_t)$ in the following Figure 1.

[4]The interpretation of $\nu_j + \pi_{gjt}$ as teacher value added relies on the assumption that, after conditioning on X_{ijt}, differences in student achievements across teachers reflect teacher contributions. This assumption can be justified if the teacher assignment is randomized.

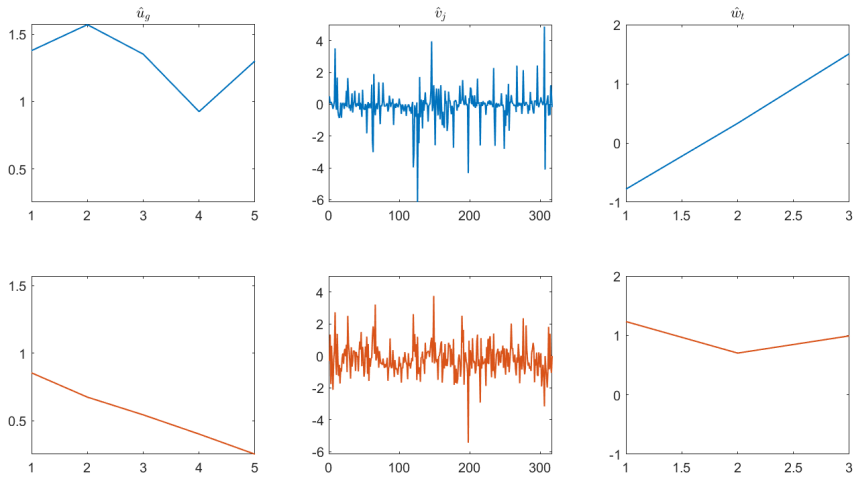


Figure 1: The estimated teacher components $(\hat{u}_g, \hat{v}_j, \hat{w}_t)$ with $\ell = 2$. The first row plots $k = 1$ and the second row plots $k = 2$.

Write $\hat{\pi}^{(k)}_{gjt} = \hat{u}_{gk}\hat{v}_{jk}\hat{w}_{tk}$. The first component combines positive group loadings with an increasing year factor. For a fixed group g and two years $t < t'$, its change is

$$\hat{\pi}^{(1)}_{gjt'} - \hat{\pi}^{(1)}_{gjt} = \hat{u}_{g1}\hat{v}_{j1}(\hat{w}_{t'1} - \hat{w}_{t1}).$$

Teachers with larger $\hat{v}_{j1}$ therefore exhibit greater increases in this component's contribution. The magnitude of the increase varies across achievement groups through $\hat{u}_{g1}$.

For the second component, the declining group loadings and positive year factors imply that, for $\hat{v}_{j2} > 0$, this component contributes more to predicted achievement among initially lower-achieving students. For a lower-achievement group g_L and a higher-achievement group g_H,

$$\hat{\pi}^{(2)}_{g_Ljt} - \hat{\pi}^{(2)}_{g_Hjt} = (\hat{u}_{g_L,2} - \hat{u}_{g_H,2})\hat{v}_{j2}\hat{w}_{t2}.$$

The differential reverses for teachers with negative $\hat{v}_{j2}$. The estimated year factor remains positive and relatively stable throughout the sample.

9.1 Variance Decomposition of Teacher Effectiveness

For achievement group g, teacher j, and period t, let $\tau_{gjt} = \nu_j + \sum_{k=1}^{\ell} u_{gk}v_{jk}w_{tk}$ denote the fitted teacher component. Let

$$\begin{aligned}
&\mathcal{O} = \{(j,t) : \text{teacher } j \text{ is observed in period } t\},\\
&\mathcal{J}_t = \{j : (j,t) \in \mathcal{O}\}, \qquad J_t = |\mathcal{J}_t|,\\
&\mathcal{T}_j = \{t : (j,t) \in \mathcal{O}\}, \qquad T_j = |\mathcal{T}_j|.
\end{aligned}$$

Let $\omega_g \geqslant 0$ denote the reference weight for group g, with $\sum_{g=1}^{G} \omega_g = 1$. Define

$$\bar{\tau}_{jt} = \sum_{g=1}^{G} \omega_g \tau_{gjt}, \qquad \bar{\tau}_{gt} = \frac{1}{J_t} \sum_{j \in \mathcal{J}_t} \tau_{gjt},$$

$$\bar{\tau}_t = \frac{1}{J_t} \sum_{j \in \mathcal{J}_t} \bar{\tau}_{jt} = \sum_{g=1}^{G} \omega_g \bar{\tau}_{gt}, \qquad \bar{\tau} = \frac{1}{|\mathcal{O}|} \sum_{t=1}^{T} J_t \bar{\tau}_t.$$

For every active teacher–year cell, τ_{gjt} is evaluated for all $g = 1, \ldots, G$, whether or not group g is observed in that cell. The reference weight ω_g is chosen by equal quintile weights, i.e., $\omega_g = 0.2, g = 1, \ldots, 5$.

Teacher value added relative to the average active teacher for group g in period t is defined as

$$\mathrm{VA}_{gjt} = \tau_{gjt} - \bar{\tau}_{gt} = L_{jt} + M_{gjt},$$

where

$$L_{jt} = \bar{\tau}_{jt} - \bar{\tau}_t,$$
$$M_{gjt} = \tau_{gjt} - \bar{\tau}_{jt} - \bar{\tau}_{gt} + \bar{\tau}_t.$$

The level effect L_{jt} measures teacher j's general effectiveness relative to the average active teacher in period t. The matching effect M_{gjt} measures the teacher's comparative advantage for achievement group g. By construction, we have

$$\sum_{g=1}^{G} \omega_g M_{gjt} = 0 \qquad \text{for every } (j,t) \in \mathcal{O},$$
$$\frac{1}{J_t} \sum_{j \in \mathcal{J}_t} M_{gjt} = 0 \qquad \text{for every } (g,t).$$

These conditions capture the economic meaning of matching. Teacher effect common to student-group is should be attributed to the level effect, not matching. For example, if a teacher is equally effective in teaching different student-groups, this is captured in the level effect. Also, group effect common to teacher is not matching. For example, if all teachers are equally more effective in some student groups, reallocating students to different teachers do not change the total teacher effectiveness.

The fitted teacher component admits the orthogonal decomposition

$$\begin{aligned}\tau_{gjt} - \bar{\tau} = {} & \underbrace{\bar{\tau}_t - \bar{\tau}}_{\text{common year component}} + \underbrace{\bar{\tau}_{gt} - \bar{\tau}_t}_{\text{common group level component}} \\ & + \underbrace{\bar{\tau}_{jt} - \bar{\tau}_t}_{\text{teacher level effect}} + \underbrace{\tau_{gjt} - \bar{\tau}_{jt} - \bar{\tau}_{gt} + \bar{\tau}_t}_{\text{teacher–group matching effect}} \\ = {} & T_t + G_{gt} + L_{jt} + M_{gjt}.\end{aligned}$$

Under the common teacher–year and group weights, the sample variance can be decomposed as

$$V_\tau = V_T + V_G + V_L + V_M,$$

where

$$V_\tau = \frac{1}{|\mathcal{O}|}\sum_{t=1}^{T}\sum_{j\in\mathcal{J}_t}\sum_{g=1}^{G}\omega_g\left(\tau_{gjt}-\bar{\tau}\right)^2,$$

$$V_T = \frac{1}{|\mathcal{O}|}\sum_{t=1}^{T}J_t\left(\bar{\tau}_t-\bar{\tau}\right)^2,$$

$$V_L = \frac{1}{|\mathcal{O}|}\sum_{t=1}^{T}\sum_{j\in\mathcal{J}_t}\left(\bar{\tau}_{jt}-\bar{\tau}_t\right)^2,$$

$$V_G = \frac{1}{|\mathcal{O}|}\sum_{t=1}^{T}J_t\sum_{g=1}^{G}\omega_g\left(\bar{\tau}_{gt}-\bar{\tau}_t\right)^2,$$

$$V_M = \frac{1}{|\mathcal{O}|}\sum_{t=1}^{T}\sum_{j\in\mathcal{J}_t}\sum_{g=1}^{G}\omega_g\left(\tau_{gjt}-\bar{\tau}_{jt}-\bar{\tau}_{gt}+\bar{\tau}_t\right)^2.$$

The estimated variance components are

$$\left(\widehat{V}_T,\widehat{V}_G,\widehat{V}_L,\widehat{V}_M\right) = (0.000,\ 0.003,\ 0.097,\ 0.036).$$

Consequently,

$$\widehat{V}_\tau = 0.136, \qquad \widehat{V}_{\mathrm{VA}} = \widehat{V}_L + \widehat{V}_M = 0.133.$$

	Grouped Three-Way		Ahn, Aucejo, and James (2025): Math	
	Variance	Share	Variance	Share
Relative teacher value added	0.133	100%	0.056	100%
Level	0.097	73%	0.049	88%
Matching	0.036	27%	0.007	12%

The matching component accounts for approximately 27% of relative teacher-value-added variation in the grouped three-way model, compared with 12% in the math estimates of Ahn et al. (2025).[5]

[5]The comparison is descriptive. Ahn et al. (2025) calculate population variance components from an estimated Gaussian-mixture distribution of teacher coefficients, accounting for sampling error in the teacher-level estimates. Their matching component incorporates multiple student characteristics, and their level component also absorbs covariance terms. The grouped three-way entries are plug-in second moments of the estimated teacher-effect surface under common reference-group and active teacher–year weights. The reported shares therefore reflect differences in samples, model specifications, variance definitions, weighting, and treatment of estimation error.

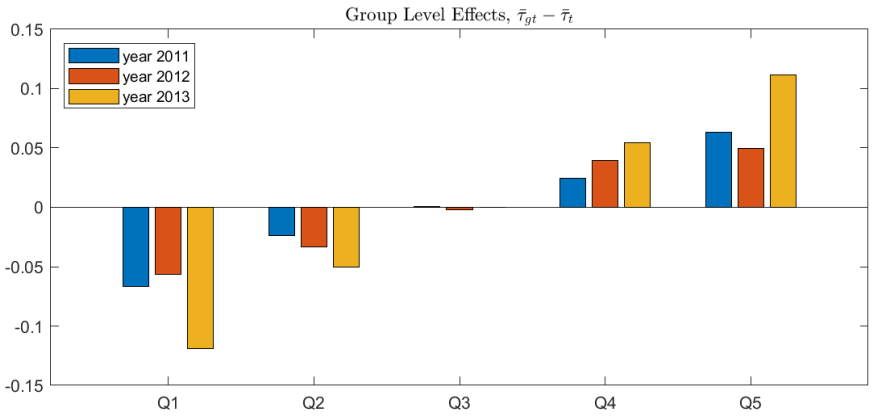
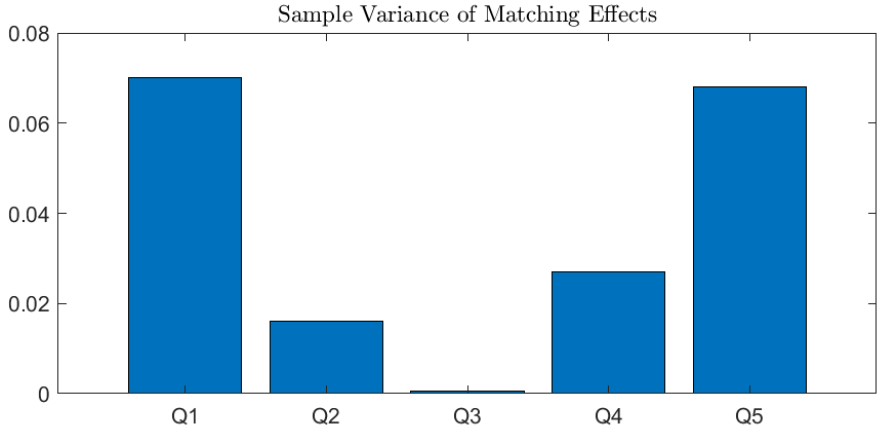


Figure 2: The average fitted teacher component, relative to its yearly reference mean, is lower for initially lower-achieving groups and higher for initially higher-achieving groups.

The group-level effect $G_{gt} = \bar{\tau}_{gt} - \bar{\tau}_t$ suggests that the average teacher effects tend to be lower for the prior low achievement groups compared to the high achievement groups. However, variation in the match effect $\hat{V}_M = 0.036$ is much larger than that of the group-level effect $0.003 = \hat{V}_G$, suggesting at matching of a student to the right teacher is more important than the group type.

To ascertain which student groups are most affected by matching effect, we define the sample variance of the matching effects for each student groups

$$\widehat{V}_{M,g} = \frac{1}{|\mathcal{O}|} \sum_{t=1}^{T} \sum_{j \in \mathcal{J}_t} \hat{M}_{gjt}^2, \qquad \widehat{V}_M = \sum_{g=1}^{G} \omega_g \widehat{V}_{M,g}.$$



The figure shows that matching dispersion is greatest for Q1 and Q5, with estimated variances of approximately 0.07, and smallest for Q3. Teachers' deviations from their general effectiveness therefore vary most for students at the extremes of the prior-achievement distribution.

9.2 Counterfactual Teacher Reallocation

We conduct two counterfactual exercises. First, we evaluate each teacher's value added when assigned to their individually best-matched classroom within their school or district. Second, we find the assignment that maximizes aggregate value added subject to every classroom receiving exactly one teacher and every teacher teaching exactly one classroom. Throughout both exercises, classroom rosters, student characteristics, and the estimated teacher-effect surface remain fixed.

For each year t, let $\mathcal{C}_t$ denote the included classrooms and $\mathcal{J}_t$ the included teachers. We restrict attention to teacher–classroom pairs for which the observed assignment is one-to-one, so $|\mathcal{C}_t| = |\mathcal{J}_t|$. Let $j_t^0(c)$ denote classroom c's observed teacher and $c_t^0(j)$ denote teacher j's observed classroom.

Let n_{gct} be the number of students from achievement group g in classroom c, and define

$$n_{ct} = \sum_{g=1}^{G} n_{gct}, \qquad p_{gct} = \frac{n_{gct}}{n_{ct}}, \qquad n_t = \sum_{c \in \mathcal{C}_t} n_{ct}.$$

The predicted matching contribution and average relative value added of teacher j in classroom c are

$$\hat{m}_{jct} = \sum_{g=1}^{G} p_{gct} \hat{M}_{gjt}, \tag{101}$$

$$\hat{q}_{jct} = \sum_{g=1}^{G} p_{gct} \widehat{\mathrm{VA}}_{gjt} = \hat{L}_{jt} + \hat{m}_{jct}. \tag{102}$$

Thus, classroom value added depends on the teacher's general effectiveness and their matching effects evaluated at the classroom's actual achievement-group composition. We retain the original estimates $\hat{L}_{jt}$ and $\hat{M}_{gjt}$ and the original reference weights throughout the counterfactuals.

Let $s(c)$, $d(c)$, and $r(c)$ denote classroom c's school, district, and grade. Define teacher j's eligible classrooms under within-school and within-district reallocation as

$$\mathcal{C}_{jt}^{S} = \{c \in \mathcal{C}_t : s(c) = s(c_t^0(j)),\ r(c) = r(c_t^0(j))\}, \tag{103}$$

$$\mathcal{C}_{jt}^{D} = \{c \in \mathcal{C}_t : d(c) = d(c_t^0(j)),\ r(c) = r(c_t^0(j))\}. \tag{104}$$

Teachers remain in their original grade and year. Within-school reallocation preserves their school, whereas within-district reallocation permits movement across schools in their original district. The observed classroom belongs to both menus, and $\mathcal{C}_{jt}^{S} \subseteq \mathcal{C}_{jt}^{D}$. These menus concern the teachers and classrooms included in the analysis.

Exercise 1: Each teacher's individually best-matched classroom. For $b \in \{S, D\}$, define

$$c_{jt}^{\mathrm{best},b} \in \operatorname*{argmax}_{c \in \mathcal{C}_{jt}^{b}} \hat{m}_{jct}.$$

Because $\hat{L}_{jt}$ is fixed for teacher j, maximizing the matching contribution also maximizes that teacher's average relative value added. The resulting value added and its change from

the observed classroom are

$$\hat{q}_{jt}^{\text{best},b} = \hat{L}_{jt} + \hat{m}_{j,c_{jt}^{\text{best},b},t}, \tag{105}$$

$$\Delta_{jt}^{\text{best},b} = \hat{q}_{jt}^{\text{best},b} - \hat{q}_{j,c_t^0(j),t} = \hat{m}_{j,c_{jt}^{\text{best},b},t} - \hat{m}_{j,c_t^0(j),t}. \tag{106}$$

Consequently,

$$0 \leqslant \Delta_{jt}^{\text{best},S} \leqslant \Delta_{jt}^{\text{best},D}.$$

We summarize these teacher-specific opportunities using their mean, median, and distribution. The mean gain across teacher–year observations is

$$\overline{\Delta}^{\text{best},b} = \frac{\sum_{t=1}^{T} \sum_{j \in \mathcal{J}_t} \Delta_{jt}^{\text{best},b}}{\sum_{t=1}^{T} |\mathcal{J}_t|}.$$

This exercise evaluates each teacher separately. Several teachers may select the same classroom, while other classrooms may receive no selections. Its results therefore measure individual opportunities rather than gains from a simultaneously feasible allocation.

Exercise 2: The optimal feasible assignment. Let $a_{jct} = 1$ indicate that teacher j is assigned to classroom c in year t, and let $A = (a_{jct})$. For $b \in \{S, D\}$, define the feasible assignment set

$$\mathcal{A}_t^b = \left\{ A : \; a_{jct} \in \{0,1\}, \; \sum_{c \in \mathcal{C}_t} a_{jct} = 1 \text{ for every } j, \; \sum_{j \in \mathcal{J}_t} a_{jct} = 1 \text{ for every } c, \; a_{jct} = 0 \text{ if } c \notin \mathcal{C}_{jt}^b \right\}.$$

The row constraints assign every teacher to exactly one classroom. The column constraints give every classroom exactly one teacher. The eligibility constraints impose the school or district restriction.

For a feasible assignment, total predicted relative value added across students is

$$\widehat{\text{TVA}}_t(A) = \sum_{c \in \mathcal{C}_t} \sum_{j \in \mathcal{J}_t} a_{jct} n_{ct} \hat{q}_{jct}.$$

Define the model-implied first-best assignment within the specified restrictions by

$$A_t^{\text{FB},b} \in \operatorname*{argmax}_{A \in \mathcal{A}_t^b} \widehat{\text{TVA}}_t(A).$$

Let A_t^0 denote the observed assignment. The maximum total value added and the average gain per student are

$$\widehat{\text{TVA}}_t^{\text{FB},b} = \widehat{\text{TVA}}_t(A_t^{\text{FB},b}), \tag{107}$$

$$\Delta_t^{\text{FB},b} = \frac{\widehat{\text{TVA}}_t(A_t^{\text{FB},b}) - \widehat{\text{TVA}}_t(A_t^0)}{n_t}. \tag{108}$$

Because the observed assignment is feasible and the district assignment set contains the school assignment set,

$$0 \leqslant \Delta_t^{\text{FB},S} \leqslant \Delta_t^{\text{FB},D}.$$

The pooled average gain across years is

$$\Delta^{\mathrm{FB},b} = \frac{\sum_{t=1}^{T} \left\{ \widehat{\mathrm{TVA}}_t(A_t^{\mathrm{FB},b}) - \widehat{\mathrm{TVA}}_t(A_t^0) \right\}}{\sum_{t=1}^{T} n_t}.$$

Computationally, we construct the teacher–classroom payoff matrix with entries $n_{ct}\hat{q}_{jct}$ and solve a maximum-weight one-to-one assignment problem separately for each school–grade–year or district–grade–year pool. The optimization accounts for the opportunity cost of assigning a teacher to a classroom that could instead receive another teacher. It can therefore assign some teachers to classrooms other than their individually preferred choices.

Holding teacher levels fixed and isolating matching gains. Holding $\hat{L}_{jt}$ fixed preserves each teacher's estimated general effectiveness. Nevertheless, with unequal classroom sizes, the student-weighted level contribution

$$\sum_{c \in \mathcal{C}_t} \sum_{j \in \mathcal{J}_t} a_{jct} n_{ct} \hat{L}_{jt}$$

can change as teachers move between classrooms. The total-value-added optimum can therefore exploit both matching and the allocation of teacher-level effectiveness across classrooms of different sizes.

To isolate the matching channel, define

$$\hat{H}_t(A) = \sum_{c \in \mathcal{C}_t} \sum_{j \in \mathcal{J}_t} a_{jct} n_{ct} \hat{m}_{jct}, \qquad A_t^{\mathrm{match},b} \in \operatorname*{argmax}_{A \in \mathcal{A}_t^b} \hat{H}_t(A).$$

The corresponding average matching gain is

$$\Delta_t^{\mathrm{match},b} = \frac{\hat{H}_t(A_t^{\mathrm{match},b}) - \hat{H}_t(A_t^0)}{n_t}.$$

This statistic measures the improvement in the matching component alone. With unequal classroom sizes, its optimizing assignment can differ from the assignment maximizing total value added.

If all classrooms in an assignment pool have the same size, the aggregate level contribution is invariant under permutation. In that case, maximizing total value added is equivalent to maximizing matching. Alternatively, under equal classroom weights,

$$\sum_{c \in \mathcal{C}_t} \sum_{j \in \mathcal{J}_t} a_{jct} \hat{q}_{jct} = \sum_{j \in \mathcal{J}_t} \hat{L}_{jt} + \sum_{c \in \mathcal{C}_t} \sum_{j \in \mathcal{J}_t} a_{jct} \hat{m}_{jct},$$

so the level contribution is constant. Equal classroom weighting targets the average classroom outcome, whereas enrollment weighting targets the average student outcome.

Distribution of gains. Let $j_t^{\mathrm{FB},b}(c)$ be classroom c's teacher under the optimal feasible assignment. The predicted gain for achievement group g in that classroom is

$$\Delta_{gct}^{\mathrm{FB},b} = \hat{L}_{j_t^{\mathrm{FB},b}(c),t} - \hat{L}_{j_t^0(c),t} + \hat{M}_{g,j_t^{\mathrm{FB},b}(c),t} - \hat{M}_{g,j_t^0(c),t}.$$

The average gain for group g is

$$\Delta_g^{\mathrm{FB},b} = \frac{\sum_{t=1}^{T} \sum_{c \in \mathcal{C}_t} n_{gct} \Delta_{gct}^{\mathrm{FB},b}}{\sum_{t=1}^{T} \sum_{c \in \mathcal{C}_t} n_{gct}},$$

provided that its denominator is positive. Although the optimized aggregate gain is nonnegative, individual classrooms or achievement groups can experience losses. We therefore report both aggregate gains and their distribution across groups and classrooms.

Interpretation. These counterfactuals change teacher assignments while preserving student rosters, classroom composition, and the fitted response functions. Common group and year components and student regression controls consequently cancel from aggregate predicted changes. The first-best is conditional on the estimated model and the specified eligibility restrictions. A causal interpretation additionally requires teacher effects to remain valid after reassignment; cross-school moves require their effectiveness to transfer across school environments. Estimated optimal gains should also be assessed for uncertainty, since optimizing estimated payoffs can select assignments whose apparent advantages partly reflect estimation error.

Empirical results. Using the fitted rank-two specification with five prior-achievement groups, the reallocation sample contains 566 teacher–classroom-year pairs and 10,621 student-year observations over 2011–2013. We exclude 17 teacher-year pairs associated with multiple classrooms and retain the original fitted effects and reference weights throughout the exercises.[6] Table 3 reports counterfactual gains in standardized state mathematics-score units and individual-best matching diagnostics.

Teachers' individual opportunities are unevenly distributed. The mean gain from a teacher's individually best classroom is 0.0644 within schools and 0.1499 within districts, whereas the corresponding medians are 0.0138 and 0.0885. Observed assignments already maximize predicted value added within the eligible menu for 206 of the 566 teacher-year observations (36.40%) under school restrictions and 26 (4.59%) under district restrictions, counting tied optima as already best matched. These opportunities are evaluated separately for each teacher and need not form a feasible joint allocation.

We also examine whether individual best matching changes the classification of teachers initially in the bottom tail of observed classroom-average relative value added. Defining the bottom 5% separately within each year and including cutoff ties yields 29 teacher-year observations. When all teachers are evaluated in their individual best classrooms and the within-year 5th-percentile cutoffs are recalculated, 5 of these observations (17.24%) leave the bottom tail under school restrictions and 8 (27.59%) under district restrictions. Relative to the fixed original within-year cutoffs, 8 observations (27.59%) improve above the threshold under school restrictions and 16 (55.17%) do so under district restrictions. The larger fixed-cutoff escape rates show that some teachers clear the original benchmark while remaining in the counterfactual bottom tail as other teachers also improve.

Under the optimal feasible allocation, pooled average total value added increases by 0.0538 per student within schools and 0.1352 within districts. Matching contributes 0.0429 and 0.0916 to these gains, respectively, accounting for approximately 80% and 68% of the

[6] For the 12 included classrooms containing students from more than one grade, eligibility is determined by the unique modal student grade.

total improvement. The remaining contributions, 0.0109 and 0.0436, reflect the allocation of fixed teacher-level effectiveness across classrooms of different sizes. Separately maximizing the matching component yields matching gains of 0.0476 within schools and 0.0989 within districts, with corresponding total value-added gains of 0.0422 and 0.0939. Thus, the two objectives favor different allocations when classroom enrollment is unequal.

Under the total-value-added optimum, pooled average gains are positive in all five achievement groups. Figure 3 shows that gains are largest in the lowest-achievement group, followed by the highest-achievement group, and smallest in the middle group. Nevertheless, classroom-average gains are negative for 27.0% of classroom-year observations under school reallocation and 36.7% under district reallocation. Positive aggregate gains therefore coexist with losses in some classrooms. These results are model-implied point estimates conditional on the fitted teacher effects and eligibility restrictions; uncertainty is not quantified in these exercises.

Table 3: Counterfactual gains and matching diagnostics

	Within school	Within district
Panel A: Individually best classroom		
Mean gain	0.0644	0.1499
Median gain	0.0138	0.0885
Panel B: Feasible total-value-added optimum		
Total value-added gain	0.0538	0.1352
Matching contribution	0.0429	0.0916
Level contribution	0.0109	0.0436
Panel C: Feasible matching-only optimum		
Total value-added gain	0.0422	0.0939
Matching contribution	0.0476	0.0989
Level contribution	-0.0054	-0.0050
Panel D: Group gains under the total-value-added optimum		
Q1 (lowest prior achievement)	0.1007	0.2245
Q2	0.0539	0.1138
Q3	0.0197	0.0649
Q4	0.0343	0.0862
Q5 (highest prior achievement)	0.0594	0.1829
Panel E: Individual-best matching diagnostics		
Already in a best classroom	206 (36.40%)	26 (4.59%)
Escape after re-ranking	5 (17.24%)	8 (27.59%)
Exceed original 5th-percentile cutoff	8 (27.59%)	16 (55.17%)

Notes: Panels A–D report changes from the observed assignment in standardized state mathematics-score units. Panel A weights the 566 teacher-year observations equally. Panels B and C report pooled gains weighted by the 10,621 student-year observations. Panel D uses student-year weights within each achievement group. Each feasible allocation assigns exactly one teacher to each classroom and preserves the teacher's grade and year. Panel C maximizes matching alone, but its total value-added gains and the group gains in Panel D include both level and matching effects. Panel E reports counts with percentages in parentheses. Already-best percentages use all 566 teacher-year observations; escape percentages use the 29 originally bottom-tail teacher-year observations. Bottom-tail membership is defined within each year, including cutoff ties. Re-ranking uses the restriction-specific individual-best value-added distribution; the original-cutoff measure holds the observed threshold fixed. Tied best classrooms count as already optimal. Panel E describes individual opportunities and does not impose a feasible joint allocation.

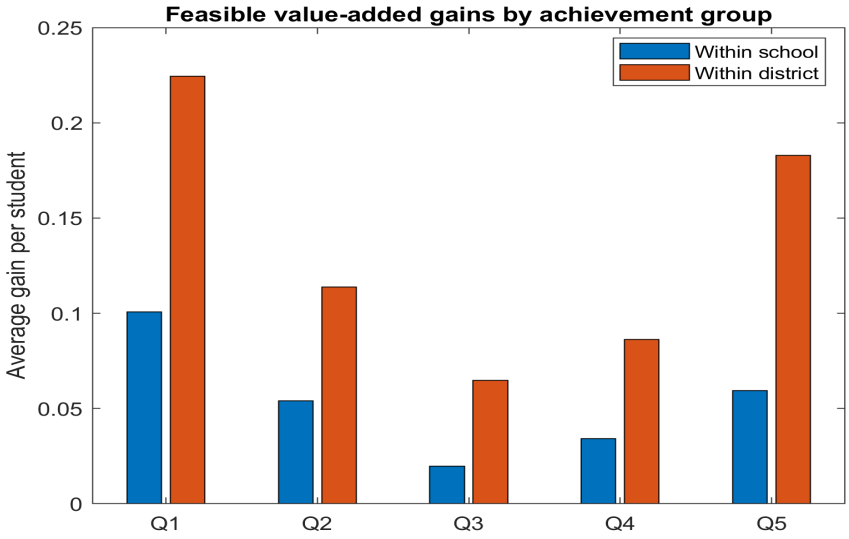


Figure 3: Under feasible allocation, gains are largest in the lowest-achievement group, followed by the highest-achievement group, and smallest in the middle group.

9.3 Comparison with Teacher Value Added in the Randomized Sample

As a final empirical exercise, we compare the final-year teacher-effect surface from the grouped three-way model with estimates from an unrestricted teacher-by-achievement-group regression fitted to the original randomized-roster sample. The comparison includes every teacher with at least one eligible teacher–group cell, including teachers without observations in all five achievement quintiles. We apply a common eligibility mask to both surfaces, calculate their group and teacher means over the available cells, and compare the resulting components.

Let T denote the final study year, corresponding to 2012–2013, and let $\mathcal{R}$ contain the usable observations from the original randomized classroom rosters that can be matched to the complete-case analysis data. For the benchmark, j denotes the originally assigned teacher. Using observations $(i, j, T) \in \mathcal{R}$, we estimate

$$y_{ijT} = x'_{ijT}\delta^B + \tau^B_{g(i),j} + \varepsilon^B_{ijT}. \tag{109}$$

The regression estimates separate effects for every observed teacher–group cell without imposing the grouped three-way factor structure. Achievement groups, the outcome scale, and the control vector are the same as in the main analysis. The benchmark retains the original assignment index for all 968 observations, including 90 whose observed teacher differs from the assigned teacher.

The grouped three-way model remains estimated on the full panel. For the rank-two specification, its final-year teacher-effect surface is

$$\widehat{\tau}^{3W}_{gj} = \widehat{\nu}_j + \sum_{k=1}^{2} \widehat{u}_{gk}\widehat{v}_{jk}\widehat{w}_{Tk}.$$

Both surfaces refer to year T, so we suppress the time index.

Let $n_{gj}^{\mathcal{R}}$ count the usable benchmark observations in cell (g, j). For a minimum cell count $m \geqslant 1$, define

$$\mathcal{S}_m = \{(g, j) : n_{gj}^{\mathcal{R}} \geqslant m\},\ \widehat{\tau}_{gj}^{B} \text{ and } \widehat{\tau}_{gj}^{3W} \text{ are available}\}.$$

Let $\mathcal{J}_{R,m}$ and $\mathcal{G}_m$ be the teachers and groups represented in $\mathcal{S}_m$, let $J_m = |\mathcal{J}_{R,m}|$, and let $J_{g,m}$ be the number of eligible teachers in group g. A teacher enters the comparison whenever at least one cell is eligible. Ineligible cells are treated as missing in both comparison surfaces before computing means, correlations, or scatter plots.

The reference quintile weights are $\omega_g = 1/5$. Define the normalized eligible-cell weights

$$Z_m = \sum_{(h,k)\in\mathcal{S}_m} \omega_h, \qquad \varpi_{gj,m} = \frac{\omega_g}{Z_m}, \quad (g, j) \in \mathcal{S}_m.$$

Thus each eligible cell has equal weight in the present application. Teachers with more eligible quintiles contribute more cells to pooled cell-level comparisons. Student observation counts determine eligibility but do not otherwise determine the correlation weights.

For each model $q \in \{B, 3W\}$, calculate

$$\overline{\tau}_j^{q,m} = \frac{\sum_{g:(g,j)\in\mathcal{S}_m} \omega_g \widehat{\tau}_{gj}^{q}}{\sum_{g:(g,j)\in\mathcal{S}_m} \omega_g}, \qquad \overline{\tau}_g^{q,m} = \frac{1}{J_{g,m}} \sum_{j:(g,j)\in\mathcal{S}_m} \widehat{\tau}_{gj}^{q},$$

and

$$\overline{\tau}^{q,m} = \sum_{(g,j)\in\mathcal{S}_m} \varpi_{gj,m} \widehat{\tau}_{gj}^{q}.$$

A teacher's mean therefore uses that teacher's available quintiles, with the reference group weights renormalized over them. A group mean uses that group's available teachers with equal weights. The grand mean averages over the common eligible cells.

Define the group, teacher-level, and matching components by

$$\widehat{G}_g^{q,m} = \overline{\tau}_g^{q,m} - \overline{\tau}^{q,m}, \qquad \widehat{L}_j^{q,m} = \overline{\tau}_j^{q,m} - \overline{\tau}^{q,m},$$

$$\widehat{M}_{gj}^{q,m} = \widehat{\tau}_{gj}^{q} - \overline{\tau}_j^{q,m} - \overline{\tau}_g^{q,m} + \overline{\tau}^{q,m}, \qquad (g, j) \in \mathcal{S}_m.$$

These definitions reconstruct every eligible effect:

$$\widehat{\tau}_{gj}^{q} - \overline{\tau}^{q,m} = \widehat{G}_g^{q,m} + \widehat{L}_j^{q,m} + \widehat{M}_{gj}^{q,m}.$$

The teacher and group means are calculated separately. On irregular support, the components generally are not orthogonal. Relative teacher value added is defined cell by cell as

$$\widehat{\mathrm{VA}}_{gj}^{q,m} = \widehat{L}_j^{q,m} + \widehat{M}_{gj}^{q,m} = \widehat{\tau}_{gj}^{q} - \overline{\tau}_g^{q,m}.$$

Its reference is the mean across available teachers in the same group. Both models use the same eligible teacher set for each group, although that set can differ across groups.

We report weighted Pearson correlations between corresponding components. Group-component correlations use the reference weights ω_g, renormalized over groups with finite paired estimates. Teacher-level correlations give each of the J_m comparable teachers weight $1/J_m$. Pooled matching and relative-value-added correlations use $\varpi_{gj,m}$ over eligible cells. Within-group correlations give each of the $J_{g,m}$ paired teachers weight $1/J_{g,m}$. The group

and teacher correlation weights need not equal the marginals of the eligible-cell weights. Each correlation uses the same paired estimates and reporting weights for the two models. Correlations are reported as missing when fewer than two paired estimates are available or either quantity has zero variance within numerical precision.

The benchmark regression uses 968 complete-case observations, and the rank-two grouped three-way model uses 11,390 full-panel observations. For $m = 1$, the comparison contains 71 teachers, 284 eligible cells, and 968 observations in those cells. For $m = 2$, the corresponding counts are 70 teachers, 215 cells, and 899 observations; for $m = 3$, they are 60 teachers, 162 cells, and 793 observations. Both fitted models are held fixed across the reported threshold comparisons. The benchmark uses all 968 observations at every threshold, including observations in cells excluded from the comparison.

Empirical results. Table 4 reports Pearson correlations between corresponding estimates. For $m = 1$, the group-level, teacher-level, matching, and relative-value-added correlations are 0.923, 0.755, 0.234, and 0.508, respectively. For $m = 2$, they are 0.938, 0.760, 0.415, and 0.647; for $m = 3$, they are 0.963, 0.809, 0.585, and 0.718. Relative value added, defined as $L_j + M_{gj}$, thus exhibits a stronger pooled association than matching effects under every threshold.

Within quintiles, matching correlations range from 0.034 to 0.385 for $m = 1$, from 0.222 to 0.507 for $m = 2$, and from 0.357 to 0.669 for $m = 3$. Relative-value-added correlations range from 0.379 to 0.672, from 0.603 to 0.680, and from 0.682 to 0.767, respectively. Relative value added has the higher correlation in every quintile at every threshold, while Q3 has the lowest correlation for both measures. Pooled correlations increase with the minimum cell count, although within-quintile matching correlations do not all increase monotonically. Each threshold changes the teachers, eligible cells, and centering populations.

The results suggest that the grouped three-way interactive fixed effects model largely agrees with the randomized roster benchmark model. Deviations can largely be explained by the estimation variance in the benchmark model resulting from having too small samples in group-teacher cells.

Table 4: All-teacher correlations with the randomized-roster benchmark

Panel A: Pooled correlations and sample sizes

Component	$m = 1$	$m = 2$	$m = 3$
Group level	0.923	0.938	0.963
Teacher level	0.755	0.760	0.809
Matching	0.234	0.415	0.585
Relative value added ($L + M$)	0.508	0.647	0.718
Paired teachers	71	70	60
Paired cells	284	215	162
Observations in paired cells	968	899	793

Panel B: Within-quintile correlations

Minimum m	Quintile	Teachers	Matching	$L + M$
1	Q1	57	0.220	0.470
	Q2	61	0.251	0.672
	Q3	57	0.034	0.379
	Q4	62	0.385	0.552
	Q5	47	0.160	0.444
2	Q1	40	0.499	0.630
	Q2	44	0.507	0.680
	Q3	48	0.222	0.603
	Q4	47	0.322	0.661
	Q5	36	0.415	0.658
3	Q1	31	0.657	0.713
	Q2	31	0.518	0.767
	Q3	34	0.357	0.682
	Q4	38	0.490	0.687
	Q5	28	0.669	0.760

Notes: m is the minimum number of usable benchmark observations per teacher–quintile cell. The benchmark regression uses all 968 observations at all three thresholds. The same eligibility mask is applied to both models before centering. Teacher means average over eligible quintiles using renormalized reference weights $\omega_g = 1/5$; group means average over eligible teachers. The grand mean uses product weights renormalized over eligible cells. The decomposition need not be orthogonal on this irregular support. Pooled correlations weight the five group estimates, paired teachers, or eligible cells equally, as applicable. Within-quintile correlations weight paired teachers equally. Q1 and Q5 denote the lowest and highest prior-achievement quintiles.

10 Conclusion

This paper develops a grouped three-way interactive fixed-effects model for panel data indexed by two cross-sectional dimensions and time. The model represents unobserved heterogeneity through the low-rank trilinear component

$$\pi_{ijt} = \sum_{k=1}^{\ell} u_{g(i)k} v_{jk} w_{tk},$$

which imposes the separable loading structure $\lambda_{ij,k} = u_{g(i)k} v_{jk}$. Relative to a flattened two-way interactive fixed-effects model with unrestricted pair-level loadings, grouping and

separability reduce the factor-parameter count from approximately $\ell(NM + T)$ to $\ell(G + M + T)$, before accounting for normalization restrictions.

We establish the large-sample properties of the global least-squares estimator and provide an alternating-least-squares implementation. We derive stochastic expansions and limiting distributions for the slope estimator and aligned factor estimates, characterize the mode-specific incidental-parameter biases, and construct a feasible bias correction under additional error and moment conditions. These results explain how the separable structure can reduce loading-side incidental-parameter bias relative to an unrestricted flattened specification. We also develop a consistent information criterion for selecting the number of interactive components and extend the consistency analysis to additive effects and unbalanced observation patterns.

Under the maintained identification and growth conditions, aggregate consistency in the balanced-panel framework can hold when any two of N, M, and T diverge, allowing the remaining dimension to remain fixed. Stronger conditions are needed for the limiting distributions and negligible asymptotic bias. Under correctly specified trilinear heterogeneity, the Monte Carlo experiments show lower slope-estimation RMSE than the unrestricted flattened two-way estimator in the designs considered.

The empirical application estimates heterogeneous teacher value added using matched student–teacher achievement data. The estimated teacher-effect surface varies across prior-achievement groups, and the matching component accounts for approximately 27% of fitted relative-value-added variation under the chosen reference weights. These estimates illustrate how a scalar teacher-level measure can conceal group-specific comparative advantage. Comparison with a randomized-roster benchmark shows stronger descriptive agreement when more samples are used to estimate the group-specific teacher effects in estimation.

The framework is relevant to other settings with naturally three-dimensional data, including worker–firm, product–market, trade, and health-care panels. Extending the model to latent group structures, dynamic effects, and more flexible loading structures would broaden its applicability and clarify the tradeoff between structural restrictions and estimation precision.

11 Appendix

11.1 Lemmas used in Proofs

Lemma 4 (Conditional contraction bounds)**.** *Let $e = \mathrm{vec}(E)$, with entries indexed by $\mathcal{I}$ in the order induced by* vec*. Let $z_{\mathcal{R}^D}$ stack the variables $\{z_{ab} : (a,b) \in \mathcal{R}^D\}$ in a fixed ordering of the relevant ordered pairs. Suppose Assumption 3 holds. The following statements hold.*

(i) For every conformable $\mathcal{F}$-measurable matrix A, we have

$$\mathbb{E}[\|Ae\|^2 \mid \mathcal{F}] \leqslant c_\varepsilon \|A\|_F^2 \tag{110}$$

with probability approaching one. Consequently,

$$\|Ae\| = O_p(\|A\|_F).$$

(ii) For every conformable $\mathcal{F}$-measurable matrix B, we have

$$\mathbb{E}[\|Bz_{\mathcal{R}^D}\|^2 \mid \mathcal{F}] \leqslant c_4 \|B\|_F^2 \tag{111}$$

with probability approaching one. Consequently,

$$\|Bz_{\mathcal{R}^D}\| = O_p(\|B\|_F).$$

(iii) For every $\mathcal{F}$-measurable square matrix K conformable with e, we have

$$\mathbb{E}[|e'Ke| \mid \mathcal{F}] \leqslant c_\varepsilon \|K\|_* \tag{112}$$

with probability approaching one, where $\|\cdot\|_*$ denotes the nuclear norm. Consequently,

$$|e'Ke| = O_p(\|K\|_*).$$

No symmetry or pair-support restriction on K is required.

The bounds in parts (i) and (iii), including their stochastic-order consequences, also hold with e replaced by $Q'e$ for any $\mathcal{F}$-measurable matrix Q satisfying $Q'Q = I$. In particular, they apply to

$$e_G = \mathrm{vec}(E^G) = Q'e, \qquad Q = \{D(D'D)^{-1/2}\} \otimes I_M \otimes I_T.$$

The coefficient matrices and vectors, including their dimensions, may vary with the panel dimensions. All stochastic-order statements with a random nonnegative scale are interpreted using the convention $0/0 = 0$; in each application above, a zero scale implies that the corresponding numerator is zero.

Proof. Let $\mathcal{E}_2$, and $\mathcal{E}_4$ be the event that

$$\sup_{a \in \mathcal{I}} \sum_{b \in \mathcal{I}} |\Gamma_{ab}| \leqslant c_\varepsilon.$$

and

$$\sup_{(a,b) \in \mathcal{R}^D} \sum_{(c,d) \in \mathcal{R}^D} |\mathrm{Cov}(z_{ab}, z_{cd} \mid \mathcal{F})| \leqslant c_4$$

holds in Assumption 3.

Part (i). Let

$$\Gamma = \mathbb{E}[ee' \mid \mathcal{F}] = (\Gamma_{ab})_{a,b\in\mathcal{I}},$$

where the rows and columns follow the ordering of e. Since $\mathbb{E}[e \mid \mathcal{F}] = 0$, the matrix Γ is the conditional covariance matrix of e. In particular, it is symmetric and positive semidefinite. Its induced matrix 1-norm and ∞-norm are equal, so

$$\|\Gamma\| \leqslant \{\|\Gamma\|_1\|\Gamma\|_\infty\}^{1/2} = \sup_{a\in\mathcal{I}} \sum_{b\in\mathcal{I}} |\Gamma_{ab}| \leqslant c_\varepsilon \qquad \text{on } \mathcal{E}_2.$$

Because A is $\mathcal{F}$-measurable,

$$\begin{aligned}\mathbb{E}[\|Ae\|^2 \mid \mathcal{F}] &= \operatorname{tr}\big(A\mathbb{E}[ee' \mid \mathcal{F}]A'\big) \\ &= \operatorname{tr}(A\Gamma A') \\ &\leqslant \|\Gamma\|_{\mathrm{op}} \operatorname{tr}(AA') \\ &\leqslant c_\varepsilon \|A\|_F^2 \qquad \text{on } \mathcal{E}_2.\end{aligned}$$

This proves (110).

For any $K > 0$, conditional Markov's inequality yields

$$\mathbf{1}_{\mathcal{E}_2}\mathbb{P}(\|Ae\| > K\|A\|_F \mid \mathcal{F}) \leqslant \frac{c_\varepsilon}{K^2}\mathbf{1}_{\mathcal{E}_2}.$$

When $\|A\|_F = 0$, the event inside the conditional probability is empty. Taking expectations and bounding the probability of $\mathcal{E}_2^c$ gives

$$\mathbb{P}(\|Ae\| > K\|A\|_F) \leqslant \frac{c_\varepsilon}{K^2} + \mathbb{P}(\mathcal{E}_2^c).$$

Since $\mathbb{P}(\mathcal{E}_2^c) \to 0$, first taking the limit superior along the asymptotic sequence and then letting $K \to \infty$ proves $\|Ae\| = O_p(\|A\|_F)$.

Part (ii). By the definition of z_{ab},

$$\mathbb{E}[z_{ab} \mid \mathcal{F}] = 0, \qquad (a,b) \in \mathcal{R}^D.$$

The finite fourth-moment condition ensures that these centered products have finite second moments. Let

$$\Sigma_{\mathcal{R}^D} = \mathbb{E}[z_{\mathcal{R}^D} z'_{\mathcal{R}^D} \mid \mathcal{F}].$$

Its entries are

$$[\Sigma_{\mathcal{R}^D}]_{(a,b),(c,d)} = \operatorname{Cov}(z_{ab}, z_{cd} \mid \mathcal{F}).$$

The matrix $\Sigma_{\mathcal{R}^D}$ is symmetric and positive semidefinite. Consequently, Assumption 3 implies

$$\|\Sigma_{\mathcal{R}^D}\|_{\mathrm{op}} \leqslant \sup_{(a,b)\in\mathcal{R}^D} \sum_{(c,d)\in\mathcal{R}^D} |\operatorname{Cov}(z_{ab}, z_{cd} \mid \mathcal{F})| \leqslant c_4 \qquad \text{on } \mathcal{E}_4.$$

Because B is $\mathcal{F}$-measurable,

$$\begin{aligned}\mathbb{E}[\|Bz_{\mathcal{R}^D}\|^2 \mid \mathcal{F}] &= \operatorname{tr}(B\Sigma_{\mathcal{R}^D}B') \\ &\leqslant \|\Sigma_{\mathcal{R}^D}\|_{\mathrm{op}}\|B\|_F^2 \\ &\leqslant c_4\|B\|_F^2 \qquad \text{on } \mathcal{E}_4.\end{aligned}$$

This proves (111). As in part (i), localization permits arbitrary finite $\mathcal{F}$-measurable B. Conditional Markov's inequality then gives, for every $K > 0$,

$$\mathbb{P}(\|Bz_{\mathcal{R}^D}\| > K\|B\|_F) \leqslant \frac{c_4}{K^2} + \mathbb{P}(\mathcal{E}_4^c).$$

Since $\mathbb{P}(\mathcal{E}_4^c) \to 0$, the asserted stochastic order follows.

Part (iii). On $\mathcal{E}_2$, write a singular value decomposition $K = \sum_h \sigma_h a_h b_h'$, where $\|a_h\| = \|b_h\| = 1$. Part (i) and conditional Cauchy–Schwarz give

$$\begin{aligned}\mathbb{E}[|e'Ke| \mid \mathcal{F}] &\leqslant \sum_h \sigma_h \left\{\mathbb{E}[(a_h'e)^2 \mid \mathcal{F}]\mathbb{E}[(b_h'e)^2 \mid \mathcal{F}]\right\}^{1/2} \\ &\leqslant c_\varepsilon \sum_h \sigma_h = c_\varepsilon \|K\|_*.\end{aligned}$$

Conditional Markov's inequality yields, for every $L > 0$,

$$\mathbb{P}\big(|e'Ke| > L\|K\|_*\big) \leqslant c_\varepsilon/L + \mathbb{P}(\mathcal{E}_2^c),$$

which proves the asserted stochastic order under the stated zero-scale convention.

For the coordinate extension, apply parts (i) and (iii) to AQ' and QKQ', respectively. Since $Q'Q = I$,

$$\|AQ'\|_F = \|A\|_F, \qquad \|QKQ'\|_* = \|K\|_*.$$

□

Lemma 5 (Low-rank tensor contraction bounds)**.** *Let $E \in \mathbb{R}^{N_1 \times N_2 \times N_3}$. Fix $m \in \{1,2,3\}$, and let $r_1 < r_2$ be the remaining mode indices, ordered consistently with the Kronecker products below. Let $R, R_0 \in \mathbb{R}^{N_{r_1} \times \ell}$ and $S, S_0 \in \mathbb{R}^{N_{r_2} \times \ell}$. If $R * S$ has full column rank, then*

$$\big\|E_{(m)}P_{R*S}\big\|_F \leqslant \ell\|E\|, \tag{113}$$

$$\big\|E_{(m)}M_{R*S}(R_0 * S_0)\big\|_F \leqslant \sqrt{\ell+1}\,\|E\|\,\|M_{R*S}(R_0 * S_0)\|_F\,. \tag{114}$$

For the grouped setting, let $E \in \mathbb{R}^{N \times M \times T}$ and let $D \in \{0,1\}^{N \times G}$ have full column rank. Define

$$Q_D = D(D'D)^{-1/2}, \qquad E^G = E \times_1 Q_D', \qquad \bar{U} = (D'D)^{1/2}U, \qquad \bar{U}_0 = (D'D)^{1/2}U_0,$$

where $U, U_0 \in \mathbb{R}^{G \times \ell}$, $V, V_0 \in \mathbb{R}^{M \times \ell}$, and $W, W_0 \in \mathbb{R}^{T \times \ell}$. Let $(m, L, L_0) \in \{(2, W, W_0), (3, V, V_0)\}$. If $(DU * L)$ has full column rank, then

$$\big\|E_{(m)}P_{DU*L}\big\|_F = \Big\|E^G_{(m)}P_{\bar{U}*L}\Big\|_F \leqslant \ell\|E^G\|, \tag{115}$$

$$\big\|E_{(m)}M_{DU*L}(DU_0 * L_0)\big\|_F = \Big\|E^G_{(m)}M_{\bar{U}*L}(\bar{U}_0 * L_0)\Big\|_F \leqslant \sqrt{\ell+1}\,\|E^G\|\,\|M_{DU*L}(DU_0 * L_0)\|_F\,. \tag{116}$$

All bounds are deterministic and uniform over the admissible loading matrices.

Let $\mathcal{L}_m \subseteq \mathbb{R}^{N_m}$ have dimension r_m, and let Q_m denote its orthogonal projector, for $m = 1,2,3$. Define

$$\mathcal{V} = (\mathbb{R}^{N_1} \otimes \mathcal{L}_2 \otimes \mathcal{L}_3) + (\mathcal{L}_1 \otimes \mathbb{R}^{N_2} \otimes \mathcal{L}_3) + (\mathcal{L}_1 \otimes \mathcal{L}_2 \otimes \mathbb{R}^{N_3}).$$

Its orthogonal projector is

$$P_{\mathcal{V}} = I_{N_1} \otimes Q_2 \otimes Q_3 + Q_1 \otimes (I_{N_2} - Q_2) \otimes Q_3 + Q_1 \otimes Q_2 \otimes (I_{N_3} - Q_3).$$

Writing $e = \mathrm{vec}(E)$ *and* $c_{\mathcal{V}} = \sqrt{r_2 r_3 + r_1 r_3 + r_1 r_2}$*, we have*

$$\|P_{\mathcal{V}} e\| \leqslant c_{\mathcal{V}} \|E\|. \tag{117}$$

Consequently,

$$|e'z| \leqslant c_{\mathcal{V}} \|E\| \|z\|, \qquad z \in \mathcal{V},$$

and every conformable matrix H *satisfying* $H = HP_{\mathcal{V}}$ *obeys*

$$\|He\| \leqslant c_{\mathcal{V}} \|H\| \|E\|.$$

These inequalities are deterministic and uniform over the subspaces and coefficient matrices. They therefore also apply when these objects depend on the errors.

Proof. For $q \in \mathbb{R}^{N_{r_1} N_{r_2}}$, let $\mathrm{mat}_{-m}(q) \in \mathbb{R}^{N_{r_1} \times N_{r_2}}$ denote its reshape, with $\mathrm{mat}_{-m}(a \otimes b) = ab'$. Let $r = \mathrm{rank}(\mathrm{mat}_{-m}(q))$, and write a singular value decomposition as

$$\mathrm{mat}_{-m}(q) = \sum_{h=1}^{r} \sigma_h a_h b_h', \qquad \|a_h\| = \|b_h\| = 1.$$

Then $q = \sum_{h=1}^{r} \sigma_h (a_h \otimes b_h)$. For every unit vector $c \in \mathbb{R}^{N_m}$, the scalar $c' E_{(m)} (a_h \otimes b_h)$ is the contraction of E against unit vectors in its three modes, so its absolute value is bounded by the tensor spectral norm $\|E\|$. Consequently,

$$\begin{aligned} \|E_{(m)} q\| &= \sup_{\|c\|=1} \left| \sum_{h=1}^{r} \sigma_h c' E_{(m)} (a_h \otimes b_h) \right| \\ &\leqslant \|E\| \sum_{h=1}^{r} \sigma_h = \|E\| \, \|\mathrm{mat}_{-m}(q)\|_* \\ &\leqslant \sqrt{r} \, \|E\| \left(\sum_{h=1}^{r} \sigma_h^2 \right)^{1/2} = \sqrt{r} \, \|E\| \|q\|. \end{aligned} \tag{118}$$

Here $\|\cdot\|_*$ denotes the nuclear norm, the last inequality follows from Cauchy–Schwarz, and the last equality uses $\|\mathrm{mat}_{-m}(q)\|_F = \|q\|$. The case $q = 0$ is immediate.

Let $Q = (q_1, \ldots, q_\ell)$ be an orthonormal basis of the column space of $R * S$, and write $M_{R*S}(R_0 * S_0) = (d_1, \ldots, d_\ell)$. Each q_k is a linear combination of the ℓ columns of $R * S$, so its reshape has rank at most ℓ. Each d_k is a rank-one column of $R_0 * S_0$ minus an element in the column space of $R * S$, so its reshape has rank at most $\ell + 1$. Since $P_{R*S} = QQ'$, (118) therefore implies

$$\left\| E_{(m)} P_{R*S} \right\|_F^2 = \sum_{k=1}^{\ell} \|E_{(m)} q_k\|^2 \leqslant \ell \|E\|^2 \sum_{k=1}^{\ell} \|q_k\|^2 = \ell^2 \|E\|^2,$$

$$\left\| E_{(m)} M_{R*S}(R_0 * S_0) \right\|_F^2 = \sum_{k=1}^{\ell} \|E_{(m)} d_k\|^2 \leqslant (\ell + 1) \|E\|^2 \sum_{k=1}^{\ell} \|d_k\|^2 = (\ell + 1) \|E\|^2 \left\| M_{R*S}(R_0 * S_0) \right\|_F^2.$$

Taking square roots proves (113)–(114).

For either grouped case, let K be the number of rows of L, and set $Q = Q_D \otimes I_K$. Since $Q_D' Q_D = I_G$, the Khatri–Rao and matricization identities give

$$Q'Q = I_{GK}, \qquad (DU * L) = Q(\bar{U} * L), \qquad (DU_0 * L_0) = Q(\bar{U}_0 * L_0), \qquad E^G_{(m)} = E_{(m)}Q.$$

In particular, $(DU * L)$ has full column rank, and

$$P_{DU*L} = QP_{\bar{U}*L}Q', \qquad M_{DU*L}(DU_0 * L_0) = QM_{\bar{U}*L}(\bar{U}_0 * L_0).$$

Consequently,

$$E_{(m)}P_{DU*L} = E^G_{(m)}P_{\bar{U}*L}Q', \qquad E_{(m)}M_{DU*L}(DU_0 * L_0) = E^G_{(m)}M_{\bar{U}*L}(\bar{U}_0 * L_0).$$

Because $Q'Q = I_{GK}$, multiplication by Q on the left or by Q' on the right preserves the Frobenius norm. Applying (113), (114) to E^G, $\bar{U} * L$, and $\bar{U}_0 * L_0$ now proves (115), (116).

The three summands defining $P_{\mathcal{V}}$ are mutually orthogonal projectors whose ranges sum to $\mathcal{V}$. Let $\{b_h\}_{h=1}^{r_2}$ and $\{c_k\}_{k=1}^{r_3}$ be orthonormal bases of $\mathcal{L}_2$ and $\mathcal{L}_3$. The definition of the tensor spectral norm gives

$$\|(I_{N_1} \otimes b_h' \otimes c_k')e\| = \sup_{\|a\|=1} |(a' \otimes b_h' \otimes c_k')e| \leqslant \|E\|.$$

Therefore,

$$\|(I_{N_1} \otimes Q_2 \otimes Q_3)e\|^2 \leqslant r_2 r_3 \|E\|^2.$$

The other two summands are bounded analogously by $r_1 r_3 \|E\|^2$ and $r_1 r_2 \|E\|^2$, since their additional complementary projectors cannot increase the norm. Orthogonality proves (117). The remaining bounds follow from $z = P_{\mathcal{V}} z$ and $H = HP_{\mathcal{V}}$. □

Lemma 6 (Convergence of the generated regressors)**.** *Under Assumptions 1, 2, 3, and 4, the global least-squares estimator defined in Assumption 1 satisfies*

$$\begin{aligned}
\frac{1}{\sqrt{MT}} \left\| M_{\hat{V}*\hat{W}}(V * W) \right\|_F &= O_p\left(\sqrt{\frac{G+M+T}{NMT}}\right), \\
\frac{1}{\sqrt{NT}} \left\| M_{D\hat{U}*\hat{W}}(DU * W) \right\|_F &= O_p\left(\sqrt{\frac{G+M+T}{NMT}}\right), \\
\frac{1}{\sqrt{NM}} \left\| M_{D\hat{U}*\hat{V}}(DU * V) \right\|_F &= O_p\left(\sqrt{\frac{G+M+T}{NMT}}\right), \\
\|\hat{\delta} - \delta\| &= O_p\left(\sqrt{\frac{G+M+T}{NMT}}\right).
\end{aligned}$$

Also, the common component satisfies

$$\frac{\|\hat{\Pi} - \Pi\|_F}{\sqrt{NMT}} = O_p\left(\sqrt{\frac{G+M+T}{NMT}}\right).$$

Proof. Let $n = NMT$, $s = G + M + T$, $\rho = \sqrt{s/n}$, $Q_D = D(D'D)^{-1/2}$, $E^G = E \times_1 Q_D'$, and $e = \mathrm{vec}(E)$. Assumption 2 and Lemma 4(i) give

$$\frac{\|X\|_F}{\sqrt{n}} = O_p(1), \qquad \left\| \frac{X_{(4)}e}{\sqrt{n}} \right\| = O_p(1), \qquad \frac{\|E^G\|}{\sqrt{n}} = O_p(\rho). \tag{119}$$

Let Π and $\hat{\Pi}$ denote the true and estimated common-component tensors, and define

$$\hat{\gamma} = \hat{\delta} - \delta, \qquad \Delta = \hat{\Pi} - \Pi, \qquad R = X \times_4 \hat{\gamma}' + \Delta, \qquad r_n = \frac{\|R\|_F}{\sqrt{n}}.$$

Global least-squares optimality implies $\|E - R\|_F^2 \leqslant \|E\|_F^2$, and hence

$$\|R\|_F^2 \leqslant 2\|\hat{\gamma}\|\|X_{(4)}e\| + 2|\langle E, \Delta\rangle|. \tag{120}$$

Let (U, V, W) and $(\hat{U}, \hat{V}, \hat{W})$ be the true and estimated fixed effects. Choose a full-column-rank submatrix K of $[(D\hat{U} * \hat{V}), (DU * V)]$ with the same column space. Since

$$\Delta_{(3)} = \hat{W}(D\hat{U} * \hat{V})' - W(DU * V)', \qquad \Delta_{(3)}P_K = \Delta_{(3)},$$

and K contains at most 2ℓ grouped rank-one columns, Lemma 5 gives

$$\begin{aligned} |\langle E, \Delta\rangle| &= |\langle E_{(3)}P_K, \Delta_{(3)}\rangle| \\ &\leqslant \left\|E_{(3)}P_K\right\|_F \|\Delta\|_F \leqslant 2\ell\|E^G\| \, \|\Delta\|_F . \end{aligned} \tag{121}$$

The contraction bound is deterministic and uniform, so it applies to the data-dependent matrix K.

Next, $M_W\Pi_{(3)} = 0$ and $\hat{\Pi}_{(3)}M_{D\hat{U}*\hat{V}} = 0$. Therefore, Assumption 4 implies that, with probability approaching one,

$$\begin{aligned} r_n^2 &\geqslant \frac{1}{n}\left\|M_W R_{(3)} M_{D\hat{U}*\hat{V}}\right\|_F^2 \\ &= \hat{\gamma}'\left[\frac{1}{n}X_{(4)}(M_{D\hat{U}*\hat{V}} \otimes M_W)X_{(4)}'\right]\hat{\gamma} \\ &\geqslant c_\delta\|\hat{\gamma}\|^2 \end{aligned}$$

for some $c_\delta > 0$. On this event,

$$\|\hat{\gamma}\| \leqslant c_\delta^{-1/2} r_n, \qquad \frac{\|\Delta\|_F}{\sqrt{n}} \leqslant r_n + \frac{\|X\|_F}{\sqrt{n}}\|\hat{\gamma}\| \leqslant \left(1 + c_\delta^{-1/2}\frac{\|X\|_F}{\sqrt{n}}\right) r_n.$$

Substituting these inequalities and (121) into (120), and using (119) and $n^{-1/2} \leqslant \rho$, yields

$$r_n^2 \leqslant \frac{2}{\sqrt{n}}\left\|\frac{X_{(4)}e}{\sqrt{n}}\right\|\|\hat{\gamma}\| + 4\ell\frac{\|E^G\|}{\sqrt{n}}\frac{\|\Delta\|_F}{\sqrt{n}} \leqslant O_p(\rho)r_n.$$

Thus $r_n = O_p(\rho)$, and consequently

$$\|\hat{\delta} - \delta\| + \frac{\left\|\hat{\Pi} - \Pi\right\|_F}{\sqrt{n}} = O_p(\rho). \tag{122}$$

It remains to obtain the three span bounds. For $m = 1, 2, 3$, write $\Pi_{(m)} = A_m C_m'$, where

$$(A_m, C_m, s_m) = \begin{cases} (DU, V * W, N), & m = 1, \\ (V, DU * W, M), & m = 2, \\ (W, DU * V, T), & m = 3, \end{cases}$$

and let $\hat{A}_m, \hat{C}_m$ denote the corresponding estimated matrices. Since $\hat{\Pi}_{(m)} = \hat{A}_m \hat{C}_m'$,

$$\Delta_{(m)} M_{\hat{C}_m} = -A_m C_m' M_{\hat{C}_m}.$$

Assumption 2 bounds $\lambda_{\min}(A_m' A_m / s_m)$ away from zero with probability approaching one. Hence, for each m,

$$\sqrt{\frac{s_m}{n}} \left\| M_{\hat{C}_m} C_m \right\|_F \leqslant \left[\lambda_{\min} \left(\frac{A_m' A_m}{s_m} \right) \right]^{-1/2} \frac{\|\Delta\|_F}{\sqrt{n}} = O_p(\rho).$$

The normalizations $\sqrt{s_m/n}$ are $1/\sqrt{MT}$, $1/\sqrt{NT}$, and $1/\sqrt{NM}$, respectively. Together with (122), these establish all four claims. □

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A Proofs for Section 5 (Asymptotic Theory)

Lemma 1 (Normalized factor alignment and asymptotic inactivity of the inequality constraint)**.** *Suppose that Assumptions 1, 2, 3, and 4 hold. Let (U, V, W) denote the true factor matrices in the normalized representation satisfying* (19)*. Let $(\hat{\delta}, \hat{U}, \hat{V}, \hat{W})$ be the global least-squares estimator defined in Assumption 1. There exist possibly random $\ell \times \ell$ permutation and diagonal sign matrices Π_ℓ, S_v, S_w, with the diagonal entries of S_v and S_w belonging to $\{-1, 1\}$, such that*

$$\frac{\|D(\hat{U}\Pi_\ell S_v S_w - U)\|_F}{\sqrt{N}} + \frac{\|\hat{V}\Pi_\ell S_v - V\|_F}{\sqrt{M}} + \frac{\|\hat{W}\Pi_\ell S_w - W\|_F}{\sqrt{T}} = O_p\left(\sqrt{\frac{G+M+T}{NMT}}\right). \tag{21}$$

Define the residual tensor

$$\hat{E} = Y - X \times_4 \hat{\delta}' - \sum_{k=1}^{\ell} (D\hat{u}_k \circ \hat{v}_k \circ \hat{w}_k).$$

With probability approaching one, the estimator satisfies the unrestricted normal equations

$$\begin{aligned} X_{(4)}\,\mathrm{vec}(\hat{E}) = 0, \quad D'\hat{E}_{(1)}(\hat{V} * \hat{W}) = 0, \\ \hat{E}_{(2)}(D\hat{U} * \hat{W}) = 0, \quad \hat{E}_{(3)}(D\hat{U} * \hat{V}) = 0. \end{aligned} \tag{22}$$

Proof. Write $n = NMT$, $s = G + M + T$, and $\rho = \sqrt{s/n}$. Assumption 1 implies $\rho \to 0$. Let Π and $\hat{\Pi}$ denote the true and estimated common-component tensors. Define

$$Q_D = D(D'D)^{-1/2}, \qquad A_1 = \frac{(D'D)^{1/2}U}{\sqrt{N}}, \qquad A_2 = \frac{V}{\sqrt{M}}, \qquad A_3 = \frac{W}{\sqrt{T}},$$

with estimated counterparts $\hat{A}_m$, and set

$$\mathcal{P} = \frac{1}{\sqrt{n}}\Pi \times_1 Q_D', \qquad \hat{\mathcal{P}} = \frac{1}{\sqrt{n}}\hat{\Pi} \times_1 Q_D', \qquad h = \hat{\mathcal{P}} - \mathcal{P}.$$

Also, define

$$C_1 = A_2 * A_3, \qquad \hat{C}_1 = \hat{A}_2 * \hat{A}_3.$$

Then

$$\mathcal{P}_{(1)} = A_1 C_1', \qquad \hat{\mathcal{P}}_{(1)} = \hat{A}_1 \hat{C}_1'.$$

The preliminary common-component bound established by the global least-squares comparison in Lemma 6 gives

$$\|h\|_F = \frac{\|\hat{\Pi} - \Pi\|_F}{\sqrt{n}} = O_p(\rho). \tag{123}$$

This bound uses global optimality and does not require the factor normal equations. The equality follows because $Q_D'Q_D = I_G$ and the first-mode fibers of both Π and $\hat{\Pi}$ belong to $\mathrm{col}(D)$. All inverses below are used on the indicated events whose probabilities tend to one; the eventual permutation and sign matrices may be defined arbitrarily on the complementary events.

1. Conditioning of the estimated joint loading space. On the strong-factor event,

$$c_0 I_\ell \preceq A_m' A_m \preceq c_1 I_\ell, \qquad m = 1, 2, 3.$$

For every $z \in \mathbb{R}^\ell$,

$$\begin{aligned} \|C_1 z\|^2 &= \|A_2 \operatorname{diag}(z) A_3'\|_F^2 \\ &\geqslant \sigma_{\min}(A_2)^2 \sigma_{\min}(A_3)^2 \|\operatorname{diag}(z)\|_F^2 \\ &\geqslant c_0^2 \|z\|^2. \end{aligned}$$

Thus $\sigma_{\min}(C_1) \geqslant c_0$, and

$$\sigma_\ell(\mathcal{P}_{(1)}) \geqslant \sigma_{\min}(A_1)\sigma_{\min}(C_1) \geqslant c_0^{3/2}.$$

By (123) and the singular-value perturbation inequality,

$$\sigma_\ell(\hat{\mathcal{P}}_{(1)}) \geqslant \sigma_\ell(\mathcal{P}_{(1)}) - \|h_{(1)}\| \geqslant \frac{c_0^{3/2}}{2}$$

with probability approaching one. Since $\hat{\mathcal{P}}_{(1)} = \hat{A}_1 \hat{C}_1'$ has rank at most ℓ, it follows that $\hat{A}_1$ and $\hat{C}_1$ have full column rank on this event. Moreover, feasibility gives $\|\hat{A}_1\| \leqslant \sqrt{c_U}$, so

$$\sigma_{\min}(\hat{C}_1) \geqslant \frac{\sigma_\ell(\hat{\mathcal{P}}_{(1)})}{\|\hat{A}_1\|} \geqslant b_*, \qquad b_* := \frac{c_0^{3/2}}{2\sqrt{c_U}} > 0. \tag{124}$$

The identity

$$h_{(1)} M_{\hat{C}_1} = -A_1 C_1' M_{\hat{C}_1}$$

and the lower bound on $\sigma_{\min}(A_1)$ imply

$$\|M_{\hat{C}_1} C_1\|_F \leqslant c_0^{-1/2} \|h\|_F.$$

Since $\operatorname{col}(C_1)$ and $\operatorname{col}(\hat{C}_1)$ both have dimension ℓ, their projector distance satisfies

$$\begin{aligned} \|P_{\hat{C}_1} - P_{C_1}\| &= \left\| M_{\hat{C}_1} C_1 (C_1' C_1)^{-1/2} \right\| \\ &\leqslant \frac{\|M_{\hat{C}_1} C_1\|_F}{\sigma_{\min}(C_1)} \\ &\leqslant c_0^{-3/2} \|h\|_F = O_p(\rho). \end{aligned} \tag{125}$$

2. Normalization-preserving alignment. For $m = 2, 3$, let $d_2 = M$ and $d_3 = T$, and complete A_m to an invertible square matrix

$$B_m = [A_m, A_{m,\perp}],$$

where the columns of $A_{m,\perp}$ form an orthonormal basis of $\operatorname{col}(A_m)^\perp$. Then

$$B_m' B_m = \begin{pmatrix} A_m' A_m & 0 \\ 0 & I_{d_m - \ell} \end{pmatrix}.$$

Because $c_0 \leqslant 1 \leqslant c_1$, the strong-factor event gives

$$\|B_m\| \leqslant \sqrt{c_1}, \qquad \|B_m^{-1}\| \leqslant c_0^{-1/2}.$$

Let

$$L = B_2^{-1} \otimes B_3^{-1}.$$

Since $B_m^{-1} a_{m,k} = e_k^{(d_m)}$ for $k = 1, \dots, \ell$, the image of $\operatorname{col}(C_1)$ under L is

$$\mathcal{H}_0 = \operatorname{span}\{e_k^{(M)} \otimes e_k^{(T)} : k = 1, \dots, \ell\},$$

where $(e_k^{(M)})$ and $(e_k^{(T)})$ are elementary basis of dimension M and T respectively. Furthermore,

$$\|L\| \leqslant c_0^{-1}, \qquad \|L^{-1}\| \leqslant c_1.$$

For any nonzero $z \in \operatorname{col}(\hat{C}_1)$, we have $P_{\hat{C}_1} z = z$ and $L P_{C_1} z \in \mathcal{H}_0$. Consequently,

$$\operatorname{dist}(Lz, \mathcal{H}_0) = \min_{z_0 \in \mathcal{H}_0} \|Lz - z_0\| \leqslant \|Lz - L P_{C_1} z\| = \|L(P_{\hat{C}_1} - P_{C_1}) z\| \leqslant \|L\| \|P_{\hat{C}_1} - P_{C_1}\| \|z\|.$$

Also,

$$\|Lz\| \geqslant \frac{\|z\|}{\|L^{-1}\|}.$$

It follows that

$$\begin{aligned} \epsilon_n &:= \sup_{\substack{z \in \operatorname{col}(\hat{C}_1) \\ \|z\| = 1}} \frac{\operatorname{dist}(Lz, \mathcal{H}_0)}{\|Lz\|} \\ &\leqslant \|L\| \|L^{-1}\| \|P_{\hat{C}_1} - P_{C_1}\| \\ &\leqslant \frac{c_1}{c_0} \|P_{\hat{C}_1} - P_{C_1}\| = O_p(\rho). \end{aligned} \tag{126}$$

For each estimated component $r = 1, \dots, \ell$, define

$$x_r = \frac{B_2^{-1} \hat{a}_{2,r}}{\|B_2^{-1} \hat{a}_{2,r}\|}, \qquad y_r = \frac{B_3^{-1} \hat{a}_{3,r}}{\|B_3^{-1} \hat{a}_{3,r}\|},$$

where $a_{m,r}$ and $\hat{a}_{m,r}$ denote columns of A_m and $\hat{A}_m$. Let $\hat{c}_r = \hat{a}_{2,r} \otimes \hat{a}_{3,r}$ be column r of $\hat{C}_1$. The column normalizations imply $\|\hat{c}_r\| = 1$, and

$$q_r := x_r \otimes y_r = \frac{L \hat{c}_r}{\|L \hat{c}_r\|}.$$

Therefore, q_r is a unit vector in $L \operatorname{col}(\hat{C}_1)$ and

$$\|M_{\mathcal{H}_0} q_r\| = \frac{\operatorname{dist}(L \hat{c}_r, \mathcal{H}_0)}{\|L \hat{c}_r\|} \leqslant \epsilon_n.$$

The vectors $e_k^{(M)} \otimes e_k^{(T)}$, $k = 1, \dots, \ell$, are orthonormal, so

$$P_{\mathcal{H}_0} q_r = \sum_{k=1}^{\ell} x_{r,k} y_{r,k} (e_k^{(M)} \otimes e_k^{(T)}).$$

By orthogonality,

$$1 - \sum_{k=1}^{\ell} x_{r,k}^2 y_{r,k}^2 = \|M_{\mathcal{H}_0} q_r\|^2 \leqslant \epsilon_n^2.$$

Choose k_r maximizing $x_{r,k}^2$ over $k = 1, \dots, \ell$, with ties broken by a fixed rule. Since x_r and y_r are unit vectors,

$$1 - \epsilon_n^2 \leqslant \sum_{k=1}^{\ell} x_{r,k}^2 y_{r,k}^2 \leqslant x_{r,k_r}^2 \sum_{k=1}^{\ell} y_{r,k}^2 \leqslant x_{r,k_r}^2.$$

Moreover,

$$\sum_{k=1}^{\ell} x_{r,k}^2 y_{r,k}^2 \leqslant y_{r,k_r}^2 + \sum_{\substack{k \leqslant \ell \\ k \neq k_r}} x_{r,k}^2 \leqslant y_{r,k_r}^2 + 1 - x_{r,k_r}^2 \leqslant y_{r,k_r}^2 + \epsilon_n^2.$$

Hence

$$x_{r,k_r}^2 \geqslant 1 - \epsilon_n^2, \qquad y_{r,k_r}^2 \geqslant 1 - 2\epsilon_n^2.$$

Choose signs $s_{v,r}, s_{w,r} \in \{-1, 1\}$ satisfying

$$s_{v,r} x_{r,k_r} = |x_{r,k_r}|, \qquad s_{w,r} y_{r,k_r} = |y_{r,k_r}|.$$

Then

$$\|x_r - s_{v,r} e_{k_r}^{(M)}\|^2 = 2(1 - |x_{r,k_r}|) \leqslant 2(1 - x_{r,k_r}^2) \leqslant 2\epsilon_n^2,$$
$$\|y_r - s_{w,r} e_{k_r}^{(T)}\|^2 = 2(1 - |y_{r,k_r}|) \leqslant 2(1 - y_{r,k_r}^2) \leqslant 4\epsilon_n^2.$$

To return to the original coordinates, note that for any invertible matrix B and unit vectors x, y,

$$\left\| \frac{Bx}{\|Bx\|} - \frac{By}{\|By\|} \right\| \leqslant \frac{2\|B(x-y)\|}{\|Bx\|} \leqslant \frac{2\|B\|}{\sigma_{\min}(B)} \|x - y\|.$$

The estimated column normalizations give

$$\hat{a}_{2,r} = \frac{B_2 x_r}{\|B_2 x_r\|}, \qquad \hat{a}_{3,r} = \frac{B_3 y_r}{\|B_3 y_r\|},$$

while the true column normalizations imply

$$\frac{B_m(s e_k^{(d_m)})}{\|B_m(s e_k^{(d_m)})\|} = s a_{m,k}, \qquad s \in \{-1, 1\}.$$

The uniform bounds on B_m and B_m^{-1} therefore yield

$$\|\hat{a}_{2,r} - s_{v,r} a_{2,k_r}\| + \|\hat{a}_{3,r} - s_{w,r} a_{3,k_r}\| \leqslant K\epsilon_n, \qquad r = 1, \dots, \ell, \tag{127}$$

where K is a finite constant depending only on c_0 and c_1.

We next show that different estimated components select different coordinates. Let $c_k = a_{2,k} \otimes a_{3,k}$ and $\eta_r = s_{v,r} s_{w,r}$. Since all columns of $A_2, A_3, \hat{A}_2, \hat{A}_3$ have unit norm, (127) implies

$$\|\hat{c}_r - \eta_r c_{k_r}\| \leqslant K\epsilon_n.$$

If $k_r = k_s$ for some $r \neq s$, then

$$\|\hat{c}_r - \eta_r \eta_s \hat{c}_s\| \leqslant 2K\epsilon_n.$$

Consequently,

$$\sigma_{\min}(\hat{C}_1) \leqslant \left\| \hat{C}_1 \frac{e_r - \eta_r \eta_s e_s}{\sqrt{2}} \right\| = \frac{\|\hat{c}_r - \eta_r \eta_s \hat{c}_s\|}{\sqrt{2}} \leqslant \sqrt{2} K \epsilon_n.$$

This contradicts (124) whenever $\sqrt{2}K\epsilon_n < b_*$. Since $\epsilon_n = o_p(1)$, the map $r \mapsto k_r$ is a permutation with probability approaching one.

Choose a permutation matrix Π_ℓ that reorders the estimated columns according to this matching, and choose diagonal sign matrices S_v, S_w that remove the corresponding signs. Define

$$\check{A}_1 = \hat{A}_1 \Pi_\ell S_v S_w, \qquad \check{A}_2 = \hat{A}_2 \Pi_\ell S_v, \qquad \check{A}_3 = \hat{A}_3 \Pi_\ell S_w.$$

The transformation preserves the fitted tensor and the column normalizations. Since ℓ is fixed, (127) and (126) give

$$\|\check{A}_2 - A_2\|_F + \|\check{A}_3 - A_3\|_F = O_p(\rho).$$

Let

$$\check{C}_1 = \check{A}_2 * \check{A}_3 = \hat{C}_1 \Pi_\ell S_v S_w.$$

The unit column norms imply

$$\|\check{C}_1 - C_1\|_F \leqslant \|\check{A}_2 - A_2\|_F + \|\check{A}_3 - A_3\|_F = O_p(\rho).$$

Moreover, $\Pi_\ell S_v S_w$ is orthogonal, so (124) yields

$$\left\| \check{C}_1 (\check{C}_1' \check{C}_1)^{-1} \right\| = \frac{1}{\sigma_{\min}(\check{C}_1)} = \frac{1}{\sigma_{\min}(\hat{C}_1)} \leqslant b_*^{-1} = \frac{2\sqrt{c_U}}{c_0^{3/2}}$$

with probability approaching one.

Because the fitted tensor is unchanged, $\hat{\mathcal{P}}_{(1)} = \check{A}_1 \check{C}_1'$. Therefore,

$$h_{(1)} + A_1 (C_1 - \check{C}_1)' = (\check{A}_1 - A_1) \check{C}_1',$$

and hence

$$\check{A}_1 - A_1 = \left[h_{(1)} + A_1 (C_1 - \check{C}_1)' \right] \check{C}_1 (\check{C}_1' \check{C}_1)^{-1}.$$

Taking Frobenius norms and using $\|A_1\| \leqslant \sqrt{c_1}$ gives

$$\|\check{A}_1 - A_1\|_F \leqslant b_*^{-1} \left\{ \|h\|_F + \sqrt{c_1} \|\check{C}_1 - C_1\|_F \right\} = O_p(\rho).$$

Thus

$$\sum_{m=1}^{3} \|\check{A}_m - A_m\|_F = O_p(\rho). \tag{128}$$

Finally, for every conformable matrix H,

$$\|(D'D)^{1/2} H\|_F = \|DH\|_F.$$

Consequently, (128) is exactly the weighted factor-alignment rate in (21).

3. Eigenvalue consistency and inactivity of the upper bound. For the inactivity of the upper bound in the parameter constraints, we show

$$\mathbb{P}\left\{ \lambda_{\max}\left(\frac{1}{N} \hat{U}' D' D \hat{U} \right) < c_U \right\} \to 1. \tag{129}$$

Since $\Pi_\ell S_v S_w$ is orthogonal,

$$\lambda_{\max}(\hat{A}_1'\hat{A}_1) = \lambda_{\max}(\check{A}_1'\check{A}_1).$$

Moreover, feasibility gives $\|\check{A}_1\| = \|\hat{A}_1\| \leqslant \sqrt{c_U}$. On the strong-factor event,

$$\begin{aligned}\|\check{A}_1'\check{A}_1 - A_1'A_1\| &\leqslant (\|\check{A}_1\| + \|A_1\|)\|\check{A}_1 - A_1\| \\ &\leqslant (\sqrt{c_U} + \sqrt{c_1})\|\check{A}_1 - A_1\| \\ &= O_p(\rho).\end{aligned}$$

The largest-eigenvalue perturbation inequality therefore gives

$$\left|\lambda_{\max}(\hat{A}_1'\hat{A}_1) - \lambda_{\max}(A_1'A_1)\right| \leqslant \|\check{A}_1'\check{A}_1 - A_1'A_1\| = O_p(\rho),$$

which proves

$$\left|\lambda_{\max}\left(\frac{1}{N}\hat{U}'D'D\hat{U}\right) - \lambda_{\max}\left(\frac{1}{N}U'D'DU\right)\right| = O_p(\rho)$$

Since $\lambda_{\max}(A_1'A_1) \leqslant c_1$ with probability approaching one and $c_U - c_1 > 0$ is fixed,

$$\begin{aligned}&\mathbb{P}\left\{\lambda_{\max}(\hat{A}_1'\hat{A}_1) > \frac{c_1 + c_U}{2}\right\} \\ &\quad\leqslant \mathbb{P}\{\lambda_{\max}(A_1'A_1) > c_1\} + \mathbb{P}\left\{\left|\lambda_{\max}(\hat{A}_1'\hat{A}_1) - \lambda_{\max}(A_1'A_1)\right| > \frac{c_U - c_1}{2}\right\} \to 0.\end{aligned}$$

This proves (129).

4. Validity of the unrestricted normal equations. Work at the original estimated factors on the strict-slack event in (129), and evaluate all gradients below at $(\hat{\delta}, \hat{U}, \hat{V}, \hat{W})$. Since δ is unrestricted and sufficiently small perturbations of U remain feasible,

$$\nabla_\delta S_{NMT} = 0, \qquad \nabla_U S_{NMT} = 0.$$

These are the slope and U-block equations in (22).

The equality constraints $\|v_k\|^2 = M$ and $\|w_k\|^2 = T$ remain in force. First-order optimality along their tangent directions implies that, for each k, there exist scalars α_k, β_k such that

$$\nabla_{v_k} S_{NMT} = \alpha_k \hat{v}_k, \qquad \nabla_{w_k} S_{NMT} = \beta_k \hat{w}_k.$$

Direct differentiation of the trilinear component gives

$$\begin{aligned}\hat{u}_k'\nabla_{u_k} S_{NMT} = \hat{v}_k'\nabla_{v_k} S_{NMT} &= \hat{w}_k'\nabla_{w_k} S_{NMT} \\ &= -\frac{2}{n}\left\langle \hat{E}, D\hat{u}_k \circ \hat{v}_k \circ \hat{w}_k \right\rangle_F.\end{aligned}$$

Since $\nabla_{u_k} S_{NMT} = 0$, all three radial derivatives vanish. Hence

$$0 = \hat{v}_k'\nabla_{v_k} S_{NMT} = \alpha_k \|\hat{v}_k\|^2 = \alpha_k M,$$

and

$$0 = \hat{w}_k'\nabla_{w_k} S_{NMT} = \beta_k \|\hat{w}_k\|^2 = \beta_k T.$$

Because $M, T > 0$, we obtain $\alpha_k = \beta_k = 0$ for every k. Thus the unrestricted gradients with respect to V and W also vanish. Differentiating the criterion with respect to these blocks gives

$$\hat{E}_{(2)}(D\hat{U} * \hat{W}) = 0, \qquad \hat{E}_{(3)}(D\hat{U} * \hat{V}) = 0.$$

Together with the slope and U-block equations, these establish (22) on an event whose probability tends to one. □

Proposition 1. *Suppose that Assumptions 1, 2, 3, and 4 hold. Let $\mathcal{T}$ be the tangent space defined in* (11) *evaluated at the true parameters (U, V, W). Let*

$$\Gamma_{NMT} = \frac{1}{NMT} X_{(4)} M_{\mathcal{T}} X_{(4)}', \qquad \mathcal{Z}_{NMT} = \frac{1}{\sqrt{NMT}} X_{(4)} M_{\mathcal{T}} \operatorname{vec}(E).$$

The least squares estimator $\hat{\delta}$ defined in Assumption 1 satisfies

$$\Gamma_{NMT}\sqrt{NMT}(\hat{\delta} - \delta) = \mathcal{Z}_{NMT} - \mathcal{B}_1 - \mathcal{B}_2 - \mathcal{B}_3 + O_p\left(\sqrt{\frac{G+M+T}{NMT}} + \frac{(G+M+T)^{3/2}}{NMT}\right), \tag{23}$$

where

$$\mathcal{B}_1 = \frac{1}{\sqrt{NMT}} X_{(4)} P_{\mathcal{T}} Q \operatorname{vec}\left(\bar{U}(\bar{U}'\bar{U})^{-1}((V*W)'(V*W))^{-1}(V*W)'\Omega_1^G M_{V*W}\right)$$
$$\mathcal{B}_2 = \frac{1}{\sqrt{NMT}} X_{(4)} P_{\mathcal{T}} Q K_{1,2}^G \operatorname{vec}\left(V(V'V)^{-1}((\bar{U}*W)'(\bar{U}*W))^{-1}(\bar{U}*W)'\Omega_2^G M_{\bar{U}*W}\right)$$
$$\mathcal{B}_3 = \frac{1}{\sqrt{NMT}} X_{(4)} P_{\mathcal{T}} Q K_{1,3}^G \operatorname{vec}\left(W(W'W)^{-1}((\bar{U}*V)'(\bar{U}*V))^{-1}(\bar{U}*V)'\Omega_3^G M_{\bar{U}*V}\right).$$

Here, $\bar{U} = (D'D)^{1/2}U$ and $\Omega_m^G = \mathbb{E}[(E_{(m)}^G)'E_{(m)}^G \mid \mathcal{F}]$ for $m = 1, 2, 3$. The matrices Q and $P_{\mathcal{T}}$ are as defined in (12) *and* (15)*, respectively, and $K_{1,m}^G$ denotes the mode-permutation matrix for tensors of dimensions $G \times M \times T$. All factor-dependent matrices and projectors in the displayed bias terms are evaluated at the true parameters. Moreover,*

$$\mathcal{B}_1 = O_p(G/\sqrt{NMT}), \qquad \mathcal{B}_2 = O_p(M/\sqrt{NMT}), \qquad \mathcal{B}_3 = O_p(T/\sqrt{NMT}).$$

Proof. Write

$$n = NMT, \qquad s = G+M+T, \qquad \rho = \sqrt{\frac{s}{n}}, \qquad d_1 = G, \quad d_2 = M, \quad d_3 = T, \qquad d_\times = GMT.$$

Assumption 1 implies $\rho \to 0$, while p and ℓ remain fixed. All ordinary inverses and unrestricted normal equations below are used on events whose probabilities tend to one.

For $v \in \mathbb{R}^{d_\times}$, let $v_{(m)}$ denote the mode-m matricization of its tensor reshape, with the remaining modes ordered increasingly, and let $\mathfrak{v}_m$ denote the inverse operation. Thus $\mathfrak{v}_m(v_{(m)}) = v$. Put $J_m = K_{m,1}^G$, so that

$$J_m v = \operatorname{vec}(v_{(m)}), \qquad \mathfrak{v}_m(B) = J_m' \operatorname{vec}(B).$$

Define

$$Q_D = D(D'D)^{-1/2}, \qquad Q = Q_D \otimes I_M \otimes I_T, \qquad \mathsf{X} = \frac{X_{(4)}}{\sqrt{n}}, \qquad \mathsf{X}_G = \mathsf{X}Q.$$

Then $Q'Q = I_{d_\times}$. Let

$$e = \mathrm{vec}(E), \qquad E^G = E \times_1 Q_D', \qquad e_G = Q'e = \mathrm{vec}(E^G),$$

and introduce the normalized factors

$$A_1 = \frac{(D'D)^{1/2}U}{\sqrt{N}}, \qquad A_2 = \frac{V}{\sqrt{M}}, \qquad A_3 = \frac{W}{\sqrt{T}}.$$

Writing $a_{m,k}$ for column k of A_m, define

$$p_0 = \sum_{k=1}^{\ell} a_{1,k} \otimes a_{2,k} \otimes a_{3,k}, \qquad \mathrm{vec}(\Pi) = \sqrt{n}\, Q p_0,$$

and

$$C_1 = A_2 * A_3, \qquad C_2 = A_1 * A_3, \qquad C_3 = A_1 * A_2, \qquad Y_m = (p_0)_{(m)} = A_m C_m'.$$

Estimated quantities are defined analogously.

For each mode, let

$$\mathcal{T}_m^G = \left\{ v \in \mathbb{R}^{d_\times} : v_{(m)} = BC_m',\ B \in \mathbb{R}^{d_m \times \ell} \right\},$$

with orthogonal projector

$$P_m = J_m'(I_{d_m} \otimes P_{C_m})J_m.$$

Let P^G project onto

$$\mathcal{T}^G = \mathcal{T}_1^G + \mathcal{T}_2^G + \mathcal{T}_3^G,$$

and put

$$M^G = I_{d_\times} - P^G, \qquad \mathcal{T} = Q\mathcal{T}^G, \qquad P_{\mathcal{T}} = QP^GQ', \qquad M_{\mathcal{T}} = I_n - P_{\mathcal{T}}.$$

Use hats for the corresponding estimated spaces and projectors, and write

$$\Delta^G = \hat{P}^G - P^G.$$

Finally, define

$$\Gamma_n = \mathsf{X} M_{\mathcal{T}} \mathsf{X}' = \Gamma_{NMT}, \qquad \mathcal{Z}_n = \mathsf{X} M_{\mathcal{T}} e = \mathcal{Z}_{NMT}.$$

The pseudoinverse of Y_m is

$$\Psi_m = Y_m^+ = C_m(C_m'C_m)^{-1}(A_m'A_m)^{-1}A_m'.$$

In these coordinates, the bias vectors in the statement can be written as

$$\mathcal{B}_m = \frac{1}{\sqrt{n}} \mathsf{X}_G P^G\, \mathfrak{v}_m\left(\Psi_m' \Omega_m^G M_{C_m}\right), \qquad \Omega_m^G = \mathbb{E}[(E_{(m)}^G)' E_{(m)}^G \mid \mathcal{F}],$$

where we've used $P_{\mathcal{T}}Q = QP^G$. Write

$$\gamma = \hat{\delta} - \delta, \qquad \theta = \sqrt{n}\,\gamma, \qquad h = \hat{p} - p_0,$$

and define the residual vectors

$$\hat{e} = e - X'_{(4)}\gamma - \sqrt{n}\, Qh, \qquad \hat{e}_G = Q'\hat{e} = e_G - \mathsf{X}'_G\theta - \sqrt{n}\, h.$$

Let $\hat{E}^G$ denote the grouped residual tensor satisfying $\mathrm{vec}(\hat{E}^G) = \hat{e}_G$, so that

$$\hat{E}^G_{(m)} = (\hat{e}_G)_{(m)}.$$

The projector $M_{DU*V} \otimes M_W$ annihilates each true tangent block, and hence $M_{DU*V} \otimes M_W \preceq M_{\mathcal{T}}$. Assumption 4 therefore gives, with probability approaching one,

$$\Gamma_n \succeq \frac{1}{n} X_{(4)}(M_{DU*V} \otimes M_W)X'_{(4)} \succeq c_\delta I_p, \qquad \|\Gamma_n^{-1}\| \leqslant c_\delta^{-1}. \tag{130}$$

Assumption 2 also gives

$$\frac{\|X_{(4)}\|_F}{\sqrt{n}} + \frac{\|X_{(4)}Q\|_F}{\sqrt{n}} = O_p(1), \qquad \|\Gamma_n\| = O_p(1).$$

Throughout the proof, conditional linear and nuclear-norm contractions follow from Lemma 4(i), (iii), while contractions involving error-dependent structured spaces follow from Lemma 5.

1. Alignment, tangent-projector control, and first-order approximations. Apply Lemma 1 and replace the estimated factors by their jointly aligned signed-permutation representatives, retaining the hat notation. This transformation leaves the fitted tensor, residuals, and estimated loading and tangent spaces unchanged. The lemma gives

$$\eta := \sum_{m=1}^{3} \|\hat{A}_m - A_m\|_F = O_p(\rho). \tag{131}$$

Trilinearity and the strong-factor bounds therefore imply

$$\|h\| + \sum_{m=1}^{3} \|\hat{C}_m - C_m\|_F = O_p(\rho).$$

The singular values of the true factors and Khatri–Rao matrices are bounded above and away from zero on the strong-factor event. For example,

$$\|C_1 z\|^2 = \|A_2 \,\mathrm{diag}(z) A'_3\|_F^2 \geqslant c_0^2 \|z\|^2.$$

Hence their estimated counterparts have full column rank with probability approaching one, and

$$\max_{m=1,2,3} \left\{ \|(\hat{A}'_m \hat{A}_m)^{-1}\| + \|(\hat{C}'_m \hat{C}_m)^{-1}\| \right\} = O_p(1).$$

The equal-rank projector identity gives

$$\begin{aligned} \|P_{\hat{C}_m} - P_{C_m}\| &\leqslant \frac{\|M_{\hat{C}_m} C_m\|_F}{\sigma_{\min}(C_m)} \\ &\leqslant \frac{\|\hat{C}_m - C_m\|_F}{\sigma_{\min}(C_m)} = O_p(\rho), \qquad m = 1, 2, 3. \end{aligned} \tag{132}$$

Lemma 1 also supplies the unrestricted normal equations. In normalized grouped coordinates, they imply

$$X_{(4)}\hat{e} = 0, \qquad \hat{E}^G_{(m)}\hat{C}_m = 0, \quad m = 1, 2, 3, \qquad \hat{P}^G\hat{e}_G = 0. \tag{133}$$

The last identity holds because $\hat{e}_G$ is orthogonal to each estimated tangent block. The slope equation can be written as

$$\mathsf{X}\mathsf{X}'\gamma = \frac{1}{\sqrt{n}}\mathsf{X}e - \mathsf{X}_G h.$$

Since $\mathsf{X}\mathsf{X}' \geq \Gamma_n \geq c_\delta I_p$, $\mathsf{X}e = O_p(1)$ by (110), and $n^{-1/2} \leqslant \rho$, we obtain

$$\|\gamma\| + \|h\| = O_p(\rho). \tag{134}$$

We next control the full tangent projector. Let

$$S = P_1 + P_2 + P_3.$$

Its range is $\mathcal{T}^G$, so $P^G = SS^+$. To bound S^+, complete each A_m to an invertible square matrix $B_m = [A_m, A_{m,\perp}]$, with orthonormal complement columns, and put $B = B_1 \otimes B_2 \otimes B_3$. On the strong-factor event,

$$\kappa(B) = \|B\|\|B^{-1}\| \leqslant (c_1/c_0)^{3/2}.$$

Under B^{-1}, the tangent blocks become coordinate subspaces supported on (i, k, k), (k, j, k), and (k, k, t), respectively. Their only shared coordinates are (k, k, k). Assigning these shared coordinates to the first block gives, for every $x \in \mathcal{T}^G$, an orthogonal coordinate decomposition

$$B^{-1}x = z_1 + z_2 + z_3, \qquad x_m = Bz_m \in \mathcal{T}^G_m, \qquad x = x_1 + x_2 + x_3.$$

Consequently,

$$\sum_{m=1}^{3} \|x_m\|^2 \leqslant \|B\|^2 \sum_{m=1}^{3} \|z_m\|^2 = \|B\|^2\|B^{-1}x\|^2 \leqslant \kappa(B)^2\|x\|^2.$$

Since

$$\|x\|^2 = \sum_{m=1}^{3} \langle P_m x, x_m\rangle \leqslant (x'Sx)^{1/2} \left(\sum_{m=1}^{3} \|x_m\|^2\right)^{1/2},$$

we obtain

$$\lambda^+_{\min}(S) \geqslant \kappa(B)^{-2}, \qquad \|S^+\| \leqslant \kappa(B)^2 = O_p(1). \tag{135}$$

The same coordinate construction gives $\dim(\mathcal{T}^G) = \ell(s-2)$. By (131), the estimated factors have full column rank with probability approaching one, so the estimated tangent space has the same dimension. With $\hat{S} = \hat{P}_1 + \hat{P}_2 + \hat{P}_3$, equation (132) implies $\|\hat{S} - S\| = O_p(\rho)$. Therefore,

$$\begin{aligned}\|\Delta^G\| &= \|(I_{d_\times} - \hat{P}^G)P^G\| \\ &= \|(I_{d_\times} - \hat{P}^G)(S - \hat{S})S^+\| \\ &\leqslant \|S - \hat{S}\|\|S^+\| = O_p(\rho).\end{aligned}$$

The true and estimated tangent spaces are contained in the space $\mathcal{V}$ generated by $\mathcal{L}_m = \mathrm{col}[A_m, \hat{A}_m]$, whose dimensions are at most 2ℓ. Hence $\Delta^G = \Delta^G P_{\mathcal{V}}$, and Lemma 5 gives

$$\|\Delta^G\| = O_p(\rho), \qquad \|\Delta^G e_G\| \leqslant 2\sqrt{3}\,\ell\,\|\Delta^G\|\|E^G\| = O_p(\sqrt{s}\,\rho). \tag{136}$$

Combining the slope equation with the definition of $\hat{e}_G$ gives the exact residual-score identity

$$\Gamma_n\theta = \mathcal{Z}_n + \mathsf{X}_G P^G \hat{e}_G - \sqrt{n}\,\mathsf{X}_G M^G h. \tag{137}$$

To use this identity, write $\Delta A_m = \hat{A}_m - A_m$ and expand

$$h = q_1 + q_2 + q_3,$$

where

$$\begin{aligned}
q_1 &= \sum_{k=1}^{\ell} \left[\Delta a_{1,k} \otimes a_{2,k} \otimes a_{3,k} + a_{1,k} \otimes \Delta a_{2,k} \otimes a_{3,k} + a_{1,k} \otimes a_{2,k} \otimes \Delta a_{3,k}\right], \\
q_2 &= \sum_{k=1}^{\ell} \left[\Delta a_{1,k} \otimes \Delta a_{2,k} \otimes a_{3,k} + \Delta a_{1,k} \otimes a_{2,k} \otimes \Delta a_{3,k} + a_{1,k} \otimes \Delta a_{2,k} \otimes \Delta a_{3,k}\right], \\
q_3 &= \sum_{k=1}^{\ell} \Delta a_{1,k} \otimes \Delta a_{2,k} \otimes \Delta a_{3,k}.
\end{aligned}$$

Then $q_1 \in \mathcal{T}^G$, and

$$\|q_1\| = O_p(\rho), \qquad \|q_2\| = O_p(\rho^2), \qquad \|q_3\| = O_p(\rho^3), \qquad \|M^G h\| = O_p(\rho^2).$$

Also, $q_1 + 2q_2 + 3q_3$ is the derivative at $t = 1$ of the tensor generated by $A_m + t\Delta A_m$, so it belongs to $\hat{\mathcal{T}}^G$. Direct subtraction consequently gives

$$2M^G h - \Delta^G h = -(I_{d_\times} - \hat{P}^G)q_3 + \Delta^G(q_2 + q_3),$$

and therefore

$$M^G h = \frac{1}{2}\Delta^G h + O_p(\rho^3). \tag{138}$$

By (133), $P^G\hat{e}_G = -\Delta^G\hat{e}_G$. Equations (134) and (136) imply

$$\begin{aligned}
\|\Delta^G\hat{e}_G\| &\leqslant \|\Delta^G e_G\| + \|\Delta^G\|\|\mathsf{X}_G\|\|\theta\| + \sqrt{n}\,\|\Delta^G\|\|h\| \\
&= O_p(\sqrt{n}\,\rho^2).
\end{aligned}$$

Since $\mathcal{Z}_n = O_p(1)$ by Lemma 4 (i), the exact score equation yields

$$\Gamma_n\theta = \mathcal{Z}_n + O_p(s/\sqrt{n}). \tag{139}$$

Define

$$\theta_1 = \Gamma_n^{-1}\mathcal{Z}_n, \qquad r_0 = e_G - \mathsf{X}_G'\theta_1, \qquad \xi = P^G r_0, \qquad q = \frac{\xi}{\sqrt{n}}. \tag{140}$$

Here, θ_1, ξ, and $M^G r_0$ are the slope change, common-component change, and grouped residual from the model obtained by replacing the nonlinear factor component with its tangent-space approximation, i.e.,

$$\min_{\vartheta\in\mathbb{R}^p,\ z\in\mathcal{T}^G} \left\|e - \mathsf{X}'\vartheta - Qz\right\|^2.$$

Then

$$\theta_1 = O_p(1), \qquad \theta - \theta_1 = O_p(s/\sqrt{n}), \qquad \|q\| = O_p(\rho).$$

The last bound uses $\|P^G e_G\| = O_p(\sqrt{s})$, which follows from Lemma 5 with the true loading spaces. In particular,

$$\|\gamma\| = O_p(n^{-1/2} + s/n).$$

Let

$$F_0 = Q' - \mathsf{X}_G' \Gamma_n^{-1} \mathsf{X} M_{\mathcal{T}}, \qquad r_0 = F_0 e, \qquad \xi = P^G F_0 e. \tag{141}$$

The matrices F_0, $P^G F_0$, and $S^+ P^G F_0$ are $\mathcal{F}$-measurable and have $O_p(1)$ operator norms. The slope-adjusted linearized error admits the representation $r_0 = F_0 e$. Consequently, the linearized common-component change $q = n^{-1/2} P^G F_0 e$ and the first-order grouped fitted residual $M^G F_0 e$ are also linear in the original errors, conditional on $\mathcal{F}$. This representation is useful because products involving these objects can be written as quadratic forms in e, with $\mathcal{F}$-measurable coefficient matrices.

The identity $\hat{P}^G \hat{e}_G = 0$ also gives

$$h - q = M^G h - \frac{1}{\sqrt{n}} P^G \mathsf{X}_G'(\theta - \theta_1) + \frac{1}{\sqrt{n}} \Delta^G \hat{e}_G.$$

Each term on the right is $O_p(\rho^2)$. Thus

$$\|h - q\| = O_p(\rho^2), \qquad \|\hat{e}_G - M^G r_0\| = O_p(\sqrt{n}\,\rho^2), \tag{142}$$

where the second bound follows from

$$\hat{e}_G - M^G r_0 = -\mathsf{X}_G'(\theta - \theta_1) - \sqrt{n}(h - q).$$

2. Graph-aligned loading spaces and the exact tangent score. For each mode, define

$$J_m^C = (C_m' C_m)^{-1} C_m' \hat{C}_m, \qquad F_m = \hat{C}_m (J_m^C)^{-1} - C_m.$$

The aligned rate gives

$$J_m^C = I_\ell + (C_m' C_m)^{-1} C_m' (\hat{C}_m - C_m) = I_\ell + O_p(\rho).$$

Hence J_m^C is nonsingular with probability approaching one. Since $C_m J_m^C = P_{C_m} \hat{C}_m$,

$$F_m = M_{C_m} (\hat{C}_m - C_m)(J_m^C)^{-1}.$$

It follows that

$$C_m' F_m = 0, \qquad \operatorname{col}(C_m + F_m) = \operatorname{col}(\hat{C}_m), \qquad \|F_m\|_F = O_p(\rho). \tag{143}$$

The loading normal equations therefore imply

$$\hat{E}_{(m)}^G C_m = -\hat{E}_{(m)}^G F_m. \tag{144}$$

Define

$$\widetilde{A}_m = \hat{p}_{(m)} C_m (C_m' C_m)^{-1}.$$

Then

$$\hat{p}_{(m)} = \widetilde{A}_m (C_m + F_m)', \qquad \widetilde{A}_m = A_m + h_{(m)} C_m (C_m' C_m)^{-1},$$

so $\|\widetilde{A}_m - A_m\|_F = O_p(\rho)$. Since $C_m' F_m = 0$,

$$F_m = M_{C_m} h'_{(m)} \widetilde{A}_m (\widetilde{A}_m' \widetilde{A}_m)^{-1}. \tag{145}$$

Let

$$F_m^{(1)} = M_{C_m} q'_{(m)} A_m (A_m' A_m)^{-1}.$$

The inverse perturbation identity, the strong-factor bounds, and (142) give

$$\|F_m - F_m^{(1)}\|_F = O_p(\rho^2). \tag{146}$$

Indeed, the difference is bounded by a constant in probability times

$$\|h - q\| + \|h\| \|\widetilde{A}_m - A_m\|_F.$$

For slope coordinate $r = 1, \dots, p$, let $\mathbf{e}_r$ denote the corresponding coordinate vector and define

$$x_r = \mathsf{X}_G' \mathbf{e}_r, \qquad t_r = S^+ x_r, \qquad L_{m,r} = (t_r)_{(m)} C_m (C_m' C_m)^{-1}.$$

These matrices are $\mathcal{F}$-measurable, and $\|L_{m,r}\|_F = O_p(1)$. Since

$$P^G x_r = \sum_{m=1}^{3} P_m t_r, \qquad (P_m t_r)_{(m)} = L_{m,r} C_m',$$

equation (144) gives

$$x_r' P^G \hat{e}_G = -\sum_{m=1}^{3} \langle \hat{E}^G_{(m)}, L_{m,r} F_m' \rangle_F.$$

Thus (137) becomes

$$[\Gamma_n \theta]_r = [\mathcal{Z}_n]_r - \sqrt{n}\, x_r' M^G h - \sum_{m=1}^{3} \langle \hat{E}^G_{(m)}, L_{m,r} F_m' \rangle_F. \tag{147}$$

3. The normal common-component contraction. We establish the sharper scalar bound

$$\sqrt{n}\, x_r' M^G h = O_p\left(\rho + \frac{s^{3/2}}{n}\right). \tag{148}$$

The preliminary bound $\|M^G h\| = O_p(\rho^2)$ alone is insufficient for this conclusion. We begin by obtaining a first-order expansion of the sum of the tangent-block projectors, with an explicit second-order remainder.

Fix $m \in \{1, 2, 3\}$ and write

$$G_m = C_m' C_m.$$

The strong-factor bounds imply that $\|C_m\|$ and $\|G_m^{-1}\|$ are bounded on an event whose probability tends to one. More precisely, if $a < b$ are the modes other than m, then

$$c_0 \|z\| \leqslant \|C_m z\| = \|A_a \operatorname{diag}(z) A_b'\|_F \leqslant c_1 \|z\|, \qquad z \in \mathbb{R}^\ell.$$

Consequently,

$$\|C_m\| \leqslant c_1, \qquad \lambda_{\min}(G_m) \geqslant c_0^2, \qquad \|G_m^{-1}\| \leqslant c_0^{-2}.$$

All inverse identities below are evaluated on this event. By (143),

$$C_m' F_m = 0, \qquad \operatorname{col}(C_m + F_m) = \operatorname{col}(\hat{C}_m), \qquad \|F_m\| \leqslant \|F_m\|_F = O_p(\rho).$$

In particular,

$$(C_m + F_m)'(C_m + F_m) = G_m + F_m' F_m.$$

Since G_m is positive definite and $F_m' F_m$ is positive semidefinite, $C_m + F_m$ has full column rank. Its column-space projector is therefore

$$P_{\hat{C}_m} = (C_m + F_m)(G_m + F_m' F_m)^{-1}(C_m + F_m)'.$$

To expand the inverse, define

$$R_m^{\text{inv}} = (G_m + F_m' F_m)^{-1} - G_m^{-1}.$$

The exact inverse identity gives

$$R_m^{\text{inv}} = -(G_m + F_m' F_m)^{-1} F_m' F_m G_m^{-1}.$$

Moreover, $G_m + F_m' F_m \succeq G_m$ implies

$$\|(G_m + F_m' F_m)^{-1}\| \leqslant \|G_m^{-1}\|.$$

Hence

$$\begin{aligned} \|R_m^{\text{inv}}\| &\leqslant \|(G_m + F_m' F_m)^{-1}\| \|F_m' F_m\| \|G_m^{-1}\| \\ &\leqslant \|G_m^{-1}\|^2 \|F_m\|^2 \\ &= O_p(\rho^2). \end{aligned}$$

Thus the inverse perturbation is second order because the graph orthogonality eliminates the linear cross terms from the Gram matrix.

Substituting $(G_m + F_m' F_m)^{-1} = G_m^{-1} + R_m^{\text{inv}}$ into the projector formula and using $P_{C_m} = C_m G_m^{-1} C_m'$ gives

$$P_{\hat{C}_m} - P_{C_m} = F_m G_m^{-1} C_m' + C_m G_m^{-1} F_m' + R_m^C,$$

where the remainder is exactly

$$R_m^C = F_m G_m^{-1} F_m' + (C_m + F_m) R_m^{\text{inv}} (C_m + F_m)'.$$

Its operator norm satisfies

$$\begin{aligned} \|R_m^C\| &\leqslant \|G_m^{-1}\| \|F_m\|^2 + \|C_m + F_m\|^2 \|R_m^{\text{inv}}\| \\ &\leqslant \left\{ \|G_m^{-1}\| + \|C_m + F_m\|^2 \|G_m^{-1}\|^2 \right\} \|F_m\|^2 \\ &= O_p(\rho^2). \end{aligned}$$

We have therefore obtained

$$P_{\hat{C}_m} - P_{C_m} = F_m (C_m' C_m)^{-1} C_m' + C_m (C_m' C_m)^{-1} F_m' + R_m^C, \qquad \|R_m^C\| = O_p(\rho^2). \tag{149}$$

We next replace F_m by its first-order approximation. Recall that

$$F_m^{(1)} = M_{C_m} q_{(m)}' A_m (A_m' A_m)^{-1},$$

and, by (146),

$$R_m^F := F_m - F_m^{(1)}, \qquad \|R_m^F\|_F = O_p(\rho^2).$$

Substituting $F_m = F_m^{(1)} + R_m^F$ into the two linear terms in (149) gives

$$P_{\hat{C}_m} - P_{C_m} = F_m^{(1)} G_m^{-1} C_m' + C_m G_m^{-1} (F_m^{(1)})' + \widetilde{R}_m^C,$$

where

$$\widetilde{R}_m^C = R_m^C + R_m^F G_m^{-1} C_m' + C_m G_m^{-1} (R_m^F)'.$$

Because

$$\|C_m G_m^{-1}\| = \frac{1}{\sigma_{\min}(C_m)},$$

the replacement error obeys

$$\begin{aligned} \|\widetilde{R}_m^C\| &\leqslant \|R_m^C\| + 2\|R_m^F\|\|C_m G_m^{-1}\| \\ &\leqslant \|R_m^C\| + \frac{2\|R_m^F\|_F}{\sigma_{\min}(C_m)} \\ &= O_p(\rho^2). \end{aligned}$$

Thus replacing F_m by $F_m^{(1)}$ in the linear projector terms preserves the second-order remainder.

Recall the normalized common-component pseudoinverse

$$\Psi_m = C_m G_m^{-1} (A_m' A_m)^{-1} A_m', \qquad \Psi_m' = A_m (A_m' A_m)^{-1} G_m^{-1} C_m'.$$

The retained first-order terms can therefore be written as

$$\begin{aligned} F_m^{(1)} G_m^{-1} C_m' &= M_{C_m} q_{(m)}' A_m (A_m' A_m)^{-1} G_m^{-1} C_m' \\ &= M_{C_m} q_{(m)}' \Psi_m', \end{aligned}$$

and

$$\begin{aligned} C_m G_m^{-1} (F_m^{(1)})' &= C_m G_m^{-1} (A_m' A_m)^{-1} A_m' q_{(m)} M_{C_m} \\ &= \Psi_m q_{(m)} M_{C_m}. \end{aligned}$$

For $v \in \mathbb{R}^{d_\times}$, define the symmetric matrix

$$\mathcal{L}_m(v) = M_{C_m} v_{(m)}' \Psi_m' + \Psi_m v_{(m)} M_{C_m}.$$

The preceding calculations show that

$$P_{\hat{C}_m} - P_{C_m} = \mathcal{L}_m(q) + \widetilde{R}_m^C, \qquad \|\widetilde{R}_m^C\| = O_p(\rho^2). \tag{150}$$

Define the associated matrix-valued linear maps

$$\mathcal{K}_m(v) = J_m' (I_{d_m} \otimes \mathcal{L}_m(v)) J_m, \qquad \mathcal{K}(v) = \sum_{m=1}^{3} \mathcal{K}_m(v).$$

For every v, the two summands defining $\mathcal{L}_m(v)$ are transposes of one another, and

$$\begin{aligned} \|\mathcal{L}_m(v)\| &\leqslant 2\|\Psi_m\|\|v_{(m)}\| \\ &\leqslant 2\|\Psi_m\|\|v_{(m)}\|_F \\ &= 2\|\Psi_m\|\|v\|. \end{aligned}$$

Since J_m is orthogonal and $\|I_{d_m} \otimes B\| = \|B\|$ for every matrix B,

$$\|\mathcal{K}_m(v)\| = \|\mathcal{L}_m(v)\|.$$

It follows that

$$\|\mathcal{K}(v)\| \leqslant \sum_{m=1}^{3} \|\mathcal{K}_m(v)\| \leqslant 2 \left(\sum_{m=1}^{3} \|\Psi_m\| \right) \|v\|.$$

The strong-factor bounds also give

$$\|\Psi_m\| \leqslant \frac{1}{\sigma_{\min}(C_m)\sigma_{\min}(A_m)} \leqslant c_0^{-3/2}$$

on the strong-factor event. Hence $\sum_m \|\Psi_m\| = O_p(1)$, and in particular $\|\mathcal{K}(q)\| = O_p(\rho)$.

Finally, recall that

$$P_m = J_m'(I_{d_m} \otimes P_{C_m})J_m, \qquad \hat{P}_m = J_m'(I_{d_m} \otimes P_{\hat{C}_m})J_m,$$

and

$$S = \sum_{m=1}^{3} P_m, \qquad \hat{S} = \sum_{m=1}^{3} \hat{P}_m.$$

Equation (150) gives the exact remainder representation

$$\hat{S} - S - \mathcal{K}(q) = \sum_{m=1}^{3} J_m'(I_{d_m} \otimes \widetilde{R}_m^C)J_m.$$

Consequently,

$$\begin{aligned} \|\hat{S} - S - \mathcal{K}(q)\| &\leqslant \sum_{m=1}^{3} \left\| J_m'(I_{d_m} \otimes \widetilde{R}_m^C)J_m \right\| \\ &= \sum_{m=1}^{3} \|\widetilde{R}_m^C\| \\ &= O_p(\rho^2). \end{aligned}$$

Thus

$$\hat{S} - S = \mathcal{K}(q) + O_p(\rho^2) \tag{151}$$

in operator norm. No dimension factor is introduced when passing from the loading-space projectors to the tangent-block projectors, because multiplication by the orthogonal mode-permutation matrices and taking a Kronecker product with an identity matrix preserve the operator norm.

The identities $\hat{S}(I_{d_\times} - \hat{P}^G) = 0$ and $(P^G + \Delta^G)^2 = P^G + \Delta^G$ imply

$$\Delta^G = M^G(\hat{S} - S)S^+ + S^+(\hat{S} - S)M^G + R_S,$$

where

$$R_S = (\Delta^G)^2 - S^+(\hat{S} - S)\Delta^G - \Delta^G(\hat{S} - S)S^+.$$

Equations (135) and (136) imply $\|R_S\| = O_p(\rho^2)$. Therefore,

$$\Delta^G = M^G\mathcal{K}(q)S^+ + S^+\mathcal{K}(q)M^G + R_\Delta^G, \qquad \|R_\Delta^G\| = O_p(\rho^2).$$

Using (138), $M^G q = 0$, and (142), we obtain

$$\begin{aligned} M^G h &= \frac{1}{2}\Delta^G q + \frac{1}{2}\Delta^G (h-q) + O_p(\rho^3) \\ &= \frac{1}{2} M^G \mathcal{K}(q) S^+ q + O_p(\rho^3). \end{aligned}$$

For fixed m, put

$$Z_m = (M^G x_r)_{(m)}, \qquad T_m = (S^+ q)_{(m)}.$$

Orthogonality to the m-th tangent block gives

$$Z_m C_m = 0, \qquad Z_m M_{C_m} = Z_m, \qquad Z_m \Psi_m = 0.$$

Since $\mathcal{L}_m(q)$ is symmetric, row-major vectorization gives

$$(\mathcal{K}_m(q) S^+ q)_{(m)} = T_m \mathcal{L}_m(q).$$

Consequently,

$$\begin{aligned} \langle M^G x_r, \mathcal{K}_m(q) S^+ q \rangle &= \langle Z_m, T_m M_{C_m} q'_{(m)} \Psi'_m \rangle_F + \langle Z_m, T_m \Psi_m q_{(m)} M_{C_m} \rangle_F \\ &= \langle Z_m \Psi_m, T_m M_{C_m} q'_{(m)} \rangle_F + \langle Z_m M_{C_m}, T_m \Psi_m q_{(m)} \rangle_F \\ &= \langle Z_m, T_m \Psi_m q_{(m)} \rangle_F. \end{aligned}$$

With $\xi = \sqrt{n}\, q$, this proves

$$\sqrt{n}\, x_r' M^G h = \frac{1}{2\sqrt{n}} \sum_{m=1}^{3} \left\langle (M^G x_r)_{(m)}, (S^+ \xi)_{(m)} \Psi_m \xi_{(m)} \right\rangle_F + O_p(\sqrt{n}\, \rho^3). \tag{152}$$

Fix m and write

$$d_{-m} = \frac{d_\times}{d_m}, \qquad Z_m = (M^G x_r)_{(m)} \in \mathbb{R}^{d_m \times d_{-m}}.$$

In addition to the mode-permutation matrix J_m, define the permutation matrix $\tilde{J}_m \in \mathbb{R}^{d_\times \times d_\times}$ by

$$J_m v = \mathrm{vec}(v_{(m)}), \qquad \tilde{J}_m v = \mathrm{vec}(v'_{(m)}), \qquad v \in \mathbb{R}^{d_\times}.$$

Thus J_m vectorizes the mode-m matricization, whereas $\tilde{J}_m$ vectorizes its transpose. Both matrices are orthogonal:

$$J_m' J_m = \tilde{J}_m' \tilde{J}_m = I_{d_\times}, \qquad \|J_m\| = \|\tilde{J}_m\| = 1.$$

Since $\Psi_m \in \mathbb{R}^{d_{-m} \times d_m}$, define

$$\mathscr{Q}_{m,r} = Z_m \otimes \Psi_m \in \mathbb{R}^{d_\times \times d_\times}.$$

The Frobenius inner-product identity $\langle A, BC\rangle_F = \langle AC', B\rangle_F$ and the row-major vectorization identity $\operatorname{vec}(ABC) = (A \otimes C')\operatorname{vec}(B)$ give

$$\begin{aligned}
\left\langle Z_m, (S^+\xi)_{(m)}\Psi_m\xi_{(m)}\right\rangle_F &= \left\langle Z_m\xi'_{(m)}, (S^+\xi)_{(m)}\Psi_m\right\rangle_F \\
&= \operatorname{vec}(Z_m\xi'_{(m)})'\operatorname{vec}((S^+\xi)_{(m)}\Psi_m) \\
&= \operatorname{vec}(\xi'_{(m)})'(Z'_m \otimes I_{d_m})(I_{d_m} \otimes \Psi'_m)\operatorname{vec}((S^+\xi)_{(m)}) \\
&= \operatorname{vec}(\xi'_{(m)})'(Z'_m \otimes \Psi'_m)\operatorname{vec}((S^+\xi)_{(m)}) \\
&= \operatorname{vec}((S^+\xi)_{(m)})'\mathscr{Q}_{m,r}\operatorname{vec}(\xi'_{(m)}) \\
&= (J_m S^+\xi)'\mathscr{Q}_{m,r}(\tilde{J}_m\xi).
\end{aligned}$$

The fourth equality uses the mixed-product identity for Kronecker products, and the fifth follows by transposing the scalar. In particular, $\mathscr{Q}_{m,r}$ is an exact Kronecker product, with the transposed matricization on the right represented by $\tilde{J}_m$.

We next bound the nuclear norm of $\mathscr{Q}_{m,r}$. Since M^G is an orthogonal projector,

$$\|Z_m\|_F = \|M^G x_r\| \leqslant \|x_r\| = O_p(1).$$

The strong-factor bounds imply $\|\Psi_m\| = O_p(1)$ and $\operatorname{rank}(\Psi_m) = \ell$ with probability approaching one. Since ℓ is fixed,

$$\|\Psi_m\|_F \leqslant \sqrt{\ell}\,\|\Psi_m\| = O_p(1).$$

The Frobenius norm and rank identities for Kronecker products therefore give

$$\|\mathscr{Q}_{m,r}\|_F = \|Z_m\|_F\|\Psi_m\|_F = O_p(1)$$

and

$$\operatorname{rank}(\mathscr{Q}_{m,r}) = \operatorname{rank}(Z_m)\operatorname{rank}(\Psi_m) \leqslant \ell d_m.$$

Applying Cauchy–Schwarz to the nonzero singular values yields

$$\|\mathscr{Q}_{m,r}\|_* \leqslant \sqrt{\operatorname{rank}(\mathscr{Q}_{m,r})}\,\|\mathscr{Q}_{m,r}\|_F \leqslant \sqrt{\ell d_m}\,\|\mathscr{Q}_{m,r}\|_F = O_p(\sqrt{d_m}).$$

By (141), $\xi = P^G F_0 e$, so

$$\operatorname{vec}((S^+\xi)_{(m)}) = J_m S^+ P^G F_0 e, \qquad \operatorname{vec}(\xi'_{(m)}) = \tilde{J}_m P^G F_0 e.$$

Consequently,

$$\left\langle Z_m, (S^+\xi)_{(m)}\Psi_m\xi_{(m)}\right\rangle_F = e'\mathscr{H}_{m,r}e,$$

where

$$\mathscr{H}_{m,r} = (J_m S^+ P^G F_0)'\mathscr{Q}_{m,r}(\tilde{J}_m P^G F_0) \in \mathbb{R}^{n\times n}.$$

All matrices defining $\mathscr{H}_{m,r}$ are $\mathcal{F}$-measurable. Moreover, J_m and $\tilde{J}_m$ are orthogonal, while $S^+P^G F_0$ and $P^G F_0$ have $O_p(1)$ operator norms. The nuclear-norm inequality $\|ABC\|_* \leqslant \|A\|\|B\|_*\|C\|$ therefore gives

$$\begin{aligned}
\|\mathscr{H}_{m,r}\|_* &\leqslant \|J_m S^+ P^G F_0\|\,\|\mathscr{Q}_{m,r}\|_*\,\|\tilde{J}_m P^G F_0\| \\
&= \|S^+ P^G F_0\|\,\|\mathscr{Q}_{m,r}\|_*\,\|P^G F_0\| \\
&= O_p(\sqrt{d_m}).
\end{aligned}$$

On the conditional covariance-summability event, Lemma 4 (iii) implies

$$\mathbb{E}[|e'\mathscr{H}_{m,r}e| \mid \mathcal{F}] \leqslant c_\varepsilon \|\mathscr{H}_{m,r}\|_*.$$

Conditional Markov's inequality, localized to events on which $\|\mathscr{H}_{m,r}\|_*/\sqrt{d_m}$ is bounded, yields

$$\big\langle Z_m, (S^+\xi)_{(m)}\Psi_m\xi_{(m)}\big\rangle_F = e'\mathscr{H}_{m,r}e = O_p(\sqrt{d_m}).$$

Substituting these bounds into (152) gives

$$\begin{aligned}\sqrt{n}\, x_r' M^G h &= \frac{1}{2\sqrt{n}} \sum_{m=1}^{3} e'\mathscr{H}_{m,r}e + O_p(\sqrt{n}\,\rho^3) \\ &= O_p\left(\frac{\sqrt{d_1}+\sqrt{d_2}+\sqrt{d_3}}{\sqrt{n}} + \sqrt{n}\,\rho^3\right).\end{aligned}$$

Finally,

$$\sqrt{d_1}+\sqrt{d_2}+\sqrt{d_3} \leqslant \sqrt{3(d_1+d_2+d_3)} = \sqrt{3s}, \qquad \sqrt{n}\,\rho^3 = \frac{s^{3/2}}{n}.$$

Therefore,

$$\sqrt{n}\, x_r' M^G h = O_p\left(\rho + \frac{s^{3/2}}{n}\right),$$

which proves (148).

4. First-order expansion of the graph score. Define

$$D_{m,r} = L_{m,r}(A_m'A_m)^{-1}A_m' = (t_r)_{(m)}\Psi_m.$$

These matrices are $\mathcal{F}$-measurable and have $O_p(1)$ operator norms. By (142) and (143),

$$\begin{aligned}\left|\langle \hat{E}^G_{(m)} - (M^G r_0)_{(m)}, L_{m,r}F_m'\rangle_F\right| &\leqslant \|\hat{e}_G - M^G r_0\|\|L_{m,r}\|\|F_m\|_F \\ &= O_p(\sqrt{n}\,\rho^3).\end{aligned}$$

Next, let

$$z_m = \mathfrak{v}_m\Big(L_{m,r}(F_m - F_m^{(1)})'\Big).$$

Then $\|z_m\| = O_p(\rho^2)$ due to (146) and $\|L_{m,r}\| = O_p(1)$. To obtain an explicit tangent-block decomposition of q, define

$$B_j^q = (S^+q)_{(j)}C_j(C_j'C_j)^{-1} \in \mathbb{R}^{d_j\times\ell}, \qquad j = 1,2,3.$$

The inverses are well defined on the strong-factor event. Multiplying by C_j' gives

$$B_j^q C_j' = (S^+q)_{(j)}C_j(C_j'C_j)^{-1}C_j' = (S^+q)_{(j)}P_{C_j}.$$

This matrix is the mode-j matricization of the projection of S^+q onto $\mathcal{T}_j^G$. Indeed, recall that

$$P_j = J_j'(I_{d_j}\otimes P_{C_j})J_j, \qquad J_j v = \mathrm{vec}(v_{(j)}).$$

For any $v \in \mathbb{R}^{d\times}$, the row-major vectorization identity and the symmetry of P_{C_j} imply

$$\begin{aligned} J_j P_j v &= (I_{d_j} \otimes P_{C_j}) J_j v \\ &= (I_{d_j} \otimes P_{C_j}) \operatorname{vec}(v_{(j)}) \\ &= \operatorname{vec}(v_{(j)} P_{C_j}). \end{aligned}$$

Consequently,

$$(P_j v)_{(j)} = v_{(j)} P_{C_j}.$$

Applying this identity to $v = S^+ q$ yields

$$B_j^q C_j' = (P_j S^+ q)_{(j)},$$

and hence, because $\mathfrak{v}_j$ reverses mode-j matricization,

$$\mathfrak{v}_j(B_j^q C_j') = P_j S^+ q \in \mathcal{T}_j^G.$$

Since $S = P_1 + P_2 + P_3$ has range $\mathcal{T}^G$, its Moore–Penrose pseudoinverse satisfies $SS^+ = P^G$. Moreover, $q \in \mathcal{T}^G$, so $P^G q = q$. Summing the preceding identities therefore gives

$$\begin{aligned} \sum_{j=1}^{3} \mathfrak{v}_j(B_j^q C_j') &= \sum_{j=1}^{3} P_j S^+ q \\ &= (P_1 + P_2 + P_3) S^+ q \\ &= S S^+ q \\ &= P^G q = q. \end{aligned}$$

Thus

$$q = \sum_{j=1}^{3} \mathfrak{v}_j(B_j^q C_j'). \tag{153}$$

This is an exact tangent-block decomposition, but its three summands need not be mutually orthogonal. The use of $S^+ q$, rather than q, accounts for the overlap among the tangent blocks: in general, $\sum_j P_j q = Sq \neq q$, whereas $\sum_j P_j S^+ q = q$. Let

$$\mathcal{L}_j = \operatorname{col}[A_j, \hat{A}_j, B_j^q], \qquad r_j = \dim(\mathcal{L}_j) \leqslant 3\ell, \qquad j = 1, 2, 3.$$

The dimension bound holds because each of the three matrices has ℓ columns. These spaces contain the true loading directions, the estimated loading directions, and the directions appearing in the tangent decomposition of q.

Fix m, let $a < b$ be the other two modes, and define

$$\mathcal{U}_{-m} = \mathcal{L}_a \otimes \mathcal{L}_b \subseteq \mathbb{R}^{d_a d_b}.$$

We first verify that the columns of both F_m and $F_m^{(1)}$ belong to $\mathcal{U}_{-m}$. Recall that

$$C_m = A_a * A_b, \qquad \hat{C}_m = \hat{A}_a * \hat{A}_b.$$

Every column of C_m is a tensor product of a column of A_a and a column of A_b, and the same statement holds for the estimated matrices. Since $\operatorname{col}(A_j), \operatorname{col}(\hat{A}_j) \subseteq \mathcal{L}_j$,

$$\operatorname{col}(C_m) \subseteq \mathcal{U}_{-m}, \qquad \operatorname{col}(\hat{C}_m) \subseteq \mathcal{U}_{-m}.$$

The graph representation

$$F_m = \hat{C}_m (J_m^C)^{-1} - C_m$$

then implies

$$\mathrm{col}(F_m) \subseteq \mathrm{col}(\hat{C}_m) + \mathrm{col}(C_m) \subseteq \mathcal{U}_{-m},$$

because right multiplication only forms linear combinations of columns.

Next, the tangent decomposition

$$q = \sum_{j=1}^{3} \mathfrak{v}_j (B_j^q C_j')$$

has the following mode-m representation:

$$q_{(m)} = B_m^q C_m' + A_m \left(B_a^q * A_b + A_a * B_b^q \right)'.$$

The first term is the contribution from the m-th tangent block, while the two terms inside the parentheses arise from the other two tangent blocks. Transposing gives

$$q_{(m)}' = C_m (B_m^q)' + \left(B_a^q * A_b + A_a * B_b^q \right) A_m'.$$

Since $\mathrm{col}(B_j^q) \subseteq \mathcal{L}_j$, each of C_m, $B_a^q * A_b$, and $A_a * B_b^q$ has columns in $\mathcal{U}_{-m}$. Consequently,

$$\mathrm{col}(q_{(m)}') \subseteq \mathrm{col}[C_m, B_a^q * A_b, A_a * B_b^q] \subseteq \mathcal{U}_{-m}.$$

The space $\mathcal{U}_{-m}$ is also preserved by M_{C_m}. Indeed, for any $u \in \mathcal{U}_{-m}$,

$$P_{C_m} u \in \mathrm{col}(C_m) \subseteq \mathcal{U}_{-m},$$

and therefore

$$M_{C_m} u = u - P_{C_m} u \in \mathcal{U}_{-m}.$$

Using

$$F_m^{(1)} = M_{C_m} q_{(m)}' A_m (A_m' A_m)^{-1},$$

we conclude that

$$\mathrm{col}(F_m^{(1)}) \subseteq \mathcal{U}_{-m}.$$

Thus

$$\mathrm{col}(F_m - F_m^{(1)}) \subseteq \mathcal{U}_{-m}.$$

Recall that

$$z_m = \mathfrak{v}_m \Big(L_{m,r} (F_m - F_m^{(1)})' \Big).$$

Its transposed mode-m matricization satisfies

$$(z_m)_{(m)}' = (F_m - F_m^{(1)}) L_{m,r}',$$

so

$$\mathrm{col}((z_m)_{(m)}') \subseteq \mathcal{U}_{-m}.$$

Equivalently, the mode-m direction of z_m is unrestricted, while its other two modes are restricted to $\mathcal{L}_a$ and $\mathcal{L}_b$. In the original tensor ordering, this gives

$$\begin{aligned} z_1 &\in \mathbb{R}^{d_1} \otimes \mathcal{L}_2 \otimes \mathcal{L}_3, \\ z_2 &\in \mathcal{L}_1 \otimes \mathbb{R}^{d_2} \otimes \mathcal{L}_3, \\ z_3 &\in \mathcal{L}_1 \otimes \mathcal{L}_2 \otimes \mathbb{R}^{d_3}. \end{aligned}$$

Hence every z_m belongs to the structured space

$$\mathcal{V} = (\mathbb{R}^{d_1} \otimes \mathcal{L}_2 \otimes \mathcal{L}_3) + (\mathcal{L}_1 \otimes \mathbb{R}^{d_2} \otimes \mathcal{L}_3) + (\mathcal{L}_1 \otimes \mathcal{L}_2 \otimes \mathbb{R}^{d_3}).$$

Because $\mathrm{col}(A_j) \subseteq \mathcal{L}_j$ for every j, each true tangent block is contained in the corresponding summand of $\mathcal{V}$. Therefore,

$$\mathcal{T}^G \subseteq \mathcal{V}.$$

In particular, $P^G z_m \in \mathcal{T}^G \subseteq \mathcal{V}$. Since $z_m \in \mathcal{V}$ and $\mathcal{V}$ is a linear space,

$$M^G z_m = z_m - P^G z_m \in \mathcal{V}.$$

Let $P_{\mathcal{V}}$ denote the orthogonal projector onto $\mathcal{V}$. Lemma 5 with $r_j = \dim(\mathcal{L}_j) \leqslant 3\ell$ gives the deterministic bound

$$\begin{aligned} \|P_{\mathcal{V}} e_G\| &\leqslant \sqrt{r_2 r_3 + r_1 r_3 + r_1 r_2}\, \|E^G\| \\ &\leqslant 3\sqrt{3}\, \ell\, \|E^G\|. \end{aligned}$$

Here $\|E^G\|$ is the tensor spectral norm. Since $M^G z_m \in \mathcal{V}$,

$$\langle e_G, M^G z_m \rangle = \langle P_{\mathcal{V}} e_G, M^G z_m \rangle.$$

Using $r_0 = e_G - \mathsf{X}_G' \theta_1$, the symmetry of M^G, and Cauchy–Schwarz, we therefore obtain

$$\begin{aligned} |\langle M^G r_0, z_m \rangle| &= |\langle r_0, M^G z_m \rangle| \\ &\leqslant |\langle e_G, M^G z_m \rangle| + |\langle \mathsf{X}_G' \theta_1, M^G z_m \rangle| \\ &= |\langle P_{\mathcal{V}} e_G, M^G z_m \rangle| + |\langle \mathsf{X}_G' \theta_1, M^G z_m \rangle| \\ &\leqslant (\|P_{\mathcal{V}} e_G\| + \|\mathsf{X}_G\| \|\theta_1\|)\, \|M^G z_m\| \\ &\leqslant \Big(3\sqrt{3}\, \ell\, \|E^G\| + \|\mathsf{X}_G\| \|\theta_1\|\Big)\, \|z_m\|. \end{aligned}$$

Since ℓ is fixed and

$$\|E^G\| = O_p(\sqrt{s}), \qquad \|\mathsf{X}_G\| = O_p(1), \qquad \|\theta_1\| = O_p(1), \qquad \|z_m\| = O_p(\rho^2),$$

it follows that

$$|\langle M^G r_0, z_m \rangle| = O_p(\sqrt{s}\, \rho^2) = O_p(\sqrt{n}\, \rho^3).$$

The subspace inclusions and the structured contraction inequality hold for every realized choice of the factors, graph errors, and mode spaces. Consequently, the argument remains valid when F_m, q, z_m, and $\mathcal{L}_j$ depend on the errors. It does not require these objects to be $\mathcal{F}$-measurable, and no conditional independence between them and e_G is used.

Finally, $(M^G r_0)_{(m)} C_m = 0$ and $q = \xi/\sqrt{n}$ imply

$$\langle (M^G r_0)_{(m)}, L_{m,r}(F_m^{(1)})' \rangle_F = \frac{1}{\sqrt{n}} \langle (M^G r_0)_{(m)}, D_{m,r} \xi_{(m)} \rangle_F.$$

Substitution into (147), together with (148), gives

$$[\Gamma_n \theta]_r = [\mathcal{Z}_n]_r - \frac{1}{\sqrt{n}} \sum_{m=1}^{3} \langle (M^G r_0)_{(m)}, D_{m,r} (P^G r_0)_{(m)} \rangle_F + O_p\left(\rho + \frac{s^{3/2}}{n}\right). \tag{154}$$

5. Reduction to error Gram matrices. Note that we have

$$\begin{aligned}
\langle (M^G r_0)_{(m)}, D_{m,r} (P^G r_0)_{(m)} \rangle_F &= \operatorname{vec}((M^G r_0)_{(m)})' \operatorname{vec}(D_{m,r} (P^G r_0)_{(m)}) \\
&= \operatorname{vec}((M^G r_0)_{(m)})' (D_{m,r} \otimes I_{d_{-m}}) \operatorname{vec}((P^G r_0)_{(m)}) \\
&= r_0' M^G J_m' (D_{m,r} \otimes I_{d_{-m}}) J_m P^G r_0.
\end{aligned}$$

Let

$$\Lambda_{m,r} = J_m' (D_{m,r} \otimes I_{d_{-m}}) J_m, \qquad \Lambda_r = \sum_{m=1}^{3} \Lambda_{m,r}.$$

Then $\|\Lambda_r\| = O_p(1)$, and the quadratic term in (154) is

$$\frac{1}{\sqrt{n}} r_0' M^G \Lambda_r P^G r_0.$$

We first remove the slope adjustment. By (141), $F_0 - Q' = -\mathsf{X}_G' \Gamma_n^{-1} \mathsf{X} M_{\mathcal{T}}$ has rank at most p and operator norm $O_p(1)$. Therefore,

$$F_0' M^G \Lambda_r P^G F_0 - Q M^G \Lambda_r P^G Q'$$

has rank at most $3p$ and operator norm $O_p(1)$, and hence has nuclear norm $O_p(1)$. Lemma 4 (iii) gives

$$\frac{1}{\sqrt{n}} r_0' M^G \Lambda_r P^G r_0 = \frac{1}{\sqrt{n}} e_G' M^G \Lambda_r P^G e_G + O_p(n^{-1/2}). \tag{155}$$

Decompose the tangent space into its component inside the product loading space and its three orthogonal exterior blocks:

$$P^G = P_0 + \sum_{h=1}^{3} P_h^{\circ}, \qquad P_h^{\circ} = J_h' (M_{A_h} \otimes P_{C_h}) J_h.$$

Here P_0 projects onto

$$\mathcal{T}^G \cap \{\operatorname{col}(A_1) \otimes \operatorname{col}(A_2) \otimes \operatorname{col}(A_3)\},$$

so $\operatorname{rank}(P_0) \leqslant \ell^3$. This decomposition follows by splitting each unrestricted loading perturbation into its projection onto the corresponding loading space and its orthogonal complement. The exterior blocks are mutually orthogonal and orthogonal to the product loading space.

Fix a mode h, and let $a < b$ be the other two modes. Define

$$D_{h,r}^{(-)} = D_{a,r} \otimes I_{d_b} + I_{d_a} \otimes D_{b,r}, \qquad B_{h,r} = M_{C_h} D_{h,r}^{(-)} P_{C_h}.$$

For every matrix Y satisfying $A_h' Y = 0$,

$$\big(M^G \mathfrak{v}_h(Y)\big)_{(h)} = Y M_{C_h}.$$

Indeed, $Y P_{C_h}$ belongs to the h-th tangent block, while $Y M_{C_h}$ is orthogonal to that block. It is also orthogonal to the other tangent blocks, whose mode-h matricizations have columns in $\mathrm{col}(A_h)$.

For an arbitrary $v \in \mathbb{R}^{d\times}$, define, for each $h = 1, 2, 3$,

$$V_h = v_{(h)}, \qquad z_h^\circ = P_h^\circ v, \qquad Z_h^\circ = (z_h^\circ)_{(h)} = M_{A_h} V_h P_{C_h}.$$

Thus V_h is the mode-h matricization of the common input v, whereas Z_h° is the mode-h matricization of its h-th exterior tangent component. In particular,

$$A_h' Z_h^\circ = 0, \qquad Z_h^\circ P_{C_h} = Z_h^\circ.$$

Fix h, and let $a < b$ be the other two modes. The transformation in mode h satisfies

$$\begin{aligned}(\Lambda_{h,r} z_h^\circ)_{(h)} &= D_{h,r} Z_h^\circ \\ &= D_{h,r} Z_h^\circ P_{C_h} \\ &= \big[D_{h,r} Z_h^\circ C_h (C_h' C_h)^{-1}\big] C_h'.\end{aligned}$$

Hence $\Lambda_{h,r} z_h^\circ \in \mathcal{T}_h^G$, and therefore

$$M^G \Lambda_{h,r} z_h^\circ = 0.$$

The transformations in the other two modes act on the same input component z_h° and satisfy

$$(\Lambda_{a,r} z_h^\circ)_{(h)} = Z_h^\circ (D_{a,r} \otimes I_{d_b})', \qquad (\Lambda_{b,r} z_h^\circ)_{(h)} = Z_h^\circ (I_{d_a} \otimes D_{b,r})'.$$

Consequently, we have

$$\big((\Lambda_{a,r} + \Lambda_{b,r}) z_h^\circ\big)_{(h)} = Z_h^\circ (D_{h,r}^{(-)})'.$$

Since

$$A_h' Z_h^\circ (D_{h,r}^{(-)})' = 0,$$

the previously established residual-projection identity applies to $Y = Z_h^\circ (D_{h,r}^{(-)})'$. Using $\Lambda_r = \Lambda_{h,r} + \Lambda_{a,r} + \Lambda_{b,r}$, we obtain

$$\begin{aligned}(M^G \Lambda_r P_h^\circ v)_{(h)} &= (M^G \Lambda_r z_h^\circ)_{(h)} \\ &= \big(M^G (\Lambda_{a,r} + \Lambda_{b,r}) z_h^\circ\big)_{(h)} \\ &= Z_h^\circ (D_{h,r}^{(-)})' M_{C_h} \\ &= M_{A_h} V_h P_{C_h} (D_{h,r}^{(-)})' M_{C_h} \\ &= M_{A_h} V_h B_{h,r}'.\end{aligned}$$

By row-major vectorization,

$$\begin{aligned}J_h M^G \Lambda_r P_h^\circ v &= \mathrm{vec}\big((M^G \Lambda_r P_h^\circ v)_{(h)}\big) \\ &= \mathrm{vec}(M_{A_h} V_h B_{h,r}') \\ &= (M_{A_h} \otimes B_{h,r}) \,\mathrm{vec}(V_h) \\ &= (M_{A_h} \otimes B_{h,r}) J_h v.\end{aligned}$$

Since v was arbitrary and J_h is orthogonal, this proves

$$M^G \Lambda_r P_h^\circ = J_h'(M_{A_h} \otimes B_{h,r}) J_h, \qquad h = 1, 2, 3.$$

Finally, the decomposition $P^G = P_0 + \sum_{h=1}^3 P_h^\circ$ gives

$$\begin{aligned} M^G \Lambda_r P^G &= M^G \Lambda_r P_0 + \sum_{h=1}^3 M^G \Lambda_r P_h^\circ \\ &= M^G \Lambda_r P_0 + \sum_{h=1}^3 J_h'(M_{A_h} \otimes B_{h,r}) J_h. \end{aligned}$$

Each $B_{h,r}$ has rank at most ℓ, and

$$\|B_{h,r}\| \leqslant \|D_{h,r}^{(-)}\| \leqslant \|D_{a,r}\| + \|D_{b,r}\| = O_p(1).$$

Replacing M_{A_h} by $I_{d_h} - P_{A_h}$ gives

$$M^G \Lambda_r P^G = \sum_{h=1}^3 J_h'(I_{d_h} \otimes B_{h,r}) J_h + \mathcal{R}_r,$$

where

$$\mathcal{R}_r = M^G \Lambda_r P_0 - \sum_{h=1}^3 J_h'(P_{A_h} \otimes B_{h,r}) J_h.$$

Since $\operatorname{rank}(P_{A_h}) = \ell$,

$$\operatorname{rank}(\mathcal{R}_r) \leqslant \ell^3 + 3\ell^2, \qquad \|\mathcal{R}_r\| \leqslant \|\Lambda_r\| + \sum_{h=1}^3 \|B_{h,r}\| = O_p(1).$$

Thus $\|\mathcal{R}_r\|_* = O_p(1)$. All these coefficients are $\mathcal{F}$-measurable, so Lemma 4 (iii) implies

$$e_G' \mathcal{R}_r e_G = O_p(1).$$

If $f_{h,i}'$ is row i of $E_{(h)}^G$, then

$$\begin{aligned} e_G' J_h'(I_{d_h} \otimes B_{h,r}) J_h e_G &= \sum_{i=1}^{d_h} f_{h,i}' B_{h,r} f_{h,i} \\ &= \operatorname{tr}\Big(B_{h,r} (E_{(h)}^G)' E_{(h)}^G \Big). \end{aligned}$$

Therefore,

$$\frac{1}{\sqrt{n}} e_G' M^G \Lambda_r P^G e_G = \frac{1}{\sqrt{n}} \sum_{h=1}^3 \operatorname{tr}\Big(B_{h,r} (E_{(h)}^G)' E_{(h)}^G \Big) + O_p(n^{-1/2}). \tag{156}$$

The coefficients admit an expression that does not involve S^+. Since $\Psi_m Y_m = P_{C_m}$ and $D_{m,r} = (t_r)_{(m)} \Psi_m$,

$$(\Lambda_{m,r} p_0)_{(m)} = D_{m,r} Y_m = (t_r)_{(m)} P_{C_m},$$

and hence

$$\Lambda_r p_0 = \sum_{m=1}^{3} \Lambda_{m,r} p_0 = \sum_{m=1}^{3} P_m t_r = S t_r = P^G x_r.$$

The mode-m version of this identity is

$$(P^G x_r)_{(m)} = D_{m,r} Y_m + Y_m (D_{m,r}^{(-)})'.$$

Premultiplying its transpose by M_{C_m} and postmultiplying by Ψ'_m gives

$$B_{m,r} = M_{C_m} (P^G x_r)'_{(m)} \Psi'_m. \tag{157}$$

Here we used $M_{C_m} Y'_m = 0$ and $Y'_m \Psi'_m = P_{C_m}$. In particular,

$$\|B_{m,r}\|_F = O_p(1), \qquad \|B_{m,r}\|_* = O_p(1).$$

Define the sample bias vectors

$$\mathcal{B}^s_m = \frac{1}{\sqrt{n}} \mathsf{X}_G P^G \, \mathfrak{v}_m \Big(\Psi'_m (E^G_{(m)})' E^G_{(m)} M_{C_m} \Big).$$

Equation (157) implies

$$[\mathcal{B}^s_m]_r = \frac{1}{\sqrt{n}} \operatorname{tr}\Big(B_{m,r} (E^G_{(m)})' E^G_{(m)} \Big).$$

Combining (154), (155), and (156), using fixed p and $n^{-1/2} \leqslant \rho$, yields

$$\Gamma_n \theta = \mathcal{Z}_n - \sum_{m=1}^{3} \mathcal{B}^s_m + O_p\left(\rho + \frac{s^{3/2}}{n} \right). \tag{158}$$

6. Conditional centering and bias orders. For each coordinate r,

$$[\mathcal{B}^s_m - \mathcal{B}_m]_r = \frac{1}{\sqrt{n}} \operatorname{tr}\Big[B_{m,r} \Big\{ (E^G_{(m)})' E^G_{(m)} - \Omega^G_m \Big\} \Big].$$

The corresponding quadratic coefficient in the original error coordinates is

$$K_{m,r} = Q J'_m (I_{d_m} \otimes B_{m,r}) J_m Q'.$$

Because $Q'Q = I_{d_\times}$ and J_m is orthogonal,

$$\|K_{m,r}\|_F^2 = d_m \|B_{m,r}\|_F^2 = O_p(d_m). \tag{159}$$

The nonzero coefficients of $K_{m,r}$ are supported on $\mathcal{R}^D$. For $m = 1$,

$$(E^G_{(1)})' E^G_{(1)} = E'_{(1)} P_D E_{(1)},$$

and $(P_D)_{ii'}$ vanishes unless $g(i) = g(i')$. For the other two modes,

$$E^G_{(2)} = E_{(2)} (Q_D \otimes I_T), \qquad E^G_{(3)} = E_{(3)} (Q_D \otimes I_M),$$

so the Gram products retain a common teacher index and a common time index, respectively. Consequently,

$$[\mathcal{B}^s_m - \mathcal{B}_m]_r = \frac{1}{\sqrt{n}} \sum_{(a,b) \in \mathcal{R}^D} [K_{m,r}]_{ab} z_{ab}.$$

The fourth-order covariance row-sum condition in Assumption 3 therefore gives

$$\mathbb{E}\Big[\,|[\mathcal{B}_m^s - \mathcal{B}_m]_r|^2 \Big| \mathcal{F}\Big] \leqslant \frac{c_4}{n}\|K_{m,r}\|_F^2$$

on an event whose probability tends to one. By (159) and conditional Markov's inequality,

$$\mathcal{B}_m^s - \mathcal{B}_m = O_p\left(\sqrt{\frac{d_m}{n}}\right).$$

Summing over the three modes produces an $O_p(\rho)$ difference. Substitution into (158) gives

$$\Gamma_n\sqrt{n}\,\gamma = \mathcal{Z}_n - \mathcal{B}_1 - \mathcal{B}_2 - \mathcal{B}_3 + O_p\left(\rho + \frac{s^{3/2}}{n}\right).$$

Since $\gamma = \hat{\delta} - \delta$, this proves (23).

Finally, each row-block conditional covariance of $E_{(m)}^G$ has operator norm at most c_ε on the covariance-summability event. Therefore,

$$\|\Omega_m^G\| \leqslant c_\varepsilon d_m.$$

Using (157),

$$\begin{aligned}
|[\mathcal{B}_m]_r| &= \frac{1}{\sqrt{n}}\left|\mathrm{tr}(B_{m,r}\Omega_m^G)\right| \\
&\leqslant \frac{1}{\sqrt{n}}\|B_{m,r}\|_*\|\Omega_m^G\| \\
&= O_p(d_m/\sqrt{n}).
\end{aligned}$$

Since p is fixed and $(d_1, d_2, d_3) = (G, M, T)$, this proves the three asserted bias orders. The stated expansion uses only $\rho \to 0$; its displayed remainder is $o_p(1)$ if, additionally, $s^{3/2}/n \to 0$. □

Proposition 2 (Fixed-index expansions with mode-specific remainders)**.** *Suppose that Assumptions 1, 2, 3, 4, and 5 hold. Let $(\hat{\delta}, \hat{U}, \hat{V}, \hat{W})$ be the global least-squares estimator defined in Assumption 1, and let H_u, H_v, H_w be the alignment matrices defined as*

$$\begin{aligned}
H_u' &= (V * W)'(\hat{V} * \hat{W})\left[(\hat{V} * \hat{W})'(\hat{V} * \hat{W})\right]^{-1}, \\
H_v' &= (DU * W)'(D\hat{U} * \hat{W})\left[(D\hat{U} * \hat{W})'(D\hat{U} * \hat{W})\right]^{-1}, \\
H_w' &= (DU * V)'(D\hat{U} * \hat{V})\left[(D\hat{U} * \hat{V})'(D\hat{U} * \hat{V})\right]^{-1}
\end{aligned}$$

and let $\Sigma_{VW} = \left(\frac{1}{M}V'V\right) \odot \left(\frac{1}{T}W'W\right)$, $\Sigma_{UW} = \left(\frac{1}{N}U'D'DU\right) \odot \left(\frac{1}{T}W'W\right)$, and $\Sigma_{UV} = \left(\frac{1}{N}U'D'DU\right) \odot \left(\frac{1}{M}V'V\right)$.

Write

$$n = NMT, \qquad s = G + M + T, \qquad \rho = \sqrt{\frac{s}{n}}, \qquad \theta_{NMT} = \sqrt{n}(\hat{\delta} - \delta).$$

For each fixed admissible group g, teacher j, and period t, define

$$S_{u,g} = \frac{1}{\sqrt{N_g MT}} \sum_{\substack{1 \leqslant i \leqslant N \\ g(i)=g}} \sum_{j'=1}^{M} \sum_{t'=1}^{T} \varepsilon_{ij't'} (v_{j'} \odot w_{t'}),$$

$$S_{v,j} = \frac{1}{\sqrt{NT}} \sum_{i=1}^{N} \sum_{t'=1}^{T} \varepsilon_{ijt'} (u_{g(i)} \odot w_{t'}),$$

$$S_{w,t} = \frac{1}{\sqrt{NM}} \sum_{i=1}^{N} \sum_{j'=1}^{M} \varepsilon_{ij't} (u_{g(i)} \odot v_{j'}),$$

and the $\ell \times p$ cross-moment matrices

$$Q_{u,g} = \frac{1}{N_g MT} \sum_{\substack{1 \leqslant i \leqslant N \\ g(i)=g}} \sum_{j'=1}^{M} \sum_{t'=1}^{T} (v_{j'} \odot w_{t'}) x'_{ij't'},$$

$$Q_{v,j} = \frac{1}{NT} \sum_{i=1}^{N} \sum_{t'=1}^{T} (u_{g(i)} \odot w_{t'}) x'_{ijt'},$$

$$Q_{w,t} = \frac{1}{NM} \sum_{i=1}^{N} \sum_{j'=1}^{M} (u_{g(i)} \odot v_{j'}) x'_{ij't}.$$

Then

$$\sqrt{NM} H_w^{-1} (\hat{w}_t - H_w w_t) = \Sigma_{UV}^{-1} \left[S_{w,t} - \frac{1}{\sqrt{T}} Q_{w,t} \theta_{NMT} \right] + R_{w,t,NMT}, \tag{27}$$

$$\sqrt{NT} H_v^{-1} (\hat{v}_j - H_v v_j) = \Sigma_{UW}^{-1} \left[S_{v,j} - \frac{1}{\sqrt{M}} Q_{v,j} \theta_{NMT} \right] + R_{v,j,NMT}, \tag{28}$$

$$\sqrt{N_g MT} H_u^{-1} (\hat{u}_g - H_u u_g) = \Sigma_{VW}^{-1} \left[S_{u,g} - \sqrt{\frac{N_g}{N}} Q_{u,g} \theta_{NMT} \right] + R_{u,g,NMT}. \tag{29}$$

The remainder vectors satisfy

$$\|R_{w,t,NMT}\| = O_p\left(\rho + \frac{\sqrt{N}}{T\sqrt{M}} + \frac{\sqrt{M}}{T\sqrt{N}} \right), \tag{30}$$

$$\|R_{v,j,NMT}\| = O_p\left(\rho + \frac{\sqrt{N}}{M\sqrt{T}} + \frac{\sqrt{T}}{M\sqrt{N}} \right), \tag{31}$$

$$\|R_{u,g,NMT}\| = O_p\left(\rho + \frac{\sqrt{N_g M}}{N\sqrt{T}} + \frac{\sqrt{N_g T}}{N\sqrt{M}} + \sqrt{N_g MT}\, \rho^4 \right). \tag{32}$$

If $N_g G/N = O_p(1)$, we have

$$\|R_{u,g,NMT}\| = O_p\left(\rho + \frac{\sqrt{N_g M}}{N\sqrt{T}} + \frac{\sqrt{N_g T}}{N\sqrt{M}} \right). \tag{33}$$

The group-size condition holds, in particular, when G remains fixed or when $N_g = O_p(N/G)$ for the group under consideration.

Proof. Write

$$n = NMT, \qquad s = G + M + T, \qquad \rho = \sqrt{\frac{s}{n}}, \qquad \gamma = \hat{\delta} - \delta, \qquad \theta = \theta_{NMT} = \sqrt{n}\,\gamma.$$

Assumption 1 implies $\rho \to 0$, while p and ℓ remain fixed. All slice arguments below are pointwise in the fixed admissible index under consideration.

1. Grouped coordinates and an exact remainder identity. By Lemma 1, the estimated factor and Khatri–Rao loading matrices have full column rank and well-conditioned normalized Gram matrices with probability approaching one, the relevant loading spaces converge, and the unrestricted block normal equations hold. All inverses and normal equations below are evaluated on these events. Inverse expressions may be assigned arbitrary values wherever they are undefined without affecting the asserted stochastic orders.

Define

$$Q_D = D(D'D)^{-1/2}, \qquad Q = Q_D \otimes I_M \otimes I_T, \qquad e = \operatorname{vec}(E), \qquad E^G = E \times_1 Q_D', \qquad e_G = Q'e.$$

Let

$$A_1 = \frac{(D'D)^{1/2}U}{\sqrt{N}}, \qquad A_2 = \frac{V}{\sqrt{M}}, \qquad A_3 = \frac{W}{\sqrt{T}}, \qquad (d_1, d_2, d_3) = (G, M, T).$$

Set $d_\times = GMT$, $d_{-m} = d_\times / d_m$, and

$$C_1 = A_2 * A_3, \qquad C_2 = A_1 * A_3, \qquad C_3 = A_1 * A_2, \qquad \Sigma_m = C_m' C_m.$$

Thus

$$(\Sigma_1, \Sigma_2, \Sigma_3) = (\Sigma_{VW}, \Sigma_{UW}, \Sigma_{UV}).$$

These are the normalized Gram matrices evaluated at the true factors. Use hats for the estimated counterparts.

For $v \in \mathbb{R}^{d_\times}$, let $v_{(m)}$ denote the mode-m matricization of its tensor reshape, with the remaining modes ordered increasingly, and let $\mathfrak{v}_m$ denote the inverse operation. Define the mode-permutation matrices by

$$J_m = K^G_{m,1}, \qquad J_m v = \operatorname{vec}(v_{(m)}), \qquad \mathfrak{v}_m(B) = J_m' \operatorname{vec}(B).$$

Writing $a_{m,\lambda}$ for column λ of A_m, define

$$p_0 = \sum_{\lambda=1}^{\ell} a_{1,\lambda} \otimes a_{2,\lambda} \otimes a_{3,\lambda}, \qquad (p_0)_{(m)} = A_m C_m',$$

and define $\hat{p}$ analogously. Put

$$h = \hat{p} - p_0.$$

Write $(H_1, H_2, H_3) = (H_u, H_v, H_w)$. The scalar normalizations and the group-coordinate isometries preserve the alignment matrices in the proposition, so

$$H_m = (\hat{C}_m' \hat{C}_m)^{-1} \hat{C}_m' C_m.$$

The true and estimated Gram matrices are well conditioned in probability, and the corresponding loading spaces converge. Consequently, $\hat{C}_m' C_m$ and H_m are nonsingular with probability approaching one, and

$$\|H_m^{-1}\| = O_p(1).$$

Define the graph-alignment matrix and graph error by

$$J_m^C = (C_m' C_m)^{-1} C_m' \hat{C}_m, \qquad F_m = \hat{C}_m (J_m^C)^{-1} - C_m.$$

Then

$$C_m' F_m = 0, \qquad \hat{C}_m = (C_m + F_m) J_m^C, \qquad \hat{C}_m' C_m = (J_m^C)' \Sigma_m.$$

The graph error is invariant to changes of basis in the estimated loading space: replacing $\hat{C}_m$ by $\hat{C}_m R$, for nonsingular R, replaces J_m^C by $J_m^C R$ and leaves F_m unchanged. Hence the graph-error bounds established using the signed-permutation alignment in Proposition 1 apply to the present representatives as well.

Define $X^G = X \times_1 Q_D'$. Let $e_{m,k}^G \in \mathbb{R}^{d_{-m}}$ be the transpose of row k of $E_{(m)}^G$, and let $X_{m,k}^G \in \mathbb{R}^{d_{-m} \times p}$ stack the corresponding regressor vectors from X^G. To distinguish factor rows from the columns $a_{m,\lambda}$, write

$$\alpha_{m,k} = A_m' \mathbf{e}_k^{(d_m)}, \qquad \hat{\alpha}_{m,k} = \hat{A}_m' \mathbf{e}_k^{(d_m)},$$

where $\mathbf{e}_k^{(d_m)}$ is the k-th coordinate vector in $\mathbb{R}^{d_m}$. Thus $\alpha_{m,k}$ and $\hat{\alpha}_{m,k}$ are the transposes of row k of the corresponding normalized factor matrices.

The block normal equations give

$$\sqrt{n}\,(\hat{\alpha}_{m,k} - H_m \alpha_{m,k}) = (\hat{C}_m' \hat{C}_m)^{-1} \hat{C}_m' \{e_{m,k}^G - X_{m,k}^G \gamma\}.$$

Premultiplying by H_m^{-1} and using the graph representation yields the exact identity

$$\begin{aligned} \sqrt{n}\, H_m^{-1}(\hat{\alpha}_{m,k} - H_m \alpha_{m,k}) &= (\hat{C}_m' C_m)^{-1} \hat{C}_m' \{e_{m,k}^G - X_{m,k}^G \gamma\} \\ &= \Sigma_m^{-1} (C_m + F_m)' \{e_{m,k}^G - X_{m,k}^G \gamma\}. \end{aligned} \tag{160}$$

Indeed,

$$(\hat{C}_m' C_m)^{-1} \hat{C}_m' = \big((J_m^C)' \Sigma_m\big)^{-1} (J_m^C)' (C_m + F_m)' = \Sigma_m^{-1} (C_m + F_m)'.$$

The normalized rows satisfy

$$\alpha_{1,g} = \sqrt{\frac{N_g}{N}}\, u_g, \qquad \alpha_{2,j} = \frac{v_j}{\sqrt{M}}, \qquad \alpha_{3,t} = \frac{w_t}{\sqrt{T}},$$

with analogous identities for their estimators. Therefore, the left-hand side of (160) has exactly the group, teacher, and period normalizations in the proposition. Moreover,

$$C_1' e_{1,g}^G = S_{u,g}, \qquad C_2' e_{2,j}^G = S_{v,j}, \qquad C_3' e_{3,t}^G = S_{w,t},$$

and

$$\begin{aligned} C_1' X_{1,g}^G \gamma &= \sqrt{\frac{N_g}{N}}\, Q_{u,g} \theta, \\ C_2' X_{2,j}^G \gamma &= \frac{1}{\sqrt{M}}\, Q_{v,j} \theta, \\ C_3' X_{3,t}^G \gamma &= \frac{1}{\sqrt{T}}\, Q_{w,t} \theta. \end{aligned}$$

It therefore remains to bound

$$R_{m,k} = \Sigma_m^{-1} F_m' \{e_{m,k}^G - X_{m,k}^G \gamma\}, \tag{161}$$

where

$$(R_{1,g}, R_{2,j}, R_{3,t}) = (R_{u,g,NMT}, R_{v,j,NMT}, R_{w,t,NMT}).$$

2. First-order graph expansion and slice moments. Retain the tangent-space notation from the proof of Proposition 1. For $m = 1, 2, 3$, define

$$\mathcal{T}_m^G = \left\{ v \in \mathbb{R}^{d_\times} : v_{(m)} = BC_m',\ B \in \mathbb{R}^{d_m \times \ell} \right\}, \qquad P_m = J_m'(I_{d_m} \otimes P_{C_m}) J_m.$$

Let P^G be the orthogonal projector onto

$$\mathcal{T}^G = \mathcal{T}_1^G + \mathcal{T}_2^G + \mathcal{T}_3^G,$$

and write

$$S = P_1 + P_2 + P_3, \qquad P^G = SS^+, \qquad M^G = I_{d_\times} - P^G, \qquad M_{\mathcal{T}} = I_n - QP^G Q'.$$

The bound $\|S^+\| = O_p(1)$ is established in the proof of Proposition 1.

Define

$$\mathsf{X} = \frac{X_{(4)}}{\sqrt{n}}, \qquad \mathsf{X}_G = \mathsf{X}Q, \qquad \Gamma_n = \mathsf{X} M_{\mathcal{T}} \mathsf{X}', \qquad \mathcal{Z}_n = \mathsf{X} M_{\mathcal{T}} e,$$

and

$$\theta_1 = \Gamma_n^{-1} \mathcal{Z}_n, \qquad F_0 = Q' - \mathsf{X}_G' \Gamma_n^{-1} \mathsf{X} M_{\mathcal{T}}.$$

Then

$$r_0 = F_0 e = e_G - \mathsf{X}_G' \theta_1, \qquad \xi = P^G r_0, \qquad q = \frac{\xi}{\sqrt{n}}.$$

For convenience, put

$$\mathscr{A} = P^G F_0, \qquad q = \frac{1}{\sqrt{n}} \mathscr{A} e.$$

The conditional regressor-moment bound gives $\|\mathsf{X}\|_F = O_p(1)$, while identification gives $\|\Gamma_n^{-1}\| = O_p(1)$. Hence $\mathscr{A}$ is $\mathcal{F}$-measurable and

$$\|\mathscr{A}\| = O_p(1).$$

The preliminary linearization bounds established in the proof of Proposition 1 give

$$\|\gamma\| = O_p(n^{-1/2} + \rho^2), \qquad \theta_1 = O_p(1), \qquad \|h\| + \|q\| = O_p(\rho), \qquad \|h - q\| = O_p(\rho^2). \tag{162}$$

In particular, no bound $\sqrt{n}\,\gamma = O_p(1)$ is used.

Define the linear map

$$\mathcal{D}_m(v) = M_{C_m} v_{(m)}' A_m (A_m' A_m)^{-1}.$$

Its operator norm between the relevant Euclidean and Frobenius spaces is $O_p(1)$. The graph expansion is

$$F_m = F_m^{(1)} + \mathcal{E}_m, \qquad F_m^{(1)} = \mathcal{D}_m(q), \qquad \|\mathcal{E}_m\|_F = O_p(\rho^2). \tag{163}$$

To recall its algebraic basis, set

$$\widetilde{A}_m = \hat{p}_{(m)} C_m \Sigma_m^{-1} = A_m + h_{(m)} C_m \Sigma_m^{-1}.$$

The graph representation implies

$$\hat{p}_{(m)} = \widetilde{A}_m (C_m + F_m)', \qquad F_m = M_{C_m} h'_{(m)} \widetilde{A}_m (\widetilde{A}'_m \widetilde{A}_m)^{-1}.$$

The strong-factor bounds, the inverse perturbation identity, and (162) therefore yield (163).

Write

$$\kappa_{1,g} = \sqrt{N_g MT}, \qquad \kappa_{2,j} = \sqrt{NT}, \qquad \kappa_{3,t} = \sqrt{NM}.$$

The conditional fourth-moment bound on the regressors implies

$$\|X^G_{m,k}\|_F = O_p(\kappa_{m,k}). \tag{164}$$

Indeed, conditional on D, each regressor second moment is at most $c_x^{1/2}$. For the group slice,

$$\|X^G_{1,g}\|_F^2 \leqslant \sum_{i:g(i)=g} \sum_{j=1}^{M} \sum_{t=1}^{T} \|x_{ijt}\|^2,$$

whose conditional expectation is at most $c_x^{1/2} N_g MT$. For teacher and period slices, multiplication by Q'_D is a Frobenius contraction, giving the corresponding conditional expectation bounds $c_x^{1/2} NT$ and $c_x^{1/2} NM$.

On the conditional covariance-summability event, Assumption 3 also gives

$$\mathbb{E}[\|e^G_{m,k}\|^2 \mid \mathcal{F}] \leqslant c_\varepsilon d_{-m}, \qquad \mathbb{E}[\|Be\|^2 \mid \mathcal{F}] \leqslant c_\varepsilon \|B\|_F^2$$

for every conformable $\mathcal{F}$-measurable matrix B. We additionally use

$$\mathbb{E}[|e'Ke| \mid \mathcal{F}] \leqslant c_\varepsilon \|K\|_* \tag{165}$$

for every $\mathcal{F}$-measurable square matrix K, where $\|\cdot\|_*$ denotes the nuclear norm. This follows from a singular value decomposition and conditional Cauchy–Schwarz, as in Lemma 4 (iii). The same inequality holds for e_G, since

$$\big\|\mathbb{E}[e_G e'_G \mid \mathcal{F}]\big\| = \big\|Q'\mathbb{E}[ee' \mid \mathcal{F}]Q\big\| \leqslant c_\varepsilon.$$

All conditional bounds below are used on these high-probability events, with localization for random coefficient norms.

3. The graph–regressor contribution. Because $q = \mathscr{A}e/\sqrt{n}$, each entry of $(F_m^{(1)})' X^G_{m,k}$ is a linear contraction of e with an $\mathcal{F}$-measurable coefficient vector. More precisely, the matrix mapping e to the vectorization of this product has Frobenius norm at most

$$\frac{\|\mathscr{A}\| \|A_m (A'_m A_m)^{-1}\|_F \|X^G_{m,k}\|_F}{\sqrt{n}}.$$

Lemma 4(i), (164), and fixed p, ℓ therefore give

$$\|(F_m^{(1)})' X^G_{m,k}\|_F = O_p\left(\frac{\kappa_{m,k}}{\sqrt{n}}\right).$$

Combining this bound with (163) and the slope rate in (162),

$$\begin{aligned}\|F'_m X^G_{m,k}\gamma\| &\leqslant \|(F^{(1)}_m)' X^G_{m,k}\|_F\|\gamma\| + \|\mathcal{E}_m\|_F\|X^G_{m,k}\|_F\|\gamma\| \\ &= O_p\left(\kappa_{m,k}(n^{-1/2}+\rho^2)^2\right) \\ &= O_p\left(\rho + \kappa_{m,k}\rho^4\right). \end{aligned} \tag{166}$$

For the last equality, use $\kappa_{m,k} \leqslant \sqrt{n}$, $n^{-1/2} \leqslant \rho$, and $\rho \to 0$:

$$\kappa_{m,k}(n^{-1/2}+\rho^2)^2 = \frac{\kappa_{m,k}}{n} + \frac{2\kappa_{m,k}\rho^2}{\sqrt{n}} + \kappa_{m,k}\rho^4 \leqslant n^{-1/2} + 2\rho^2 + \kappa_{m,k}\rho^4.$$

The conditional linear-contraction argument is what avoids the larger bound obtained by replacing $\|(F^{(1)}_m)' X^G_{m,k}\|_F$ with $\|F^{(1)}_m\|_F\|X^G_{m,k}\|_F$.

4. The higher-order graph–error contribution. Use the tangent decomposition from the proof of Proposition 1. Define

$$B^q_a = (S^+ q)_{(a)} C_a (C'_a C_a)^{-1}, \qquad a = 1,2,3.$$

Then

$$\mathfrak{v}_a(B^q_a C'_a) = P_a S^+ q.$$

Since $q \in \mathcal{T}^G$ and $SS^+ = P^G$,

$$q = \sum_{a=1}^{3} \mathfrak{v}_a(B^q_a C'_a).$$

For each mode a, let

$$\mathcal{L}_a = \operatorname{col}[A_a, \hat{A}_a, B^q_a], \qquad \dim(\mathcal{L}_a) \leqslant 3\ell.$$

Fix m, and let $a < b$ be the other two modes. Every column of F_m belongs to $\mathcal{L}_a \otimes \mathcal{L}_b$, because

$$F_m = \hat{C}_m (J^C_m)^{-1} - C_m, \qquad C_m = A_a * A_b, \qquad \hat{C}_m = \hat{A}_a * \hat{A}_b.$$

The tangent representation of q gives

$$q'_{(m)} = C_m (B^q_m)' + (B^q_a * A_b + A_a * B^q_b)\, A'_m,$$

so every column of $q'_{(m)}$ also lies in $\mathcal{L}_a \otimes \mathcal{L}_b$. Since $\operatorname{col}(C_m)$ is contained in that product space, applying M_{C_m} preserves it. Thus the columns of

$$F^{(1)}_m = M_{C_m} q'_{(m)} A_m (A'_m A_m)^{-1}$$

belong to the same product space. Consequently, each column of $\mathcal{E}_m = F_m - F^{(1)}_m$, reshaped as a $d_a \times d_b$ matrix, has rank at most 3ℓ.

The matrix spectral norm of a fixed mode-m slice of E^G is at most the tensor spectral norm $\|E^G\|$. Applying the nuclear/Frobenius inequality to each rank-at-most-3ℓ column reshape, and also using ordinary Cauchy–Schwarz, gives the deterministic bound

$$\|\mathcal{E}'_m e^G_{m,k}\| \leqslant \min\{\sqrt{3\ell}\|E^G\|, \|e^G_{m,k}\|\}\|\mathcal{E}_m\|_F.$$

The spectral-error assumption, the slice second-moment bound, and (163) therefore imply

$$\|\mathcal{E}'_m e^G_{m,k}\| = O_p\left(\rho^2 \min\{\sqrt{s}, \sqrt{d_{-m}}\}\right). \tag{167}$$

This argument holds for every realized choice of the estimated loading spaces; it does not treat these spaces or $\mathcal{E}_m$ as $\mathcal{F}$-measurable.

5. The first-order graph–error contribution. We first separate the overlap among the tangent blocks. Let P_0 denote the orthogonal projector onto

$$\mathcal{T}^G \cap \{\mathrm{col}(A_1) \otimes \mathrm{col}(A_2) \otimes \mathrm{col}(A_3)\},$$

and let

$$P_a^\circ = J_a'(M_{A_a} \otimes P_{C_a})J_a, \qquad a = 1, 2, 3.$$

Splitting each tangent direction into its component in $\mathrm{col}(A_a)$ and its orthogonal complement gives

$$P^G = P_0 + \sum_{a=1}^{3} P_a^\circ, \qquad P_a = P_a^\circ + J_a'(P_{A_a} \otimes P_{C_a})J_a.$$

Consequently,

$$P^G = \sum_{a=1}^{3} P_a + \mathcal{R}_P, \qquad \mathcal{R}_P := P^G - S = P_0 - \sum_{a=1}^{3} J_a'(P_{A_a} \otimes P_{C_a})J_a. \tag{168}$$

Every term in the last expression has range contained in the product loading space. Hence

$$\mathrm{rank}(\mathcal{R}_P) \leqslant \ell^3, \qquad \|\mathcal{R}_P\| \leqslant \|P^G\| + \sum_{a=1}^{3} \|P_a\| \leqslant 4.$$

The mode-m block makes no contribution to the graph map. Indeed,

$$(P_m v)_{(m)} = v_{(m)} P_{C_m}$$

implies

$$\mathcal{D}_m(P_m v) = M_{C_m} P_{C_m} v_{(m)}' A_m (A_m' A_m)^{-1} = 0.$$

The term generated by $\mathcal{R}_P$ contributes $O_p(n^{-1/2})$ to $(F_m^{(1)})' e_{m,k}^G$. For each coordinate, its quadratic-form coefficient has rank at most ℓ^3 and operator norm $O_p(1)$ before multiplication by $n^{-1/2}$, so (165) applies.

The part of q equal to

$$-\frac{1}{\sqrt{n}} P^G \mathsf{X}_G' \theta_1$$

also contributes $O_p(n^{-1/2})$. To see this, expand over the fixed number p of coordinates of θ_1. Each remaining coefficient vector is $\mathcal{F}$-measurable and has $O_p(1)$ norm, so its contraction with $e_{m,k}^G$ is $O_p(1)$, while $\theta_1 = O_p(1)$.

For every $a \neq m$, replacing $M_{C_m} e_{m,k}^G$ by $e_{m,k}^G$ in the remaining quadratic term introduces an additional $O_p(n^{-1/2})$ error. The difference contains $P_{C_m} e_{m,k}^G$, and its quadratic-form coefficient has rank at most ℓ and operator norm $O_p(1)$. Therefore,

$$(F_m^{(1)})' e_{m,k}^G = \frac{1}{\sqrt{n}} \sum_{a \neq m} (A_m' A_m)^{-1} A_m' (P_a e_G)_{(m)} e_{m,k}^G + O_p(n^{-1/2}). \tag{169}$$

Fix distinct m, a, and let b be the remaining mode. Write $\mathscr{E}^{mab}$ for E^G with its modes ordered as (m, a, b). To distinguish factor-component indices from the slope-coordinate index used in Proposition 1, use $\mu, \nu \in \{1, \ldots, \ell\}$. Since

$$(A_m' A_m)^{-1} A_m' A_m = I_\ell,$$

and P_a projects the other two modes onto $\mathrm{col}(C_a)$, the μ-th coordinate of the mode-a summand satisfies

$$\left[\frac{1}{\sqrt{n}}(A_m'A_m)^{-1}A_m'(P_a e_G)_{(m)}e_{m,k}^G\right]_\mu = \sum_{\nu=1}^{\ell}[\Sigma_a^{-1}]_{\mu\nu}T_{ma,k}^{\mu\nu},$$

where

$$T_{ma,k}^{\mu\nu} = \frac{1}{\sqrt{n}}\sum_{u=1}^{d_m}\sum_{v=1}^{d_a}\sum_{w,w'=1}^{d_b}\mathscr{E}_{kvw}^{mab}\mathscr{E}_{uvw'}^{mab}[A_m]_{u\nu}[A_b]_{w\mu}[A_b]_{w'\nu}. \tag{170}$$

The strong-factor bounds give $\|\Sigma_a^{-1}\| = O_p(1)$.

We first bound the centered contraction. Let $K_{ma,k}^{G,\mu\nu}$ be the coefficient matrix satisfying

$$T_{ma,k}^{\mu\nu} = e_G'K_{ma,k}^{G,\mu\nu}e_G.$$

Its squared Frobenius norm is

$$\|K_{ma,k}^{G,\mu\nu}\|_F^2 = \frac{d_a}{n}\|[A_m]_{\cdot\nu}\|^2\|[A_b]_{\cdot\mu}\|^2\|[A_b]_{\cdot\nu}\|^2 = O_p(d_a/n).$$

The coefficient in the original error coordinates is

$$K_{ma,k}^{\mu\nu} = QK_{ma,k}^{G,\mu\nu}Q'.$$

Since $Q'Q = I_{d_\times}$,

$$\|K_{ma,k}^{\mu\nu}\|_F = \|K_{ma,k}^{G,\mu\nu}\|_F.$$

Its nonzero ordered error pairs share the mode-a index in grouped coordinates. In the original coordinates, they therefore have a common group, teacher, or period, according as $a = 1, 2, 3$. Thus the coefficient is supported on $\mathcal{R}^D$. Lemma 4(ii) yields

$$T_{ma,k}^{\mu\nu} - \mathbb{E}[T_{ma,k}^{\mu\nu} \mid \mathcal{F}] = O_p\left(\sqrt{\frac{d_a}{n}}\right). \tag{171}$$

For the conditional means, let

$$\mathcal{I}_{1,g} = \mathcal{I}_g^{(g)}, \qquad \mathcal{I}_{2,j} = \mathcal{I}_j^{(j)}, \qquad \mathcal{I}_{3,t} = \mathcal{I}_t^{(t)}$$

be the source slices in Assumption 5. Their cardinalities are $\kappa_{m,k}^2$. For an original error index α, abbreviate

$$u_\alpha = u_{g(i_\alpha)}, \qquad v_\alpha = v_{j_\alpha}, \qquad w_\alpha = w_{t_\alpha}.$$

Also define

$$\chi_J(\alpha,\beta) = \mathbf{1}\{j_\alpha = j_\beta\}, \qquad \chi_T(\alpha,\beta) = \mathbf{1}\{t_\alpha = t_\beta\}, \qquad p_{\alpha\beta} = (P_D)_{i_\alpha i_\beta},$$

where

$$p_{\alpha\beta} = \frac{\mathbf{1}\{g(i_\alpha) = g(i_\beta)\}}{N_{g(i_\alpha)}}.$$

Expanding the normalized grouped errors and factors in (170) gives

$$T_{ma,k}^{\mu\nu} = \frac{1}{\kappa_{m,k}c_{ma}}\sum_{\alpha\in\mathcal{I}_{m,k}}\sum_{\beta\in\mathcal{I}}\chi_{ma}(\alpha,\beta)\phi_\alpha\psi_\beta\varepsilon_\alpha\varepsilon_\beta, \tag{172}$$

with the following coefficients, where the arguments of χ_J, χ_T, p are suppressed:

(m,a)	ϕ_α	ψ_β	χ_{ma}	c_{ma}
$(1,2)$	$w_{\alpha,\mu}$	$u_{\beta,\nu}w_{\beta,\nu}$	χ_J	NT
$(1,3)$	$v_{\alpha,\mu}$	$u_{\beta,\nu}v_{\beta,\nu}$	χ_T	NM
$(2,1)$	$w_{\alpha,\mu}$	$v_{\beta,\nu}w_{\beta,\nu}$	p	MT
$(2,3)$	$u_{\alpha,\mu}$	$u_{\beta,\nu}v_{\beta,\nu}$	χ_T	NM
$(3,1)$	$v_{\alpha,\mu}$	$v_{\beta,\nu}w_{\beta,\nu}$	p	MT
$(3,2)$	$u_{\alpha,\mu}$	$u_{\beta,\nu}w_{\beta,\nu}$	χ_J	NT

For example,

$$T^{\mu\nu}_{12,g} = \frac{1}{NT\sqrt{N_g MT}} \sum_{i:g(i)=g} \sum_{i'=1}^{N} \sum_{j=1}^{M} \sum_{t,t'=1}^{T} \varepsilon_{ijt}\varepsilon_{i'jt'} u_{g(i'),\nu} w_{t,\mu} w_{t',\nu}.$$

The source grouped error supplies the factor $N_g^{-1/2}$, while the destination grouped-error normalization cancels the corresponding group-size normalization in A_1.

In every row of the table,

$$0 \leqslant \chi_{ma}(\alpha,\beta) \leqslant 1, \qquad |\psi_\beta|^2 \leqslant \omega_\beta,$$

where ω_β is defined in Assumption 5. Let

$$\Gamma_{\alpha\beta} = \mathbb{E}[\varepsilon_\alpha \varepsilon_\beta \mid \mathcal{F}].$$

The conditional covariance row-sum bound and the factor Gram bounds imply

$$\begin{aligned} \sum_{\alpha\in\mathcal{I}_{m,k}} \sum_{\beta\in\mathcal{I}} |\Gamma_{\alpha\beta}|\,|\phi_\alpha|^2 &\leqslant c_\varepsilon \sum_{\alpha\in\mathcal{I}_{m,k}} |\phi_\alpha|^2 \\ &= O_p(\kappa^2_{m,k}). \end{aligned}$$

Assumption 5 also gives

$$\begin{aligned} \sum_{\alpha\in\mathcal{I}_{m,k}} \sum_{\beta\in\mathcal{I}} |\Gamma_{\alpha\beta}|\,|\psi_\beta|^2 &\leqslant \sum_{\alpha\in\mathcal{I}_{m,k}} \sum_{\beta\in\mathcal{I}} |\Gamma_{\alpha\beta}|\omega_\beta \\ &= O_p(\kappa^2_{m,k}). \end{aligned}$$

Weighted Cauchy–Schwarz in (172) therefore yields

$$\begin{aligned} \left|\mathbb{E}[T^{\mu\nu}_{ma,k} \mid \mathcal{F}]\right| &\leqslant \frac{1}{\kappa_{m,k} c_{ma}} \left(\sum_{\alpha\in\mathcal{I}_{m,k}} \sum_{\beta\in\mathcal{I}} |\Gamma_{\alpha\beta}|\,|\phi_\alpha|^2 \right)^{1/2} \\ &\quad \times \left(\sum_{\alpha\in\mathcal{I}_{m,k}} \sum_{\beta\in\mathcal{I}} |\Gamma_{\alpha\beta}|\,|\psi_\beta|^2 \right)^{1/2} \\ &= O_p\left(\frac{\kappa_{m,k}}{c_{ma}} \right). \end{aligned} \tag{173}$$

Define

$$b_{1,g} = \frac{\sqrt{N_g M}}{N\sqrt{T}} + \frac{\sqrt{N_g T}}{N\sqrt{M}},$$
$$b_{2,j} = \frac{\sqrt{N}}{M\sqrt{T}} + \frac{\sqrt{T}}{M\sqrt{N}},$$
$$b_{3,t} = \frac{\sqrt{N}}{T\sqrt{M}} + \frac{\sqrt{M}}{T\sqrt{N}}.$$

These are exactly

$$b_{m,k} = \sum_{a \neq m} \frac{\kappa_{m,k}}{c_{ma}}.$$

Combining (169)–(173), using fixed ℓ, $\|\Sigma_a^{-1}\| = O_p(1)$, and $\sqrt{d_a/n} \leqslant \rho$, gives

$$\|(F_m^{(1)})' e_{m,k}^G\| = O_p(\rho + b_{m,k}). \tag{174}$$

6. Rate comparisons and conclusion. The higher-order error term in (167) is absorbed by the same rate:

$$\rho^2 \min\{\sqrt{s}, \sqrt{d_{-m}}\} \lesssim \rho + b_{m,k}. \tag{175}$$

To verify this, first suppose $d_m = \max_{a=1,2,3} d_a$. Since $s \leqslant 3d_m$ and $n = (N/G)d_m d_{-m}$,

$$\rho^2 \sqrt{d_{-m}} \leqslant \frac{3d_m\sqrt{d_{-m}}}{n} \leqslant 3\sqrt{\frac{G}{N}}\,\rho \leqslant 3\rho.$$

Otherwise, one of the other two modes attains the largest dimension, so

$$\rho^2\sqrt{s} = \frac{s^{3/2}}{n} \leqslant \frac{3\sqrt{3}}{n} \sum_{a \neq m} d_a^{3/2} \leqslant 3\sqrt{3}\, b_{m,k}.$$

The last inequality follows from the displayed expressions for $b_{m,k}$, using $G \leqslant N$, $N_g \geqslant 1$, and $N, M, T \geqslant 1$.

Equations (163), (167), (174), and (175) imply

$$\|F_m' e_{m,k}^G\| \leqslant \|(F_m^{(1)})' e_{m,k}^G\| + \|\mathcal{E}_m' e_{m,k}^G\| = O_p(\rho + b_{m,k}).$$

Combining this with (166), (161), and $\|\Sigma_m^{-1}\| = O_p(1)$ proves the unified remainder bound

$$\|R_{m,k}\| = O_p(\rho + b_{m,k} + \kappa_{m,k}\rho^4). \tag{176}$$

For the period block, $G \leqslant N$ gives

$$\kappa_{3,t}\rho^4 = \rho^2 \frac{G + M + T}{T\sqrt{NM}}$$
$$\leqslant \rho^2 \left(\frac{\sqrt{N}}{T\sqrt{M}} + \frac{\sqrt{M}}{T\sqrt{N}} + \frac{1}{\sqrt{NM}} \right).$$

Since $1/\sqrt{NM} \leqslant \rho$ and $\rho \to 0$, this term is absorbed into $\rho + b_{3,t}$. Similarly,

$$\kappa_{2,j}\rho^4 \leqslant \rho^2 \left(\frac{\sqrt{N}}{M\sqrt{T}} + \frac{\sqrt{T}}{M\sqrt{N}} + \frac{1}{\sqrt{NT}} \right)$$

is absorbed into $\rho + b_{2,j}$. Thus (176) yields

$$\|R_{w,t,NMT}\| = O_p\left(\rho + \frac{\sqrt{N}}{T\sqrt{M}} + \frac{\sqrt{M}}{T\sqrt{N}}\right)$$

and

$$\|R_{v,j,NMT}\| = O_p\left(\rho + \frac{\sqrt{N}}{M\sqrt{T}} + \frac{\sqrt{T}}{M\sqrt{N}}\right).$$

For the group block, retaining $\kappa_{1,g}\rho^4$ gives

$$\|R_{u,g,NMT}\| = O_p\left(\rho + \frac{\sqrt{N_g M}}{N\sqrt{T}} + \frac{\sqrt{N_g T}}{N\sqrt{M}} + \sqrt{N_g MT}\,\rho^4\right).$$

Finally, suppose $N_g G/N = O_p(1)$. Then

$$\begin{aligned}\sqrt{N_g MT}\,\rho^4 &= \rho^2\left[\sqrt{\frac{N_g G}{N}}\sqrt{\frac{G}{n}} + \frac{\sqrt{N_g M}}{N\sqrt{T}} + \frac{\sqrt{N_g T}}{N\sqrt{M}}\right] \\ &= \rho^2\left[O_p(\rho) + b_{1,g}\right] \\ &= o_p(\rho + b_{1,g}).\end{aligned}$$

Hence the group remainder simplifies to

$$\|R_{u,g,NMT}\| = O_p\left(\rho + \frac{\sqrt{N_g M}}{N\sqrt{T}} + \frac{\sqrt{N_g T}}{N\sqrt{M}}\right).$$

The group-size condition holds when G remains fixed, since $N_g \leqslant N$, and when $N_g = O_p(N/G)$ for the group under consideration. Together with the exact identity (160) and the score and regressor identities in Step 1, these bounds establish all three expansions and the stated remainder rates. The remainder vectors are $o_p(1)$ whenever their corresponding displayed rates tend to zero. □

Theorem 1 (Joint asymptotic normality of the slope and aligned factor errors)**.** *Suppose that Assumptions 1, 2, 3, 4, 5, and 6 hold. Let $(\hat{\delta}, \hat{U}, \hat{V}, \hat{W})$ be the global least-squares estimator defined in Assumption 1. Write*

$$n = NMT, \qquad s = G + M + T, \qquad \rho = \sqrt{\frac{s}{n}},$$

and suppose that

$$\frac{s^{3/2}}{n} \to 0. \tag{40}$$

Let $b_\delta = b_1 + b_2 + b_3$, and use the limiting matrices and vectors defined in Assumption 6.

The estimated factors are expressed in the true-factor column labeling and sign orientation. Specifically, among the equivalent least-squares triples

$$(\hat{U}\Pi_\ell S_v S_w,\ \hat{V}\Pi_\ell S_v,\ \hat{W}\Pi_\ell S_w),$$

where Π_ℓ ranges over $\ell \times \ell$ permutation matrices and S_v, S_w range over diagonal matrices with entries in $\{-1, 1\}$, choose a representative minimizing

$$\frac{1}{M}\|\hat{V}\Pi_\ell S_v - V\|_F^2 + \frac{1}{T}\|\hat{W}\Pi_\ell S_w - W\|_F^2.$$

Retain the hat notation for the selected representative, and define H_u, H_v, H_w from it as in Proposition 2. The followings hold.

1. *(Identity limits of the alignment matrices.)*
   The matrices H_u, H_v, H_w are nonsingular with probability approaching one and satisfy

$$H_u \xrightarrow{p} I_\ell, \qquad H_v \xrightarrow{p} I_\ell, \qquad H_w \xrightarrow{p} I_\ell.$$

2. *(Slope central limit theorem.)*
   The slope estimator satisfies

$$\sqrt{n}(\hat{\delta} - \delta) \xrightarrow{d} \mathcal{N}\big(-\Gamma_\delta^{-1} b_\delta,\ \Gamma_\delta^{-1}\Omega_Z\Gamma_\delta^{-1}\big)\,. \tag{41}$$

3. *(Joint fixed-index central limit theorem.)*
   Fix an admissible triple (g, j, t) and suppose additionally that

$$\frac{N+M}{T\sqrt{NM}} \to 0, \qquad \frac{N+T}{M\sqrt{NT}} \to 0, \qquad \frac{\sqrt{N_g}(M+T)}{N\sqrt{MT}} \xrightarrow{p} 0. \tag{42}$$

   Let

$$C_{u,g} = \sqrt{\pi_g}\,\overline{Q}_{u,g}.$$

   Then the aligned factor errors satisfy

$$\begin{pmatrix} \sqrt{N_g MT}(\hat{u}_g - H_u u_g) \\ \sqrt{NT}(\hat{v}_j - H_v v_j) \\ \sqrt{NM}(\hat{w}_t - H_w w_t) \\ \sqrt{n}(\hat{\delta} - \delta) \end{pmatrix} \xrightarrow{d} \mathcal{N}\big(\mu_{gjt},\ \mathcal{A}_{gjt}\Sigma_{gjt}\mathcal{A}_{gjt}'\big)\,, \tag{43}$$

   where

$$\mathcal{A}_{gjt} = \begin{pmatrix} \overline{\Sigma}_{VW}^{-1} & 0 & 0 & -\overline{\Sigma}_{VW}^{-1} C_{u,g}\Gamma_\delta^{-1} \\ 0 & \overline{\Sigma}_{UW}^{-1} & 0 & 0 \\ 0 & 0 & \overline{\Sigma}_{UV}^{-1} & 0 \\ 0 & 0 & 0 & \Gamma_\delta^{-1} \end{pmatrix}$$

   and

$$\mu_{gjt} = \begin{pmatrix} \overline{\Sigma}_{VW}^{-1} C_{u,g}\Gamma_\delta^{-1} b_\delta \\ 0_\ell \\ 0_\ell \\ -\Gamma_\delta^{-1} b_\delta \end{pmatrix}.$$

   For completeness, define

$$V_{u,g} = \overline{\Sigma}_{VW}^{-1}\Big[\Gamma_g^{vw} + C_{u,g}\Gamma_\delta^{-1}\Omega_Z\Gamma_\delta^{-1}C_{u,g}' - \Xi_{gZ}\Gamma_\delta^{-1}C_{u,g}' - C_{u,g}\Gamma_\delta^{-1}\Xi_{gZ}'\Big]\overline{\Sigma}_{VW}^{-1},$$
$$V_{v,j} = \overline{\Sigma}_{UW}^{-1}\Gamma_j^{uw}\overline{\Sigma}_{UW}^{-1},$$
$$V_{w,t} = \overline{\Sigma}_{UV}^{-1}\Gamma_t^{uv}\overline{\Sigma}_{UV}^{-1}.$$

The marginal fixed-index limits are

$$\begin{aligned}\sqrt{N_g MT}(\hat{u}_g - H_u u_g) &\xrightarrow{d} \mathcal{N}\Big(\overline{\Sigma}_{VW}^{-1} C_{u,g} \Gamma_\delta^{-1} b_\delta,\ V_{u,g}\Big),\\ \sqrt{NT}(\hat{v}_j - H_v v_j) &\xrightarrow{d} \mathcal{N}(0, V_{v,j}),\\ \sqrt{NM}(\hat{w}_t - H_w w_t) &\xrightarrow{d} \mathcal{N}(0, V_{w,t}).\end{aligned} \tag{44}$$

The joint covariance matrix in (43) *may be singular, as may* $V_{u,g}$*. All fixed-index conclusions are pointwise in* (g, j, t)*, and no additional condition on* $N_g G/N$ *is required. The slope conclusion in part (ii) does not require* (42)*. The factor conclusions retain the finite-sample alignment centers* $H_u u_g, H_v v_j, H_w w_t$*; they do not assert the same limits after replacing those centers by* u_g, v_j, w_t*.*

Proof. Write

$$n = NMT, \qquad s = G + M + T, \qquad \rho = \sqrt{\frac{s}{n}}, \qquad \gamma = \hat{\delta} - \delta, \qquad \theta = \theta_{NMT} = \sqrt{n}\,\gamma.$$

Assumption 1 implies $\rho \to 0$. Throughout, the estimated factors are represented using the column matching and compensating signs specified in the theorem. All inverse expressions are evaluated on events whose probabilities tend to one and may be extended arbitrarily elsewhere.

1. Identity limits of the alignment matrices. Introduce the normalized grouped factors

$$Q_D = D(D'D)^{-1/2}, \qquad A_1 = \frac{(D'D)^{1/2}U}{\sqrt{N}}, \qquad A_2 = \frac{V}{\sqrt{M}}, \qquad A_3 = \frac{W}{\sqrt{T}},$$

with estimated counterparts $\hat{A}_m$.

Lemma 1 supplies an equivalent signed-permuted representative whose sum of squared normalized V- and W-errors is $O_p(\rho^2)$. The representative specified in the theorem minimizes this sum over the same finite collection. Therefore,

$$\|\hat{A}_2 - A_2\|_F^2 + \|\hat{A}_3 - A_3\|_F^2 = O_p(\rho^2),$$

and hence

$$\|\hat{A}_2 - A_2\|_F + \|\hat{A}_3 - A_3\|_F = O_p(\rho).$$

The minimum is well defined because the collection is finite, and the stated tie-breaking rule defines a particular representative.

Let Π and $\hat{\Pi}$ denote the true and estimated common-component tensors. Define their normalized grouped vectors by

$$p_0 = \frac{1}{\sqrt{n}} \operatorname{vec}(\Pi \times_1 Q_D'), \qquad \hat{p} = \frac{1}{\sqrt{n}} \operatorname{vec}(\hat{\Pi} \times_1 Q_D'), \qquad h = \hat{p} - p_0.$$

Also, put

$$C_1 = A_2 * A_3, \qquad C_2 = A_1 * A_3, \qquad C_3 = A_1 * A_2,$$

and define $\hat{C}_m$ analogously. The factor-alignment rate in Lemma 1, the bounded normalized factor norms, and trilinearity imply the common-component rate

$$\|h\| = \frac{\|\hat{\Pi} - \Pi\|_F}{\sqrt{n}} = O_p(\rho).$$

The equality follows because $Q_D' Q_D = I_G$ and the first-mode fibers of both common components belong to $\mathrm{col}(D)$. The common-component difference is unchanged by the choice of signed-permuted representative.

The true and estimated columns of A_2 and A_3 have unit norm. Expanding the Khatri–Rao difference gives

$$\hat{C}_1 - C_1 = (\hat{A}_2 - A_2) * \hat{A}_3 + A_2 * (\hat{A}_3 - A_3),$$

so

$$\|\hat{C}_1 - C_1\|_F \leqslant \|\hat{A}_2 - A_2\|_F + \|\hat{A}_3 - A_3\|_F = O_p(\rho).$$

On the strong-factor event,

$$\sigma_{\min}(C_1) \geqslant c_0.$$

Consequently, $\sigma_{\min}(\hat{C}_1)$ is bounded away from zero with probability approaching one, and

$$\left\| \hat{C}_1 (\hat{C}_1' \hat{C}_1)^{-1} \right\| = \frac{1}{\sigma_{\min}(\hat{C}_1)} = O_p(1).$$

As in the preceding proofs, $h_{(1)}$ denotes the mode-1 matricization of the tensor reshape of h. Since

$$h_{(1)} = \hat{A}_1 \hat{C}_1' - A_1 C_1',$$

we have the exact identity

$$\hat{A}_1 - A_1 = \left[h_{(1)} + A_1 (C_1 - \hat{C}_1)' \right] \hat{C}_1 (\hat{C}_1' \hat{C}_1)^{-1}.$$

The preceding bounds and $\|A_1\| = O_p(1)$ imply

$$\|\hat{A}_1 - A_1\|_F = O_p(\rho).$$

Expanding the other two Khatri–Rao differences in the same way therefore gives

$$\sum_{m=1}^{3} \|\hat{A}_m - A_m\|_F = O_p(\rho), \qquad \sum_{m=1}^{3} \|\hat{C}_m - C_m\|_F = O_p(\rho). \tag{177}$$

The strong-factor bounds imply that the singular values of each C_m are bounded above and away from zero. Hence

$$\max_{m=1,2,3} \left\{ \|\hat{C}_m\| + \|(\hat{C}_m' \hat{C}_m)^{-1}\| \right\} = O_p(1).$$

Writing $(H_1, H_2, H_3) = (H_u, H_v, H_w)$, the definitions in Proposition 2 become

$$H_m = (\hat{C}_m' \hat{C}_m)^{-1} \hat{C}_m' C_m, \qquad m = 1, 2, 3.$$

The scalar normalizations and isometric group embeddings cancel from these projection coefficients. Thus

$$H_m - I_\ell = (\hat{C}'_m \hat{C}_m)^{-1} \hat{C}'_m (C_m - \hat{C}_m).$$

Equation (177) implies

$$\sum_{m=1}^{3} \|H_m - I_\ell\| = O_p(\rho).$$

Because $\rho \to 0$, the event

$$\max_{m=1,2,3} \|H_m - I_\ell\| < \frac{1}{2}$$

has probability tending to one. On this event, every H_m is nonsingular, $\|H_m^{-1}\| \leqslant 2$, and

$$H_m^{-1} - I_\ell = -H_m^{-1}(H_m - I_\ell).$$

Therefore,

$$\sum_{m=1}^{3} \left\{ \|H_m - I_\ell\| + \|H_m^{-1} - I_\ell\| \right\} = O_p(\rho). \tag{178}$$

This proves part (i) without imposing any alignment-limit condition in Assumption 6.

2. Slope linearization and marginal limit. Proposition 1 gives

$$\Gamma_{NMT}\theta = \mathcal{Z}_{NMT} - \mathcal{B}_1 - \mathcal{B}_2 - \mathcal{B}_3 + O_p\left(\rho + \frac{s^{3/2}}{n}\right).$$

The remainder is $o_p(1)$ by (40) and $\rho \to 0$. Assumption 6 implies

$$\mathcal{Z}_{NMT} = O_p(1), \qquad \sum_{m=1}^{3} \mathcal{B}_m \xrightarrow{p} b_\delta, \qquad \Gamma_{NMT}^{-1} \xrightarrow{p} \Gamma_\delta^{-1}.$$

Consequently,

$$\theta = \Gamma_\delta^{-1}(\mathcal{Z}_{NMT} - b_\delta) + o_p(1) = O_p(1). \tag{179}$$

The marginal score CLT in Assumption 6 gives

$$\mathcal{Z}_{NMT} \xrightarrow{d} \mathcal{N}(0, \Omega_Z).$$

The continuous mapping theorem and Slutsky's theorem therefore imply

$$\theta \xrightarrow{d} \mathcal{N}\left(-\Gamma_\delta^{-1} b_\delta,\ \Gamma_\delta^{-1} \Omega_Z \Gamma_\delta^{-1}\right).$$

This proves part (ii), without using (42).

3. Fixed-index linearizations and removal of the inverse alignments. Fix an admissible triple (g, j, t) and impose (42). Define

$$a_{u,g,n} = \frac{\sqrt{N_g}(M+T)}{N\sqrt{MT}} = \frac{\sqrt{N_g M}}{N\sqrt{T}} + \frac{\sqrt{N_g T}}{N\sqrt{M}},$$
$$a_{v,n} = \frac{N+T}{M\sqrt{NT}} = \frac{\sqrt{N}}{M\sqrt{T}} + \frac{\sqrt{T}}{M\sqrt{N}},$$
$$a_{w,n} = \frac{N+M}{T\sqrt{NM}} = \frac{\sqrt{N}}{T\sqrt{M}} + \frac{\sqrt{M}}{T\sqrt{N}}.$$

Proposition 2 gives

$$\|R_{u,g,NMT}\| = O_p\left(\rho + a_{u,g,n} + \sqrt{N_g MT}\,\rho^4\right),$$
$$\|R_{v,j,NMT}\| = O_p(\rho + a_{v,n}),$$
$$\|R_{w,t,NMT}\| = O_p(\rho + a_{w,n}).$$

The three quantities $a_{u,g,n}, a_{v,n}, a_{w,n}$ tend to zero under (42). Moreover, $N_g \leqslant N$ implies

$$\sqrt{N_g MT}\,\rho^4 \leqslant \sqrt{n}\,\rho^4 = \rho\frac{s^{3/2}}{n} = o(\rho).$$

Hence

$$\|R_{u,g,NMT}\| + \|R_{v,j,NMT}\| + \|R_{w,t,NMT}\| = o_p(1). \tag{180}$$

In particular, no additional condition on $N_g G/N$ is needed.

The first two conditions in (42) imply $T \to \infty$ and $M \to \infty$, respectively, because

$$\frac{N+M}{T\sqrt{NM}} \geqslant \frac{2}{T}, \qquad \frac{N+T}{M\sqrt{NT}} \geqslant \frac{2}{M}.$$

Define the effective slope coefficients

$$C_{u,g,n} = \sqrt{\frac{N_g}{N}}Q_{u,g}, \qquad C_{v,j,n} = \frac{1}{\sqrt{M}}Q_{v,j}, \qquad C_{w,t,n} = \frac{1}{\sqrt{T}}Q_{w,t}.$$

The group-share and group cross-moment limits in Assumption 6, together with continuity of the square-root function on $[0,1]$, give

$$C_{u,g,n} \xrightarrow{p} \sqrt{\pi_g}\,\overline{Q}_{u,g} = C_{u,g}.$$

For the other two modes, the boundedness assumptions $Q_{v,j} = O_p(1)$ and $Q_{w,t} = O_p(1)$, together with $M, T \to \infty$, imply

$$C_{u,g,n} \xrightarrow{p} C_{u,g}, \qquad C_{v,j,n} = o_p(1), \qquad C_{w,t,n} = o_p(1). \tag{181}$$

Since $\theta = O_p(1)$ by (179), the teacher and period slope-estimation contributions are $o_p(1)$, while the group contribution need not vanish.

Put

$$D_u = \overline{\Sigma}_{VW}^{-1}, \qquad D_v = \overline{\Sigma}_{UW}^{-1}, \qquad D_w = \overline{\Sigma}_{UV}^{-1}, \qquad D_\delta = \Gamma_\delta^{-1},$$

and define the inverse-alignment-adjusted errors

$$\begin{aligned}\zeta_{u,g,n} &= \sqrt{N_g MT}\, H_u^{-1}(\hat{u}_g - H_u u_g),\\ \zeta_{v,j,n} &= \sqrt{NT}\, H_v^{-1}(\hat{v}_j - H_v v_j),\\ \zeta_{w,t,n} &= \sqrt{NM}\, H_w^{-1}(\hat{w}_t - H_w w_t).\end{aligned}$$

Proposition 2 gives

$$\begin{aligned}\zeta_{u,g,n} &= \Sigma_{VW}^{-1}\left[S_{u,g} - C_{u,g,n}\theta\right] + R_{u,g,NMT},\\ \zeta_{v,j,n} &= \Sigma_{UW}^{-1}\left[S_{v,j} - C_{v,j,n}\theta\right] + R_{v,j,NMT},\\ \zeta_{w,t,n} &= \Sigma_{UV}^{-1}\left[S_{w,t} - C_{w,t,n}\theta\right] + R_{w,t,NMT}.\end{aligned}$$

The joint score CLT implies that the three loading scores and $\mathcal{Z}_{NMT}$ are jointly bounded in probability. The loading Gram limits, (180), and (181) therefore imply

$$\begin{aligned}\zeta_{u,g,n} &= D_u\left[S_{u,g} - C_{u,g}\theta\right] + o_p(1),\\ \zeta_{v,j,n} &= D_v S_{v,j} + o_p(1),\\ \zeta_{w,t,n} &= D_w S_{w,t} + o_p(1).\end{aligned}$$

For the group block, replacing $C_{u,g,n}$ by $C_{u,g}$ is valid because

$$(C_{u,g,n} - C_{u,g})\theta = o_p(1).$$

Substituting (179) into the group expansion gives

$$\begin{aligned}\zeta_{u,g,n} &= D_u\left[S_{u,g} - C_{u,g}D_\delta(\mathcal{Z}_{NMT} - b_\delta)\right] + o_p(1),\\ \zeta_{v,j,n} &= D_v S_{v,j} + o_p(1),\\ \zeta_{w,t,n} &= D_w S_{w,t} + o_p(1).\end{aligned} \tag{182}$$

These three vectors are jointly $O_p(1)$. Thus (178) gives

$$\begin{aligned}\sqrt{N_g MT}(\hat{u}_g - H_u u_g) - \zeta_{u,g,n} &= (H_u - I_\ell)\zeta_{u,g,n} = o_p(1),\\ \sqrt{NT}(\hat{v}_j - H_v v_j) - \zeta_{v,j,n} &= (H_v - I_\ell)\zeta_{v,j,n} = o_p(1),\\ \sqrt{NM}(\hat{w}_t - H_w w_t) - \zeta_{w,t,n} &= (H_w - I_\ell)\zeta_{w,t,n} = o_p(1).\end{aligned}$$

Consequently, (182) remains valid without the premultiplying inverse alignment matrices. This proves

$$\begin{aligned}\sqrt{N_g MT}(\hat{u}_g - H_u u_g) &= \overline{\Sigma}_{VW}^{-1}\left[S_{u,g} - C_{u,g}\Gamma_\delta^{-1}(\mathcal{Z}_{NMT} - b_\delta)\right] + o_p(1),\\ \sqrt{NT}(\hat{v}_j - H_v v_j) &= \overline{\Sigma}_{UW}^{-1} S_{v,j} + o_p(1),\\ \sqrt{NM}(\hat{w}_t - H_w w_t) &= \overline{\Sigma}_{UV}^{-1} S_{w,t} + o_p(1).\end{aligned} \tag{183}$$

The finite-sample alignment matrices remain inside the centers throughout this argument. For example, replacing $H_u u_g$ by u_g would additionally require

$$\sqrt{N_g MT}(H_u - I_\ell)u_g = o_p(1),$$

which does not follow from (178) alone. No such replacement is made here.

4. Joint Gaussian limit and marginal covariance matrices. Define the score and estimator stacks

$$\mathsf{S}_n = \begin{pmatrix} S_{u,g} \\ S_{v,j} \\ S_{w,t} \\ \mathcal{Z}_{NMT} \end{pmatrix}, \qquad \mathsf{L}_n = \begin{pmatrix} \sqrt{N_g MT}(\hat{u}_g - H_u u_g) \\ \sqrt{NT}(\hat{v}_j - H_v v_j) \\ \sqrt{NM}(\hat{w}_t - H_w w_t) \\ \theta \end{pmatrix}.$$

Equations (179) and (182), together with the preceding removal of the inverse alignment matrices, give

$$\mathsf{L}_n = \mathcal{A}_{gjt}\mathsf{S}_n + \mu_{gjt} + o_p(1),$$

where $\mathcal{A}_{gjt}$ and μ_{gjt} are the deterministic matrix and vector specified in the theorem. In particular,

$$D_u\left[S_{u,g} - C_{u,g}D_\delta(\mathcal{Z}_{NMT} - b_\delta)\right] = D_u S_{u,g} - D_u C_{u,g} D_\delta \mathcal{Z}_{NMT} + D_u C_{u,g} D_\delta b_\delta.$$

Thus the group-loading mean has the displayed plus sign, while the slope mean is $-D_\delta b_\delta$. Neither mean vector is asserted to have componentwise positive or negative entries.

Assumption 6 gives

$$\mathsf{S}_n \xrightarrow{d} \mathcal{N}(0, \Sigma_{gjt}).$$

The continuous mapping theorem and Slutsky's theorem therefore imply

$$\mathsf{L}_n \xrightarrow{d} \mathcal{N}\big(\mu_{gjt},\ \mathcal{A}_{gjt}\Sigma_{gjt}\mathcal{A}_{gjt}'\big)\,.$$

This proves (43). The stack with the first three blocks premultiplied by $H_u^{-1}, H_v^{-1}, H_w^{-1}$ differs from L_n by $o_p(1)$ and therefore has the same joint limit.

To obtain the marginal covariance formulas, let

$$\begin{pmatrix} S_u^\infty \\ S_v^\infty \\ S_w^\infty \\ Z^\infty \end{pmatrix} \sim \mathcal{N}(0, \Sigma_{gjt}).$$

The random part of the group-loading limit is

$$D_u(S_u^\infty - C_{u,g}D_\delta Z^\infty).$$

Since

$$\mathrm{Var}(S_u^\infty) = \Gamma_g^{vw}, \qquad \mathrm{Var}(Z^\infty) = \Omega_Z, \qquad \mathrm{Cov}(S_u^\infty, Z^\infty) = \Xi_{gZ},$$

and D_u, D_δ are symmetric, its covariance is

$$\begin{aligned}\mathrm{Var}[D_u(S_u^\infty - C_{u,g}D_\delta Z^\infty)] &= D_u\Big[\Gamma_g^{vw} + C_{u,g}D_\delta\Omega_Z D_\delta C_{u,g}' - \Xi_{gZ}D_\delta C_{u,g}' - C_{u,g}D_\delta\Xi_{gZ}'\Big]D_u \\ &= V_{u,g}.\end{aligned}$$

Its mean is $D_u C_{u,g} D_\delta b_\delta$. The teacher and period limits are $D_v S_v^\infty$ and $D_w S_w^\infty$, respectively, so their means are zero and their covariance matrices are

$$D_v\Gamma_j^{uw}D_v = V_{v,j}, \qquad D_w\Gamma_t^{uv}D_w = V_{w,t}.$$

These calculations establish (44). All cross-covariance blocks in Assumption 6 remain represented in the full covariance matrix $\mathcal{A}_{gjt}\Sigma_{gjt}\mathcal{A}'_{gjt}$; none is assumed to vanish.

Finally, $\mathcal{A}_{gjt}$ is invertible because it is block upper triangular with invertible diagonal blocks. Therefore,

$$\operatorname{rank}(\mathcal{A}_{gjt}\Sigma_{gjt}\mathcal{A}'_{gjt}) = \operatorname{rank}(\Sigma_{gjt}).$$

The joint limiting distribution may thus be degenerate when Σ_{gjt} is singular. This does not affect the asserted convergence in distribution. All applications of the score CLT and the fixed-index expansions concern the selected triple (g, j, t), so the conclusions are pointwise, as stated. □

Proposition 3. *Suppose that Assumptions 1, 2, 3, and 4 hold. Suppose also that* $\mathbb{E}[\varepsilon_a\varepsilon_b \mid \mathcal{F}] = 0$ *for all distinct* $a, b \in \mathcal{I}$. *The true factor rows satisfy*

$$\frac{1}{N}\sum_{g=1}^{G} N_g\|u_g\|^4 + \frac{1}{M}\sum_{j=1}^{M}\|v_j\|^4 + \frac{1}{T}\sum_{t=1}^{T}\|w_t\|^4 = O_p(1)$$

and the group sizes satisfies

$$\sum_{g=1}^{G}\frac{1}{N_g} = O_p\left(\frac{G^2}{N}\right). \tag{46}$$

Let $\mathcal{T}$ *be the tangent space defined in* (11) *evaluated at the true parameters* (U, V, W). *Let*

$$\Gamma_{NMT} = \frac{1}{NMT}X_{(4)}M_{\mathcal{T}}X'_{(4)}, \qquad \mathcal{Z}_{NMT} = \frac{1}{\sqrt{NMT}}X_{(4)}M_{\mathcal{T}}\operatorname{vec}(E).$$

Set $\widehat{\widetilde{U}} = (D'D)^{1/2}\hat{U}$ *and*

$$(\widehat{F}_1, \widehat{F}_2, \widehat{F}_3) = (\widehat{\widetilde{U}}, \hat{V}, \hat{W}), \qquad (\widehat{C}_1, \widehat{C}_2, \widehat{C}_3) = (\hat{V} * \hat{W}, \widehat{\widetilde{U}} * \hat{W}, \widehat{\widetilde{U}} * \hat{V}).$$

Let $\widehat{\Omega}^G_m$, $m = 1, 2, 3$, *be data-based estimators of the conditional covariance defined in* (45), *of dimensions* $MT \times MT$, $GT \times GT$, *and* $GM \times GM$, *respectively. Put* $K^G_{1,1} = I_{GMT}$ *and define*

$$\widehat{\mathcal{B}}_m = \frac{1}{\sqrt{n}}X_{(4)}P_{\widehat{\mathcal{T}}}QK^G_{1,m}\operatorname{vec}\left(\widehat{F}_m(\widehat{F}'_m\widehat{F}_m)^{-1}(\widehat{C}'_m\widehat{C}_m)^{-1}\widehat{C}'_m\widehat{\Omega}^G_m M_{\widehat{C}_m}\right), \qquad m = 1, 2, 3. \tag{47}$$

The bias-corrected estimator $\hat{\delta}_{bc}$ *is defined as*

$$\hat{\delta}_{\mathrm{bc}} = \hat{\delta} + \frac{1}{\sqrt{NMT}}\widehat{\Gamma}^{-1}_{NMT}\left(\widehat{\mathcal{B}}_1 + \widehat{\mathcal{B}}_2 + \widehat{\mathcal{B}}_3\right). \tag{48}$$

Then, we have

$$\Gamma_{NMT}\sqrt{NMT}(\hat{\delta}_{bc} - \delta) = \mathcal{Z}_{NMT} + O_p\left(\sqrt{\frac{G+M+T}{NMT}} + \frac{(G+M+T)^{3/2}}{NMT}\right). \tag{49}$$

Proof. Write $n = NMT$, $s = G + M + T$, $\rho = \sqrt{s/n}$, and $\gamma = \hat{\delta} - \delta$. Assumption 1 implies $\rho \to 0$. Let $e = \operatorname{vec}(E)$ and $\hat{e} = \operatorname{vec}(\hat{E})$, and put

$$B_{NMT} = \sum_{m=1}^{3}\mathcal{B}_m, \qquad \widehat{B}_{NMT} = \sum_{m=1}^{3}\widehat{\mathcal{B}}_m.$$

We use the signed-permutation-aligned estimates supplied by Lemma 1. These relabelings leave the fitted common component, the slope estimate, and the displayed bias formulas unchanged. Unless a weighted norm or operator norm is specified, matrix norms below are Frobenius norms. The conditional inequalities are applied on the covariance-bound events from Assumption 3, whose probabilities approach one.

Step 1: Normalized coordinates and preliminary bounds. Put $(d_1, d_2, d_3) = (G, M, T)$ and define

$$\mu_1(g) = \frac{N_g}{N}, \qquad \mu_2(j) = \frac{1}{M}, \qquad \mu_3(t) = \frac{1}{T}, \qquad \mu = \mu_1 \otimes \mu_2 \otimes \mu_3.$$

For a probability-weight vector ν, write $D_\nu = \mathrm{diag}(\nu)$. For a matrix B with rows b_q', define

$$\|B\|_{\alpha,\nu} = \left(\sum_q \nu(q) \|b_q\|_2^\alpha \right)^{1/\alpha}, \qquad \alpha \geqslant 1.$$

For a square matrix P, write

$$\|P\|_{4,\nu \to 4,\nu} = \sup_{z \neq 0} \frac{\|Pz\|_{4,\nu}}{\|z\|_{4,\nu}}.$$

All weights are positive because D has full column rank.

Let

$$(B_1, B_2, B_3) = (U, V, W), \qquad A_m = D_{\mu_m}^{1/2} B_m.$$

Thus

$$A_1 = \frac{(D'D)^{1/2} U}{\sqrt{N}}, \qquad A_2 = \frac{V}{\sqrt{M}}, \qquad A_3 = \frac{W}{\sqrt{T}}.$$

For complementary modes $a < b$, put

$$B_{-m} = B_a * B_b, \qquad \mu_{-m} = \mu_a \otimes \mu_b, \qquad L_m = A_a * A_b = D_{\mu_{-m}}^{1/2} B_{-m}.$$

Here L_m denotes a normalized pair matrix, whereas the matrices C_m in the proposition are unnormalized. Define

$$G_m = A_m' A_m = B_m' D_{\mu_m} B_m, \qquad J_m = L_m' L_m = G_a \odot G_b,$$

and use hats for the estimated counterparts.

Lemma 1 gives

$$\sum_{m=1}^{3} \|\hat{A}_m - A_m\|_F = O_p(\rho).$$

Since ℓ is fixed and the normalized factor norms are bounded, expanding the Khatri–Rao products gives

$$\sum_{m=1}^{3} \|\hat{L}_m - L_m\|_F = O_p(\rho).$$

Consequently,

$$\|\hat{G}_m - G_m\|_{\mathrm{op}} + \|\hat{J}_m - J_m\|_{\mathrm{op}} = O_p(\rho).$$

Assumption 2 gives $c_0 I_\ell \preceq G_m \preceq c_1 I_\ell$. Moreover, J_m is a principal submatrix of $G_a \otimes G_b$, so

$$c_0^2 I_\ell \preceq J_m \preceq c_1^2 I_\ell.$$

The true and estimated Gram inverses are therefore $O_p(1)$ in operator norm, and the inverse identity gives

$$\|\hat{G}_m^{-1} - G_m^{-1}\|_{\mathrm{op}} + \|\hat{J}_m^{-1} - J_m^{-1}\|_{\mathrm{op}} = O_p(\rho). \tag{184}$$

Write $(H_1, H_2, H_3) = (H_u, H_v, H_w)$ and define

$$H'_m = L'_m \hat{L}_m \hat{J}_m^{-1}, \qquad \Delta_m = \hat{A}_m - A_m H'_m.$$

Since

$$H'_m - I_\ell = (L_m - \hat{L}_m)' \hat{L}_m \hat{J}_m^{-1},$$

we obtain

$$\sum_m \|H_m - I_\ell\|_{\mathrm{op}} + \sum_m \|\Delta_m\|_F = O_p(\rho).$$

The conditional regressor fourth-moment bound implies

$$\frac{1}{n} \sum_{i,j,t} \|x_{ijt}\|_2^4 = O_p(1), \qquad \frac{\|X_{(4)}\|_F}{\sqrt{n}} = O_p(1).$$

Assumption 3 implies $\|\mathbb{E}[ee' \mid \mathcal{F}]\|_{\mathrm{op}} \leqslant c_\varepsilon$ on its covariance-bound event. Hence

$$\mathbb{E}[\|\mathcal{Z}_{NMT}\|_2^2 \mid \mathcal{F}] \leqslant \frac{c_\varepsilon}{n} \|X_{(4)} M_{\mathcal{T}}\|_F^2 \leqslant \frac{c_\varepsilon}{n} \|X_{(4)}\|_F^2,$$

and conditional Markov's inequality gives $\mathcal{Z}_{NMT} = O_p(1)$.

Every CP tangent variation is annihilated by $M_{DU*V} \otimes M_W$: variations in U or V are annihilated by M_W, and variations in W are annihilated by M_{DU*V}. Thus

$$M_{\mathcal{T}} \succeq M_{DU*V} \otimes M_W.$$

Assumption 4, evaluated at the true factors, consequently gives $\lambda_{\min}(\Gamma_{NMT}) \geqslant c_\delta$ with probability approaching one. Proposition 1 and $B_{NMT} = O_p(s/\sqrt{n})$ now imply

$$\begin{aligned} \|\gamma\|_2 &= O_p\left(n^{-1/2} + \frac{s}{n} + \frac{\rho}{\sqrt{n}} + \frac{s^{3/2}}{n^{3/2}}\right) \\ &= O_p(n^{-1/2} + \rho^2) = O_p(\rho). \end{aligned} \tag{185}$$

Here $\rho/\sqrt{n} = \sqrt{s}/n \leqslant s/n$ and $s^{3/2}/n^{3/2} = \rho^3 \leqslant \rho^2$ eventually. No additional condition on $s^{3/2}/n$ is needed for this preliminary bound.

Step 2: Fourth moments of factor errors and residual squares. Define the source slices

$$\mathcal{I}_g^{(1)} = \{(i,j,t) \in \mathcal{I} : g(i) = g\}, \qquad \mathcal{I}_j^{(2)} = \{(i,j',t) \in \mathcal{I} : j' = j\}, \qquad \mathcal{I}_t^{(3)} = \{(i,j,t') \in \mathcal{I} : t' = t\}.$$

Their cardinalities are $N_g MT$, NT, and NM, respectively. Let S be any member of $\{\mathcal{I}_r^{(m)} : m = 1, 2, 3,\ 1 \leqslant r \leqslant d_m\}$, and let $L = \sum_{a \in S} b_a \varepsilon_a$ for $\mathcal{F}$-measurable scalar coefficients. Since $S^2 \subseteq \mathcal{R}^D$, Assumption 3 yields

$$\operatorname{Var}(L^2 \mid \mathcal{F}) \leqslant c_4 \sum_{a,b \in S} b_a^2 b_b^2 = c_4 \left(\sum_{a \in S} b_a^2 \right)^2,$$

while its second-order bound gives $\mathbb{E}[L^2 \mid \mathcal{F}] \leqslant c_\varepsilon \sum_{a \in S} b_a^2$. Therefore,

$$\mathbb{E}[L^4 \mid \mathcal{F}] \leqslant (c_4 + c_\varepsilon^2) \left(\sum_{a \in S} b_a^2 \right)^2. \tag{186}$$

In particular, taking a single nonzero coefficient gives a uniform conditional bound on $\mathbb{E}[\varepsilon_a^4 \mid \mathcal{F}]$. Conditional Markov's inequality implies

$$\frac{1}{n} \sum_a \varepsilon_a^4 = O_p(1), \qquad \|e\|_4 = O_p(n^{1/4}), \qquad \|e \odot e\|_2 = O_p(\sqrt{n}). \tag{187}$$

Let $Q_D = D(D'D)^{-1/2}$, $E^G = E \times_1 Q_D'$, and $X_\gamma^G = (X \times_4 \gamma') \times_1 Q_D'$. Constraint inactivity gives the unrestricted block equations

$$\Delta_m = \frac{1}{\sqrt{n}} (E^G - X_\gamma^G)_{(m)} \hat{L}_m \hat{J}_m^{-1}.$$

For complementary modes $a < b$, expand

$$\hat{L}_m = (A_a \otimes A_b)(H_a' * H_b') + R_m,$$

where

$$R_m = \Delta_a * (A_b H_b') + (A_a H_a') * \Delta_b + \Delta_a * \Delta_b, \qquad \|R_m\|_F = O_p(\rho).$$

Define the true-factor score vectors

$$\begin{aligned} z_{1,g}^\varepsilon &= \frac{1}{N_g MT} \sum_{i : g(i) = g} \sum_{j', t'} \varepsilon_{ij't'} (v_{j'} \otimes w_{t'}), \\ z_{2,j}^\varepsilon &= \frac{1}{NT} \sum_{i, t'} \varepsilon_{ijt'} (u_{g(i)} \otimes w_{t'}), \\ z_{3,t}^\varepsilon &= \frac{1}{NM} \sum_{i, j'} \varepsilon_{ij't} (u_{g(i)} \otimes v_{j'}). \end{aligned}$$

Let Z_m^ε have rows $(z_{m,r}^\varepsilon)'$. The normalized Gram bounds imply that the squared coefficient norms of each scalar score coordinate are bounded by constants times $(N_g MT)^{-1}$, $(NT)^{-1}$, and $(NM)^{-1}$, respectively. Equation (186) therefore gives

$$\mathbb{E}[\|z_{1,g}^\varepsilon\|_2^4 \mid \mathcal{F}] \leqslant \frac{C}{(N_g MT)^2}, \qquad \mathbb{E}[\|z_{2,j}^\varepsilon\|_2^4 \mid \mathcal{F}] \leqslant \frac{C}{(NT)^2}, \qquad \mathbb{E}[\|z_{3,t}^\varepsilon\|_2^4 \mid \mathcal{F}] \leqslant \frac{C}{(NM)^2}.$$

Writing $H_N = \sum_g N_g^{-1}$, the corresponding weighted averages satisfy

$$\mathbb{E}[\|Z_1^\varepsilon\|_{4,\mu_1}^4 \mid \mathcal{F}] \leqslant \frac{CNH_N}{n^2},$$
$$\mathbb{E}[\|Z_2^\varepsilon\|_{4,\mu_2}^4 \mid \mathcal{F}] \leqslant \frac{CM^2}{n^2},$$
$$\mathbb{E}[\|Z_3^\varepsilon\|_{4,\mu_3}^4 \mid \mathcal{F}] \leqslant \frac{CT^2}{n^2}.$$

By (46) and conditional Markov's inequality,

$$\|Z_m^\varepsilon\|_{4,\mu_m} = O_p\left(\sqrt{\frac{d_m}{n}}\right).$$

Undoing the normalizations in the block equations gives the exact decomposition

$$D_{\mu_m}^{-1/2}\Delta_m = Z_m^\varepsilon(H_a' * H_b')\hat{J}_m^{-1} + E_m^R - X_m^\gamma,$$

where

$$E_m^R = \frac{1}{\sqrt{n}} D_{\mu_m}^{-1/2} E_{(m)}^G R_m \hat{J}_m^{-1}, \qquad X_m^\gamma = \frac{1}{\sqrt{n}} D_{\mu_m}^{-1/2} (X_\gamma^G)_{(m)} \hat{L}_m \hat{J}_m^{-1}.$$

The first contribution has weighted fourth norm $O_p(\sqrt{d_m/n})$, since its right multiplier is bounded in operator norm.

For the adaptive error contribution, define

$$\eta_{1,g} = \frac{\|E_{g,:,:}^G\|_F}{\sqrt{N_g MT}}, \qquad \eta_{2,j} = \frac{\|E_{:,j,:}^G\|_F}{\sqrt{NT}}, \qquad \eta_{3,t} = \frac{\|E_{:,:,t}^G\|_F}{\sqrt{NM}}.$$

Rowwise Cauchy–Schwarz gives

$$\|(E_m^R)_{r,:}\|_2 \leqslant \|\hat{J}_m^{-1}\|_{\mathrm{op}} \|R_m\|_F \eta_{m,r}.$$

Equation (186), applied to $E_{gjt}^G = N_g^{-1/2} \sum_{i:g(i)=g} \varepsilon_{ijt}$, gives $\mathbb{E}[|E_{gjt}^G|^4 \mid \mathcal{F}] \leqslant C$. Using $(\sum_{k=1}^K e_k^2)^2 \leqslant K \sum_{k=1}^K e_k^4$ within each grouped-error slice therefore yields

$$\mathbb{E}\left[\sum_g \mu_1(g)\eta_{1,g}^4 \mid \mathcal{F}\right] \leqslant \frac{CH_N}{N} \leqslant C,$$
$$\mathbb{E}\left[\frac{1}{M}\sum_j \eta_{2,j}^4 \mid \mathcal{F}\right] \leqslant C(G/N)^2 \leqslant C,$$
$$\mathbb{E}\left[\frac{1}{T}\sum_t \eta_{3,t}^4 \mid \mathcal{F}\right] \leqslant C(G/N)^2 \leqslant C.$$

Here $N_g \geqslant 1$ implies $H_N \leqslant G \leqslant N$. Consequently,

$$\|E_m^R\|_{4,\mu_m} \leqslant \|\hat{J}_m^{-1}\|_{\mathrm{op}} \|R_m\|_F \left(\sum_r \mu_m(r)\eta_{m,r}^4\right)^{1/4} = O_p(\rho).$$

This deterministic bound permits R_m to depend on the errors.

For the regressor contribution, source-slice Cauchy–Schwarz and $\|\hat{L}_m \hat{J}_m^{-1}\|_{\mathrm{op}} = O_p(1)$ give a common $K_{NMT} = O_p(1)$ such that

$$\|(X_m^\gamma)_{r,:}\|_2 \leqslant K_{NMT}\|\gamma\|_2 \left(\frac{1}{|\mathcal{I}_r^{(m)}|} \sum_{a \in \mathcal{I}_r^{(m)}} \|x_a\|_2^2 \right)^{1/2}.$$

For modes 2 and 3, the grouped contraction cannot increase the source-slice Euclidean norm; for mode 1, the same statement follows directly from Cauchy–Schwarz over the N_g observations. Raising the bound to the fourth power and applying Jensen's inequality gives

$$\|X_m^\gamma\|_{4,\mu_m}^4 \leqslant K_{NMT}^4 \|\gamma\|_2^4 \sum_r \frac{\mu_m(r)}{|\mathcal{I}_r^{(m)}|} \sum_{a \in \mathcal{I}_r^{(m)}} \|x_a\|_2^4 = K_{NMT}^4 \|\gamma\|_2^4 \frac{1}{n} \sum_a \|x_a\|_2^4.$$

Thus $\|X_m^\gamma\|_{4,\mu_m} = O_p(\|\gamma\|_2) = O_p(\rho)$.

Combining the three contributions gives

$$\|D_{\mu_m}^{-1/2} \Delta_m\|_{4,\mu_m} = O_p(\rho).$$

Since

$$\hat{B}_m - B_m = D_{\mu_m}^{-1/2} \Delta_m + B_m(H_m' - I_\ell),$$

the assumed true-factor fourth moments and the alignment bounds imply

$$\sum_{m=1}^3 \|\hat{B}_m - B_m\|_{4,\mu_m} = O_p(\rho), \qquad \sum_{m=1}^3 \left(\|B_m\|_{4,\mu_m} + \|\hat{B}_m\|_{4,\mu_m} \right) = O_p(1). \tag{188}$$

Now let $h_{ijt} = x_{ijt}'\gamma + \hat{\pi}_{ijt} - \pi_{ijt}$, so $\hat{\varepsilon}_{ijt} = \varepsilon_{ijt} - h_{ijt}$. For scalar column vectors on separate modes, their weighted tensor fourth norm is the product of their mode-specific fourth norms. For each k, use the exact expansion

$$\begin{aligned} \hat{u}_{gk}\hat{v}_{jk}\hat{w}_{tk} - u_{gk}v_{jk}w_{tk} = {} & (\hat{u}_{gk} - u_{gk})\hat{v}_{jk}\hat{w}_{tk} \\ & + u_{gk}(\hat{v}_{jk} - v_{jk})\hat{w}_{tk} \\ & + u_{gk}v_{jk}(\hat{w}_{tk} - w_{tk}). \end{aligned}$$

Equation (188), fixed ℓ, and the triangle inequality imply

$$\left(\frac{1}{n} \sum_{i,j,t} |\hat{\pi}_{ijt} - \pi_{ijt}|^4 \right)^{1/4} = O_p(\rho).$$

The raw observation average equals the product-weighted grouped average because the common-component error depends on i only through $g(i)$. Also,

$$\left(\frac{1}{n} \sum_{i,j,t} |x_{ijt}'\gamma|^4 \right)^{1/4} \leqslant \|\gamma\|_2 \left(\frac{1}{n} \sum_{i,j,t} \|x_{ijt}\|_2^4 \right)^{1/4} = O_p(\rho).$$

Consequently, identifying h with its vectorization,

$$\|h\|_4 = O_p(n^{1/4}\rho), \qquad \|e \odot h\|_2 \leqslant \|e\|_4 \|h\|_4 = O_p(\sqrt{s}), \qquad \|h \odot h\|_2 = \|h\|_4^2 = O_p(s/\sqrt{n}).$$

Using $\hat{e} \odot \hat{e} - e \odot e = -2e \odot h + h \odot h$ gives

$$\|\hat{e} \odot \hat{e} - e \odot e\|_2 = O_p\left(\sqrt{s} + \frac{s}{\sqrt{n}} \right). \tag{189}$$

Step 3: Tangent projectors and projected regressors. For a full-column-rank matrix B and probability weights ν, put $G_B = B' D_\nu B$ and

$$P_B^\nu = B G_B^{-1} B' D_\nu.$$

This is the column-space projector for the weighted inner product induced by D_ν. Hölder's inequality gives

$$\|P_B^\nu\|_{4,\nu\to 4,\nu} \leqslant \|B\|_{4,\nu} \|G_B^{-1}\|_{\mathrm{op}} \|B\|_{4/3,\nu}. \tag{190}$$

If $\|\hat{B} - B\|_{4,\nu} = O_p(\rho)$, the true and estimated fourth norms are $O_p(1)$, and the Gram inverses are bounded, then $\|\hat{G}_B - G_B\|_{\mathrm{op}} = O_p(\rho)$ and $\|\hat{G}_B^{-1} - G_B^{-1}\|_{\mathrm{op}} = O_p(\rho)$. The exact expansion

$$\begin{aligned} P_{\hat{B}}^\nu - P_B^\nu = {} & (\hat{B} - B)\hat{G}_B^{-1}\hat{B}' D_\nu \\ & + B(\hat{G}_B^{-1} - G_B^{-1})\hat{B}' D_\nu \\ & + B G_B^{-1}(\hat{B} - B)' D_\nu \end{aligned}$$

and $\|\hat{B} - B\|_{4/3,\nu} \leqslant \|\hat{B} - B\|_{4,\nu}$ imply

$$\|P_{\hat{B}}^\nu - P_B^\nu\|_{4,\nu\to 4,\nu} = O_p(\rho). \tag{191}$$

Define

$$P_m = B_m G_m^{-1} B_m' D_{\mu_m}, \qquad P_{-m} = B_{-m} J_m^{-1} B_{-m}' D_{\mu_{-m}},$$

with analogous estimated projectors. The rowwise inequality $\|b_a \odot b_b\|_2 \leqslant \|b_a\|_2 \|b_b\|_2$ implies

$$\|B_{-m}\|_{4,\mu_{-m}} + \|\hat{B}_{-m}\|_{4,\mu_{-m}} = O_p(1).$$

Expanding $\hat{B}_{-m} - B_{-m} = (\hat{B}_a - B_a) * \hat{B}_b + B_a * (\hat{B}_b - B_b)$ also gives

$$\|\hat{B}_{-m} - B_{-m}\|_{4,\mu_{-m}} = O_p(\rho).$$

Thus the mode and pair projectors have bounded weighted fourth-norm operator norms, and their estimated-minus-true differences have order $O_p(\rho)$.

Let R_ℓ select the canonical coordinates (r,k,k), (k,r,k), and (k,k,r) in $\mathbb{R}^{\ell^3}$, with duplicates removed. It has $r_0 = 3\ell^2 - 2\ell$ orthonormal columns. Define

$$\Phi = (B_1 \otimes B_2 \otimes B_3) R_\ell, \qquad G_{\mathrm{core}} = \Phi' D_\mu \Phi = R_\ell'(G_1 \otimes G_2 \otimes G_3) R_\ell.$$

The normalized Gram bounds imply

$$c_0^3 I_{r_0} \preceq G_{\mathrm{core}} \preceq c_1^3 I_{r_0}.$$

The product weights and the telescoping expansion of the full Kronecker product give

$$\|\Phi\|_{4,\mu} + \|\hat{\Phi}\|_{4,\mu} = O_p(1), \qquad \|\hat{\Phi} - \Phi\|_{4,\mu} = O_p(\rho).$$

Equations (190) and (191) therefore apply to the core projectors P_Φ^μ and $P_{\hat{\Phi}}^\mu$.

The weighted projector onto the grouped physical tangent space has the orthogonal decomposition

$$P_{\mathcal{T}}^\mu = \sum_{m=1}^{3} K_{1,m}^G \big[(I_{d_m} - P_m) \otimes P_{-m}\big](K_{1,m}^G)' + P_\Phi^\mu. \tag{192}$$

Indeed, split each mode variation into its weighted projection onto $\mathrm{col}(B_m)$ and its orthogonal complement. The complementary variations generate $\mathrm{col}(B_m)^{\perp_{\mu_m}} \otimes \mathrm{col}(B_{-m})$. These three spaces are mutually orthogonal and orthogonal to the core space. The remaining variations generate $\mathrm{col}(\Phi)$, since their coefficient coordinates are exactly those selected by R_ℓ.

Applying a projector to each complementary slice and summing the fourth powers shows that tensor extensions preserve the weighted fourth-norm bounds. Expanding each tensor-product projector difference in (192) consequently gives

$$\|P^\mu_{\mathcal{T}}\|_{4,\mu\to 4,\mu} + \|P^\mu_{\widehat{\mathcal{T}}}\|_{4,\mu\to 4,\mu} = O_p(1), \qquad \|P^\mu_{\widehat{\mathcal{T}}} - P^\mu_{\mathcal{T}}\|_{4,\mu\to 4,\mu} = O_p(\rho). \tag{193}$$

We also need the Euclidean projector bound. In normalized grouped coordinates, the same tangent decomposition uses A_m, L_m, and the core matrix $\Psi = (A_1 \otimes A_2 \otimes A_3)R_\ell$. All these matrices have bounded norms, Gram matrices bounded away from singularity, and Frobenius perturbations $O_p(\rho)$. The ordinary column-projector expansion therefore gives an $O_p(\rho)$ operator-norm perturbation for each component and hence for their sum. Writing $Q = Q_D \otimes I_M \otimes I_T$, the original tangent projectors are obtained by the isometric embedding Q. Consequently,

$$\|P_{\widehat{\mathcal{T}}} - P_{\mathcal{T}}\|_{\mathrm{op}} = O_p(\rho). \tag{194}$$

This argument uses normalized coordinates directly.

Let $\bar{X}^G \in \mathbb{R}^{GMT\times p}$ have rows $\bar{x}'_{gjt}$, where $\bar{x}_{gjt} = N_g^{-1}\sum_{i:g(i)=g} x_{ijt}$, and put

$$\mathcal{X} = P^\mu_{\mathcal{T}}\bar{X}^G, \qquad \widehat{\mathcal{X}} = P^\mu_{\widehat{\mathcal{T}}}\bar{X}^G.$$

Jensen's inequality gives

$$\|\bar{X}^G\|^4_{4,\mu} \leqslant \frac{1}{n}\sum_{i,j,t}\|x_{ijt}\|_2^4 = O_p(1).$$

Since the projections act columnwise and p is fixed, (193) implies

$$\|\mathcal{X}\|_{4,\mu} + \|\widehat{\mathcal{X}}\|_{4,\mu} = O_p(1), \qquad \|\widehat{\mathcal{X}} - \mathcal{X}\|_{4,\mu} = O_p(\rho). \tag{195}$$

The relation between physical and normalized grouped coordinates is

$$Q'P_{\mathcal{T}}X'_{(4)} = \sqrt{n}\,D_\mu^{1/2}\mathcal{X}, \qquad Q'P_{\widehat{\mathcal{T}}}X'_{(4)} = \sqrt{n}\,D_\mu^{1/2}\widehat{\mathcal{X}}. \tag{196}$$

Step 4: Exact bias coefficients and coefficient rates. Write $b'_{m,r}$ for row r of B_m and $c'_{m,q}$ for row q of B_{-m}. Let $\xi'_{r,q}$ be the row of $\mathcal{X}$ at source index r and complementary index q. Define

$$Z_{m,q} = \sum_r \mu_m(r) b_{m,r}\xi'_{r,q} \in \mathbb{R}^{\ell\times p}.$$

Let $Z_m \in \mathbb{R}^{d_a d_b\times \ell p}$ have row q equal to $\mathrm{vec}(Z_{m,q})'$, put $T_m = (I_{d_a d_b} - P_{-m})Z_m$, and let $T_{m,q}$ denote the $\ell \times p$ reshaping of row q. Define

$$k_{m,q} = T'_{m,q}G_m^{-1}J_m^{-1}c_{m,q}, \tag{197}$$

and let K_m have rows $k'_{m,q}$. Define estimated quantities by replacing the factors, projected regressors, Grams, and projectors with their estimated counterparts.

Hölder's inequality in the source weights gives

$$\|Z_m\|_{4,\mu_{-m}} \leqslant \|B_m\|_{4/3,\mu_m}\|\mathcal{X}\|_{4,\mu} = O_p(1).$$

The blockwise identity

$$\hat{Z}_{m,q} - Z_{m,q} = \sum_r \mu_m(r)(\hat{b}_{m,r} - b_{m,r})\hat{\xi}'_{r,q} + \sum_r \mu_m(r) b_{m,r}(\hat{\xi}_{r,q} - \xi_{r,q})'$$

and (188) and (195) give $\|\hat{Z}_m - Z_m\|_{4,\mu_{-m}} = O_p(\rho)$. The pair-projector bounds and

$$\hat{T}_m - T_m = (I - \hat{P}_{-m})(\hat{Z}_m - Z_m) - (\hat{P}_{-m} - P_{-m})Z_m$$

then imply

$$\|T_m\|_{4,\mu_{-m}} + \|\hat{T}_m\|_{4,\mu_{-m}} = O_p(1), \qquad \|\hat{T}_m - T_m\|_{4,\mu_{-m}} = O_p(\rho).$$

Hölder's inequality for the product of two weighted fourth-norm matrices gives

$$\|K_m\|_{2,\mu_{-m}} \leqslant \|G_m^{-1}J_m^{-1}\|_{\mathrm{op}}\|T_m\|_{4,\mu_{-m}}\|B_{-m}\|_{4,\mu_{-m}} = O_p(1).$$

Expanding $\hat{k}_{m,q} - k_{m,q}$ one factor at a time, keeping the inverse-Gram order $G_m^{-1}J_m^{-1}$, and using (184) gives

$$\|K_m\|_{2,\mu_{-m}} + \|\hat{K}_m\|_{2,\mu_{-m}} = O_p(1), \qquad \|\hat{K}_m - K_m\|_{2,\mu_{-m}} = O_p(\rho). \tag{198}$$

To identify the bias coefficients exactly, note the unnormalized scalings

$$(F_1, F_2, F_3) = (\bar{U}, V, W) = (\sqrt{N}A_1, \sqrt{M}A_2, \sqrt{T}A_3),$$

and

$$(C_1, C_2, C_3) = (V * W, \bar{U} * W, \bar{U} * V) = (\sqrt{MT}L_1, \sqrt{NT}L_2, \sqrt{NM}L_3).$$

Thus the matrix inside the mode-m bias formula is

$$\frac{1}{\sqrt{n}} A_m G_m^{-1} J_m^{-1} L_m' \Omega_m^G M_{L_m}.$$

Fix a regressor coordinate r, and let $e_r^{(p)}$ be the rth canonical basis vector of $\mathbb{R}^p$. Let $Y_m^{(r)}$ unfold the grouped tensor with entries $\sqrt{\mu(g,j,t)}\xi_{gjt}^{(r)}$. Let $Z_m^{(r)}$ have row q equal to $(Z_{m,q}e_r^{(p)})'$, and define $T_m^{(r)}$ analogously. Then

$$(Y_m^{(r)})'A_m = D_{\mu_{-m}}^{1/2} Z_m^{(r)}, \qquad M_{L_m} D_{\mu_{-m}}^{1/2} = D_{\mu_{-m}}^{1/2}(I - P_{-m}).$$

Equation (196) and cyclicity of the trace give

$$\begin{aligned}
\mathcal{B}_m^{(r)} &= \frac{1}{\sqrt{n}} \operatorname{tr}\left((Y_m^{(r)})' A_m G_m^{-1} J_m^{-1} L_m' \Omega_m^G M_{L_m}\right) \\
&= \frac{1}{\sqrt{n}} \operatorname{tr}\left(D_{\mu_{-m}}^{1/2} T_m^{(r)} G_m^{-1} J_m^{-1} B_{-m}' D_{\mu_{-m}}^{1/2} \Omega_m^G\right) \\
&= \frac{1}{\sqrt{n}} \sum_q \mu_{-m}(q)(\Omega_m^G)_{qq} k_{m,q}^{(r)}.
\end{aligned}$$

The last equality uses conditional uncorrelatedness, which makes Ω_m^G diagonal. Stacking coordinates, and repeating the calculation for estimated quantities, gives

$$\mathcal{B}_m = \frac{1}{\sqrt{n}} \sum_q \mu_{-m}(q)(\Omega_m^G)_{qq} k_{m,q}, \qquad \widehat{\mathcal{B}}_m = \frac{1}{\sqrt{n}} \sum_q \mu_{-m}(q)(\widehat{\Omega}_m^G)_{qq} \hat{k}_{m,q}.$$

Put $\sigma_a^2 = \mathbb{E}[\varepsilon_a^2 \mid \mathcal{F}]$ and $\sigma_{gjt}^{G,2} = N_g^{-1} \sum_{i:g(i)=g} \sigma_{ijt}^2$. Under conditional uncorrelatedness,

$$(\Omega_1^G)_{(j,t),(j,t)} = \sum_g \sigma_{gjt}^{G,2}, \qquad (\Omega_2^G)_{(g,t),(g,t)} = \sum_j \sigma_{gjt}^{G,2}, \qquad (\Omega_3^G)_{(g,j),(g,j)} = \sum_t \sigma_{gjt}^{G,2}.$$

Substituting these formulas and the residual-based estimators in (45) yields the exact representations

$$\mathcal{B}_m = \sum_a \lambda_{m,a} \sigma_a^2, \qquad \widehat{\mathcal{B}}_m = \sum_a \hat{\lambda}_{m,a} \hat{\varepsilon}_a^2,$$

where

$$\lambda_{1,ijt} = \frac{N}{N_{g(i)} n^{3/2}} k_{1,(j,t)}, \qquad \lambda_{2,ijt} = \frac{M}{n^{3/2}} k_{2,(g(i),t)}, \qquad \lambda_{3,ijt} = \frac{T}{n^{3/2}} k_{3,(g(i),j)}. \tag{199}$$

The estimated coefficients replace $k_{m,q}$ by $\hat{k}_{m,q}$.

Let $\mathcal{L}_m \in \mathbb{R}^{n \times p}$ have rows $\lambda_{m,a}'$, and define $\widehat{\mathcal{L}}_m$ similarly. Direct summation gives

$$\|\mathcal{L}_1\|_F^2 = \frac{N H_N}{n^2} \|K_1\|_{2,\mu_2 \otimes \mu_3}^2, \qquad \|\mathcal{L}_2\|_F^2 = \frac{M^2}{n^2} \|K_2\|_{2,\mu_1 \otimes \mu_3}^2, \qquad \|\mathcal{L}_3\|_F^2 = \frac{T^2}{n^2} \|K_3\|_{2,\mu_1 \otimes \mu_2}^2.$$

The same identities apply to estimated coefficients and their differences. Equations (46) and (198) consequently imply

$$\|\mathcal{L}_m\|_F + \|\widehat{\mathcal{L}}_m\|_F = O_p(d_m/n), \qquad \|\widehat{\mathcal{L}}_m - \mathcal{L}_m\|_F = O_p(d_m \rho/n). \tag{200}$$

Step 5: The bias-estimation rate. Let q have entries $q_a = \varepsilon_a^2 - \sigma_a^2$. The exact coefficient representations give

$$\widehat{\mathcal{B}}_m - \mathcal{B}_m = \mathcal{L}_m' q + (\widehat{\mathcal{L}}_m - \mathcal{L}_m)'(e \odot e) + \widehat{\mathcal{L}}_m'(\hat{e} \odot \hat{e} - e \odot e).$$

The true coefficients are $\mathcal{F}$-measurable. Since all diagonal pairs (a, a) belong to $\mathcal{R}^D$, Assumption 3 implies

$$\operatorname{Var}\left(\sum_a \lambda_{m,a}^{(r)} q_a \mid \mathcal{F} \right) \leqslant c_4 \sum_a |\lambda_{m,a}^{(r)}|^2.$$

Conditional Chebyshev's inequality and fixed p therefore give

$$\|\mathcal{L}_m' q\|_2 = O_p(d_m/n).$$

For the other terms, deterministic Cauchy–Schwarz inequalities, (187), (189), and (200) give

$$\|(\widehat{\mathcal{L}}_m - \mathcal{L}_m)'(e \odot e)\|_2 \leqslant \|\widehat{\mathcal{L}}_m - \mathcal{L}_m\|_F \|e \odot e\|_2 = O_p(d_m \rho/\sqrt{n})$$

and

$$\|\widehat{\mathcal{L}}_m'(\hat{e} \odot \hat{e} - e \odot e)\|_2 \leqslant \|\widehat{\mathcal{L}}_m\|_F \|\hat{e} \odot \hat{e} - e \odot e\|_2 = O_p\left(\frac{d_m \sqrt{s}}{n} + \frac{d_m s}{n\sqrt{n}} \right).$$

No independence involving the estimated coefficients is used. Summing over modes and using $\sum_m d_m = s$ gives

$$\widehat{B}_{NMT} - B_{NMT} = O_p\left(\frac{s}{n} + \frac{s^{3/2}}{n} + \frac{s^2}{n^{3/2}}\right).$$

Because $s/n = \rho^2$, $s^2/n^{3/2} = \rho s^{3/2}/n$, and $\rho \to 0$, this proves

$$\widehat{B}_{NMT} - B_{NMT} = O_p\left(\rho^2 + \frac{s^{3/2}}{n}\right) = O_p\left(\rho + \frac{s^{3/2}}{n}\right). \tag{201}$$

Step 6: The bias-corrected expansion. Equation (194) and $\|X_{(4)}\|_{\mathrm{op}}^2/n = O_p(1)$ imply

$$\|\widehat{\Gamma}_{NMT} - \Gamma_{NMT}\|_{\mathrm{op}} \leqslant \frac{\|X_{(4)}\|_{\mathrm{op}}^2}{n}\|P_{\widehat{\mathcal{T}}} - P_{\mathcal{T}}\|_{\mathrm{op}} = O_p(\rho).$$

The lower eigenvalue bound established in Step 1 therefore gives

$$\|\widehat{\Gamma}_{NMT}^{-1}\|_{\mathrm{op}} = O_p(1), \qquad \|\Gamma_{NMT}\widehat{\Gamma}_{NMT}^{-1} - I_p\|_{\mathrm{op}} = O_p(\rho).$$

All inverses are well defined with probability approaching one; the estimator may be extended arbitrarily outside that event.

Let $R_{NMT} = O_p(\rho + s^{3/2}/n)$ denote the remainder in Proposition 1. Substituting the definition of $\hat{\delta}_{\mathrm{bc}}$ into that expansion gives

$$\begin{aligned}\Gamma_{NMT}\sqrt{n}(\hat{\delta}_{\mathrm{bc}} - \delta) = \mathcal{Z}_{NMT} + R_{NMT} \\ + (\Gamma_{NMT}\widehat{\Gamma}_{NMT}^{-1} - I_p)B_{NMT} \\ + \Gamma_{NMT}\widehat{\Gamma}_{NMT}^{-1}(\widehat{B}_{NMT} - B_{NMT}).\end{aligned}$$

Since $B_{NMT} = O_p(s/\sqrt{n})$,

$$(\Gamma_{NMT}\widehat{\Gamma}_{NMT}^{-1} - I_p)B_{NMT} = O_p(\rho s/\sqrt{n}) = O_p(s^{3/2}/n).$$

The last term is $O_p(\rho + s^{3/2}/n)$ by (201). Consequently,

$$\Gamma_{NMT}\sqrt{NMT}(\hat{\delta}_{\mathrm{bc}} - \delta) = \mathcal{Z}_{NMT} + O_p\left(\sqrt{\frac{G+M+T}{NMT}} + \frac{(G+M+T)^{3/2}}{NMT}\right),$$

which proves (49). □

B Proofs for Section 6 (Selecting the Number of Interactive Components)

Proposition 4 (Consistency of the information criterion)**.** *Suppose that Assumptions 1, 2, 3, and 7 hold, with the true-factor restrictions evaluated at rank ℓ_0. Let $\ell_{\max}$ be fixed, with $1 \leqslant \ell_0 \leqslant \ell_{\max}$, and define $\widehat{\ell}$ by* (53)*. If the positive deterministic penalty satisfies*

$$g(G,N,M,T) \to 0, \qquad \frac{NMT}{G+M+T}g(G,N,M,T) \to \infty, \tag{55}$$

then, along the asymptotic sequence in Assumption 1,

$$\mathbb{P}\{\widehat{\ell} = \ell_0\} \to 1.$$

Proof. Let $g_n = g(G, N, M, T)$. Retain the notation

$$n = NMT, \qquad s = G + M + T, \qquad \rho = \sqrt{\frac{s}{n}}, \qquad \mathsf{X} = \frac{X_{(4)}}{\sqrt{n}},$$

and define

$$e = \mathrm{vec}(E), \qquad V_E = \frac{\|E\|_F^2}{n}, \qquad Q_D = D(D'D)^{-1/2}, \qquad E^G = E \times_1 Q_D'.$$

Vector norms are Euclidean norms, matrix norms without subscripts are operator norms, and $\|E^G\|$ denotes the tensor spectral norm. The asymptotic framework gives

$$\rho^2 = \frac{G}{NMT} + \frac{1}{NT} + \frac{1}{NM} \to 0.$$

Assumption 2 and Lemma 4(i) imply

$$\|\mathsf{X}\|_F = \frac{\|X\|_F}{\sqrt{n}} = O_p(1), \qquad \|\mathsf{X}e\| = \frac{\|X_{(4)}e\|}{\sqrt{n}} = O_p(1), \qquad \|E^G\| = O_p(\sqrt{s}). \tag{202}$$

Only the conditional mean-zero and second-order covariance restrictions in Assumption 3 are used in this proof.

1. A uniform criterion bound. For an arbitrary candidate rank $0 \leqslant \ell \leqslant \ell_{\max}$ and an admissible pair $(\widetilde{\delta}, \Pi)$, define

$$\gamma = \widetilde{\delta} - \delta_0, \qquad \Delta_\Pi = \Pi - \Pi_0, \qquad R = X \times_4 \gamma' + \Delta_\Pi, \qquad r = \frac{\|R\|_F}{\sqrt{n}}.$$

Here R is the displacement of the fitted conditional mean from the true conditional mean. Consequently,

$$Y - X \times_4 \widetilde{\delta}' - \Pi = E - R.$$

For $\ell \geqslant 1$, choose any admissible representation of Π with candidate factors $(\widetilde{U}, \widetilde{V}, \widetilde{W})$, and put

$$\mathcal{M}_{\widetilde{U},\widetilde{V}} = M_{D\widetilde{U} * \widetilde{V}} \otimes M_{W_0}.$$

For $\ell = 0$, use $\mathcal{M}_{\widetilde{U},\widetilde{V}} = I_{NM} \otimes M_{W_0}$. This is an orthogonal projector. It annihilates $\mathrm{vec}(\Pi)$ because the candidate mode-3 matricization satisfies

$$\Pi_{(3)} = \widetilde{W}(D\widetilde{U} * \widetilde{V})',$$

and it annihilates $\mathrm{vec}(\Pi_0)$ because

$$\Pi_{0,(3)} = W_0(DU_0 * V_0)'.$$

Therefore,

$$\mathcal{M}_{\widetilde{U},\widetilde{V}} \,\mathrm{vec}(R) = \mathcal{M}_{\widetilde{U},\widetilde{V}} X_{(4)}' \gamma.$$

On the identification event in Assumption 7,

$$\begin{aligned} r^2 &\geqslant \frac{1}{n} \left\| \mathcal{M}_{\widetilde{U},\widetilde{V}} \,\mathrm{vec}(R) \right\|^2 \\ &= \gamma' \left[\mathsf{X} \mathcal{M}_{\widetilde{U},\widetilde{V}} \mathsf{X}' \right] \gamma \\ &\geqslant c_{\mathrm{rank}} \|\gamma\|^2. \end{aligned}$$

This inequality is uniform over all candidate ranks and admissible representations. Hence

$$\|\gamma\| \leqslant c_{\text{rank}}^{-1/2} r.$$

Also,

$$\begin{aligned}\frac{\|\Delta_\Pi\|_F}{\sqrt{n}} &\leqslant \frac{\|R\|_F}{\sqrt{n}} + \frac{\|X \times_4 \gamma'\|_F}{\sqrt{n}} \\ &= r + \|\mathsf{X}'\gamma\| \\ &\leqslant r + \|\mathsf{X}\|_F\|\gamma\|.\end{aligned}$$

Consequently,

$$\|\gamma\| \leqslant c_{\text{rank}}^{-1/2} r, \qquad \frac{\|\Delta_\Pi\|_F}{\sqrt{n}} \leqslant a_n r, \qquad a_n := 1 + c_{\text{rank}}^{-1/2}\|\mathsf{X}\|_F = O_p(1). \tag{203}$$

The same random bound a_n applies to every candidate.

We next establish a deterministic error-contraction bound. Let

$$L = \ell_{\max} + \ell_0.$$

The difference Δ_Π admits a grouped CP representation with at most $\ell + \ell_0 \leqslant L$ terms. We claim that

$$|\langle E, \Delta_\Pi\rangle_F| \leqslant L\|E^G\|\|\Delta_\Pi\|_F. \tag{204}$$

To verify this, define the grouped tensor difference

$$\Delta_\Pi^G = \Delta_\Pi \times_1 Q_D'.$$

Because the first-mode fibers of Δ_Π belong to $\text{col}(D) = \text{col}(Q_D)$ and $Q_D'Q_D = I_G$,

$$\langle E, \Delta_\Pi\rangle_F = \langle E^G, \Delta_\Pi^G\rangle_F, \qquad \|\Delta_\Pi^G\|_F = \|\Delta_\Pi\|_F.$$

For $\Delta_\Pi \neq 0$, choose a representation of Δ_Π^G with at most L terms. Let

$$O_1 \in \mathbb{R}^{G\times r_1}, \qquad O_2 \in \mathbb{R}^{M\times r_2}$$

be orthonormal bases of the spans of its first-mode and second-mode factor vectors. Then

$$O_1'O_1 = I_{r_1}, \qquad O_2'O_2 = I_{r_2}, \qquad r_1, r_2 \leqslant L,$$

and

$$(\Delta_\Pi^G)_{(3)} = (\Delta_\Pi^G)_{(3)}(O_1 \otimes O_2)(O_1 \otimes O_2)'.$$

If $o_{1,a}$ and $o_{2,b}$ are columns of O_1 and O_2, respectively, then

$$\begin{aligned}\left\|E_{(3)}^G(o_{1,a} \otimes o_{2,b})\right\| &= \sup_{\substack{w\in\mathbb{R}^T \\ \|w\|=1}} \left|\left\langle E^G, o_{1,a} \circ o_{2,b} \circ w\right\rangle_F\right| \\ &\leqslant \|E^G\|.\end{aligned}$$

Since there are r_1r_2 such columns,

$$\|E_{(3)}^G(O_1 \otimes O_2)\|_F \leqslant \sqrt{r_1r_2}\,\|E^G\|.$$

Using the displayed support identity and Cauchy–Schwarz,

$$\begin{aligned}|\langle E, \Delta_\Pi\rangle_F| &= \left|\left\langle E^G_{(3)}(O_1 \otimes O_2), (\Delta^G_\Pi)_{(3)}(O_1 \otimes O_2)\right\rangle_F\right| \\ &\leqslant \|E^G_{(3)}(O_1 \otimes O_2)\|_F \|(\Delta^G_\Pi)_{(3)}(O_1 \otimes O_2)\|_F \\ &\leqslant \sqrt{r_1 r_2}\, \|E^G\| \|\Delta^G_\Pi\|_F \\ &\leqslant L\|E^G\|\|\Delta_\Pi\|_F.\end{aligned}$$

The bound is immediate when $\Delta_\Pi = 0$. Thus (204) holds for every realization and every admissible candidate. In particular, it applies to data-dependent candidate factors without conditioning on them, and it does not require candidate factor matrices to satisfy strong-factor lower bounds.

Since $Y = X \times_4 \delta_0' + \Pi_0 + E$, equations (203) and (204) give

$$\begin{aligned}\frac{1}{n}\|Y - X \times_4 \widetilde{\delta}' - \Pi\|_F^2 - V_E &= r^2 - \frac{2}{n}\langle E, R\rangle_F \\ &\geqslant r^2 - \frac{2}{\sqrt{n}}\|\gamma\|\|\mathsf{X}e\| - \frac{2L}{n}\|E^G\|\|\Delta_\Pi\|_F \\ &\geqslant r^2 - b_n r \\ &= \left(r - \frac{b_n}{2}\right)^2 - \frac{b_n^2}{4} \\ &\geqslant -\frac{1}{4}b_n^2, \end{aligned} \tag{205}$$

where

$$b_n = \frac{2c_{\mathrm{rank}}^{-1/2}}{\sqrt{n}}\|\mathsf{X}e\| + \frac{2La_n}{\sqrt{n}}\|E^G\|.$$

By (202), fixed L, and $n^{-1/2} \leqslant \rho$,

$$b_n = O_p(\rho), \qquad b_n \geqslant 0, \qquad b_n = o_p(1).$$

The criterion inequality holds simultaneously over all candidate ranks and admissible parameter values on the identification event.

2. Overfitted ranks. For every $\ell \geqslant \ell_0$, the true common component Π_0 belongs to $\mathcal{C}^D_{\ell,c_U}$ by zero padding. Evaluating the criterion at (δ_0, Π_0) therefore gives $\mathcal{V}(\ell) \leqslant V_E$. Together with (205),

$$V_E - \frac{1}{4}b_n^2 \leqslant \mathcal{V}(\ell) \leqslant V_E, \qquad \ell \geqslant \ell_0,$$

simultaneously on an event whose probability tends to one.

The candidate classes are nested, so $\mathcal{V}(\ell) \leqslant \mathcal{V}(\ell_0)$ when $\ell > \ell_0$. Consequently,

$$0 \leqslant \mathcal{V}(\ell_0) - \mathcal{V}(\ell) \leqslant \frac{1}{4}b_n^2 = O_p(s/n), \qquad \ell > \ell_0. \tag{206}$$

It follows that

$$PC(\ell) - PC(\ell_0) \geqslant (\ell - \ell_0)g_n - \frac{1}{4}b_n^2.$$

The penalty condition gives

$$\frac{b_n^2}{g_n} = O_p\left(\frac{s/n}{g_n}\right) = o_p(1).$$

Therefore,

$$PC(\ell) > PC(\ell_0)$$

with probability approaching one, simultaneously over all overfitted candidate ranks.

3. Underfitted ranks. On the strong-factor event,

$$\lambda_{\min}\left(\frac{1}{N}U_0'D'DU_0\right) \geqslant c_0, \qquad \lambda_{\min}\left(\frac{1}{M}V_0'V_0\right) \geqslant c_0, \qquad \lambda_{\min}\left(\frac{1}{T}W_0'W_0\right) \geqslant c_0.$$

For every $z \in \mathbb{R}^{\ell_0}$,

$$\begin{aligned}\frac{1}{NM}\|(DU_0 * V_0)z\|^2 &= \left\|\frac{DU_0}{\sqrt{N}}\operatorname{diag}(z)\left(\frac{V_0}{\sqrt{M}}\right)'\right\|_F^2 \\ &\geqslant \lambda_{\min}\left(\frac{1}{N}U_0'D'DU_0\right)\lambda_{\min}\left(\frac{1}{M}V_0'V_0\right)\|\operatorname{diag}(z)\|_F^2 \\ &\geqslant c_0^2\|z\|^2.\end{aligned}$$

Hence

$$\sigma_{\min}(DU_0 * V_0) \geqslant c_0\sqrt{NM}, \qquad \sigma_{\min}(W_0) \geqslant \sqrt{c_0 T}.$$

Since

$$\Pi_{0,(3)} = W_0(DU_0 * V_0)',$$

the smallest nonzero singular value satisfies

$$\sigma_{\ell_0}(\Pi_{0,(3)}) \geqslant \sigma_{\min}(W_0)\sigma_{\min}(DU_0 * V_0) \geqslant c_0^{3/2}\sqrt{n}.$$

Every candidate $\Pi \in \mathcal{C}^D_{\ell,c_U}$ has $\operatorname{rank}(\Pi_{(3)}) \leqslant \ell$. The matrix rank-approximation inequality therefore gives, for $\ell < \ell_0$,

$$\begin{aligned}\frac{1}{n}\|\Pi - \Pi_0\|_F^2 &= \frac{1}{n}\|\Pi_{(3)} - \Pi_{0,(3)}\|_F^2 \\ &\geqslant \frac{1}{n}\sum_{k=\ell+1}^{\ell_0}\sigma_k(\Pi_{0,(3)})^2 \\ &\geqslant (\ell_0 - \ell)c_0^3.\end{aligned} \tag{207}$$

Combining (203) and (207), every admissible candidate at rank $\ell < \ell_0$ satisfies

$$r \geqslant \underline{r}_{\ell,n}, \qquad \underline{r}_{\ell,n} := \frac{c_0^{3/2}\sqrt{\ell_0 - \ell}}{a_n}.$$

Because $a_n = O_p(1)$ and $b_n = o_p(1)$,

$$a_n b_n = o_p(1).$$

In particular, with probability approaching one,

$$b_n \leqslant \frac{1}{2}\underline{r}_{\ell,n}$$

simultaneously for every $\ell < \ell_0$. On this event, $r \geqslant \underline{r}_{\ell,n}$ implies $b_n \leqslant r/2$, and (205) gives

$$\begin{aligned}\frac{1}{n}\|Y - X \times_4 \widetilde{\delta}' - \Pi\|_F^2 - V_E &\geqslant r^2 - b_n r \\ &\geqslant \frac{1}{2}r^2 \\ &\geqslant \frac{(\ell_0 - \ell)c_0^3}{2a_n^2}.\end{aligned}$$

Taking the minimum over all admissible candidates at rank ℓ, and using $\mathcal{V}(\ell_0) \leqslant V_E$, yields

$$\mathcal{V}(\ell) - \mathcal{V}(\ell_0) \geqslant \frac{(\ell_0 - \ell)c_0^3}{2a_n^2}, \qquad \ell < \ell_0, \tag{208}$$

simultaneously with probability approaching one.

Therefore,

$$\begin{aligned}PC(\ell) - PC(\ell_0) &= \mathcal{V}(\ell) - \mathcal{V}(\ell_0) - (\ell_0 - \ell)g_n \\ &\geqslant (\ell_0 - \ell)\left(\frac{c_0^3}{2a_n^2} - g_n\right).\end{aligned}$$

The lower bound may be random, but its positivity follows from

$$a_n^2 g_n = o_p(1),$$

because $a_n = O_p(1)$ and $g_n \to 0$. Thus

$$PC(\ell) > PC(\ell_0)$$

with probability approaching one, simultaneously over all underfitted candidate ranks.

Combining the underfitting and overfitting comparisons over the finite candidate set gives

$$\mathbb{P}\{PC(\ell) > PC(\ell_0) \text{ for every } \ell \neq \ell_0,\ 0 \leqslant \ell \leqslant \ell_{\max}\} \to 1.$$

On this event, ℓ_0 is the unique minimizer of the penalized criterion. Hence

$$\mathbb{P}\{\widehat{\ell} = \ell_0\} \to 1.$$

□

C Proofs for Section 7 (Additive Fixed Effects and Unbalanced Panels)

Lemma 2 (Consistency with group, teacher, and period additive effects)**.** *Suppose Assumptions 1, 2, 3, and 8 hold. Both the true and candidate additive effects satisfy the*

observation-weighted normalizations in (58)*. Define the global least-squares estimator by* (59)*. Then the slope and the interactive component satisfy*

$$\|\hat{\delta} - \delta\| + \frac{\|\hat{\Pi} - \Pi\|_F}{\sqrt{n}} = O_p\left(\sqrt{\frac{G+M+T}{NMT}}\right). \tag{68}$$

The uncentered joint loading spaces satisfy

$$\begin{aligned}
\frac{1}{\sqrt{MT}}\|M_{\hat{V}*\hat{W}}(V*W)\|_F &= O_p\left(\sqrt{\frac{G+M+T}{NMT}}\right),\\
\frac{1}{\sqrt{NT}}\|M_{D\hat{U}*\hat{W}}(DU*W)\|_F &= O_p\left(\sqrt{\frac{G+M+T}{NMT}}\right),\\
\frac{1}{\sqrt{NM}}\|M_{D\hat{U}*\hat{V}}(DU*V)\|_F &= O_p\left(\sqrt{\frac{G+M+T}{NMT}}\right).
\end{aligned} \tag{69}$$

Moreover, the additive-effect estimators satisfy

$$\begin{aligned}
|\hat{\kappa} - \kappa| &= O_p\left(\frac{1}{\sqrt{NMT}} + \sqrt{\frac{G+M+T}{NMT}}\right),\\
\frac{1}{\sqrt{N}}\|D(\hat{\mu} - \mu)\| = \left[\frac{1}{N}\sum_{g=1}^{G} N_g(\hat{\mu}_g - \mu_g)^2\right]^{1/2} &= O_p\left(\sqrt{\frac{G}{NMT}} + \sqrt{\frac{G+M+T}{NMT}}\right),\\
\frac{1}{\sqrt{M}}\|\hat{\nu} - \nu\| &= O_p\left(\frac{1}{\sqrt{NT}} + \sqrt{\frac{G+M+T}{NMT}}\right),\\
\frac{1}{\sqrt{T}}\|\hat{\xi} - \xi\| &= O_p\left(\frac{1}{\sqrt{NM}} + \sqrt{\frac{G+M+T}{NMT}}\right).
\end{aligned} \tag{70}$$

Proof. Write $n = NMT$, $s = G + M + T$, and $\rho = \sqrt{s/n}$. Assumption 1 gives $\rho \to 0$. Let $\mathcal{A}$ and $\hat{\mathcal{A}}$ denote the true and estimated additive-effect tensors. All stochastic bounds involving conditional covariances are understood on the events in Assumption 3, whose probabilities tend to one, with localization for random coefficient norms. Only the conditional mean-zero and second-order covariance restrictions in that assumption are needed below.

The conditional regressor-moment bound implies $\|X\|_F = O_p(\sqrt{n})$. Let $e = \mathrm{vec}(E)$. Assumption 3 gives $\|\mathbb{E}[ee' \mid \mathcal{F}]\| \leqslant C_\varepsilon$ with probability approaching one. Therefore,

$$\mathbb{E}[\|X_{(4)}M_{\mathcal{A}_D}e\|^2 \mid \mathcal{F}] \leqslant C_\varepsilon\|X_{(4)}M_{\mathcal{A}_D}\|_F^2 \leqslant C_\varepsilon\|X\|_F^2.$$

Conditional Markov's inequality and Assumption 2 yield

$$\frac{\|X\|_F}{\sqrt{n}} = O_p(1), \qquad \|X_{(4)}M_{\mathcal{A}_D}e\| = O_p(\sqrt{n}), \qquad \|E^G\| = O_p(\sqrt{s}). \tag{209}$$

1. The slope and additive-within common component. For any factor triple,

$$\begin{aligned}
\Pi(U,V,W)^{\mathcal{A}_D} = \sum_{k=1}^{\ell}\Big[&Du_k^c \circ v_k^c \circ w_k^c + \bar{u}_k\,\iota_N \circ v_k^c \circ w_k^c\\
&+ \bar{v}_k\,Du_k^c \circ \iota_M \circ w_k^c + \bar{w}_k\,Du_k^c \circ v_k^c \circ \iota_T\Big].
\end{aligned} \tag{210}$$

Each first-mode vector belongs to $\mathrm{col}(D)$. Hence every transformed common component has a grouped CP representation with at most 4ℓ terms, and the difference of two such components has at most 8ℓ terms.

For any grouped tensor Z with a representation of at most r terms,

$$|\langle E, Z\rangle| \leqslant r\|E^G\|\|Z\|_F. \tag{211}$$

Indeed, let $Z^G = Z \times_1 Q_D'$, where $Q_D = D(D'D)^{-1/2}$. Then $\langle E, Z\rangle = \langle E^G, Z^G\rangle$ and $\|Z^G\|_F = \|Z\|_F$. Choose orthonormal bases B_1, B_2 for the spans of the first- and second-mode vectors in a representation of Z^G with at most r terms. Their column counts r_1, r_2 are at most r, and

$$Z^G_{(3)} = Z^G_{(3)}(B_1 \otimes B_2)(B_1 \otimes B_2)'.$$

Each column of $E^G_{(3)}(B_1 \otimes B_2)$ has Euclidean norm at most $\|E^G\|$. Frobenius Cauchy–Schwarz therefore proves (211), with $\sqrt{r_1 r_2} \leqslant r$. The bound is pathwise and applies to data-dependent tensors.

Set

$$\gamma = \hat{\delta} - \delta_0, \qquad \Delta = \hat{\Pi} - \Pi_0, \qquad \Delta^{\mathcal{A}_D} = \hat{\Pi}^{\mathcal{A}_D} - \Pi_0^{\mathcal{A}_D},$$

$$R_{\mathcal{A}_D} = M_{\mathcal{A}_D}\{X'_{(4)}\gamma + \mathrm{vec}(\Delta)\}, \qquad r_n = \frac{\|R_{\mathcal{A}_D}\|}{\sqrt{n}}.$$

Global optimality of the profiled estimator

$$(\hat{\delta}, \hat{U}, \hat{V}, \hat{W}) \in \operatorname*{argmin}_{\substack{\delta\in\mathbb{R}^p \\ (U,V,W)\in\Theta_\ell}} \frac{1}{n}\left\|M_{\mathcal{A}_D}\{y - X'_{(4)}\delta - \pi(U,V,W)\}\right\|^2.$$

implies $\|M_{\mathcal{A}_D}e - R_{\mathcal{A}_D}\|^2 \leqslant \|M_{\mathcal{A}_D}e\|^2$ so that

$$\begin{aligned}\|R_{\mathcal{A}_D}\|^2 &\leqslant 2\langle M_{\mathcal{A}_D}e, R_{\mathcal{A}_D}\rangle \\ &\leqslant 2\|\gamma\|\|X_{(4)}M_{\mathcal{A}_D}e\| + 2|\langle E, \Delta^{\mathcal{A}_D}\rangle| \\ &\leqslant 2\|\gamma\|\|X_{(4)}M_{\mathcal{A}_D}e\| + 16\ell\|E^G\|\|\Delta^{\mathcal{A}_D}\|_F. \end{aligned} \tag{212}$$

The last line follows from (210)–(211).

Let

$$\mathcal{L}_{\hat{U},\hat{V}} = M_{[\iota_N\otimes\iota_M,\, D\hat{U}*\hat{V}]} \otimes M_{[\iota_T, W_0]}.$$

This orthogonal projector annihilates $\mathcal{A}_D$, $\mathrm{vec}(\Pi_0)$, and $\mathrm{vec}(\hat{\Pi})$. Thus $\mathcal{L}_{\hat{U},\hat{V}}M_{\mathcal{A}_D} = \mathcal{L}_{\hat{U},\hat{V}}$, and the uniform identification condition gives

$$r_n^2 \geqslant \frac{1}{n}\|\mathcal{L}_{\hat{U},\hat{V}}R_A\|^2 = \gamma'\left[\frac{1}{n}X_{(4)}\mathcal{L}_{\hat{U},\hat{V}}X'_{(4)}\right]\gamma \geqslant c_{\delta,A}\|\gamma\|^2$$

with probability approaching one. Consequently,

$$\|\gamma\| \leqslant c_{\delta,A}^{-1/2}r_n, \qquad \frac{\|\Delta^{\mathcal{A}_D}\|_F}{\sqrt{n}} \leqslant a_n r_n, \qquad a_n = 1 + c_{\delta,A}^{-1/2}\frac{\|X\|_F}{\sqrt{n}} = O_p(1).$$

Dividing (212) by n gives $r_n^2 \leqslant b_n r_n$, where

$$b_n = \frac{2c_{\delta,A}^{-1/2}}{n}\|X_{(4)}M_{\mathcal{A}_D}e\| + 16\ell a_n\frac{\|E^G\|}{\sqrt{n}} = O_p(\rho).$$

Since $r_n, b_n \geqslant 0$, this implies $r_n \leqslant b_n$ on the identification event. Therefore,

$$\|\hat{\delta} - \delta_0\| + \frac{\|\hat{\Pi}^{\mathcal{A}_D} - \Pi_0^{\mathcal{A}_D}\|_F}{\sqrt{n}} = O_p(\rho). \tag{213}$$

2. Recovering the untransformed interactive component. We recover the full component from its fully centered tensor and its three pairwise contractions.

Let $(d_1, d_2, d_3) = (G, M, T)$ and define the unit constant directions in grouped coordinates by

$$q_1 = \frac{(D'D)^{1/2}\iota_G}{\sqrt{N}}, \qquad q_2 = \frac{\iota_M}{\sqrt{M}}, \qquad q_3 = \frac{\iota_T}{\sqrt{T}}, \qquad M_{q_m} = I_{d_m} - q_m q_m'.$$

Introduce normalized true factors and their centered parts,

$$A_1 = \frac{(D'D)^{1/2}U_0}{\sqrt{N}}, \qquad A_2 = \frac{V_0}{\sqrt{M}}, \qquad A_3 = \frac{W_0}{\sqrt{T}}, \qquad B_m = M_{q_m}A_m, \qquad \alpha_m = A_m'q_m.$$

Then $A_m = B_m + q_m\alpha_m'$. Use hats for the estimated counterparts. In particular,

$$B_1 = \frac{(D'D)^{1/2}U_0^c}{\sqrt{N}}, \qquad B_2 = \frac{V_0^c}{\sqrt{M}}, \qquad B_3 = \frac{W_0^c}{\sqrt{T}},$$

so the centered-strength condition gives $c_c I_\ell \preceq B_m'B_m \preceq c_1 I_\ell$ with probability approaching one.

Since $A_m'A_m = B_m'B_m + \alpha_m\alpha_m'$, Assumptions 2 and 8(i) imply

$$c_c I_\ell \preceq B_m'B_m \preceq A_m'A_m \preceq c_1 I_\ell, \qquad m = 1, 2, 3, \tag{214}$$

with probability approaching one. Below, $\lesssim$ suppresses constants depending only on ℓ, c_c, c_1, c_U.

Define the normalized grouped tensors and their difference by

$$\mathsf{C} = \frac{1}{\sqrt{n}}\Pi_0 \times_1 Q_D', \qquad \widehat{\mathsf{C}} = \frac{1}{\sqrt{n}}\hat{\Pi} \times_1 Q_D', \qquad z = \mathrm{vec}(\widehat{\mathsf{C}} - \mathsf{C}).$$

Thus $\mathsf{C} = \sum_{k=1}^{\ell} a_{1,k} \circ a_{2,k} \circ a_{3,k}$, with the analogous representation for $\widehat{\mathsf{C}}$. Let

$$Q = Q_D \otimes I_M \otimes I_T, \qquad M_{\mathcal{A}_D}^G = Q' M_{\mathcal{A}_D} Q.$$

Since $Q'Q = I_{GMT}$ and the additive space is contained in $\mathrm{col}(Q)$, we have $M_{\mathcal{A}_D}Q = QM_{\mathcal{A}_D}^G$, and $M_{\mathcal{A}_D}^G$ is the additive residual-maker in grouped coordinates. Consequently,

$$\|z\| = \frac{\|\Delta\|_F}{\sqrt{n}}, \qquad \eta_n := \|M_{\mathcal{A}_D}^G z\| = \frac{\|\Delta^{\mathcal{A}_D}\|_F}{\sqrt{n}} = O_p(\rho) = o_p(1). \tag{215}$$

We first recover the raw loading spaces. Define

$$\mathsf{S}_1 = \mathsf{C} \times_2 M_{q_2} \times_3 M_{q_3}, \qquad \widehat{\mathsf{S}}_1 = \widehat{\mathsf{C}} \times_2 M_{q_2} \times_3 M_{q_3}.$$

Every grouped additive term has at least two modes proportional to the corresponding q_m. Hence centering in modes two and three annihilates the grouped additive space, giving

$$(I_G \otimes M_{q_2} \otimes M_{q_3})M_{\mathcal{A}_D}^G = I_G \otimes M_{q_2} \otimes M_{q_3}.$$

It follows that

$$\|\widehat{\mathsf{S}}_1 - \mathsf{S}_1\|_F \leqslant \eta_n, \qquad (\mathsf{S}_1)_{(1)} = A_1(B_2 * B_3)'.$$

The Schur-product identity and (214) imply

$$(B_2 * B_3)'(B_2 * B_3) = (B_2'B_2) \odot (B_3'B_3) \geq c_c^2 I_\ell,$$

and therefore $\sigma_\ell((\mathsf{S}_1)_{(1)}) \geqslant c_c^{3/2}$. Singular-value perturbation gives

$$\sigma_\ell((\widehat{\mathsf{S}}_1)_{(1)}) \geqslant c_c^{3/2} - \eta_n > 0$$

with probability approaching one. Since $(\widehat{\mathsf{S}}_1)_{(1)} = \hat{A}_1(\hat{B}_2 * \hat{B}_3)'$ has rank at most ℓ, it has rank exactly ℓ, as does $\hat{A}_1$. Moreover,

$$c_c\|M_{\hat{A}_1}A_1\|_F \leqslant \|M_{\hat{A}_1}(\mathsf{S}_1)_{(1)}\|_F = \|M_{\hat{A}_1}\{(\mathsf{S}_1)_{(1)} - (\widehat{\mathsf{S}}_1)_{(1)}\}\|_F \leqslant \eta_n.$$

Repeating this argument cyclically yields

$$\operatorname{rank}(\hat{A}_m) = \ell, \qquad \|M_{\hat{A}_m}A_m\|_F \leqslant c_c^{-1}\eta_n, \qquad m = 1, 2, 3,$$

with probability approaching one.

Let $O_m = A_m(A_m'A_m)^{-1/2}$, so that $O_m'O_m = I_\ell$ and $P_{A_m} = O_mO_m'$. Equality of the true and estimated loading-space dimensions gives

$$\|P_{\hat{A}_m} - P_{A_m}\|_F^2 = 2\|M_{\hat{A}_m}O_m\|_F^2.$$

Thus, by (214),

$$\|P_{\hat{A}_m} - P_{A_m}\| \leqslant \sqrt{2}\|M_{\hat{A}_m}O_m\|_F \leqslant \sqrt{2}\, c_c^{-3/2}\eta_n, \qquad m = 1, 2, 3. \tag{216}$$

Next, define

$$\mathcal{P} = P_{A_1} \otimes P_{A_2} \otimes P_{A_3}, \qquad \widehat{\mathcal{P}} = P_{\hat{A}_1} \otimes P_{\hat{A}_2} \otimes P_{\hat{A}_3}, \qquad \mathcal{J} = M_{q_1} \otimes M_{q_2} \otimes M_{q_3}.$$

A telescoping expansion of the Kronecker products gives

$$\|\widehat{\mathcal{P}} - \mathcal{P}\| \leqslant \sum_{m=1}^{3} \|P_{\hat{A}_m} - P_{A_m}\| \lesssim \eta_n.$$

The parameter-space restrictions imply

$$\|\widehat{\mathsf{C}}\|_F \leqslant \sum_{k=1}^{\ell} \|\hat{a}_{1,k}\|\|\hat{a}_{2,k}\|\|\hat{a}_{3,k}\| \leqslant \ell\sqrt{C_U}.$$

Since $\mathcal{P}\operatorname{vec}(\mathsf{C}) = \operatorname{vec}(\mathsf{C})$ and $\widehat{\mathcal{P}}\operatorname{vec}(\widehat{\mathsf{C}}) = \operatorname{vec}(\widehat{\mathsf{C}})$,

$$\begin{aligned}\|(I_{GMT} - \mathcal{P})z\| &= \|(I_{GMT} - \mathcal{P})(\widehat{\mathcal{P}} - \mathcal{P})\operatorname{vec}(\widehat{\mathsf{C}})\| \\ &\leqslant \|\widehat{\mathcal{P}} - \mathcal{P}\|\|\widehat{\mathsf{C}}\|_F \lesssim \eta_n.\end{aligned} \tag{217}$$

To control the remaining component $\|\mathcal{P}z\|$, observe that

$$O_m'M_{q_m}O_m = (A_m'A_m)^{-1/2}B_m'B_m(A_m'A_m)^{-1/2} \geq \frac{c_c}{c_1}I_\ell.$$

Let $O = O_1 \otimes O_2 \otimes O_3$. Then $O'O = I_{\ell^3}$, $\mathcal{P} = OO'$, and

$$(\mathcal{J}O)'(\mathcal{J}O) = \bigotimes_{m=1}^{3} (O_m' M_{q_m} O_m) \succeq \left(\frac{c_c}{c_1}\right)^3 I_{\ell^3}.$$

Consequently,

$$\|\mathcal{J}v\| \geqslant \left(\frac{c_c}{c_1}\right)^{3/2} \|v\|, \qquad v \in \text{range}(\mathcal{P}).$$

Full centering annihilates the grouped additive space, so $\mathcal{J}M_{\mathcal{A}_D}^G = \mathcal{J}$ and $\|\mathcal{J}z\| \leqslant \|M_{\mathcal{A}_D}^G z\| = \eta_n$. Therefore, using (217),

$$\|\mathcal{P}z\| \leqslant \left(\frac{c_1}{c_c}\right)^{3/2} \|\mathcal{J}\mathcal{P}z\| \leqslant \left(\frac{c_1}{c_c}\right)^{3/2} \{\eta_n + \|(I_{GMT} - \mathcal{P})z\|\} \lesssim \eta_n.$$

where we've used

$$\|\mathcal{J}\mathcal{P}z\| = \|\mathcal{J}(z - (I_{GMT} - \mathcal{P})z)\| = \|\mathcal{J}z - \mathcal{J}(I_{GMT} - \mathcal{P})z\| \leqslant \|\mathcal{J}z\| + \|\mathcal{J}(I_{GMT} - \mathcal{P})z\|.$$

Together with (217), this proves

$$\frac{\|\hat{\Pi} - \Pi_0\|_F}{\sqrt{n}} = \|z\| \leqslant \|\mathcal{P}z\| + \|(I_{GMT} - \mathcal{P})z\| \lesssim \eta_n = O_p(\rho). \tag{218}$$

Combining (213) and (218) establishes (68).

3. The uncentered joint loading spaces. The first-mode matricizations satisfy

$$(\Pi_{0,(1)} - \hat{\Pi}_{(1)}) M_{\hat{V}*\hat{W}} = DU_0(V_0 * W_0)' M_{\hat{V}*\hat{W}}.$$

Since $U_0' D' D U_0 \succeq N c_c I_\ell$, it follows that

$$\sqrt{N c_c} \|M_{\hat{V}*\hat{W}}(V_0 * W_0)\|_F \leqslant \|(\Pi_{0,(1)} - \hat{\Pi}_{(1)}) M_{\hat{V}*\hat{W}}\|_F \leqslant \|\Delta\|_F.$$

Hence

$$\frac{\|M_{\hat{V}*\hat{W}}(V_0 * W_0)\|_F}{\sqrt{MT}} \leqslant c_c^{-1/2} \frac{\|\Delta\|_F}{\sqrt{n}} = O_p(\rho).$$

The other two bounds in (69) follow by cyclic permutation.

4. The additive effects. Let $\Delta_a = \text{vec}(\hat{\mathcal{A}} - \mathcal{A})$. The additive profiling identity gives

$$\Delta_a = P_{\mathcal{A}_D}\{e - X_{(4)}'\gamma - \text{vec}(\Delta)\}.$$

Denote the four mutually orthogonal summands of $P_{\mathcal{A}_D}$ by $P_\kappa, P_\mu, P_\nu, P_\xi$, with respective ranks

$$r_\kappa = 1, \qquad r_\mu = G - 1, \qquad r_\nu = M - 1, \qquad r_\xi = T - 1.$$

These projectors are $\mathcal{F}$-measurable. The conditional covariance bound implies

$$\mathbb{E}[\|P_b e\|^2 \mid \mathcal{F}] \leqslant C_\varepsilon r_b, \qquad b \in \{\kappa, \mu, \nu, \xi\}.$$

By conditional Markov's inequality, (209), (213), and (218),

$$\begin{aligned}\frac{\|P_b\Delta_a\|}{\sqrt{n}} &\leqslant \frac{\|P_b e\|}{\sqrt{n}} + \frac{\|X\|_F}{\sqrt{n}}\|\gamma\| + \frac{\|\Delta\|_F}{\sqrt{n}} \\ &= O_p\left(\sqrt{\frac{r_b}{n}} + \rho\right), \qquad b \in \{\kappa, \mu, \nu, \xi\}.\end{aligned}$$

The observation-weighted additive normalizations give

$$\begin{aligned}\frac{\|P_\kappa\Delta_a\|}{\sqrt{n}} &= |\hat{\kappa} - \kappa|, & \frac{\|P_\mu\Delta_a\|}{\sqrt{n}} &= \frac{\|D(\hat{\mu} - \mu)\|}{\sqrt{N}}, \\ \frac{\|P_\nu\Delta_a\|}{\sqrt{n}} &= \frac{\|\hat{\nu} - \nu\|}{\sqrt{M}}, & \frac{\|P_\xi\Delta_a\|}{\sqrt{n}} &= \frac{\|\hat{\xi} - \xi\|}{\sqrt{T}}.\end{aligned}$$

Substitution of the four ranks proves (70). □

Lemma 3 (Convergence with an unbalanced sample)**.** *Maintain the model, membership conditions, normalized parameter space, and true-parameter feasibility specified above. Let p, ℓ be fixed, and suppose that*

$$\rho_\Omega = \sqrt{\frac{G + M + T}{n_\Omega}}, \quad \rho_\Omega \to_p 0. \tag{82}$$

Let $(\hat{\delta}, \hat{U}, \hat{V}, \hat{W})$ be a global minimizer in (74)*, and suppose that Assumptions 9 and 10 hold.*

(i) Observed-sample convergence. *The slope and observed interactive component satisfy*

$$\|\hat{\delta} - \delta\| + \frac{\|R_\Omega(\hat{\pi} - \pi)\|}{\sqrt{n_\Omega}} = O_p(\rho_\Omega) = o_p(1). \tag{83}$$

(ii) Full-component convergence. *If Assumption 11 also holds, then*

$$\frac{\|\hat{\Pi} - \Pi\|_F}{\sqrt{NMT}} = O_p(\rho_\Omega) = o_p(1). \tag{84}$$

(iii) Full joint loading-space convergence. *Suppose Assumption 11 holds and, for a constant $c_f > 0$ independent of the panel dimensions, with probability approaching one,*

$$\frac{1}{N}U'D'DU \succeq c_f I_\ell, \qquad \frac{1}{M}V'V \succeq c_f I_\ell, \qquad \frac{1}{T}W'W \succeq c_f I_\ell. \tag{85}$$

Then

$$\begin{aligned}\frac{1}{\sqrt{MT}}\|M_{\hat{V}*\hat{W}}(V * W)\|_F &= O_p(\rho_\Omega), \\ \frac{1}{\sqrt{NT}}\|M_{D\hat{U}*\hat{W}}(DU * W)\|_F &= O_p(\rho_\Omega), \\ \frac{1}{\sqrt{NM}}\|M_{D\hat{U}*\hat{V}}(DU * V)\|_F &= O_p(\rho_\Omega).\end{aligned} \tag{86}$$

The observed-sample conclusion requires neither Assumption 11 nor the strong-factor bounds in (85)*. All conclusions are aggregate-norm statements, not uniform fixed-index consistency results.*

Proof. In the proof, write $(\delta_0, U_0, V_0, W_0, \Pi_0)$ for the true parameters and common component denoted by (δ, U, V, W, Π) in the statement. Write $m = n_\Omega$, $n = NMT$, $s = G+M+T$, and $\rho_\Omega = \sqrt{s/m}$. All stochastic orders involving the random sample size are relative to its realized positive value.

1. The observed regressor score. On the conditional covariance event in Assumption 10,

$$\mathbb{E}\left[\left\|\frac{X_\Omega' e_\Omega}{\sqrt{m}}\right\|^2 \middle| \mathcal{F}_\Omega\right] = \frac{1}{m}\operatorname{tr}\left(X_\Omega' \mathbb{E}[e_\Omega e_\Omega' \mid \mathcal{F}_\Omega] X_\Omega\right) \leqslant \frac{c_\varepsilon}{m}\|X_\Omega\|_F^2.$$

Conditional Markov's inequality, the regressor bound in Assumption 9, and the probability of the covariance event imply

$$\left\|\frac{X_\Omega' e_\Omega}{\sqrt{m}}\right\| = O_p(1), \qquad \frac{\|X_\Omega\|_F}{\sqrt{m}} = O_p(1). \tag{219}$$

For the first assertion, one may first restrict to $\{\|X_\Omega\|_F^2/m \leqslant A\}$ for a fixed A, apply the conditional probability bound there, and then make the complementary probability arbitrarily small by choosing A sufficiently large.

2. Identification and the basic inequality. Set

$$\gamma = \hat{\delta}-\delta_0, \qquad \Delta = \hat{\Pi}-\Pi_0, \qquad r_\Omega = X_\Omega\gamma + R_\Omega \operatorname{vec}(\Delta), \qquad t_\Omega = \frac{\|r_\Omega\|}{\sqrt{m}}, \qquad \widehat{B}_\Omega = B_\Omega(\hat{U}, \hat{V}).$$

The candidate component lies in the first block of $\widehat{B}_\Omega$, because

$$\hat{\pi}_\Omega = R_\Omega\{(D\hat{U} * \hat{V}) \otimes I_T\} \operatorname{vec}(\hat{W}').$$

The true component lies in its second block. Specifically, define $b_0 \in \mathbb{R}^{GM\ell}$ by stacking the vectors $u_{0g} \odot v_{0j}$ in increasing group–teacher order. Then

$$\pi_{0,\Omega} = R_\Omega(D \otimes I_M \otimes W_0) b_0.$$

Hence $M_{\widehat{B}_\Omega} R_\Omega \operatorname{vec}(\Delta) = 0$. On the uniform identification event,

$$t_\Omega^2 \geqslant \frac{1}{m}\|M_{\widehat{B}_\Omega} r_\Omega\|^2 = \gamma'\left(\frac{1}{m} X_\Omega' M_{\widehat{B}_\Omega} X_\Omega\right)\gamma \geqslant c_{\delta,\Omega}\|\gamma\|^2. \tag{220}$$

Although $\widehat{B}_\Omega$ depends on the errors, the inequality applies because the identification bound is uniform over all candidate matrices. It follows that

$$\|\gamma\| \leqslant c_{\delta,\Omega}^{-1/2} t_\Omega, \qquad \frac{\|R_\Omega \operatorname{vec}(\Delta)\|}{\sqrt{m}} \leqslant a_\Omega t_\Omega, \qquad a_\Omega = 1 + c_{\delta,\Omega}^{-1/2}\frac{\|X_\Omega\|_F}{\sqrt{m}} = O_p(1). \tag{221}$$

The true parameters are feasible, so global least-squares optimality gives $\|e_\Omega - r_\Omega\|^2 \leqslant \|e_\Omega\|^2$. Moreover, $\Delta \in \mathcal{L}_{2\ell}^D$, and therefore

$$|e_\Omega' R_\Omega \operatorname{vec}(\Delta)| \leqslant \|E\|_{\Omega,2\ell}\|R_\Omega \operatorname{vec}(\Delta)\|.$$

This inequality is also valid when $R_\Omega \operatorname{vec}(\Delta) = 0$, in which case both sides are zero. Consequently,

$$\begin{aligned} m t_\Omega^2 &\leqslant 2 e_\Omega' r_\Omega \\ &\leqslant 2\|\gamma\|\|X_\Omega' e_\Omega\| + 2\|E\|_{\Omega,2\ell}\|R_\Omega \operatorname{vec}(\Delta)\|. \end{aligned} \tag{222}$$

Using (221), divide by m to obtain $t_\Omega^2 \leqslant b_\Omega t_\Omega$, where

$$b_\Omega = \frac{2c_{\delta,\Omega}^{-1/2}}{m}\|X_\Omega' e_\Omega\| + \frac{2a_\Omega}{\sqrt{m}}\|E\|_{\Omega,2\ell} = O_p\left(m^{-1/2} + \sqrt{\frac{s}{m}}\right) = O_p(\rho_\Omega).$$

Both t_Ω and b_Ω are nonnegative, so $t_\Omega \leqslant b_\Omega$ on the identification event. Thus $t_\Omega = O_p(\rho_\Omega)$, and (221) proves (83). The convergence to zero follows from (82).

3. Recovery of the full component. On the sampling-stability event, uniformity of (81) over the feasible parameter space permits evaluation at $(\hat{U}, \hat{V}, \hat{W})$ and gives

$$\frac{\|\Delta\|_F}{\sqrt{n}} \leqslant c_\Pi^{-1/2}\frac{\|R_\Omega \operatorname{vec}(\Delta)\|}{\sqrt{m}} = O_p(\rho_\Omega).$$

This proves (84). No true-factor lower eigenvalue bound is used in this step.

4. The three joint loading spaces. For $h = 1, 2, 3$, write $\Pi_{0,(h)} = A_{h0}C_{h0}'$, where

$$(A_{h0}, C_{h0}, d_h) = \begin{cases} (DU_0, V_0 * W_0, N), & h = 1, \\ (V_0, DU_0 * W_0, M), & h = 2, \\ (W_0, DU_0 * V_0, T), & h = 3. \end{cases}$$

Let $\hat{A}_h, \hat{C}_h$ denote the corresponding estimated matrices. Since $\hat{\Pi}_{(h)} = \hat{A}_h\hat{C}_h'$, we have

$$\Delta_{(h)} M_{\hat{C}_h} = -A_{h0}C_{h0}' M_{\hat{C}_h}.$$

The Frobenius norm is unchanged by matricization and contracts under orthogonal projection. Hence

$$\|\Delta\|_F \geqslant \|A_{h0}C_{h0}' M_{\hat{C}_h}\|_F \geqslant \{\lambda_{\min}(A_{h0}'A_{h0})\}^{1/2}\|M_{\hat{C}_h}C_{h0}\|_F.$$

On the true-factor strength event, it follows that

$$\sqrt{\frac{d_h}{n}}\|M_{\hat{C}_h}C_{h0}\|_F \leqslant \left\{\lambda_{\min}\left(\frac{A_{h0}'A_{h0}}{d_h}\right)\right\}^{-1/2}\frac{\|\Delta\|_F}{\sqrt{n}} \leqslant c_f^{-1/2}\frac{\|\Delta\|_F}{\sqrt{n}} = O_p(\rho_\Omega).$$

The factors $\sqrt{d_h/n}$ equal $1/\sqrt{MT}$, $1/\sqrt{NT}$, and $1/\sqrt{NM}$, respectively. This proves (86). The argument uses Moore–Penrose projectors and does not assume that the estimated loading matrices or observed slice Gram matrices are nonsingular. □